AK High-Dimensional Projection Structural Theory

Appendix SDR

Separation, Discovery, and Rigidity
Target-Driven Structural Discovery, Enriched Invariant Synthesis,
Rigidity Reconstruction, and Finite-to-Infinite Completion

Post-v21 Structural-Discovery Research Supplement
SDR 1.0 — Construction Draft
Sections SDR.1–SDR.14

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SDR.1 Scope, Authority, and Constitutional Boundary

SDR.1.1 Role, release relation, and local authority

Declaration SDR.1.1 (SDR document role)**.** Appendix SDR, *Separation, Discovery, and Rigidity*, is a post-v21 structural-discovery research supplement for AK High-Dimensional Projection Structural Theory (AK-HPDST). Its purpose is to develop a target-driven mathematical research layer in which target-relevant indistinguishability is exposed, missing structure is isolated, candidate observables or enriched probes are synthesized, rigidity or reflection is proved at an explicitly declared strength, and finite-to-infinite completion is invoked only through a separately certified theorem.

Appendix SDR is not part of the frozen AK-HPDST v21.0.0 canonical source set. It therefore creates no retroactive v21 theorem identity, no v21 Claim UID, no migration of a frozen v21 statement, and no modification of the final v21 Claim Register. Its baseline relation is

$$\begin{aligned}&\mathrm{parent_baseline} = \mathrm{AK\text{-}HPDST\text{-}v21.0.0},\\ &\mathrm{release_relation} = \mathrm{post\text{-}v21\text{-}sdr\text{-}supplement}.\end{aligned}$$

Remark SDR.1.2 (Why SDR is external to the frozen v21 release)**.** The v21 source set is already frozen with distinct theorem identities, owners, dependency edges, source bindings, and Claim-Register states. SDR therefore begins in a fresh local research-supplement namespace. Any later integration of an SDR result into a successor release requires a new migration, ownership, dependency, source-binding, consumer, and Claim-Register audit. Construction inside SDR is not such an integration.

Definition SDR.1.3 (SDR primary layer)**.** The primary layer of Appendix SDR is

$$\mathrm{TargetDrivenStructuralDiscoveryResearchSupplement}.$$

Its permitted local roles include:

(i) source-class and target binding;

(ii) observable, invariant, probe, information-order, and loss-profile definition;

(iii) separation, indistinguishability, and no-go theorem;

(iv) adversarial witness and missing-structure diagnosis;

(v) candidate-probe synthesis and strict-information-gain theorem;

(vi) rigidity, reflection, reconstruction, and target-factorization theorem;

(vii) dynamic-defect and finite-to-infinite completion theorem or research program;

(viii) source-specific flagship theorem and cross-domain substitution;

(ix) counterexample, sharpness theorem, and failure witness;

(x) Search/Platform consumer contract and formalization contract;

(xi) promotion or non-promotion assessment; and

(xii) explicit non-claim.

No local SDR role is a stable-Core role merely by appearing in this appendix.

Definition SDR.1.4 (SDR statement record and axis separation)**.** Every theorem-facing SDR item is represented by a local record

$$\mathfrak{C}_{\mathrm{SDR}} = (\mathrm{id}, \mathrm{role}, \mathrm{maturity}, \mathrm{owner}, C_{\mathrm{src}}, C_{\mathrm{tgt}}, \mathrm{Hyp}, \mathrm{Data}, \mathrm{Out}, \Sigma, \mathcal{L}, \mathrm{Fail}, \mathcal{A}),$$

where $\mathcal{L}$ is the declared information/loss profile and $\mathcal{A}$ is the immutable proof, computation, source, and replay bundle when applicable. Mathematical truth, theorem ownership, maturity, invocation role, verifier status, workflow status, implementation status, and promotion status are distinct coordinates. No favorable value on one coordinate promotes another.

Proposition SDR.1.5 (Conservative post-release attachment)**.** *Suppose every SDR statement uses a fresh local identity, consumes each frozen v21 premise at no stronger than its registered source-supported strength, and exports no SDR conclusion backward into the frozen v21 dependency DAG. Then adjoining Appendix SDR as a post-v21 research supplement is conservative with respect to AK-HPDST v21.0.0.*

Proof. The frozen theorem identities, hypotheses, target strengths, source artifacts, dependency edges, ownership assignments, and Claim-Register states are unchanged. Every SDR statement is downstream of those records and carries a fresh local identity. Since no SDR conclusion is inserted as a premise of a frozen theorem, neither the truth conditions nor the admissible-use conditions of the frozen release are altered. □

SDR.1.2 Target sovereignty and mathematical mission

Definition SDR.1.6 (Target-driven SDR mission)**.** Fix a declared source class $\mathcal{S}$ and a target record

$$\Theta : \mathcal{S} \longrightarrow \mathcal{T},$$

where $\mathcal{T}$ may encode a predicate, invariant, object class, morphism class, exact or derived structure, or another explicitly typed Return target. SDR studies research routes of the schematic form

$$\begin{aligned} \text{target discrepancy} &\longrightarrow \text{indistinguishable source pair} \longrightarrow \text{missing structure} \\ &\longrightarrow \text{candidate probe} \longrightarrow \text{rigidity/reflection} \\ &\longrightarrow \text{completion} \longrightarrow \text{typed Return}. \end{aligned}$$

Each arrow is a separate theorem-facing obligation. No arrow is discharged merely by naming the next stage.

Policy 1 (SDR-TGT-01: target sovereignty)**.** *Every SDR discovery claim is relative to an exactly declared target Θ, source class $\mathcal{S}$, equivalence or comparison relation, and requested Return strength. An observable may be complete for one target and insufficient for another. SDR therefore forbids unqualified language such as "complete invariant", "lossless", "reconstructs", or "solves" unless the relevant source class and target relation are explicitly stated and proved.*

Policy 2 (SDR-DISC-01: no discovery by renaming)**.** *Relabeling an existing scalar, persistence, Ext, diagram, tail, descent, or source-specific datum as a "discovery invariant" does not increase information and does not create a new theorem. A datum is admitted as a genuinely stronger SDR probe only after its source type, target type, dependence on prior data, mathematical meaning, and target-relevant separation gain are established at the declared scope.*

Policy 3 (SDR-DISC-02: discovery is not proof)**.** *The production of a candidate formula, invariant, probe, conjectural relation, machine-generated lemma, symbolic identity, or experimentally separating statistic is a discovery event, not a theorem event. The candidate becomes theorem-bearing only after a proof establishes the exact statement consumed downstream and after its failure boundary and artifact obligations are recorded.*

SDR.1.3 Frozen-v21 ownership and Core firewall

Definition SDR.1.7 (Frozen owner partition consumed by SDR)**.** SDR consumes the frozen v21 architecture through the following ownership partition at the registered strengths:

(i) Core-P owns the repaired-persistence, detector, higher-Ext eligibility, Type-IV, certificate-complexity, atlas, Restart, and registered finite-to-infinite statements assigned to it by v21;

(ii) Core-DG owns finite-category/finite-poset diagram reconstruction, actual morphisms, cones, finite projective-generator and Morita reconstruction, and effective finite descent assigned to it by v21;

(iii) Appendix HT retains the HC, PE, SS, RL, and IW/Return research-spine details and source-specific refinements at their exact registered strengths;

(iv) Appendix UB retains its reusable components, derived theorems, validation results, finite-state tail results, Pareto/frontier results, and source-bounded arithmetic results;

(v) Technical Extension retains non-field, dg, A_∞, sheaf/stack, multiparameter, higher-descent, and other enhancement material not promoted into the stable Core;

(vi) Problem Interfaces retain source-specific semantics and external Return obligations;

(vii) Search/Platform retains retrieval, orchestration, storage, search, and replay functions without theorem promotion; and

(viii) the final v21 Claim Register retains final v21 theorem identity and governance authority.

An SDR restatement of a frozen theorem is a consumer statement or proof locator and does not create a competing owner.

Policy 4 (SDR-OWN-01: no ornamental Core invocation)**.** *Persistence, Ext, Morita, descent, dg, A_∞, or other AK terminology may be inserted into an SDR route only when a theorem proves that the inserted object preserves, reflects, reconstructs, or otherwise controls the target-relevant information actually consumed. Terminology without a typed theorem edge is mathematically ornamental and cannot strengthen the route.*

Declaration SDR.1.8 (No third stable Core is created)**.** Appendix SDR does not introduce a stable Core-SDR, does not identify Core-P with Core-DG, and does not assert a universal discovery functor, universal invariant, or universal decision procedure. A later successor-Core proposal requires an independently proved source-general theorem, nonredundancy relative to frozen v21, explicit failure boundaries, cross-domain evidence, and successor-release governance.

SDR.1.4 Projection loss and admissible information gain

Policy 5 (SDR-INF-01: no bridge-only information gain)**.** *Additional downstream syntax, routing, serialization, search coverage, replay, or Bridge composition is not treated as a source of new mathematical information. On a fixed source class, if two source objects are identified by the complete observable packet already consumed by a route, then no deterministic post-processing of that same packet can distinguish them. A genuine shrinkage of the observable indistinguishability relation therefore requires new source data or a genuinely enriched probe. By contrast, a rigidity, restriction, or reconstruction theorem may strengthen what can be* deduced *from an unchanged packet, but it does not retroactively create an observed distinction that the packet itself did not contain.*

Definition SDR.1.9 (Admissible route-strengthening modes)**.** Every claimed SDR strengthening is assigned to one of two noninterchangeable axes.

(i) *Observational refinement*: new_source_data or enriched_probe, where the new packet can strictly refine the old observable indistinguishability relation on the same source class.

(ii) *Inferential strengthening*: source_restriction_plus_rigidity or reconstruction_or_factorization_theorem, where a theorem proves that the existing packet is sufficient for a declared target on a stated source class or under stated hypotheses without changing the packet's observed values.

The two axes may coexist in one route, but they are never identified. In particular, post-processing that factors only through the old packet is neither a new observational refinement nor a substitute for a missing rigidity theorem.

Audit Obligation SDR.1.10 (Route-strengthening audit)**.** *Every later SDR claim of a "stronger invariant", "newly detected structure", "recovered information", "rigidity gain", or "strictly stronger route" must identify: the old packet; any genuinely new packet; the fixed or restricted source class; the requested target distinction; whether the strengthening is observational, inferential, or both; the theorem or source datum supporting it; and a witness or factorization theorem showing that the claimed gain is not merely a notational re-encoding of the old packet.*

SDR.1.5 Rigidity, converse, and completion firewall

Policy 6 (SDR-RIG-01: rigidity is an independent edge)**.** *A separating example does not establish a rigidity theorem. A rigidity or reflection claim requires a quantified theorem over the declared source class showing that the consumed observable packet determines, reflects, or controls the requested target relation. Benchmark separation alone is insufficient.*

Policy 7 (SDR-RIG-02: no hidden converse)**.** *A proved implication, preservation theorem, or sufficient criterion is never inverted merely because the reverse statement would close the desired Return. Every reverse implication, zero-reflection statement, reconstruction theorem, or target-faithfulness claim requires separate evidence.*

Policy 8 (SDR-COMP-01: no finite-to-infinite shortcut)**.** *Finite observation, a finite prefix, increasing defect rank, numerical convergence, stabilization on tested cases, or a stored apex does not create an infinite theorem. Passage to an infinite object, limit, tail, completion, or global Return requires a separately proved finite-state, Noetherian, Mittag–Leffler, complete-filter, compactness, convergence, descent, or other explicitly typed reduction theorem sufficient for the requested conclusion.*

SDR.1.6 Validation quarantine and failure semantics

Policy 9 (SDR-VAL-01: benchmark and validation non-backflow)**.** *A benchmark truth value, successful validation case, known target answer, or conclusion-derived witness may test an SDR construction but may not be imported as an undeclared premise into a route advertised as discovering or proving that same target. Validation remains downstream evidence unless the route is explicitly classified as reconstruction of known mathematics.*

Policy 10 (SDR-VAL-02: target-oracle firewall)**.** *If a normalized target value $\widehat{\Theta}(X)$, a target comparison $\widehat{\Theta}(X) \neq \widehat{\Theta}(Y)$, or an equivalent conclusion-derived label is not already available from the admissible premise set of a discovery run, SDR may not query, infer from the benchmark solution, or otherwise use that value as an upstream selector for witnesses, probes, hypotheses, or theorem candidates while advertising the route as independent discovery. Such information may be used only in an explicitly benchmark-assisted reconstruction or validation lane, or after a separately proved surrogate-to-target theorem licenses the exact use. Oracle provenance is therefore a theorem-facing field rather than informal metadata.*

Definition SDR.1.11 (SDR theorem-facing failure classes)**.** At the constitutional level SDR distinguishes at least:

(i) $\mathrm{undefined}$: required source, type, map, theorem, or artifact data are missing;

(ii) $\mathrm{obstructed}$: a well-typed nonzero obstruction or certified indistinguishability witness blocks the requested route;

(iii) reject: a supplied theorem-facing assertion or artifact is proved false or inconsistent at the invoked type;

(iv) $\mathrm{partial_return}$: a weaker exact conclusion remains valid although the requested strength is unavailable;

(v) return: every theorem-bearing obligation for the requested target strength is closed.

Failure of one candidate route is not, by itself, falsity of the external target.

Proposition SDR.1.12 (Typed non-compensation)**.** *A passing SDR component cannot fill a missing theorem-bearing field of a different type merely by being strong, redundant, numerically favorable, or independently verified.*

Proof. The required obligations inhabit declared source/target types. Evidence in a different field has no well-typed substitution into the missing obligation unless a conversion theorem is supplied. Hence unrelated success cannot compensate for a missing or false theorem edge. □

SDR.1.7 Promotion ladder and novelty discipline

Definition SDR.1.13 (SDR promotion ladder)**.** The permitted assessment ladder is

$$\text{SDR-local} \longrightarrow \text{Technical-Extension-candidate} \longrightarrow \text{Successor-Core-candidate}.$$

These arrows are governance transitions, not mathematical implications and not automatic promotions.

Policy 11 (SDR-PROM-01: minimum promotion burden)**.** *A theorem family is not nominated beyond SDR-local status merely because it is useful, elegant, machine checked, successful on the flagship arithmetic program, or expressed in source-independent notation. A Technical Extension candidate requires at least an AK-owned theorem at a declared abstract scope, explicit failure or sharpness boundaries, and an independently typed substitution supporting the claimed scope. A Successor-Core candidate additionally requires demonstrated nonredundancy relative to frozen v21, cross-domain necessity or a genuinely uniform ambient theorem, literature-overlap analysis, consumer and migration audits, and successor-release governance.*

Policy 12 (SDR-NOV-01: abstraction by erasure is forbidden)**.** *Deleting arithmetic, analytic, geometric, or other source nouns from a source-specific theorem does not create a generic SDR theorem. Removed source data must be replaced by abstract hypotheses sufficient for an independent proof and meaningful in at least the declared ambient class.*

SDR.1.8 Local identities, frozen-input discipline, and governance

Definition SDR.1.14 (SDR 1.0 local UID namespace)**.** During construction of Appendix SDR 1.0, local theorem-facing records use identities of the form

$$\text{ak-hpdst:rs-sdr:1.0:SDR-}\langle\text{token}\rangle.$$

The suffix 1.0 identifies the appendix construction line, not an independently released sub-version of an individual section. At the SDR 1.0 freeze, surviving local identities become immutable for that release. A post-freeze mathematical change altering theorem-facing semantics requires a successor SDR release and an explicit predecessor relation.

Policy 13 (SDR-INPUT-01: frozen theorem-facing input identity)**.** *Every replayable external or post-v21 input consumed theorem-facing by SDR must be bound to an exact statement, source identity, version or immutable locator when available, permitted-use role, and evidence status. Freshness does not silently replace identity. A changed source theorem, artifact, source class, coefficient regime, or target semantics produces a new run identity unless a proved equivalence licenses substitution.*

Policy 14 (SDR-GOV-01: no backflow into frozen v21)**.** *No SDR result—including a successful separation theorem, synthesized probe, rigidity theorem, completion theorem, arithmetic reconstruction theorem, formal replay, or cross-domain validation—is imported backward as a premise of a frozen v21 theorem. A successor release may consume an SDR result only through a fresh migration, ownership, dependency, source-binding, consumer, and registration record.*

SDR.1.9 Explicit non-claims

Declaration SDR.1.15 (SDR.1 explicit non-claims)**.** At the completion of SDR.1, none of the following is asserted:

(i) SDR already provides a universal invariant-discovery theorem;

(ii) SDR already proves that every target admits a finite separating family;

(iii) candidate-probe synthesis is algorithmically complete;

(iv) a separating probe is automatically stable, functorial, complete, or minimal;

(v) a local separation theorem automatically yields source reconstruction;

(vi) a rigidity theorem automatically supplies finite-to-infinite completion;

(vii) defect-rank improvement alone implies convergence or exact zero at a limit;

(viii) higher Bockstein or higher-Fitting data already reconstruct every nonsplit singular finite layer;

(ix) dg or A_∞ enhancement automatically strengthens a problem route;

(x) successful Search/Platform retrieval or AI generation is theorem evidence;

(xi) the semantic definition of a target-separation defect supplies an executable oracle for unknown target values;

(xii) Appendix SDR creates a third stable Core;

(xiii) Appendix SDR modifies the frozen AK-HPDST v21.0.0 release; or

(xiv) Appendix SDR, at this stage, solves a named open problem.

SDR.1.10 Dependency and downstream section contract

Definition SDR.1.16 (SDR incoming dependency baseline)**.** The SDR 1.0 construction may consume, only at exact proved or registered strengths, AK-HPDST v21.0.0 Main, Foundation Appendices A–G, Appendix DG, Appendix HT, Appendix UB, Appendices CM and TB at their individually proved strengths, Technical Extension Appendices H–N, Problem Interface and Search/Platform appendices in their typed roles, the Companion as an operational projection only, and the final v21 Claim Register as the frozen ownership record. Post-v21 Appendix BFA may be consumed only for its separately proved local lemmas, counterexamples, promotion discipline, audit semantics, and explicitly research-only successor questions. No BFA source-specific Navier–Stokes theorem is reclassified as generic SDR mathematics by inclusion in this baseline.

Definition SDR.1.17 (SDR downstream contract)**.** Later SDR sections shall preserve the following order of mathematical responsibility:

$$\begin{aligned}\text{target typing} &\longrightarrow \text{information/separation calculus} \longrightarrow \text{adversarial defect}\\ &\longrightarrow \text{structural discovery} \longrightarrow \text{enriched probe} \longrightarrow \text{rigidity}\\ &\longrightarrow \text{minimality/completion} \longrightarrow \text{source-specific Return.}\end{aligned}$$

A later section may omit stages that are not invoked by its declared theorem, but it may not silently identify two distinct stages or borrow a conclusion from an unproved later stage.

Theorem SDR.1.18 (SDR.1 constitutional closure)**.** *Subject to the fresh-identity, exact-strength, no-backflow, target-sovereignty, non-compensation, and owner-sovereignty rules above, SDR.1 defines Appendix SDR as a conservative post-v21 research supplement whose admissible mathematical purpose is to discover, test, prove, falsify, and certify target-relevant information refinements without mutating the frozen v21 theorem graph or inflating external Return strength.*

Proof. Declaration SDR.1.1 fixes the release boundary; Proposition SDR.1.5 proves conservative attachment under fresh identity and no backflow; Definitions SDR.1.7 and SDR.1.4 preserve owner and statement typing; Policies 5, 6, and 8 block information, rigidity, and completion shortcuts; Proposition SDR.1.12 blocks cross-field compensation; and Policies 11 and 14 prevent local success from becoming unlicensed Core or frozen-v21 strength. The explicit non-claims fix the remaining ceiling. □

Remark SDR.1.19 (Next consumer: SDR.2)**.** SDR.2 shall construct the target-relative information and separation calculus. In particular, it must define the information preorder, induced indistinguishability relations, target separation, factorization, and downstream information monotonicity without assuming that those notions are already supplied by the constitutional policies of SDR.1.

SDR.2 Target-Relative Information and Separation Calculus

SDR.2.1 Purpose, mathematical scope, and theorem mode

Declaration SDR.2.1 (SDR.2 objective)**.** SDR.2 constructs the target-relative information kernel consumed by every later separation-defect, discovery, rigidity, and no-go argument in Appendix SDR. The section formalizes four distinctions that must remain separate:

(i) the source distinctions retained by a declared observable;

(ii) the source distinctions required by a declared target;

(iii) observational refinement, in which genuinely new data shrink the observable indistinguishability relation; and

(iv) inferential strengthening, in which a rigidity, restriction, reconstruction, or factorization theorem strengthens what can be deduced from an unchanged observable packet without creating new observed data.

The central results identify deterministic informativeness with reverse inclusion of equality kernels, identify target sufficiency with factorization of the normalized target through the observable, prove downstream information monotonicity, characterize joint recording, and isolate the exact role of source restriction and rigidity.

Policy 15 (SDR-INF-02: equality-normal-form discipline)**.** *The calculus below is written in equality normal form. Whenever an application is stated only up to isomorphism, quasi-isomorphism, pseudo-isomorphism, gauge equivalence, derived equivalence, or another declared equivalence relation, the application must first supply an exact comparison quotient or an equivalent normalized record on which literal equality represents precisely the requested comparison strength. SDR.2 does not manufacture such a quotient, identify distinct comparison relations, or strengthen a relation by notation.*

Remark SDR.2.2 (Set-level theorem mode)**.** For the foundational results of SDR.2, $\mathcal{S}$ is a declared set-sized source family, or a frozen set-sized slice of a larger mathematical environment. The factorization theorems are statements about normalized set maps. They do not by themselves produce a natural transformation, a functor, an actual source morphism, a canonical quotient object inside a category, or a derived equivalence. Any such categorical strengthening requires its own theorem and artifacts.

Policy 16 (SDR-INF-03: semantic factorization is not categorical reconstruction)**.** *A set-level factorization*

$$\widehat{\Theta} = \overline{\Theta}_I \circ \widehat{I}$$

certifies only that the normalized target value is determined by the normalized observable value on the declared source class. It does not imply that I is full, faithful, exact, conservative, naturally invertible, or object- or morphism-reconstructing. Those stronger statements remain governed by their own source categories and theorems.

SDR.2.2 Targets, observables, and normalized comparison maps

Definition SDR.2.3 (Target datum)**.** A *target datum* is a tuple

$$\mathfrak{T} = (\mathcal{S}, \mathcal{T}, \Theta, \approx_\Theta, q_\Theta),$$

where $\mathcal{S}$ is the declared source family, $\Theta : \mathcal{S} \to \mathcal{T}$ is the raw target map, $\approx_\Theta$ is the declared equivalence relation at the requested Return strength, and

$$q_\Theta : \mathcal{T} \longrightarrow \overline{\mathcal{T}}_\Theta$$

is a supplied quotient or normalized comparison map satisfying

$$q_\Theta(u) = q_\Theta(v) \quad \Longleftrightarrow \quad u \approx_\Theta v.$$

The *normalized target map* is

$$\widehat{\Theta} := q_\Theta \circ \Theta : \mathcal{S} \longrightarrow \overline{\mathcal{T}}_\Theta.$$

When literal equality is the requested target relation, q_Θ is the identity.

Definition SDR.2.4 (Observable datum)**.** An *observable datum* on the same source family is a tuple

$$\mathfrak{I} = (\mathcal{S}, \mathcal{Y}, I, \approx_I, q_I),$$

where $I : \mathcal{S} \to \mathcal{Y}$ is the raw observable, $\approx_I$ is the declared equivalence relation on its output, and

$$q_I : \mathcal{Y} \longrightarrow \overline{\mathcal{Y}}_I$$

is a supplied normalized comparison map satisfying

$$q_I(u) = q_I(v) \quad \Longleftrightarrow \quad u \approx_I v.$$

The normalized observable is

$$\widehat{I} := q_I \circ I : \mathcal{S} \longrightarrow \overline{\mathcal{Y}}_I.$$

The normalized codomain may encode a scalar invariant, a tuple, an object class, a morphism class, an extension class, a derived class, a higher-structure record, or another exactly declared output type.

Definition SDR.2.5 (Observable packet)**.** For a set-indexed family of normalized observables $\{\widehat{I}_a\}_{a \in A}$ on $\mathcal{S}$, the associated *observable packet* is

$$\widehat{\mathbf{I}}_A : \mathcal{S} \longrightarrow \prod_{a \in A} \overline{\mathcal{Y}}_{I_a}, \qquad X \longmapsto \big(\widehat{I}_a(X)\big)_{a \in A}.$$

A finite packet is one for which A is finite. Finiteness of the packet does not imply completeness for any target without a theorem proving such completeness.

SDR.2.3 Indistinguishability relations, fibers, and partitions

Definition SDR.2.6 (Equality kernel and observable indistinguishability)**.** For a normalized map $F : \mathcal{S} \to \mathcal{Z}$, define its *equality kernel*

$$\mathrm{EqKer}(F) := \{(X, Y) \in \mathcal{S} \times \mathcal{S} : F(X) = F(Y)\}.$$

For an observable I, write

$$X \sim_I Y \quad \Longleftrightarrow \quad (X, Y) \in \mathrm{EqKer}(\widehat{I}),$$

and for the target write

$$X \sim_\Theta Y \quad \Longleftrightarrow \quad (X, Y) \in \mathrm{EqKer}(\widehat{\Theta}).$$

Thus $\sim_I$ records exactly the distinctions forgotten by the normalized observable, while $\sim_\Theta$ records distinctions irrelevant at the requested target strength.

Proposition SDR.2.7 (Equality kernels are equivalence relations)**.** *For every normalized map $F : \mathcal{S} \to \mathcal{Z}$,* $\mathrm{EqKer}(F)$ *is an equivalence relation on $\mathcal{S}$.*

Proof. Reflexivity, symmetry, and transitivity follow directly from the corresponding properties of equality in $\mathcal{Z}$. □

Definition SDR.2.8 (Information-loss fiber and fiber partition)**.** For $y \in \mathrm{im}(\widehat{I})$, define

$$\mathcal{F}_I(y) := \widehat{I}^{-1}(y).$$

The set of nonempty fibers

$$\Pi(I) := \{\mathcal{F}_I(y) : y \in \mathrm{im}(\widehat{I})\}$$

is the *fiber partition* of the source induced by I.

SDR.2.4 Deterministic information preorder and exact factorization

Definition SDR.2.9 (Deterministic information preorder)**.** Let I and J be observable data on the same source family $\mathcal{S}$. Write

$$I \preceq J$$

and say that J is *at least as informative as* I in the deterministic factorization sense if there exists a map

$$h_{J\to I} : \mathrm{im}(\widehat{J}) \longrightarrow \mathrm{im}(\widehat{I})$$

such that

$$\widehat{I} = h_{J\to I} \circ \widehat{J}.$$

The factorization map is required only on the realized image of $\widehat{J}$. This is a deterministic functional order; no stochastic or statistical comparison order is asserted.

Theorem SDR.2.10 (SDR-INF-04: factorization–kernel criterion)**.** *For observables I, J on $\mathcal{S}$, the following are equivalent:*

(i) $I \preceq J$*;*

(ii)

$$\mathrm{EqKer}(\widehat{J}) \subseteq \mathrm{EqKer}(\widehat{I});$$

(iii) every J-indistinguishable pair is also I-indistinguishable.

Proof. If $\widehat{I} = h_{J\to I} \circ \widehat{J}$ and $\widehat{J}(X) = \widehat{J}(Y)$, then applying $h_{J\to I}$ gives $\widehat{I}(X) = \widehat{I}(Y)$. Hence (i) implies (ii), and (ii) and (iii) are identical statements.

Assume (ii). For $z \in \mathrm{im}(\widehat{J})$, there is a unique value $y \in \mathrm{im}(\widehat{I})$ such that $\widehat{I}(X) = y$ for every X with $\widehat{J}(X) = z$: existence follows because z is realized, and uniqueness follows from the kernel inclusion. Define $h_{J\to I}(z) = y$. Then $\widehat{I} = h_{J\to I} \circ \widehat{J}$, proving (i). □

Corollary SDR.2.11 (Preorder law)**.** *The relation $\preceq$ is reflexive and transitive.*

Definition SDR.2.12 (Information equivalence and strict refinement)**.** Write

$$I \simeq_{\mathrm{info}} J \quad \Longleftrightarrow \quad I \preceq J \text{ and } J \preceq I.$$

Write

$$I \prec J$$

when $I \preceq J$ but $J \not\preceq I$. Thus $I \prec J$ means that J is a strict observational refinement of I on the declared source family and comparison strengths.

Proposition SDR.2.13 (SDR-INF-05: partition semantics of informativeness)**.** *For observables I, J on $\mathcal{S}$,*

$$I \preceq J \quad \Longleftrightarrow \quad \Pi(J) \textit{ refines } \Pi(I).$$

Moreover,

$$I \simeq_{\mathrm{info}} J \quad \Longleftrightarrow \quad \Pi(I) = \Pi(J),$$

and

$$I \prec J \quad \Longleftrightarrow \quad \Pi(J) \textit{ is a strict refinement of } \Pi(I).$$

Proof. A fiber of J is contained in a fiber of I exactly when every pair equal under $\widehat{J}$ is equal under $\widehat{I}$. Apply Theorem SDR.2.10. The remaining statements follow by equality or strict inclusion of the corresponding equality kernels. □

Definition SDR.2.14 (Abstract deterministic information class)**.** For the fixed source set $\mathcal{S}$, define

$$\mathfrak{I}_{\mathrm{det}}(\mathcal{S}) := \left\{E \subseteq \mathcal{S}^2 : E = \mathrm{EqKer}(F) \text{ for some set-valued map } F : \mathcal{S} \to \mathcal{Z}\right\}.$$

An observable I represents the element $\mathrm{EqKer}(\widehat{I}) \in \mathfrak{I}_{\mathrm{det}}(\mathcal{S})$. Two observables represent the same abstract information class exactly when they are information-equivalent.

Theorem SDR.2.15 (SDR-INF-05A: representation of deterministic information classes)**.** *For every set $\mathcal{S}$,*

$$\mathfrak{I}_{\mathrm{det}}(\mathcal{S}) = \mathrm{EqRel}(\mathcal{S}),$$

the set of all equivalence relations on $\mathcal{S}$. Moreover, for observables I, J,

$$I \simeq_{\mathrm{info}} J \quad \Longleftrightarrow \quad \mathrm{EqKer}(\widehat{I}) = \mathrm{EqKer}(\widehat{J}),$$

and

$$I \preceq J \quad \Longleftrightarrow \quad \mathrm{EqKer}(\widehat{J}) \subseteq \mathrm{EqKer}(\widehat{I}).$$

Thus information-equivalence classes are canonically represented by source equivalence relations without requiring a set of all possible observable codomains.

Proof. Every equality kernel is an equivalence relation by Proposition SDR.2.7, giving $\mathfrak{I}_{\mathrm{det}}(\mathcal{S}) \subseteq \mathrm{EqRel}(\mathcal{S})$. Conversely, if E is any equivalence relation on $\mathcal{S}$, the quotient map

$$q_E : \mathcal{S} \longrightarrow \mathcal{S}/E$$

has equality kernel exactly E, so every equivalence relation is realized. The final two equivalences are Theorem SDR.2.10 in both directions. □

Theorem SDR.2.16 (SDR-INF-05B: complete lattice of abstract deterministic information)**.** *The information-equivalence classes of normalized set maps on $\mathcal{S}$, ordered by $\preceq$, form a complete lattice. For every set-indexed family $\{I_a\}_{a\in A}$, the supremum has equality kernel*

$$\mathrm{EqKer}\left(\bigvee_{a\in A} [I_a]_{\mathrm{info}}\right) = \bigcap_{a\in A} \mathrm{EqKer}(\widehat{I}_a),$$

and is represented by the joint packet $\widehat{\mathbf{I}}_A$. The infimum has equality kernel

$$\mathrm{EqKer}\left(\bigwedge_{a\in A} [I_a]_{\mathrm{info}}\right) = \mathrm{EqCl}\left(\bigcup_{a\in A} \mathrm{EqKer}(\widehat{I}_a)\right),$$

where EqCl *denotes equivalence-relation closure on $\mathcal{S}$. The infimum can be represented by the quotient map of $\mathcal{S}$ by that equivalence relation.*

Proof. The equivalence relations on a fixed set form a complete lattice under inclusion: arbitrary meets are intersections and arbitrary joins are equivalence closures of unions. Theorem SDR.2.15 reverses that order. Hence information suprema correspond to intersections of equality kernels and information infima to equivalence closures of unions. For the supremum, two source objects have equal joint-packet values exactly when every coordinate value agrees, so the packet equality kernel is the displayed intersection directly. For the infimum, quotient $\mathcal{S}$ by the displayed equivalence closure. □

Remark SDR.2.17 (Novelty ceiling for the SDR.2 information kernel)**.** The factorization criterion, partition-refinement semantics, quotient representation, complete lattice structure, and deterministic data-processing monotonicity in SDR.2 are elementary set-theoretic facts. SDR makes no external-literature novelty claim for them. Their role is architectural: they provide a typed normal form against which later, potentially nontrivial probe-synthesis, rigidity, completion, and source-specific theorems can be measured without confusing new mathematics with repackaged information.

SDR.2.5 Target separation and exact target factorization

Definition SDR.2.18 (Target separation and target sufficiency)**.** An observable I is *target-separating* or *target-sufficient* for Θ on $\mathcal{S}$ if

$$X \sim_I Y \quad \Longrightarrow \quad X \sim_\Theta Y \qquad \text{for all } X, Y \in \mathcal{S}.$$

Equivalently, every information-loss fiber of I is contained in one target fiber.

Theorem SDR.2.19 (SDR-INF-06: target-separation factorization theorem)**.** *Let Θ be a target datum and I an observable datum on the same source family $\mathcal{S}$. The following are equivalent:*

(i) *I is target-sufficient for Θ;*

(ii)
$$\mathrm{EqKer}(\widehat{I}) \subseteq \mathrm{EqKer}(\widehat{\Theta});$$

(iii) *$\Theta \preceq I$ in the deterministic information preorder;*

(iv) *there exists a unique map*
$$\overline{\Theta}_I : \mathrm{im}(\widehat{I}) \longrightarrow \mathrm{im}(\widehat{\Theta})$$
such that
$$\widehat{\Theta} = \overline{\Theta}_I \circ \widehat{I}.$$

Proof. The equivalence of (i) and (ii) is the definition of target sufficiency. The equivalence of (ii) and (iii), and the existence in (iv), follow from Theorem SDR.2.10 with the weaker datum there taken to be Θ and the stronger datum taken to be I. For uniqueness, every $y \in \mathrm{im}(\widehat{I})$ has the form $y = \widehat{I}(X)$, so any factorization map must satisfy $\overline{\Theta}_I(y) = \widehat{\Theta}(X)$. □

Corollary SDR.2.20 (Exact target decision from a sufficient observable)**.** *If I is target-sufficient for Θ, then the normalized target value is a well-defined function of the normalized observable value alone:*
$$\widehat{\Theta}(X) = \overline{\Theta}_I(\widehat{I}(X)).$$

This is a theorem only at the declared target comparison strength and does not imply reconstruction of X.

Corollary SDR.2.21 (SDR-INF-07: indistinguishable-pair no-go)**.** *Suppose there exist $X, Y \in \mathcal{S}$ such that*
$$\widehat{I}(X) = \widehat{I}(Y) \qquad \textit{but} \qquad \widehat{\Theta}(X) \neq \widehat{\Theta}(Y).$$
Then I is not target-sufficient for Θ. In particular, no deterministic map defined only on $\mathrm{im}(\widehat{I})$ can recover $\widehat{\Theta}$ on all of $\mathcal{S}$.

Proof. The displayed pair belongs to $\mathrm{EqKer}(\widehat{I})$ but not to $\mathrm{EqKer}(\widehat{\Theta})$, contradicting Theorem SDR.2.19. Any deterministic function of $\widehat{I}$ takes the same value on the pair and therefore cannot reproduce two distinct target values. □

Remark SDR.2.22 (Relation to the frozen v21 loss principle)**.** Corollary SDR.2.21 is the target-relative equality-kernel form of the frozen v21 rule that distinctions erased by every field consumed by a route cannot be recreated by downstream syntax alone. SDR.2 claims no external novelty for this elementary principle; it packages the rule into a reusable factorization calculus for later SDR discovery and rigidity arguments.

SDR.2.6 Downstream processing and the information ceiling

Definition SDR.2.23 (Downstream observable generated from an existing packet)**.** Let I be an observable and let
$$f : \mathrm{im}(\widehat{I}) \longrightarrow \mathcal{Z}$$
be a deterministic map. Define the normalized downstream observable
$$\widehat{K}_f := f \circ \widehat{I} : \mathcal{S} \longrightarrow \mathrm{im}(f \circ \widehat{I}),$$
with literal equality as its output comparison relation. The datum K_f uses no source information beyond I.

Theorem SDR.2.24 (SDR-INF-08: downstream information monotonicity)**.** *For every downstream observable K_f generated from I,*

$$K_f \preceq I,$$

or equivalently

$$\mathrm{EqKer}(\widehat{I}) \subseteq \mathrm{EqKer}(\widehat{K}_f).$$

Thus deterministic post-processing may preserve or discard source distinctions but cannot create a distinction absent from the consumed observable.

Proof. The defining factorization $\widehat{K}_f = f \circ \widehat{I}$ is exactly the witness required for $K_f \preceq I$. Apply Theorem SDR.2.10 for the kernel form. □

Corollary SDR.2.25 (No bridge-only repair on a fixed source class)**.** *If I is not target-sufficient for Θ on $\mathcal{S}$, then no downstream observable K_f generated only from I is target-sufficient for Θ on the same source class.*

Proof. A pair witnessing failure of target sufficiency for I remains indistinguishable under every K_f by Theorem SDR.2.24, while its target values remain distinct. □

Proposition SDR.2.26 (SDR-INF-09: lossless post-processing criterion)**.** *With the preceding notation, the following are equivalent:*

(i) $K_f \simeq_{\mathrm{info}} I$*;*

(ii) $I \preceq K_f$*;*

(iii) f *is injective on* $\mathrm{im}(\widehat{I})$*;*

(iv)

$$\mathrm{EqKer}(\widehat{K}_f) = \mathrm{EqKer}(\widehat{I}).$$

Proof. Theorem SDR.2.24 gives $K_f \preceq I$ automatically, so (i) and (ii) are equivalent. If f is injective on the realized image, its inverse on $\mathrm{im}(\widehat{K}_f)$ factors $\widehat{I}$ through $\widehat{K}_f$. Conversely, if $I \preceq K_f$, then equality of $f(y_1)$ and $f(y_2)$ for realized y_i forces equality of y_1 and y_2. The equivalence with (iv) follows from Theorem SDR.2.10 in both directions. □

SDR.2.7 Source-faithful observables and target-relative economy

Definition SDR.2.27 (Source-separating observable)**.** An observable I is *source-separating* on $\mathcal{S}$ if

$$\mathrm{EqKer}(\widehat{I}) = \Delta_{\mathcal{S}}, \qquad \Delta_{\mathcal{S}} := \{(X, X) : X \in \mathcal{S}\}.$$

Equivalently, $\widehat{I}$ is injective on $\mathcal{S}$.

Remark SDR.2.28 (Source-presentation ceiling)**.** Source separation in Definition SDR.2.27 is literal injectivity on the chosen set-level source presentation. If the mathematical source is meaningful only up to a declared equivalence—for example isomorphism, quasi-isomorphism, gauge equivalence, or another quotient semantics—that source relation must be normalized first before "source reconstruction" is interpreted. Literal separation of two different presentations of the same source-equivalence class is not a stronger mathematical Return.

Proposition SDR.2.29 (SDR-INF-10: universal target sufficiency of source separation)**.** *If I is source-separating on $\mathcal{S}$, then I is target-sufficient for every normalized target datum on $\mathcal{S}$. The converse is false in general for any fixed target.*

Proof. The diagonal relation $\Delta_{\mathcal{S}}$ is contained in every equivalence relation on $\mathcal{S}$, hence in every target equality kernel. Apply Theorem SDR.2.19. For the converse, any nontrivial constant-on-fibers target may be determined by an observable that still identifies source objects lying in the same target fiber. □

Remark SDR.2.30 (Why SDR is target-relative rather than universally reconstructive)**.** Source separation is sufficient for every target but is often much stronger than necessary. SDR therefore seeks observables whose fibers are no finer than required for the declared target, subject to robustness, certifiability, and later rigidity constraints. This target-relative economy is mathematically distinct from full source reconstruction.

SDR.2.8 Joint observables and genuine observational enrichment

Definition SDR.2.31 (Joint observable)**.** For normalized observables

$$\widehat{I} : \mathcal{S} \to \overline{\mathcal{Y}}_I, \qquad \widehat{J} : \mathcal{S} \to \overline{\mathcal{Y}}_J,$$

define their *joint observable*

$$\widehat{I \boxtimes J} : \mathcal{S} \longrightarrow \overline{\mathcal{Y}}_I \times \overline{\mathcal{Y}}_J, \qquad X \longmapsto (\widehat{I}(X), \widehat{J}(X)).$$

The symbol $\boxtimes$ denotes joint recording only; it asserts no tensor product, product object in the source category, compatibility theorem, or interaction law between the two observables.

Theorem SDR.2.32 (SDR-INF-11: packet-kernel intersection)**.** *For every set-indexed observable packet* $\widehat{\mathbf{I}}_A$*,*

$$\mathrm{EqKer}(\widehat{\mathbf{I}}_A) = \bigcap_{a \in A} \mathrm{EqKer}(\widehat{I}_a).$$

In particular,

$$\mathrm{EqKer}(\widehat{I \boxtimes J}) = \mathrm{EqKer}(\widehat{I}) \cap \mathrm{EqKer}(\widehat{J}).$$

Consequently

$$I \preceq I \boxtimes J, \qquad J \preceq I \boxtimes J.$$

Proof. Two source objects have equal packet values exactly when every coordinate has equal value. The binary consequences follow by projecting the joint value to either coordinate and applying Theorem SDR.2.10. □

Theorem SDR.2.33 (SDR-INF-12: least-upper-bound property of joint recording)**.** *Let K be another observable on $\mathcal{S}$. If*

$$I \preceq K \qquad \text{and} \qquad J \preceq K,$$

then

$$I \boxtimes J \preceq K.$$

Hence the information-equivalence class of $I \boxtimes J$ is the least upper bound of the information-equivalence classes of I and J in the preorder quotient.

Proof. Choose image-level factorizations $\widehat{I} = h_I \circ \widehat{K}$ and $\widehat{J} = h_J \circ \widehat{K}$. Then

$$h(z) := (h_I(z), h_J(z))$$

factors $\widehat{I \boxtimes J}$ through $\widehat{K}$. Theorem SDR.2.32 gives the two lower-bound relations. □

Definition SDR.2.34 (Relative redundancy and strict observational gain)**.** An observable J is *information-redundant relative to I* if

$$I \boxtimes J \simeq_{\mathrm{info}} I.$$

It is a *strict observational enrichment* of I if

$$I \prec I \boxtimes J.$$

Proposition SDR.2.35 (Redundancy and strict-gain criteria)**.** *For observables I, J on $\mathcal{S}$:*

(i) J is information-redundant relative to I if and only if $J \preceq I$;

(ii) J is a strict observational enrichment of I if and only if there exist $X, Y \in \mathcal{S}$ such that

$$\widehat{I}(X) = \widehat{I}(Y) \quad \text{and} \quad \widehat{J}(X) \neq \widehat{J}(Y).$$

Proof. For (i), if $J \preceq I$, both coordinates of the joint packet factor through I, so $I \boxtimes J \preceq I$; the reverse relation follows from coordinate projection. Conversely, if $I \boxtimes J \preceq I$, then $J \preceq I \boxtimes J \preceq I$. For (ii), use the strict equality-kernel inclusion characterized by Theorem SDR.2.10 and Theorem SDR.2.32. □

Theorem SDR.2.36 (SDR-INF-13: target sufficiency is monotone under observational refinement)**.** *If*

$$\Theta \preceq I \preceq J,$$

then J is target-sufficient for Θ. Equivalently, once a target is determined by an observable packet, adding genuinely richer observational data cannot destroy target sufficiency.

Proof. Transitivity gives $\Theta \preceq J$. Apply Theorem SDR.2.19. □

Remark SDR.2.37 (Source information gain is not yet target-relevant gain)**.** A strict observational enrichment may distinguish only source pairs that already share the same target value. Therefore

$$I \prec I \boxtimes J$$

does not by itself show that J helps decide Θ. SDR.3 introduces the target-relative separation defect and records exactly which unresolved target-discordant pairs are removed by an enriched packet.

SDR.2.9 Target-strength order and Return ceilings

Definition SDR.2.38 (Target-strength preorder)**.** For two normalized target data Θ_1, Θ_2 on the same source family, write

$$\Theta_1 \preceq \Theta_2$$

when $\widehat{\Theta}_1$ factors through $\widehat{\Theta}_2$. Then Θ_2 is at least as discriminating as Θ_1 in the present deterministic sense. No universal hierarchy among predicates, objects, morphisms, exact structures, or derived structures is assumed unless the required factorization maps are actually supplied.

Theorem SDR.2.39 (SDR-INF-14: monotonicity in target strength)**.** *Assume*

$$\Theta_{\text{weak}} \preceq \Theta_{\text{strong}}.$$

If an observable I is target-sufficient for Θ_{strong}, then I is target-sufficient for Θ_{weak}.

Proof. Target sufficiency gives $\Theta_{\text{strong}} \preceq I$. Transitivity yields

$$\Theta_{\text{weak}} \preceq \Theta_{\text{strong}} \preceq I.$$

Apply Theorem SDR.2.19. □

Example SDR.2.40 (Failure of the converse)**.** Let $\mathcal{S} = \{a, b\}$. Let Θ_{weak} be constant and let Θ_{strong} distinguish a from b. A constant observable is sufficient for the weaker target but not for the stronger one. Success at a weaker Return strength therefore cannot be promoted to a stronger target without additional information or a theorem that changes the admissible source class or target relation.

SDR.2.10 Source restriction, rigidity, and inferential strengthening

Definition SDR.2.41 (Restricted source problem)**.** For $\mathcal{S}' \subseteq \mathcal{S}$, define

$$\widehat{I}|_{\mathcal{S}'} : \mathcal{S}' \to \overline{\mathcal{Y}}_I, \qquad \widehat{\Theta}|_{\mathcal{S}'} : \mathcal{S}' \to \overline{\mathcal{T}}_\Theta.$$

The restriction changes the admissible source class but does not alter the observable value attached to any retained source object.

Proposition SDR.2.42 (Restriction law for equality kernels)**.** *For every $\mathcal{S}' \subseteq \mathcal{S}$,*

$$\mathrm{EqKer}(\widehat{I}|_{\mathcal{S}'}) = \mathrm{EqKer}(\widehat{I}) \cap (\mathcal{S}' \times \mathcal{S}'),$$

and likewise for $\widehat{\Theta}$.

Theorem SDR.2.43 (SDR-RIG-03: restriction-induced target-rigidity criterion)**.** *For a restricted source class $\mathcal{S}' \subseteq \mathcal{S}$, the following are equivalent:*

(i) $I|_{\mathcal{S}'}$ *is target-sufficient for* $\Theta|_{\mathcal{S}'}$*;*

(ii) every I*-fiber meets* $\mathcal{S}'$ *only in points having a single normalized target value;*

(iii)

$$\mathrm{EqKer}(\widehat{I}) \cap (\mathcal{S}' \times \mathcal{S}') \subseteq \mathrm{EqKer}(\widehat{\Theta});$$

(iv) there is a unique factorization

$$\widehat{\Theta}|_{\mathcal{S}'} = \overline{\Theta}_{I,\mathcal{S}'} \circ \widehat{I}|_{\mathcal{S}'}.$$

Thus an independently proved source restriction or rigidity theorem can establish target sufficiency on $\mathcal{S}'$ *by eliminating target-discordant pairs from admissible observable fibers.*

Proof. Apply Theorem SDR.2.19 to the restricted source family and use Proposition SDR.2.42. The fiber statement is the pointwise reading of the same restricted kernel inclusion. □

Corollary SDR.2.44 (Rigidity is inferential strengthening, not observational refinement)**.** *A source restriction or rigidity theorem may make an unchanged observable target-sufficient on a smaller or more structured source class, but it does not shrink* $\mathrm{EqKer}(\widehat{I})$ *as a relation on the original source class. Its gain is inferential: it proves that previously ambiguous pairs are not simultaneously admissible for the restricted problem.*

Example SDR.2.45 (Restricted success does not globalize automatically)**.** Let $\mathcal{S} = \{a, b, c\}$, let I identify a and b, and suppose $\widehat{\Theta}(a) \neq \widehat{\Theta}(b)$. On $\mathcal{S}' = \{a, c\}$, the restricted observable may be target-sufficient, while on all of $\mathcal{S}$ it is not. Restricted rigidity therefore cannot be exported to the ambient source class without a theorem controlling the excluded objects.

SDR.2.11 Finite observation and certification boundary

Policy 17 (SDR-INF-15: finite testing is not universal separation)**.** *Testing finitely many pairs can establish target separation only on the tested finite source family. It does not establish*

$$\mathrm{EqKer}(\widehat{I}) \subseteq \mathrm{EqKer}(\widehat{\Theta})$$

on a larger source class unless a soundness/completeness, finite-generation, reconstruction, compactness, Noetherian, finite-state, or other theorem reduces the universal statement to the tested finite record.

Proposition SDR.2.46 (Exact finite-family certification)**.** *If* $\mathcal{S}_0 \subseteq \mathcal{S}$ *is finite and every pair* $X, Y \in \mathcal{S}_0$ *satisfying* $\widehat{I}(X) = \widehat{I}(Y)$ *also satisfies* $\widehat{\Theta}(X) = \widehat{\Theta}(Y)$*, then* $I|_{\mathcal{S}_0}$ *is target-sufficient for* $\Theta|_{\mathcal{S}_0}$*. No conclusion about* $\mathcal{S} \setminus \mathcal{S}_0$ *follows without an additional theorem.*

Proof. The hypothesis is exactly the equality-kernel inclusion of Theorem SDR.2.19 on the restricted finite source family. □

SDR.2.12 SDR.2 semantic record, closure, and handoff

Definition SDR.2.47 (SDR.2 semantic information-kernel record)**.** For a theorem-facing target/observable pair, the abstract SDR.2 semantic record is

$$\mathfrak{R}_{\mathrm{SDR2}} = (\mathcal{S}, \widehat{\Theta}, \widehat{I}, \mathcal{R}_\Theta, \mathcal{R}_I, \preceq, \Sigma, \mathcal{A}),$$

where $\mathcal{R}_\Theta$ and $\mathcal{R}_I$ are exact relation objects, predicates, theorem locators, or finite encodings representing the corresponding equality kernels; Σ is the requested target strength; and $\mathcal{A}$ contains the source, comparison, theorem, and proof artifacts required by the consumer. The definition does not require explicit materialization of an infinite equality-kernel set. A machine-verifier record is a later implementation object and must provide a finite representation theorem when finiteness is claimed.

Theorem SDR.2.48 (SDR.2 information and separation kernel closure)**.** *Within the equality-normal-form and set-level conventions of this section, SDR.2 establishes the following source-independent deterministic calculus:*

(i) *informativeness is exactly reverse inclusion of observable equality kernels, equivalently refinement of fiber partitions;*

(ii) *abstract information-equivalence classes are represented exactly by equivalence relations on the source and form a complete lattice, with joint packets realizing arbitrary information suprema;*

(iii) *target sufficiency is exactly factorization of the normalized target through the normalized observable;*

(iv) *an indistinguishable target-discordant pair is a complete no-go witness for deterministic routes that consume only that observable on the same source class;*

(v) *deterministic downstream processing cannot increase observational information;*

(vi) *source-separating observables are sufficient for every normalized target, while target-specific sufficiency may be strictly weaker than source reconstruction;*

(vii) *joint recording intersects equality kernels and yields the least upper bound in the deterministic information preorder modulo information equivalence;*

(viii) *target sufficiency is monotone under observational refinement;*

(ix) *sufficiency for a stronger target implies sufficiency for every weaker target connected by a proved factorization, but not conversely;*

(x) *source restriction and rigidity can strengthen inference without creating observational information on the original source class; and*

(xi) *finite testing certifies only the declared finite source family unless a separate reduction theorem extends the quantifier domain.*

None of these statements asserts that a target-separating invariant exists for an arbitrary problem, that a useful enriched probe can always be synthesized, that set-level factorization upgrades to categorical reconstruction, or that this elementary calculus is itself a successor-Core theorem family.

Proof. Item (i) is Theorem SDR.2.10 and Proposition SDR.2.13. Item (ii) is Theorems SDR.2.15 and SDR.2.16. Item (iii) is Theorem SDR.2.19. Item (iv) is Corollary SDR.2.21; item (v) is Theorem SDR.2.24; item (vi) is Proposition SDR.2.29; item (vii) is Theorems SDR.2.32 and SDR.2.33; item (viii) is Theorem SDR.2.36; item (ix) is Theorem SDR.2.39 with Example SDR.2.40; item (x) is Theorem SDR.2.43 and Corollary SDR.2.44; and item (xi) is Policy 17 together with Proposition SDR.2.46. The final ceiling follows from Policies 15 and 16 and from the explicitly set-level theorem mode. □

Remark SDR.2.49 (Next consumer: SDR.3)**.** SDR.3 shall convert the binary sufficiency/no-go boundary of SDR.2 into a target-relative defect object. It will define the target-discordant pair space, the separation defect $\Delta_\Theta(\mathbf{I})$, strict target-relevant information gain, adversarial witness packets, and the exact relation between enriched probes and reduction of unresolved target distinctions.

SDR.3 Separation Defect and Adversarial Witness Theory

SDR.3.1 Purpose, inherited scope, and theorem mode

Declaration SDR.3.1 (SDR.3 objective)**.** SDR.3 converts the binary target-sufficiency criterion of SDR.2 into a target-relative defect object that can be localized, reduced by enriched observation, preserved under downstream coarsening, and certified by adversarial witness pairs. It also separates the *semantic defect*—which is defined whether or not target values are currently knowable—from the *evidence-certified defect* that may actually be consumed by a theorem-facing discovery run. Its governing question is not merely whether an observable packet is insufficient, but *which exact target-relevant distinctions remain unresolved, which of those distinctions are admissibly certified, and what additional structure would be required to remove them.*

Throughout this section, $\mathcal{S}$ is the declared set-level source family, $\widehat{\Theta} : \mathcal{S} \to \overline{\mathcal{T}}_\Theta$ is the normalized target map of Definition SDR.2.3, and $\widehat{\mathbf{I}}_A$ is an observable packet in the sense of Definition SDR.2.5. No categorical reconstruction, probabilistic semantics, metric geometry, or finite-to-infinite conclusion is inferred unless separately stated.

Policy 18 (SDR-SEP-01: defect is target-relative)**.** *A separation defect is defined relative to a declared target map, target equivalence relation, source class, and observable packet. The same source pair may be defective for one target and irrelevant for another. Consequently no scalar, rank table, barcode, Ext profile, enriched module, higher operation, or other observable is called "complete" without naming the target strength and source class for which completeness is asserted.*

Policy 19 (SDR-SEP-02: target-oracle and benchmark firewall)**.** *The semantic relation $\widehat{\Theta}(X) \neq \widehat{\Theta}(Y)$ may be mathematically well defined even when its truth is not available to the discovery procedure. An SDR lane advertised as independent discovery may use target-discordance information upstream only when that information is already certified by the lane's admissible premise set. If the distinction is imported from a benchmark answer, conclusion-derived label, hidden oracle, or equivalent downstream knowledge, the run must be classified as benchmark-assisted reconstruction or validation and may not be cited as independent discovery of the separating structure. A surrogate target may be used only through a separately proved theorem specifying exactly which surrogate distinctions imply distinctions for Θ.*

Definition SDR.3.2 (Evidence-relative certified separation defect)**.** Fix an admissible target-evidence policy $\mathcal{E}$. Define

$$\Delta_\Theta^{\mathrm{cert},\mathcal{E}}(\mathbf{I}_A) := \big\{(X,Y) \in \Delta_\Theta(\mathbf{I}_A) : \text{an } \mathcal{E}\text{-admissible certificate proves } \widehat{\Theta}(X) \neq \widehat{\Theta}(Y)\big\}.$$

The certificate provenance records whether the evidence is independent-admissible, conditional-admissible, benchmark-quarantined, or surrogate-bound. A conditional-admissible record carries its hypotheses into every downstream defect, gain, witness, and no-go conclusion. A surrogate-bound record enters the certified defect for Θ only after the declared surrogate-to-target implication has itself been proved.

Proposition SDR.3.3 (SDR-SEP-02A: certified-defect soundness and coverage ceiling)**.** *For every admissible evidence policy $\mathcal{E}$,*

$$\Delta_\Theta^{\mathrm{cert},\mathcal{E}}(\mathbf{I}_A) \subseteq \Delta_\Theta(\mathbf{I}_A).$$

Equality holds only when the evidence regime is complete for target discordance on every pair in the semantic defect. In particular,

$$\Delta_\Theta^{\mathrm{cert},\mathcal{E}}(\mathbf{I}_A) = \varnothing$$

does not imply $\Delta_\Theta(\mathbf{I}_A) = \varnothing$ without a separately proved target-evidence completeness theorem.

Proof. Membership in the certified defect includes, by definition, membership in the semantic defect together with admissible target-discordance evidence, proving the inclusion. Equality requires that every semantically defective pair admit such evidence. The final nonimplication is immediate when the evidence regime covers only a proper subset of the semantic defect. □

SDR.3.2 Target-discordant relation and separation defect

Definition SDR.3.4 (Target-discordant relation)**.** The *target-discordant relation* is

$$C_\Theta := (\mathcal{S} \times \mathcal{S}) \setminus \mathrm{EqKer}(\widehat{\Theta}) = \{(X,Y) : \widehat{\Theta}(X) \neq \widehat{\Theta}(Y)\}.$$

It records exactly those ordered source pairs whose distinction matters at the requested target strength.

Proposition SDR.3.5 (Structure of the target-discordant relation)**.** *The relation C_Θ is symmetric and irreflexive. Moreover, if $\Theta_{\mathrm{weak}} \preceq \Theta_{\mathrm{strong}}$, then*

$$C_{\Theta_{\mathrm{weak}}} \subseteq C_{\Theta_{\mathrm{strong}}}.$$

Proof. Symmetry and irreflexivity follow from inequality in the normalized target codomain. The target-strength relation gives $\mathrm{EqKer}(\widehat{\Theta}_{\mathrm{strong}}) \subseteq \mathrm{EqKer}(\widehat{\Theta}_{\mathrm{weak}})$ by Theorem SDR.2.10; taking complements in $\mathcal{S}^2$ reverses the inclusion. □

Definition SDR.3.6 (Target-relative separation defect)**.** The *separation defect* of the packet $\mathbf{I}_A$ relative to Θ is

$$\Delta_\Theta(\mathbf{I}_A) := \mathrm{EqKer}(\widehat{\mathbf{I}}_A) \cap C_\Theta.$$

Equivalently,

$$\Delta_\Theta(\mathbf{I}_A) = \{(X, Y) \in \mathcal{S}^2 : \widehat{\mathbf{I}}_A(X) = \widehat{\mathbf{I}}_A(Y),\ \widehat{\Theta}(X) \neq \widehat{\Theta}(Y)\}.$$

For a single observable I, write $\Delta_\Theta(I)$. This is a semantic relation: its definition does not assert that the target values are computable, known, or admissible as search labels in the current discovery lane.

Proposition SDR.3.7 (Structure of the separation-defect relation)**.** *For every observable packet, $\Delta_\Theta(\mathbf{I}_A)$ is symmetric and irreflexive, but it need not be transitive.*

Proof. Symmetry and irreflexivity follow from intersection of the symmetric equality kernel with the symmetric, irreflexive relation C_Θ. For nontransitivity, let $\mathcal{S} = \{a, b, c\}$, let the observable be constant, and take normalized target values $0, 1, 0$ on a, b, c, respectively. Then (a, b) and (b, c) lie in the defect while (a, c) does not. □

Theorem SDR.3.8 (SDR-SEP-03: exact defect criterion for target sufficiency)**.** *For every observable packet $\mathbf{I}_A$, the following are equivalent:*

(i) $\mathbf{I}_A$ *is target-sufficient for* Θ*;*

(ii) $\mathrm{EqKer}(\widehat{\mathbf{I}}_A) \subseteq \mathrm{EqKer}(\widehat{\Theta})$*;*

(iii) $\Delta_\Theta(\mathbf{I}_A) = \varnothing$*;*

(iv) no target-discordant source pair lies in a common observable-packet fiber.

Proof. The equivalence of (i) and (ii) is Theorem SDR.2.19. By Definition SDR.3.6, the defect is exactly the portion of the observable equality kernel lying outside the target equality kernel, so (ii) is equivalent to (iii), and (iv) is the pointwise reading of (iii). □

Corollary SDR.3.9 (One certified defect pair refutes universal sufficiency)**.** *If*

$$(X, Y) \in \Delta_\Theta(\mathbf{I}_A),$$

then $\mathbf{I}_A$ is not target-sufficient for Θ on $\mathcal{S}$. Thus a single certified pair is enough to refute a universal target-sufficiency claim for the declared packet and source class.

SDR.3.3 Fiber ambiguity and defect localization

Definition SDR.3.10 (Observable fiber and target ambiguity set)**.** For a value $z \in \mathrm{im}(\widehat{\mathbf{I}}_A)$, define the observable fiber

$$\mathcal{F}_z(\mathbf{I}_A) := \{X \in \mathcal{S} : \widehat{\mathbf{I}}_A(X) = z\}$$

and its *target ambiguity set*

$$\mathcal{A}_\Theta(z \mid \mathbf{I}_A) := \{\widehat{\Theta}(X) : X \in \mathcal{F}_z(\mathbf{I}_A)\}.$$

The *ambiguity locus* is

$$\mathrm{Amb}_\Theta(\mathbf{I}_A) := \{z \in \mathrm{im}(\widehat{\mathbf{I}}_A) : |\mathcal{A}_\Theta(z \mid \mathbf{I}_A)| > 1\}.$$

Theorem SDR.3.11 (SDR-SEP-04: fiber-localization theorem)**.** *The separation defect decomposes fiberwise as*

$$\Delta_\Theta(\mathbf{I}_A) = \bigcup_{z \in \mathrm{Amb}_\Theta(\mathbf{I}_A)} \left\{(X, Y) \in \mathcal{F}_z(\mathbf{I}_A)^2 : \widehat{\Theta}(X) \neq \widehat{\Theta}(Y)\right\}.$$

Consequently,

$$\Delta_\Theta(\mathbf{I}_A) = \varnothing \quad \Longleftrightarrow \quad \mathrm{Amb}_\Theta(\mathbf{I}_A) = \varnothing.$$

Proof. A pair lies in the defect exactly when its two members have the same packet value z and different target values. Such a value z belongs precisely to the ambiguity locus. The final equivalence follows because a nonempty ambiguity set of cardinality greater than one supplies a target-discordant pair in that fiber, and every defect pair lies in such a fiber. □

Remark SDR.3.12 (Semantic versus operational ambiguity localization)**.** The ambiguity locus is a semantic object. Materializing it as a search table requires admissible target labels or a complete target-discordance certificate on the relevant fiber. When those labels are unavailable, SDR.4 may use structural surrogates or independently certified subclasses, but it may not silently treat the semantic ambiguity locus as an executable target oracle.

Definition SDR.3.13 (Target-colored fiber partition)**.** For fixed $z \in \mathrm{im}(\widehat{\mathbf{I}}_A)$ and target value $t \in \mathcal{A}_\Theta(z \mid \mathbf{I}_A)$, define

$$\mathcal{F}_{z,t} := \{X \in \mathcal{S} : \widehat{\mathbf{I}}_A(X) = z, \widehat{\Theta}(X) = t\}.$$

Then $\mathcal{F}_z(\mathbf{I}_A) = \bigsqcup_t \mathcal{F}_{z,t}$. The defect inside the fiber consists exactly of ordered pairs lying in different target-colored blocks.

Proposition SDR.3.14 (Complete multipartite form inside one ambiguous fiber)**.** *For every observable value z,*

$$\Delta_\Theta(\mathbf{I}_A) \cap \mathcal{F}_z(\mathbf{I}_A)^2 = \bigcup_{t \neq t'} \mathcal{F}_{z,t} \times \mathcal{F}_{z,t'}.$$

Thus the local defect is determined entirely by the target partition induced inside the observable fiber.

Remark SDR.3.15 (Why fiber localization matters for discovery)**.** SDR does not need to search the whole source space uniformly. A candidate new probe can be evaluated first on the ambiguous fibers, and more specifically on pairs of target-colored blocks inside those fibers. This localization is a discovery heuristic only; a theorem claiming global defect elimination must still quantify over the full declared source class or invoke a separate finite-reduction theorem.

SDR.3.4 Defect monotonicity under information refinement

Theorem SDR.3.16 (SDR-SEP-05: observational refinement can only shrink target defect)**.** *If*

$$I \preceq J,$$

so that J is at least as informative as I, then

$$\Delta_\Theta(J) \subseteq \Delta_\Theta(I).$$

More generally, the same holds for observable packets ordered by the SDR.2 information preorder.

Proof. Theorem SDR.2.10 gives $\mathrm{EqKer}(\widehat{J}) \subseteq \mathrm{EqKer}(\widehat{I})$. Intersect both sides with the fixed target-discordant relation C_Θ. □

Theorem SDR.3.17 (SDR-SEP-06: joint-defect intersection law)**.** *For normalized observables I, J,*

$$\Delta_\Theta(I \boxtimes J) = \Delta_\Theta(I) \cap \Delta_\Theta(J).$$

For any set-indexed packet $\mathbf{I}_A$,

$$\Delta_\Theta(\mathbf{I}_A) = C_\Theta \cap \bigcap_{a \in A} \mathrm{EqKer}(\widehat{I}_a).$$

Proof. Apply Theorem SDR.2.32 and distribute intersection with C_Θ. Since $\Delta_\Theta(I) = C_\Theta \cap \mathrm{EqKer}(\widehat{I})$ and likewise for J, the binary formula follows. □

Corollary SDR.3.18 (No compensation by unrelated success)**.** *A target-discordant pair remains unresolved by a joint packet exactly when every invoked coordinate still identifies that pair. Passing values, strong certificates, or exact reconstruction on other coordinates do not compensate for one residual pair unless at least one coordinate actually separates it or a separate rigidity theorem removes it from the admissible target ambiguity.*

Theorem SDR.3.19 (SDR-SEP-07: downstream coarsening preserves every existing defect witness)**.** *If $K \preceq I$, in particular if $\widehat{K} = f \circ \widehat{I}$ is deterministic downstream processing of I, then*

$$\Delta_\Theta(I) \subseteq \Delta_\Theta(K).$$

Hence no deterministic post-processing of an unchanged observable can discharge a defect pair already present for that observable.

Proof. The relation $K \preceq I$ gives $\mathrm{EqKer}(\widehat{I}) \subseteq \mathrm{EqKer}(\widehat{K})$. Intersect with C_Θ. The downstream case is Theorem SDR.2.24. □

SDR.3.5 Target-relevant gain and defect-reduction sets

Definition SDR.3.20 (Defect-reduction set)**.** Given a current observable I and a candidate enrichment J, define the *target-relevant defect-reduction set*

$$\mathcal{R}_\Theta(J \mid I) := \Delta_\Theta(I) \setminus \Delta_\Theta(I \boxtimes J).$$

Equivalently,

$$\mathcal{R}_\Theta(J \mid I) = \{(X,Y) \in \Delta_\Theta(I) : \widehat{J}(X) \neq \widehat{J}(Y)\}.$$

Theorem SDR.3.21 (SDR-SEP-08: exact target-relevant gain criterion)**.** *For a current observable I and enrichment J, the following are equivalent:*

(i) $\Delta_\Theta(I \boxtimes J) \subsetneq \Delta_\Theta(I)$*;*

(ii) $\mathcal{R}_\Theta(J \mid I) \neq \varnothing$*;*

(iii) there exist $X, Y \in \mathcal{S}$ *such that*

$$\widehat{I}(X) = \widehat{I}(Y), \qquad \widehat{\Theta}(X) \neq \widehat{\Theta}(Y), \qquad \widehat{J}(X) \neq \widehat{J}(Y).$$

When these conditions hold, J has strict target-relevant gain relative to I.

Proof. By Theorem SDR.3.16, joint enrichment cannot enlarge the defect. Strict inclusion is therefore equivalent to a nonempty set difference, which is (ii). The second displayed form of Definition SDR.3.20 identifies that set difference with the pairs satisfying (iii). □

Corollary SDR.3.22 (Evidence-certified positive gain witness)**.** *Let* $(X,Y) \in \Delta_\Theta^{\mathrm{cert},\mathcal{E}}(I)$*. If a candidate enrichment J has a certified comparison*

$$\widehat{J}(X) \neq \widehat{J}(Y),$$

then J has strict target-relevant gain relative to I. The mathematical gain statement is sound independently of how the witness was found, but the discovery-provenance claim is limited by the evidence mode recorded in $\mathcal{E}$: benchmark-quarantined labels certify benchmark-assisted gain, not independent discovery.

Proof. The certified witness lies in the semantic defect by Proposition SDR.3.3. The displayed inequality gives condition (iii) of Theorem SDR.3.21. The provenance ceiling is Policy 19. □

Definition SDR.3.23 (Target-redundant enrichment)**.** A candidate enrichment J is *target-redundant relative to* I if

$$\mathcal{R}_\Theta(J \mid I) = \varnothing.$$

This is weaker than information redundancy: J may distinguish source pairs and hence be a strict observational enrichment while still separating no pair in the current target defect.

Proposition SDR.3.24 (Source gain versus target-relevant gain)**.** *The implication*

$$\mathcal{R}_\Theta(J \mid I) \neq \varnothing \quad \Longrightarrow \quad I \prec I \boxtimes J$$

holds, but the converse fails in general.

Proof. Any pair in the reduction set has equal I-value and unequal J-value, so Proposition SDR.2.35 gives strict observational enrichment. Conversely, J may separate only pairs having equal target value, in which case it strictly refines source information while leaving $\Delta_\Theta(I)$ unchanged. □

Theorem SDR.3.25 (SDR-SEP-09: reduction-set union law)**.** *For candidate enrichments J, K,*

$$\mathcal{R}_\Theta(J \boxtimes K \mid I) = \mathcal{R}_\Theta(J \mid I) \cup \mathcal{R}_\Theta(K \mid I).$$

More generally, for a family $\{J_b\}_{b \in B}$,

$$\mathcal{R}_\Theta(\mathbf{J}_B \mid I) = \bigcup_{b \in B} \mathcal{R}_\Theta(J_b \mid I).$$

Proof. A pair in $\Delta_\Theta(I)$ is removed by the joint packet precisely when at least one added coordinate distinguishes the pair. This is the complement, inside $\Delta_\Theta(I)$, of the intersection of the equality kernels of all added coordinates. De Morgan's law gives the stated union. □

Corollary SDR.3.26 (Exact elimination criterion by an enrichment family)**.** *A family $\{J_b\}_{b\in B}$ eliminates the entire residual target defect of I if and only if*

$$\Delta_\Theta(I) = \bigcup_{b\in B} \mathcal{R}_\Theta(J_b \mid I).$$

Equivalently,

$$\Delta_\Theta(I \boxtimes \mathbf{J}_B) = \varnothing.$$

In that case the enriched packet is target-sufficient for Θ.

SDR.3.6 Adversarial witness packets and certification

Definition SDR.3.27 (Atomic separation witness)**.** An *atomic separation witness* against target sufficiency of $\mathbf{I}_A$ is an ordered pair

$$w = (X, Y)$$

together with certified statements

$$X, Y \in \mathcal{S}, \qquad \widehat{\mathbf{I}}_A(X) = \widehat{\mathbf{I}}_A(Y), \qquad \widehat{\Theta}(X) \neq \widehat{\Theta}(Y).$$

Equivalently, $w \in \Delta_\Theta(\mathbf{I}_A)$.

Definition SDR.3.28 (Canonical SDR.3 witness record)**.** A theorem-facing witness record has the form

$$\mathfrak{W}_{\mathrm{SDR3}} = (\mathcal{S}, \Theta, \Sigma, \mathbf{I}_A, X, Y, V_I, V_\Theta, C_I, C_\Theta, \mathsf{EvidenceMode}, \mathsf{DiscoveryUse}, \mathsf{Status}, \mathcal{A}),$$

where:

(i) Σ is the requested target strength;

(ii) V_I records the normalized observable values and V_Θ the normalized target values;

(iii) C_I is the proof or verifier certificate for observable equality;

(iv) C_Θ is the proof or verifier certificate for target discordance;

(v) EvidenceMode records whether the target evidence is independent-admissible, conditional-admissible, benchmark-quarantined, or surrogate-bound;

(vi) DiscoveryUse records whether the witness may be used upstream in independent discovery, only in conditional discovery, or only in benchmark-assisted validation/reconstruction;

(vii) Status records candidate, certified, discharged-relative-to-a-new-packet, excluded-relative-to-a-new-source-class, stale, or rejected status; and

(viii) $\mathcal{A}$ contains immutable identities, source bindings, comparison conventions, theorem locators, and replay artifacts required by the consuming claim.

No field may be inferred from a visually similar benchmark example or from a downstream conclusion.

Policy 20 (SDR-SEP-10: candidate witness is not certified witness)**.** *A pair produced by search, AI generation, numerical exploration, a toy model, or heuristic analogy has status* candidate *until source membership, observable equality, target discordance, scope, comparison conventions, and required artifacts are individually certified. Discovery value does not substitute for theorem-facing evidence. Conversely, a mathematically valid benchmark-derived target label does not become independent-discovery evidence merely because it has been serialized into a witness record; its permitted use remains controlled by Policy 19.*

Theorem SDR.3.29 (SDR-SEP-11: certified witness no-go theorem)**.** *A certified witness record for* $(X, Y) \in \Delta_\Theta(\mathbf{I}_A)$ *proves that no deterministic route whose complete source-dependent input factors through* $\widehat{\mathbf{I}}_A$ *can return different normalized target values for* X *and* Y*. Therefore such a route cannot be universally correct for* Θ *on the declared source class.*

Proof. Any route whose source-dependent input factors through $\widehat{\mathbf{I}}_A$ assigns the same internal input record to X and Y. Determinism therefore gives the same downstream value. Since the certified target values differ, that downstream value cannot equal both target values. This is Corollary SDR.2.21 expressed with an explicit witness record and consumer boundary. □

Policy 21 (Witness no-go ceiling)**.** *A certified separation witness refutes the declared target-sufficiency or route-completeness claim. It does not by itself refute the external target, prove that no different observable can solve the problem, prove that a source theorem is false, or establish novelty of any proposed repair.*

Policy 22 (Absence of found witnesses is not completeness)**.** *Failure to find a witness does not prove* $\Delta_\Theta(\mathbf{I}_A) = \varnothing$ *unless the search domain is known to be complete for the quantified source class through a separately proved exhaustive-enumeration, finite-reduction, generation, reconstruction, compactness, Noetherian, finite-state, or equivalent theorem.*

SDR.3.7 Witness discharge, restriction, and immutable history

Definition SDR.3.30 (Relative witness discharge)**.** Let $w = (X, Y) \in \Delta_\Theta(I)$ be a certified witness.

(i) An enrichment J *observationally discharges* w relative to I if $\widehat{J}(X) \neq \widehat{J}(Y)$.

(ii) A restricted source class $\mathcal{S}' \subseteq \mathcal{S}$ *excludes* w if $X \notin \mathcal{S}'$ or $Y \notin \mathcal{S}'$.

(iii) A weaker target Θ' *neutralizes* w if $\widehat{\Theta}'(X) = \widehat{\Theta}'(Y)$.

These are relative dispositions. They do not change the truth of the historical statement $w \in \Delta_\Theta(I)$ for the original source class, target, and packet.

Theorem SDR.3.31 (SDR-SEP-12: exact witness-discharge criterion)**.** *For a certified witness* $w = (X, Y) \in \Delta_\Theta(I)$ *and enrichment* J*,*

$$w \notin \Delta_\Theta(I \boxtimes J) \quad \Longleftrightarrow \quad \widehat{J}(X) \neq \widehat{J}(Y).$$

Thus an enrichment discharges that witness exactly when it separates the witnessed pair.

Proof. The pair is already target-discordant. By Theorem SDR.3.17, it survives in the joint defect exactly when it also lies in $\mathrm{EqKer}(\widehat{J})$. □

Policy 23 (SDR-SEP-13: immutable witness history)**.** *A later enrichment, source restriction, target weakening, or stronger theorem may change the current applicability of a certified witness, but it shall not rewrite the historical witness as though it had never existed. A witness record is invalidated only if one of its theorem-facing premises, identities, comparison conventions, or certificates is itself shown to be false or stale for the original run.*

Proposition SDR.3.32 (Restriction law for separation defect)**.** *For* $\mathcal{S}' \subseteq \mathcal{S}$*,*

$$\Delta_{\Theta|_{\mathcal{S}'}}(I|_{\mathcal{S}'}) = \Delta_\Theta(I) \cap (\mathcal{S}' \times \mathcal{S}').$$

Thus source restriction removes exactly the defect pairs no longer simultaneously admissible.

Proof. Combine Definition SDR.3.6 with Proposition SDR.2.42 for both observable and target equality kernels. □

Theorem SDR.3.33 (SDR-SEP-14: target-strength law for defects)**.** *If*

$$\Theta_{\text{weak}} \preceq \Theta_{\text{strong}},$$

then for every observable packet $\mathbf{I}_A$*,*

$$\Delta_{\Theta_{\text{weak}}}(\mathbf{I}_A) \subseteq \Delta_{\Theta_{\text{strong}}}(\mathbf{I}_A).$$

A packet may therefore be complete for a weaker target while retaining defects for a stronger target, but not conversely.

Proof. Use Proposition SDR.3.5 and intersect the inclusion of target-discordant relations with the fixed observable equality kernel. □

SDR.3.8 Finite defect graphs and exact finite certification

Definition SDR.3.34 (Finite separation-defect graph)**.** Let $\mathcal{S}_0 \subseteq \mathcal{S}$ be finite. The *finite separation-defect graph*

$$G_{\Theta,\mathbf{I}}(\mathcal{S}_0)$$

has vertex set $\mathcal{S}_0$ and an undirected edge $\{X, Y\}$ for each distinct pair satisfying

$$(X, Y) \in \Delta_{\Theta|_{\mathcal{S}_0}}(\mathbf{I}|_{\mathcal{S}_0}).$$

Symmetry of the defect relation makes the undirected encoding exact on the finite family.

Policy 24 (Finite defect-graph materialization contract)**.** *The graph* $G_{\Theta,\mathbf{I}}(\mathcal{S}_0)$ *is always a well-defined semantic finite graph once the maps are fixed. A theorem-facing executed* complete *graph, however, requires sufficient admissible evidence to decide the edge/nonedge status of every unordered pair of distinct vertices, or a separate theorem that produces exactly the same complete edge set. A partially evaluated pair table is not a complete defect graph. A graph whose target labels come from a benchmark remains benchmark-assisted and cannot be reclassified as an independently discovered defect graph.*

Proposition SDR.3.35 (Finite graph criterion)**.** *For finite* $\mathcal{S}_0$*, the restricted packet* $\mathbf{I}|_{\mathcal{S}_0}$ *is target-sufficient for* $\Theta|_{\mathcal{S}_0}$ *if and only if*

$$E\big(G_{\Theta,\mathbf{I}}(\mathcal{S}_0)\big) = \varnothing.$$

Proof. The graph has an edge exactly for each unordered representative of a nonempty restricted separation defect. Apply Theorem SDR.3.8. □

Definition SDR.3.36 (Finite candidate-probe coverage)**.** For a candidate probe J, define its edge-coverage set on the finite defect graph by

$$E_J := \big\{\{X, Y\} \in E(G_{\Theta,\mathbf{I}}(\mathcal{S}_0)) : \widehat{J}(X) \neq \widehat{J}(Y)\big\}.$$

Then a family of candidate probes eliminates the finite benchmark defect exactly when the union of their coverage sets equals the full edge set. Optimization of such families is deferred to SDR.7.

Policy 25 (Finite graph ceiling)**.** *Empty defect graph, full edge coverage, or minimal probe coverage on* $\mathcal{S}_0$ *proves only the corresponding finite restricted statement. It does not certify global target sufficiency, global minimality, or cross-domain completeness without a separate theorem reducing the ambient source class to the finite graph.*

SDR.3.9 SDR.3 defect record, closure, and handoff

Definition SDR.3.37 (Canonical SDR.3 defect record)**.** A theorem-facing SDR.3 defect record is

$$\mathfrak{D}_{\text{SDR3}} = (\mathcal{S}, \Theta, \Sigma, \mathbf{I}_A, C_\Theta, \Delta_\Theta(\mathbf{I}_A), \mathcal{E}, \Delta_\Theta^{\text{cert},\mathcal{E}}(\mathbf{I}_A), \text{Amb}_\Theta(\mathbf{I}_A), \mathfrak{W}, \mathfrak{R}, \mathsf{Status}, \mathcal{A}),$$

where $\mathcal{E}$ is the target-evidence/provenance policy, $\mathfrak{W}$ is the certified-witness registry, $\mathfrak{R}$ is the registry of target-relevant reduction sets for invoked candidate enrichments, and $\mathcal{A}$ contains the source identities, comparison conventions, theorem locators, and artifacts needed by downstream consumers. Infinite relations may be represented intensionally by proved predicates or theorem locators; finite materialization requires a finite representation theorem when claimed.

Theorem SDR.3.38 (SDR.3 separation-defect closure theorem)**.** *Within the deterministic set-level conventions inherited from SDR.2, SDR.3 establishes:*

(i) target insufficiency is represented exactly by the nonempty semantic relation $\Delta_\Theta(\mathbf{I}_A)$, while evidence-relative certified defect is a sound, potentially proper subset whose emptiness has no completeness meaning without an evidence-coverage theorem;

(ii) the target-oracle firewall separates semantic target discordance from target labels that are admissible for upstream discovery, and benchmark-derived labels retain benchmark-assisted provenance;

(iii) the defect is symmetric and irreflexive, need not be transitive, and localizes exactly to target-colored blocks inside ambiguous observable fibers;

(iv) observational refinement can only shrink the semantic defect, while deterministic downstream coarsening preserves every pre-existing defect witness;

(v) joint recording intersects defects exactly;

(vi) strict target-relevant gain is equivalent to removal of at least one previously unresolved target-discordant pair, and an admissibly certified witness plus a certified probe separation gives a sound positive gain certificate;

(vii) defect reduction by multiple candidate probes is the union of their pairwise coverage sets relative to the current defect;

(viii) a single certified witness refutes universal target sufficiency for the declared packet and source class, but its permitted use in independent discovery is controlled by its target-evidence provenance;

(ix) witness discharge is relative to an enrichment, source restriction, or target weakening and does not rewrite historical witness validity;

(x) source restriction and target-strength change obey exact monotonicity laws; and

(xi) finite defect graphs provide exact finite-family semantics, while executed graph completeness and globalization each require their own evidence or reduction theorem.

No statement above asserts that target labels are algorithmically available, that a useful separating probe can always be found, that candidate generation is complete, that every defect admits a low-cost repair, or that eliminating benchmark defects proves a new domain theorem.

Proof. Item (i) is Theorem SDR.3.8 together with Proposition SDR.3.3. Item (ii) is Policy 19. Item (iii) is Proposition SDR.3.7, Theorem SDR.3.11, and Proposition SDR.3.14. Item (iv) is Theorems SDR.3.16 and SDR.3.19. Item (v) is Theorem SDR.3.17. Item (vi) is Theorem SDR.3.21 and Corollary SDR.3.22. Item (vii) is Theorem SDR.3.25. Item (viii) is Theorem SDR.3.29 together with Policies 19 and 20. Item (ix) is Definition SDR.3.30, Theorem SDR.3.31, and Policy 23. Item (x) is Proposition SDR.3.32 and Theorem SDR.3.33. Item (xi) is Proposition SDR.3.35 with Policies 24 and 25. The final ceiling follows from the witness-certification, absence-of-witness, and frozen SDR.1 promotion policies. □

Remark SDR.3.39 (Next consumer: SDR.4)**.** SDR.4 shall treat a certified nonempty defect not as the end of a failed route but as a structural-discovery input. It will define missing-structure classes, counterexample-guided probe synthesis, candidate admissibility, discovery improvement, and the iterative refinement loop

$$\mathbf{I}_0 \longrightarrow \mathfrak{W}_0 \longrightarrow J_1 \longrightarrow \mathbf{I}_1 \longrightarrow \mathfrak{W}_1 \longrightarrow \cdots,$$

while preserving the rule that pair separation is discovery evidence rather than a general rigidity theorem. SDR.4 must explicitly separate a *label-available* lane, in which admissible target-discordance certificates may guide synthesis, from a *label-free or surrogate* lane, in which unknown target values cannot be queried and candidate generation must proceed from independently available structural discrepancies or proved surrogate targets.

SDR.4 Counterexample-Guided Structural Discovery

SDR.4.1 Purpose, discovery lanes, and theorem mode

Declaration SDR.4.1 (SDR.4 objective)**.** SDR.4 converts the separation-defect semantics of SDR.3 into a controlled structural-discovery process. Its purpose is not to assert that every target admits a useful new invariant, but to define when a certified unresolved distinction may legitimately drive the construction of a new observable, how a label-free run may proceed without querying the unknown target, and when a newly synthesized probe has produced a mathematically certified reduction of target ambiguity.

The canonical discovery spine is

$$\begin{aligned}\text{admissible discrepancy input} &\longrightarrow \text{structural contrast} \longrightarrow \text{candidate probe}\\ &\longrightarrow \text{probe admissibility} \longrightarrow \text{certified separation} \longrightarrow \text{defect reduction.}\end{aligned}$$

Every arrow is an independent theorem-facing obligation. Candidate generation, including AI-assisted generation, is intentionally separated from proof and from literature novelty.

Definition SDR.4.2 (SDR.4 discovery lanes)**.** A discovery run has exactly one primary lane at each theorem-bearing step.

(i) *Label-available lane*: the run consumes a certified witness

$$w = (X, Y) \in \Delta_\Theta(\mathbf{I})$$

whose DiscoveryUse field permits upstream discovery.

(ii) *Surrogate-reflected lane*: direct target values are not queried, but a separately proved surrogate Ψ certifies target discordance in the direction required by Theorem SDR.4.6.

(iii) *Label-free exploratory lane*: neither target values nor a target-reflecting surrogate are available. Candidate generation may use independently available source structure, internal inconsistencies, deformation directions, morphism data, extension data, tail data, coherence data, or other declared source observables, but no target-relevant gain is certified until a later theorem supplies admissible target relevance.

The three lanes may share candidate-generation machinery, but their theorem ceilings are different.

Policy 26 (SDR-DISC-03: lane separation and oracle provenance)**.** *A run shall not be relabeled from benchmark-assisted, surrogate-assisted, or exploratory discovery to a predeclared label-available lane after the fact. A label-available lane is not thereby an independent-recovery lane: independence depends on the provenance and admissibility of the label evidence itself. Every target label, surrogate theorem, benchmark comparison, and generated candidate retains its provenance through all downstream records. In particular, an exploratory candidate may later become theorem-bearing, but its historical generation mode is immutable.*

Definition SDR.4.3 (Canonical SDR.4 discovery input)**.** A theorem-facing discovery input is the record

$$\mathfrak{P}_{\mathrm{SDR4}} = (\mathcal{S}, \Theta, \Sigma, \mathbf{I}, \mathsf{Lane}, \mathfrak{W}, \Psi, \mathfrak{C}, \mathsf{PremiseSet}, \mathsf{OraclePolicy}, \mathcal{A}),$$

where $\mathfrak{W}$ is the admissible witness registry, Ψ is an optional surrogate datum, $\mathfrak{C}$ is an optional source-structural contrast packet, PremiseSet records exactly what mathematics is available before candidate synthesis, OraclePolicy records forbidden target information, and $\mathcal{A}$ contains immutable identities and replay artifacts. A missing field may be marked $\mathrm{not_invoked}$; it is never filled optimistically from a benchmark conclusion.

Definition SDR.4.4 (Precursor-frozen discovery context)**.** Before a run is allowed to claim independent or benchmark-quarantined discovery, it freezes a precursor context

$$\mathfrak{Q}^- = (\mathcal{S}, \Theta, \Sigma, \mathbf{I}, \mathsf{Lane}, \mathfrak{W}^-, \Psi^-, \mathfrak{C}^-, \mathsf{PremiseSet}^-, \mathsf{OraclePolicy}, \mathcal{A}^-).$$

The superscript "−" means "available before candidate generation", not mathematical negation. A theorem or artifact obtained only after a candidate has been proposed may validate, reject, or reinterpret that candidate, but it is not silently inserted into $\mathfrak{Q}^-$ when the provenance of the discovery step is later reported.

Policy 27 (SDR-DISC-03A: no retroactive premise closure)**.** *A candidate may not be advertised as independently discovered by using, after the fact, a target theorem, benchmark witness, surrogate theorem, source classification, or other theorem-bearing fact that was not contained in the precursor-frozen context actually used for synthesis. Post-candidate mathematics may strengthen the current theorem status of the candidate, but the historical discovery provenance is immutable. This is a dependency-DAG requirement, not a claim that post hoc proofs are mathematically invalid.*

SDR.4.2 Target-reflecting surrogates and label-free witness extraction

Definition SDR.4.5 (Target-reflecting surrogate)**.** A normalized observable Ψ on $\mathcal{S}$ is a *target-reflecting surrogate for discordance* if

$$\Psi \preceq \Theta.$$

Equivalently, by Theorem SDR.2.10,

$$\mathrm{EqKer}(\widehat{\Theta}) \subseteq \mathrm{EqKer}(\widehat{\Psi}).$$

Thus target equality forces surrogate equality. The terminology "reflecting" here refers only to the implication from surrogate inequality back to target inequality; it does not assert source reconstruction.

Theorem SDR.4.6 (SDR-DISC-04: surrogate discordance reflection)**.** *Let $\Psi \preceq \Theta$. Then for all $X, Y \in \mathcal{S}$,*

$$\widehat{\Psi}(X) \neq \widehat{\Psi}(Y) \quad \Longrightarrow \quad \widehat{\Theta}(X) \neq \widehat{\Theta}(Y).$$

Consequently, if an observable packet $\mathbf{I}$ also satisfies

$$\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y),$$

then

$$(X, Y) \in \Delta_{\Theta}(\mathbf{I})$$

without querying either target value directly.

Proof. Since $\Psi \preceq \Theta$, there exists $h : \mathrm{im}(\widehat{\Theta}) \to \mathrm{im}(\widehat{\Psi})$ with $\widehat{\Psi} = h \circ \widehat{\Theta}$. If the target values were equal, applying h would make the surrogate values equal. Contraposition proves the first implication. Combining surrogate inequality with packet equality gives membership in the target-discordant relation and in the packet equality kernel, hence in their intersection $\Delta_{\Theta}(\mathbf{I})$. □

Corollary SDR.4.7 (Label-free certified-witness extraction)**.** *Suppose the theorem $\Psi \preceq \Theta$, the packet equality $\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y)$, and the surrogate inequality $\widehat{\Psi}(X) \neq \widehat{\Psi}(Y)$ are all certified from premises allowed by the run's oracle policy. Then (X, Y) may be entered as a certified target-defect witness with* EvidenceMode = surrogate_reflected*, without exposing $\widehat{\Theta}(X)$ or $\widehat{\Theta}(Y)$ to the candidate generator.*

Proposition SDR.4.8 (SDR-DISC-05: wrong-way surrogate warning)**.** *The reverse information relation*

$$\Theta \preceq \Psi$$

does not in general license the implication

$$\widehat{\Psi}(X) \neq \widehat{\Psi}(Y) \Longrightarrow \widehat{\Theta}(X) \neq \widehat{\Theta}(Y).$$

Proof. Let $\mathcal{S} = \{0, 1\}$, let Θ be constant, and let Ψ be the identity map. Then $\Theta \preceq \Psi$ because a constant map factors through every observable, while $\Psi(0) \neq \Psi(1)$ and $\Theta(0) = \Theta(1)$. □

Theorem SDR.4.9 (SDR-DISC-06: exact surrogate replacement)**.** *If*

$$\Psi \preceq \Theta \qquad \textit{and} \qquad \Theta \preceq \Psi,$$

then

$$\mathrm{EqKer}(\widehat{\Psi}) = \mathrm{EqKer}(\widehat{\Theta}),$$

and for every observable packet $\mathbf{I}$,

$$\Delta_{\Psi}(\mathbf{I}) = \Delta_{\Theta}(\mathbf{I}).$$

Hence an information-equivalent surrogate may replace the target exactly for SDR.3 defect semantics at the declared set-level strength.

Proof. The two preorder relations give the two opposite equality-kernel inclusions by Theorem SDR.2.10. Equality of target-discordant relations follows by taking complements of the equality kernels inside $\mathcal{S} \times \mathcal{S}$, and intersecting with the fixed packet equality kernel gives equality of separation defects. □

Policy 28 (Surrogate theorem ceiling)**.** *A target-reflecting surrogate transfers only the exact discordance implication proved above. It does not automatically supply the target value, a decoder for the target, a categorical equivalence, an external Return theorem, or novelty of the surrogate. Exact target replacement requires the two-sided information equivalence of Theorem SDR.4.9 or another theorem at the requested strength.*

SDR.4.3 Structural contrasts and missing-structure profiles

Definition SDR.4.10 (Raw structural contrast)**.** For source objects $X, Y \in \mathcal{S}$, a *raw structural contrast* is a typed record

$$\mathfrak{c}(X, Y) = (\mathsf{Field}, \mathsf{SourceType}, \mathsf{Value}_X, \mathsf{Value}_Y, \mathsf{Relation}, \mathsf{Evidence}, \mathsf{Provenance})$$

asserting a source-side difference, obstruction, or unresolved structural asymmetry independently of any claim that the field is a valid invariant or is target-relevant. Examples of admissible field labels include scalar, object, morphism, exact/Yoneda, derived, higher-composition, tail, coherence, coefficient, and source-semantic fields.

Definition SDR.4.11 (Missing-structure profile)**.** Relative to a current packet $\mathbf{I}$ and pair (X, Y), an SDR missing-structure profile is a finite or theorem-indexed record

$$\mathfrak{M}_{\mathbf{I}}(X, Y) = (m_{\mathrm{obj}}, m_{\mathrm{mor}}, m_{\mathrm{ext}}, m_{\mathrm{der}}, m_{\mathrm{high}}, m_{\mathrm{tail}}, m_{\mathrm{coh}}, m_{\mathrm{sem}}, \ldots),$$

whose entries take values in

$$\{\mathrm{equal}, \mathrm{different}, \mathrm{unresolved}, \mathrm{not_invoked}\}.$$

An entry records only the status of the declared structural field under the supplied evidence. The profile is a search coordinate, not an assertion that the listed axes form a complete taxonomy of mathematical structure.

Policy 29 (SDR-DISC-07: raw contrast is not target evidence)**.** *A raw structural contrast may guide candidate generation, but it does not prove target discordance, target relevance, or the existence of a useful invariant. In the label-free exploratory lane, a contrast remains exploratory until a theorem connects it to an admissible target witness or proves target sufficiency of the resulting probe.*

Definition SDR.4.12 (Candidate-synthesis relation)**.** Let $\mathfrak{Q}$ denote the permitted discovery context, consisting of the current packet, admissible witnesses or contrasts, source theorems, structural profiles, and oracle policy. A *candidate-synthesis relation* is a possibly multivalued relation

$$\mathfrak{S} : \mathfrak{Q} \rightsquigarrow \mathfrak{J}_{\mathrm{cand}}$$

whose outputs are formulas, functors, probes, derived records, filtrations, higher operations, tail summaries, coherence observables, or other typed candidate observables. No totality, optimality, completeness, or uniqueness of $\mathfrak{S}$ is assumed.

Policy 30 (SDR-DISC-08: synthesis is nonproof)**.** *An output of $\mathfrak{S}$, whether produced by a human, symbolic search, numerical experiment, AI model, proof-search heuristic, or literature mining, has status* candidate*. Candidate status does not establish that the formula is well-defined, invariant, functorial, independent of forbidden target information, nonredundant with the current packet, or useful for the requested target.*

Definition SDR.4.13 (Declared source semantics and intrinsic probe)**.** Let

$$\pi_{\mathrm{src}} : \mathcal{S} \longrightarrow \overline{\mathcal{S}}_{\mathrm{src}}$$

be the declared source normalization for the invoked problem: literal identity when literal presentations are mathematically meaningful, or the quotient by the declared source equivalence when objects are meaningful only up to isomorphism, quasi-isomorphism, gauge equivalence, relabelling, or another registered relation. An

admitted probe J is *source-semantic* at the claimed strength if its normalized value factors through this source normalization,

$$\widehat{J} = j \circ \pi_{\mathrm{src}}$$

for a well-defined map j on the declared source semantics, or if an explicitly stronger theorem proves the exact invariance property consumed downstream.

Definition SDR.4.14 (Memorization separator)**.** A candidate is a *memorization separator* when its separating power depends on raw object identity, file hash, benchmark index, enumeration order, witness identifier, or a lookup table not justified as part of the declared source semantics. Such a candidate may be useful for debugging, finite classification, or implementation tests, but it is not counted as structural invariant discovery merely because it separates a defect pair.

Theorem SDR.4.15 (SDR-DISC-08A: finite memorization trivialization)**.** *Let $\mathcal{S}_0$ be a finite literal set. If arbitrary identity-dependent probes are permitted, then the family of singleton indicators*

$$\chi_X(Y) = \begin{cases} 1, & Y = X, \\ 0, & Y \neq X, \end{cases} \qquad X \in \mathcal{S}_0,$$

source-separates $\mathcal{S}_0$. Consequently its joint packet is target-sufficient for every target datum on $\mathcal{S}_0$.

Proof. For distinct $Y, Z \in \mathcal{S}_0$, the coordinate χ_Y has $\chi_Y(Y) = 1$ and $\chi_Y(Z) = 0$. Hence the joint singleton-indicator packet is injective, so Proposition SDR.2.29 applies. □

Policy 31 (SDR-DISC-08B: anti-memorization and semantic nontriviality)**.** *A memorization separator may certify a literal finite-set separation fact, but it shall not count as evidence for a structural-discovery, rigidity, source-general, Technical-Extension, or successor-Core claim unless a separate theorem shows that the allegedly identifying field is itself part of the declared mathematical source semantics and is uniformly available on the claimed source class. In particular, finite target sufficiency is not mathematically substantive merely because a finite lookup table can encode the answer.*

SDR.4.4 Probe admissibility and witness-directed effectiveness

Definition SDR.4.16 (Admissible synthesized probe)**.** A synthesized candidate J is admitted as an *SDR observable probe* only after the following fields are closed at the invoked strength:

(i) a declared source class and source type;

(ii) a declared output type and comparison/equivalence relation, with normalized map $\widehat{J}$;

(iii) a proof that J is well-defined on the declared source representatives and respects the registered source semantics at every strength consumed downstream, or a precise choice-relative policy that prevents a stronger invariant reading;

(iv) any invariance, functoriality, naturality, exactness, stability, or higher-coherence property actually consumed downstream;

(v) a source/evaluation certificate or theorem locator sufficient to replay the claimed values;

(vi) oracle provenance showing that forbidden target information is not hidden inside the definition, parameter choice, source enumeration, or training/selection channel;

(vii) a semantic-nontriviality classification distinguishing source-semantic structure from memorization or raw identifier lookup;

(viii) a failure boundary and scope ceiling; and

(ix) artifact identities required by the consuming claim.

A field not consumed by a later theorem need not be proved merely for decoration, but a consumed field may not be omitted.

Definition SDR.4.17 (Witness-effective probe)**.** Let $w = (X, Y) \in \Delta_\Theta(\mathbf{I})$ be an admissible certified witness. An admitted probe J is *effective on* w if

$$\widehat{J}(X) \neq \widehat{J}(Y).$$

It is *strictly target-relevant relative to* $\mathbf{I}$ if its reduction set $\mathcal{R}_\Theta(J \mid \mathbf{I})$ from Definition SDR.3.20 is nonempty.

Theorem SDR.4.18 (SDR-DISC-09: witness-effective probe gives strict target gain)**.** *Let $w = (X, Y) \in \Delta_\Theta(\mathbf{I})$ be an admissible certified witness and let J be an admitted probe effective on w. Then*

$$\Delta_\Theta(\mathbf{I} \boxtimes J) \subsetneq \Delta_\Theta(\mathbf{I}).$$

In particular, J is strictly target-relevant relative to $\mathbf{I}$.

Proof. By Theorem SDR.3.17, adjoining J can only shrink the defect. Since (X, Y) belongs to the old defect and J separates the pair, Theorem SDR.3.31 removes (X, Y) from the new defect. Therefore the inclusion is strict. □

Policy 32 (SDR-DISC-10: pair effectiveness is not general rigidity)**.** *Theorem SDR.4.18 certifies strict progress relative to the declared packet and witness. It does not prove that J separates every defect pair, that J is globally minimal, that J alone is target-sufficient, or that a pair-specific construction extends uniformly to a source class. Each stronger statement requires a separately quantified theorem.*

Definition SDR.4.19 (Discovery-improvement certificate)**.** A one-step SDR discovery-improvement certificate is

$$\mathfrak{G} = (\mathbf{I}, w, J, C_{\mathrm{adm}}, C_w, C_{J(w)}, \Delta\text{-relation}, \mathsf{Lane}, \mathsf{Provenance}, \mathcal{A}),$$

where C_{adm} proves probe admissibility, C_w certifies the old witness at its allowed discovery use, $C_{J(w)}$ certifies separation by J, and Δ-relation records the exact old/new defect relation. The record certifies progress, not final solution.

SDR.4.5 Label-available and label-free discovery contracts

Theorem SDR.4.20 (Label-available one-step discovery theorem)**.** *In the label-available lane, suppose $w \in \Delta_\Theta(\mathbf{I})$ is a certified witness whose* DiscoveryUse *permits upstream synthesis, and suppose the synthesis process returns an admitted probe J effective on w. Then the resulting packet $\mathbf{I} \boxtimes J$ is a certified strict target-relevant refinement of $\mathbf{I}$.*

Proof. The hypotheses make the witness admissible for synthesis and close the probe-admissibility edge. The conclusion is Theorem SDR.4.18 with provenance retained. □

Theorem SDR.4.21 (Surrogate-reflected one-step discovery theorem)**.** *In the surrogate-reflected lane, suppose $\Psi \preceq \Theta$ is certified without target-oracle leakage, and suppose*

$$\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y), \qquad \widehat{\Psi}(X) \neq \widehat{\Psi}(Y).$$

If an admitted synthesized probe J separates X and Y, then

$$\Delta_\Theta(\mathbf{I} \boxtimes J) \subsetneq \Delta_\Theta(\mathbf{I})$$

without direct use of the unknown target values in candidate generation.

Proof. Theorem SDR.4.6 converts the certified surrogate inequality into a target-defect witness. Apply Theorem SDR.4.18. □

Proposition SDR.4.22 (Exploratory-lane ceiling)**.** *In the purely label-free exploratory lane, a source contrast and an admitted probe J may prove a strict observational refinement*

$$\mathbf{I} \prec \mathbf{I} \boxtimes J,$$

but they do not by themselves prove

$$\Delta_\Theta(\mathbf{I} \boxtimes J) \subsetneq \Delta_\Theta(\mathbf{I}).$$

Proof. Strict observational refinement requires only a source pair separated by J that was not separated by $\mathbf{I}$. Target-relevant gain additionally requires that some removed pair be target-discordant. Without target evidence or a target-reflecting surrogate, that premise is unavailable. □

Policy 33 (Exploratory promotion rule)**.** *An exploratory probe may later be reclassified as target-relevant only after an admissible theorem or witness establishes the missing target relation. The later theorem changes the probe's current mathematical status but does not rewrite the historical fact that candidate generation occurred without target labels.*

SDR.4.6 Iterative refinement and exact defect dynamics

Definition SDR.4.23 (SDR discovery sequence)**.** An SDR discovery sequence is a sequence of admitted packets

$$\mathbf{I}_0, \mathbf{I}_1, \mathbf{I}_2, \ldots$$

with

$$\mathbf{I}_{n+1} = \mathbf{I}_n \boxtimes J_{n+1},$$

where each J_{n+1} is an admitted probe. Write

$$D_n := \Delta_\Theta(\mathbf{I}_n).$$

A step is *certified strict* if an admissible witness $w_n \in D_n$ is separated by J_{n+1}.

Theorem SDR.4.24 (SDR-DISC-11: exact defect recurrence)**.** *Every SDR discovery sequence satisfies*

$$D_{n+1} = D_n \cap \mathrm{EqKer}(\widehat{J}_{n+1}).$$

Equivalently,

$$D_n \setminus D_{n+1} = \mathcal{R}_\Theta(J_{n+1} \mid \mathbf{I}_n).$$

Hence (D_n) is a descending sequence of target defects.

Proof. Apply Theorem SDR.3.17 to $\mathbf{I}_n$ and J_{n+1}. The set-difference identity is Definition SDR.3.20. □

Theorem SDR.4.25 (SDR-DISC-12: certified strict progress)**.** *If the n-th step is certified strict, then*

$$D_{n+1} \subsetneq D_n.$$

Consequently, a certified strict discovery sequence cannot return to an earlier target-defect state: once a defect pair is removed, no later probe adjunction can restore it.

Proof. The selected witness belongs to D_n and is absent from D_{n+1}. Thus inclusion is strict. If a later packet had the same target defect as the pre-step packet, monotonicity of the descending chain would force restoration of a removed pair, contradicting Theorem SDR.4.24. □

Definition SDR.4.26 (Semantic defect-complete probe family)**.** A family $\mathcal{J}$ of admitted source-semantic probes is *semantically defect-complete relative to* $(\mathcal{S}, \Theta, \mathbf{I}_0)$ if for every packet $\mathbf{I}$ reachable from $\mathbf{I}_0$ by adjoining finitely many members of $\mathcal{J}$,

$$\Delta_\Theta(\mathbf{I}) \neq \varnothing$$

implies the existence of $J \in \mathcal{J}$ with

$$\mathcal{R}_\Theta(J \mid \mathbf{I}) \neq \varnothing.$$

This is a semantic existence property of the probe family. It does not assert that the unresolved pair can be certified, that a synthesis procedure can find the useful probe, or that the family is finite or computable.

Definition SDR.4.27 (Witness-complete evidence system)**.** Fix an evidence system $\mathcal{E}$ and oracle policy. It is *witness-complete on the reachable defect family* if every reachable nonempty semantic defect $\Delta_\Theta(\mathbf{I})$ contains at least one pair that $\mathcal{E}$ can certify as an admissible discovery witness without violating the oracle policy.

Definition SDR.4.28 (Certified-progress-complete discovery mechanism)**.** A discovery mechanism $(\mathcal{E}, \mathfrak{S}, \mathcal{J})$ is *certified-progress complete* relative to $(\mathcal{S}, \Theta, \mathbf{I}_0)$ if, for every reachable packet with nonempty defect, it produces an admissible certified witness w and an admitted source-semantic probe $J \in \mathcal{J}$ effective on w. This combines evidence availability, candidate selection, and probe effectiveness and is strictly stronger than semantic defect-completeness of $\mathcal{J}$ alone.

Policy 34 (SDR-DISC-12A: completeness-axis separation)**.** *Semantic probe completeness, witness certifiability, synthesis/search completeness, and executable discovery progress are distinct axes. None is inferred from another, and none is inferred from a finite collection of successful runs. In particular, a semantically defect-complete family may be unusable if the relevant witnesses are not certifiable or if no procedure can select an effective member.*

Theorem SDR.4.29 (SDR-DISC-13: finite strict-discovery termination)**.** *Let $\mathcal{S}_0$ be finite and let*

$$D_0 = \Delta_{\Theta|_{\mathcal{S}_0}}(\mathbf{I}_0|_{\mathcal{S}_0}).$$

Suppose every nonempty D_n admits a certified strict next step. Then after at most

$$|E(G_{\Theta,\mathbf{I}_0}(\mathcal{S}_0))| \le \binom{|\mathcal{S}_0|}{2}$$

strict steps, the restricted defect is empty.

Proof. Each certified strict step removes at least one undirected edge of the current finite defect graph, and Theorem SDR.4.24 prevents any removed edge from returning. The initial graph has exactly $|E(G_{\Theta,\mathbf{I}_0}(\mathcal{S}_0))|$ edges. Therefore no more than that many strict removals are possible before the edge set becomes empty. The binomial bound is the total number of unordered pairs of distinct vertices. □

Corollary SDR.4.30 (Finite discovery completeness under certified-progress completeness)**.** *On a finite source family $\mathcal{S}_0$, if the invoked discovery mechanism is certified-progress complete for the restricted problem, then it produces after finitely many strict steps a joint packet that is target-sufficient on $\mathcal{S}_0$. The same conclusion follows from the separately proved conjunction of semantic defect-completeness, witness-complete evidence, and a complete selector that returns an effective probe whenever one exists.*

Proof. Certified-progress completeness supplies a certified strict next step whenever the defect is nonempty. Apply Theorem SDR.4.29 and then Theorem SDR.3.8. The second formulation merely decomposes the same operational premise into its three noncompensating axes. □

Definition SDR.4.31 (Maximal certified-strict discovery sequence)**.** Relative to a fixed evidence system, synthesis rule, and admitted probe family, a certified-strict discovery sequence is *maximal* if it cannot be extended by one further certified strict step under those same frozen rules. Maximality is an operational property of the invoked discovery mechanism; it is not implied merely because an implementation chose to stop.

Proposition SDR.4.32 (SDR-DISC-14: descending-chain termination criterion)**.** *Let $\mathfrak{D}$ be the family of target defects reachable from $\mathbf{I}_0$ by admitted probe adjunction. If $\mathfrak{D}$, ordered by inclusion, satisfies the descending-chain condition, then every maximal discovery sequence all of whose executed steps are certified strict is finite. Its terminal defect admits no further certified strict refinement under the frozen discovery mechanism. If that mechanism is additionally certified-progress complete, the terminal defect is empty.*

Proof. A certified strict step is a strict descent in $\mathfrak{D}$ by Theorem SDR.4.25. The descending-chain condition forbids an infinite sequence of such steps, so a maximal strict sequence is finite. By maximality its terminal defect admits no further certified strict refinement under the frozen mechanism. If the terminal defect were nonempty and the mechanism were certified-progress complete, another certified strict step would exist, contradicting maximality. □

Policy 35 (SDR-DISC-15: no infinite-discovery shortcut)**.** *Outside a finite source family, and in the absence of a proved descending-chain condition, Noetherian reduction, compactness theorem, finite-generation theorem, or another explicit completion principle, an indefinitely improving discovery sequence does not imply eventual target sufficiency. In particular,*

$$D_0 \supsetneq D_1 \supsetneq D_2 \supsetneq \cdots$$

need not stabilize, and asymptotic reduction of sampled or measured defect does not imply $\bigcap_n D_n = \varnothing$.

SDR.4.7 Generalization ladder, nonredundancy, and theorem promotion

Definition SDR.4.33 (Discovery generalization ladder)**.** SDR distinguishes the following increasing theorem burdens:

(i) *pair separator*: J separates one certified witness;

(ii) *finite-family separator*: a packet eliminates all defects on one declared finite source family;

(iii) *class separator*: a quantified theorem eliminates all defects on a declared infinite or parametric source class;

(iv) *target-rigidity theorem*: the packet is proved target-sufficient, equivalently $\Theta \preceq \mathbf{I}$, on that class;

(v) *source-general structural theorem*: the result is parameterized away from its origin, shown nonredundant with frozen AK, and survives the required independent substitutions and failure tests.

No rung is inherited from a lower rung merely by repeating successful examples.

Policy 36 (SDR-DISC-16: no generalization by sample accumulation)**.** *A large number of pair separations, benchmark successes, numerical cases, or finite-family closures does not by itself prove a class separator or rigidity theorem. Generalization requires a quantified proof with explicit hypotheses and a declared source class.*

Policy 37 (SDR-DISC-17: no abstraction by erasure)**.** *Deleting problem-specific nouns, replacing a source symbol by a generic letter, or presenting a source-derived formula in category-theoretic language does not create a source-general theorem. A generic claim requires an independently stated hypothesis class, AK-owned proof, failure boundaries, and the promotion evidence required by SDR.1.*

Policy 38 (SDR-DISC-18: mathematical discovery and literature novelty are separate)**.** *An SDR run may correctly rediscover a known invariant or theorem. Mathematical validity, target relevance, AK-owner status, and literature novelty are separate coordinates. No candidate is advertised as novel until the later literature-overlap and promotion audits close that question.*

Remark SDR.4.34 (Novelty ceiling for SDR.4 set-level machinery)**.** The surrogate-contraposition theorem, finite-set termination bound, descending-chain criterion, singleton-memorization counterexample, and monotone defect recurrence are elementary consequences of the SDR.2–SDR.3 set-level calculus. SDR claims no literature novelty for those formal facts. Potentially substantive new mathematics begins only when a nontrivial source-semantic probe, rigidity theorem, completion theorem, or cross-domain structural principle is proved beyond these generic wrappers.

SDR.4.8 Failure semantics and candidate disposition

Definition SDR.4.35 (SDR.4 candidate dispositions)**.** A synthesized candidate may receive one of the following local dispositions:

(i) $\mathrm{ill_typed}$: the proposed construction has no valid declared source/output type;

(ii) $\mathrm{undefined}$: a required definition, map, theorem, or artifact is missing;

(iii) $\mathrm{circular_or_oracle_contaminated}$: the claimed discovery provenance uses a forbidden or retroactively inserted target/benchmark premise;

(iv) memorization_only: the candidate separates through raw identity, enumeration, or lookup rather than declared source-semantic structure;

(v) redundant: the candidate factors through the current packet at the claimed strength and adds no observational information;

(vi) nonseparating_for_witness: the candidate is admissible but fails to separate the invoked witness;

(vii) observational_gain_only: the candidate strictly refines source information but target relevance is unproved;

(viii) target_gain_certified: at least one admissible target defect is strictly removed;

(ix) finite_closure_certified: a declared finite-family defect is eliminated;

(x) rigidity_candidate: evidence motivates a quantified target-sufficiency theorem, not yet proved;

(xi) rejected: a theorem-facing assertion required by the candidate is proved false at the invoked scope.

These statuses are noncompensating and may change only through new evidence with preserved history.

Proposition SDR.4.36 (Candidate failure ceiling)**.** *Failure of a candidate probe to separate one witness proves only that the candidate is ineffective on that witness at the declared comparison strength. It does not prove that the mathematical structure represented by the candidate is irrelevant to every target, nor that no different parameterization or enrichment can be useful.*

Proof. Witness effectiveness is a relation among one candidate, one pair, one source class, one comparison convention, and one target context. Negating that relation does not quantify over other pairs, targets, parameterizations, or enriched constructions. □

SDR.4.9 SDR.4 canonical record, closure, and handoff

Definition SDR.4.37 (Canonical SDR.4 discovery record)**.** A theorem-facing SDR.4 record is

$$\begin{aligned}\mathfrak{R}_{\mathrm{SDR4}} := \big(&\mathfrak{P}_{\mathrm{SDR4}}, \mathfrak{Q}^{-}, \mathfrak{M}, \mathfrak{S}, \mathfrak{J}_{\mathrm{cand}}, \mathfrak{J}_{\mathrm{adm}}, \mathfrak{G},\\ &(D_n)_{n\in N}, \mathsf{SemanticClass}, \mathsf{CompletenessAxes},\\ &\mathsf{Disposition}, \mathsf{NoveltyStatus}, \mathcal{A}\big).\end{aligned}$$

where $\mathfrak{Q}^{-}$ freezes the actual precursor context used for synthesis, $\mathfrak{M}$ is the structural-contrast/missing-structure registry, $\mathfrak{S}$ is the synthesis relation actually used, $\mathfrak{J}_{\mathrm{cand}}$ and $\mathfrak{J}_{\mathrm{adm}}$ separate generated candidates from admitted probes, $\mathfrak{G}$ stores certified improvement records, SemanticClass records source-semantic versus memorization-only status, CompletenessAxes records semantic-probe, witness, synthesis, and operational-progress completeness separately, and (D_n) is included only to the extent actually materialized or theorem-defined. The index set N may be finite; no infinite execution is inferred from notation.

Theorem SDR.4.38 (SDR.4 counterexample-guided discovery closure theorem)**.** *Within the deterministic set-level scope inherited from SDR.2–SDR.3, SDR.4 establishes:*

(i) target-labelled, surrogate-reflected, and purely exploratory discovery lanes are distinct, retain immutable oracle provenance, and cannot acquire a stronger historical independence status by retroactively importing post-candidate premises;

(ii) a target-reflecting surrogate $\Psi \preceq \Theta$ converts certified surrogate discordance into certified target discordance without querying target values, while the reverse information direction is insufficient in general;

(iii) information-equivalent surrogates induce exactly the same separation defect as the target;

(iv) raw structural contrasts and missing-structure profiles may guide synthesis but are not target evidence by themselves;

(v) *synthesized formulas become SDR probes only after their declared typing, well-definedness, source-semantic status, consumed structure, oracle provenance, scope, failure, and replay obligations are closed;*

(vi) *unrestricted identity-dependent probes trivialize finite separation, so memorization-only success is quarantined from structural-discovery and promotion claims;*

(vii) *separating one admissible target-defect witness by an admitted source-semantic probe certifies a strict target-relevant refinement of the current packet;*

(viii) *in a purely exploratory lane, strict observational refinement does not imply strict target-relevant gain;*

(ix) *iterative probe adjunction obeys the exact recurrence*

$$D_{n+1} = D_n \cap \mathrm{EqKer}(\widehat{J}_{n+1}),$$

and every certified strict step removes at least one current defect pair permanently from the descending sequence;

(x) *on a finite source family, repeated certified strict progress terminates after at most the number of initial defect edges, while general infinite termination requires an independent descending-chain or completion theorem;*

(xi) *semantic probe completeness, witness certifiability, synthesis completeness, and certified-progress completeness are nonidentical quantified hypotheses and are never inferred from successful examples; and*

(xii) *pair separation, finite-family closure, class separation, target rigidity, source-general theorem status, and literature novelty remain distinct promotion levels.*

No item above asserts that a useful candidate always exists, that the synthesis relation is computable or complete, that AI generation constitutes proof, that finite discovery globalizes automatically, or that a discovered probe is new to the literature.

Proof. Items (i)–(iii) are Definition SDR.4.2, Policies 26 and 27, Theorem SDR.4.6, Proposition SDR.4.8, and Theorem SDR.4.9. Items (iv)–(v) are Definitions SDR.4.10, SDR.4.11, SDR.4.12, SDR.4.13, and SDR.4.16 together with Policies 29 and 30. Item (vi) is Theorem SDR.4.15 and Policy 31. Item (vii) is Theorem SDR.4.18. Item (viii) is Proposition SDR.4.22. Item (ix) is Theorems SDR.4.24 and SDR.4.25. Item (x) is Theorem SDR.4.29, Proposition SDR.4.32, and Policy 35. Item (xi) is Definitions SDR.4.26, SDR.4.27, and SDR.4.28 together with Policy 34. Item (xii) is Definition SDR.4.33 and Policies 36, 37, and 38. The final nonclaims follow from SDR.1's discovery, validation, no-backflow, and promotion firewalls. □

Remark SDR.4.39 (Next consumer: SDR.5)**.** SDR.5 shall classify how a missing-structure profile chooses an enriched structural level without assuming that the levels form one universal total order. It will connect the SDR discovery record to the frozen AK hierarchy of scalar, object, morphism, exact/Yoneda, derived, dg/A_∞, tail, and coherence data; prove the permitted escalation and non-escalation rules; and isolate when a stronger structural carrier is genuinely necessary rather than ornamental. The first internal positive controls shall come from existing AK cases in which object-level scalar data fail to distinguish morphisms but enriched probe action restores the missing distinction.

SDR.5 Enriched Structural Escalation

SDR.5.1 Purpose, inherited authority, and the non-total-order rule

Declaration SDR.5.1 (SDR.5 objective)**.** SDR.5 converts the missing-structure and witness records of SDR.4 into a typed decision about what additional mathematical structure, theorem strength, or semantic interface must be supplied before a target-relevant distinction can be certified. Its task is not to reward stronger terminology. It is to determine whether an unresolved target obligation requires additional scalar data, object data, morphism action, exact/Yoneda data, derived data, higher or enhanced data, tail information, coherence data, source-semantic control, or an independently justified combination of these.

The section consumes the frozen AK-HPDST v21 and Appendix HT results only at their registered strengths. In particular, the PE ladder supplies internal positive and negative controls for object-, morphism-, exact/Yoneda-, derived-, and enhanced-strength Returns. Those results remain owned by their frozen v21 sources; SDR uses them as calibration theorems and does not re-own or promote them.

Policy 39 (SDR-ENR-01: no universal total structural order)**.** *The labels*

$$\mathsf{scalar},\ \mathsf{object},\ \mathsf{morphism},\ \mathsf{exact/Yoneda},\ \mathsf{derived},\ \mathsf{enhanced},\ \mathsf{tail},\ \mathsf{coherence},\ \mathsf{source\text{-}semantic}$$

are not declared to form one universal total order. Some routes admit theorem-backed local progressions, such as the frozen finite type-A PE object–morphism–exact/Yoneda–derived chain. Other requirements, notably tail control and descent coherence, are orthogonal unless a separate theorem relates them. A stronger-sounding label therefore creates no comparison edge by itself.

Definition SDR.5.2 (Structural capability axis)**.** A *structural capability axis* is a declared theorem-facing requirement type

$$c \in \mathsf{Cap}_{\mathrm{SDR}}$$

whose semantics specify:

(i) the source class and equivalence convention on which the capability is meaningful;

(ii) the mathematical structure that must be observed, reconstructed, reflected, or certified;

(iii) the target strength exported when the capability is discharged;

(iv) every prerequisite theorem, interface, variance, coefficient, finiteness, amplitude, tail, or coherence condition; and

(v) the failure state when the requirement is absent, false, incompatible, or out of scope.

The canonical initial registry contains the capability families

$$\mathsf{pred},\ \mathsf{scalar},\ \mathsf{object},\ \mathsf{morphism},\ \mathsf{exact/Yoneda},\ \mathsf{derived},\ \mathsf{enhanced},\ \mathsf{tail},\ \mathsf{coherence},\ \mathsf{source\text{-}semantic}.$$

The registry is extensible, but adding a label does not prove a theorem relating it to any existing label.

Definition SDR.5.3 (Theorem-backed capability dependency hyperedge)**.** A theorem-backed capability dependency has the form

$$C \overset{T}{\Longrightarrow} d, \qquad C = \{c_1, \dots, c_r\} \subseteq \mathsf{Cap}_{\mathrm{SDR}},$$

where the named theorem or verified interface T proves, on its exact declared source class, that jointly available carriers discharging every capability in C supply the prerequisites needed to construct or certify capability d at the stated strength. The record stores the source class, hypotheses, direction, compatibility conditions, loss profile, target strength, and failure boundary of T. The unary case $r = 1$ may be written $c \overset{T}{\longrightarrow} d$.

Remark SDR.5.4 (Why a hypergraph rather than a simple ladder)**.** Many theorem-facing upgrades require several incomparable inputs simultaneously: for example, a descent Return may require local object data, actual morphisms, cocycle or higher-coherence cells, and a separate effectivity theorem. A binary or total-order picture can hide this conjunction. The capability architecture is therefore a directed hypergraph of typed theorem dependencies, not a universal chain.

Policy 40 (SDR-ENR-02: no terminology-to-edge promotion)**.** *Words such as* derived*,* Morita*,* dg*,* A_∞*,* stable*,* coherent*, or* descent *do not create a dependency hyperedge. Every edge requires an actual theorem or verified interface at the exact invoked source and target types.*

SDR.5.2 Structural carriers and three different meanings of escalation

Definition SDR.5.5 (Structural carrier)**.** A theorem-facing *structural carrier* is a record

$$\mathfrak{E} = (\mathsf{Obs}, \mathsf{Cap}, \mathcal{S}_{\mathfrak{E}}, \mathsf{Sem}, \mathsf{Thm}, \mathsf{Art}),$$

where:

(i) Obs is either an SDR-admissible observational payload J or the distinguished value none for a theorem-only carrier;

(ii) Cap is the set of capabilities actually proved for the carrier;

(iii) $\mathcal{S}_{\mathfrak{E}}$ is its exact source scope and equivalence regime;

(iv) Sem is its source-semantic binding, including every interface needed to reach the carrier's layer;

(v) Thm lists the theorem locators justifying the advertised capabilities; and

(vi) Art stores the replayable artifacts required by those theorems.

If $\mathsf{Obs} = J$, write $\mathfrak{E}_J$ and let $\widehat{J}$ denote its normalized observable. If $\mathsf{Obs} = \mathrm{none}$, the carrier contributes no equality-kernel refinement merely by existing.

Definition SDR.5.6 (Observational, inferential, and semantic-layer escalation)**.** Let $\mathbf{I}$ be the current packet and $\mathfrak{E}$ a candidate carrier.

(i) *Observational escalation* occurs only when $\mathsf{Obs} = J$ and

$$\mathbf{I} \prec \mathbf{I} \boxtimes J,$$

so the new observational payload strictly refines the source partition.

(ii) *Inferential or certificate escalation* occurs when the consumed observable information need not become strictly finer, but a new theorem certifies a stronger admissible conclusion, such as exactness reflection, a rigidity implication, a tail theorem, or an effectivity theorem. Such a gain is theorem-relative and is not renamed observational information gain.

(iii) *Semantic-layer escalation* occurs when the proposed carrier changes the source category, enhancement, coefficient regime, amplitude regime, tail object, or coherence model. It requires a separately proved source-to-new-layer interface and Return semantics; it is not an ordinary within-layer refinement.

The three modes may co-occur, but every theorem-facing record reports them separately.

Proposition SDR.5.7 (Ornamental escalation gives no observational gain)**.** *Suppose $\mathfrak{E}_J$ has observational payload J that is information-equivalent to a deterministic post-processing of the current packet $\mathbf{I}$. Then*

$$\mathrm{EqKer}(\widehat{\mathbf{I} \boxtimes J}) = \mathrm{EqKer}(\widehat{\mathbf{I}})$$

and hence

$$\Delta_\Theta(\mathbf{I} \boxtimes J) = \Delta_\Theta(\mathbf{I}).$$

In particular, replacing a record by a more sophisticated representation of the same consumed information does not remove a separation defect.

Proof. Information equivalence means that J factors through $\mathbf{I}$ and contributes no strictly finer equality kernel on the declared comparison quotient. By the joint-kernel intersection theorem of SDR.2, adjoining J leaves the equality kernel unchanged. SDR.3 then identifies the separation defect as the intersection of that kernel with target discordance. □

Policy 41 (SDR-ENR-03: capability strength is not information strength)**.** *A carrier may support a stronger theorem-facing capability without being strictly more informative in the SDR equality-kernel preorder, and a carrier may be strictly more informative without supporting the target capability requested by the route. Every escalation record therefore reports observational strength and theorem/capability strength separately.*

SDR.5.3 Witness visibility and when escalation is mathematically necessary

Definition SDR.5.8 (Observational capability family)**.** For a capability specification c, let

$$\mathfrak{C}_c^{\mathrm{obs}}$$

denote the declared family of admitted carriers $\mathfrak{E}_J$ whose observational payloads are defined on the invoked source scope and whose proved capability records satisfy c. A theorem-only carrier with $\mathsf{Obs} = \mathrm{none}$ is not an element of an observational family for blindness purposes. A cross-layer carrier enters such a family only after a proved interface pulls its observational payload back to the original source semantics.

Definition SDR.5.9 (Witness visibility)**.** Let

$$w = (X, Y, \Theta, \mathbf{I}, \mathcal{E})$$

be an admissible certified SDR.3 defect witness. An observational carrier family $\mathfrak{C}$ is *visible on* w if there exists $\mathfrak{E}_J \in \mathfrak{C}$ such that

$$\widehat{J}(X) \neq \widehat{J}(Y).$$

Otherwise it is *blind on* w.

Theorem SDR.5.10 (Family-blindness escalation necessity)**.** *Let $w = (X, Y, \Theta, \mathbf{I}, \mathcal{E})$ be a certified target-defect witness. Let $\mathfrak{C}$ be any family of admitted observational carriers such that*

$$\widehat{J}(X) = \widehat{J}(Y) \qquad \textit{for every } \mathfrak{E}_J \in \mathfrak{C}.$$

Then no joint observational packet formed from any subfamily $A \subseteq \mathfrak{C}$ separates w. Consequently, no deterministic route whose complete source-dependent observational input factors through $\mathbf{I}$ together with payloads from $\mathfrak{C}$ can exactly Return the witnessed target distinction.

Proof. Equality holds in every admitted observational coordinate, hence in every set-indexed product packet formed from those coordinates. Jointing such a packet with $\mathbf{I}$ still identifies X and Y, because w already satisfies $\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y)$. Since $\widehat{\Theta}(X) \neq \widehat{\Theta}(Y)$, SDR.3's indistinguishable-pair no-go applies. □

Corollary SDR.5.11 (Necessity of leaving a blind structural regime)**.** *If every admitted observational carrier in a declared regime $\mathfrak{R}$ is blind on a certified defect witness, then exact target recovery for that witness requires at least one of:*

(i) an observational carrier outside $\mathfrak{R}$ that supplies new source-semantic information;

(ii) a theorem that legitimately restricts or refines the admissible source semantics so that the certified witness pair is no longer jointly admissible for the quantified claim; or

(iii) a justified weakening or change of target strength.

Merely recombining observational carriers already in $\mathfrak{R}$ is insufficient. A theorem-only carrier does not retrospectively make the old observational packet distinguish X from Y; its role is inferential and must be typed as such.

Policy 42 (SDR-ENR-12: failed search is not certified family blindness)**.** *Failure of a finite search, heuristic synthesis run, AI proposal set, literature query, benchmark sweep, or incomplete carrier enumeration to find a separator does not establish that a structural regime is blind. The disposition* $\mathrm{blind_regime_certified}$ *requires a theorem quantifying over the declared carrier family, or a separately proved exhaustive finite reduction that makes the search complete for that family.*

Definition SDR.5.12 (Escalation request)**.** An SDR.5 escalation request is

$$\mathfrak{Q}_{\mathrm{esc}} = (w, \mathsf{EvidenceMode}, \mathbf{I}, \mathfrak{M}, \mathsf{Blind}, \mathsf{Needed}, \mathsf{Scope}, \mathsf{Budget}),$$

where w is a certified witness, a surrogate-certified witness, or an explicitly exploratory pair; EvidenceMode records which of these cases applies; $\mathfrak{M}$ is the SDR.4 missing-structure profile; Blind lists structural families already certified blind or inadequate at a declared quantifier strength; Needed is the exact target-facing capability not yet discharged; Scope fixes the source and equivalence regime; and Budget records finite certificate or complexity constraints when invoked.

Policy 43 (SDR-ENR-11: exploratory separation is not target-relevant escalation)**.** *If* EvidenceMode = exploratory*, separating the pair* $w = (X, Y)$ *certifies only a structural distinction between* X *and* Y*. It does not certify reduction of* Δ_Θ*, target relevance, or target-facing adequacy until a permitted target witness, target-reflecting surrogate, or class-wide target theorem is supplied. Such a result receives exploratory status and is excluded from the theorem-facing escalation frontier defined below.*

SDR.5.4 Escalation frontiers and minimal structural burden

Definition SDR.5.13 (Adequate carrier for an escalation request)**.** A carrier $\mathfrak{E}$ is *target-facing adequate* for $\mathfrak{Q}_{\mathrm{esc}}$ when:

(i) it is admissible on the declared source scope;

(ii) its theorem-backed capabilities discharge the requested Needed field;

(iii) if observational target relevance is invoked, the request has certified or surrogate-certified target evidence and the observational payload separates the invoked witness or strictly reduces the certified target defect at the stated strength;

(iv) if the gain is inferential, every theorem premise and direction required for the stronger conclusion is explicit;

(v) all new interfaces, coefficients, variance conventions, choice-dependence conditions, and artifacts are closed; and

(vi) its exported target strength does not exceed the strength proved by the carrier and its interfaces.

An exploratory separator may be retained as a candidate carrier but is not target-facing adequate merely because it separates an unlabeled pair.

Definition SDR.5.14 (Declared burden product)**.** Fix a finite set K of burden coordinates chosen before comparing adequate carriers. For each $k \in K$, let $(B_k, \leq_k)$ be a declared preorder and let

$$b_k : \mathfrak{A}(\mathfrak{Q}_{\mathrm{esc}}) \to B_k$$

be the corresponding burden map. Examples include assumption strength, certificate cost, computational cost, source-scope restriction, amplitude burden, interface count, or enhancement depth, provided the direction $\leq_k$ is explicitly defined. The product burden preorder is

$$\mathfrak{E} \sqsubseteq_{\mathbf{b}} \mathfrak{F} \quad \Longleftrightarrow \quad b_k(\mathfrak{E}) \leq_k b_k(\mathfrak{F}) \quad \text{for every } k \in K.$$

No scalarization, weighted sum, or total order is implicit. Observational informativeness is not treated as a cost coordinate unless a separate, explicitly oriented burden map is declared.

Definition SDR.5.15 (Escalation frontier)**.** Let $\mathfrak{A}(\mathfrak{Q}_{\mathrm{esc}})$ be the set of target-facing adequate carriers for one escalation request. Relative to a declared burden product $\mathbf{b}$, the *escalation frontier* is the set of $\sqsubseteq_{\mathbf{b}}$-minimal equivalence classes in the preorder quotient

$$\mathrm{Front}_{\mathrm{esc}}(\mathfrak{Q}_{\mathrm{esc}}; \mathbf{b}) := \min\big(\mathfrak{A}(\mathfrak{Q}_{\mathrm{esc}})/{\sim_{\mathbf{b}}}\big),$$

where

$$\mathfrak{E} \sim_{\mathbf{b}} \mathfrak{F} \quad \Longleftrightarrow \quad \mathfrak{E} \sqsubseteq_{\mathbf{b}} \mathfrak{F} \text{ and } \mathfrak{F} \sqsubseteq_{\mathbf{b}} \mathfrak{E}.$$

Representatives may be chosen for implementation, but the mathematical frontier is the set of minimal burden classes.

Proposition SDR.5.16 (Finite frontier existence)**.** *If* $\mathfrak{A}(\mathfrak{Q}_{\mathrm{esc}})$ *is finite and nonempty, then for every declared finite burden product* $\mathbf{b}$ *the escalation frontier is nonempty.*

Proof. Quotienting the finite nonempty preorder by $\sim_{\mathbf{b}}$ gives a finite nonempty partially ordered set. Every finite nonempty poset has at least one minimal element. □

Policy 44 (SDR-ENR-04: no deepest-structure default)**.** *SDR does not default to the most elaborate available mathematical carrier. If a lower-burden carrier already separates the certified witness and discharges the requested target capability, an A_∞, derived, higher-coherent, or other enhanced replacement requires an independent reason. Unneeded structure is not counted as mathematical progress.*

Policy 45 (SDR-ENR-05: incomparable minima remain incomparable)**.** *If two adequate carriers lie on different structural axes and neither dominates the other in the declared product burden preorder, SDR retains both on the frontier. It does not fabricate a total ranking by terminology, dimension, category level, or aesthetic preference.*

SDR.5.5 Positive control I: object data to morphism data in the frozen PE lane

Definition SDR.5.17 (Frozen PE arrow source and object-only observable)**.** Fix the frozen finite type-A sentinel regime in which the HT PE-2 and PE-3 theorems apply. Let

$$\mathrm{Arr}(\mathcal{S}_n) := \bigsqcup_{M,N\in\mathcal{S}_n} \mathrm{Hom}_{\mathcal{S}_n}(M,N)$$

be the tagged arrow source set. For $f : M \to N$, define the object-only arrow observable

$$I_{\mathrm{obj}}(f) := \big(\Sigma_{\mathrm{PE2}}(M), \Sigma_{\mathrm{PE2}}(N)\big),$$

where Σ_{PE2} denotes the exact frozen PE-2 object signature at its registered strength. Define the morphism target

$$\Theta_{\mathrm{mor}}(f) := [\![M, N, f]\!]_{\mathrm{mor}},$$

where the comparison relation is the exact source-morphism equivalence convention consumed by the frozen PE-3 full-faithfulness theorem. No coordinate representative or matrix basis is implied.

Proposition SDR.5.18 (PE-2 object data are blind to scalar-distinct arrows)**.** *For every nonzero indecomposable sentinel object M, the arrows*

$$0_M, \qquad \mathrm{id}_M : M \to M$$

have identical I_{obj} values but distinct Θ_{mor} values. More generally, if $\lambda \neq \mu$ in the coefficient field, then the distinct scalar endomorphisms $\lambda\,\mathrm{id}_M$ and $\mu\,\mathrm{id}_M$ have the same PE-2 object-only arrow observable.

Proof. The frozen HT PE-2 stage returns object/barcode information but no source-arrow data. Hence every endomorphism of the same source object has the same ordered pair of PE-2 object signatures. Distinct scalar endomorphisms are distinct source morphisms in the frozen sentinel category. □

Definition SDR.5.19 (PE-3 enriched arrow carrier)**.** On the same frozen sentinel source class, let $J_{\mathrm{PE3}}(f)$ be the normalized PE-3 probe-module morphism produced by the contravariant fully faithful embedding. Thus for $f : M \to N$ its arrow orientation is the frozen PE-3 orientation

$$J_{\mathrm{PE3}}(f) : \mathcal{P}_n(N) \longrightarrow \mathcal{P}_n(M),$$

with the induced module action and naturality data required by the source theorem. The reversal is part of the typed observable and is not treated as information loss.

Theorem SDR.5.20 (PE-2 to PE-3 strict target-relevant escalation)**.** *On the exact frozen PE-3 sentinel scope,*

$$I_{\mathrm{obj}} \prec I_{\mathrm{obj}} \boxtimes J_{\mathrm{PE3}}$$

on the tagged arrow source class, and the PE-3 carrier eliminates the scalar-arrow defect witnessed by 0_M and id_M. Hence the passage from PE-2 object signatures to PE-3 enriched morphism action is a genuine observational and target-relevant escalation for the morphism target.

Proof. By Proposition SDR.5.18, the object-only packet identifies 0_M and id_M although the morphism target distinguishes them. The frozen PE-3 faithfulness theorem sends distinct scalar source arrows to distinct probe-module morphisms, with the registered contravariant orientation. Therefore J_{PE3} separates the certified witness. SDR.3's witness-effective gain theorem gives strict target-defect reduction, and SDR.2's information preorder gives strict observational refinement. □

Remark SDR.5.21 (What the positive control does and does not show)**.** The theorem above is an SDR calibration of an already proved HT phenomenon. It is not a new proof of PE-3 full faithfulness, does not extend PE-3 beyond its sentinel, coefficient, finite-type, or grid-complete hypotheses, and does not imply exact/Yoneda or derived Return merely from morphism faithfulness.

SDR.5.6 Positive control II: exact/Yoneda and derived capabilities are separate gates

Proposition SDR.5.22 (Morphism Return does not silently become exact/Yoneda Return)**.** *Within the frozen HT PE regime, a PE-3 carrier that Returns source morphisms at full-faithfulness strength does not, by that fact alone, certify every short exact sequence or classify every Yoneda extension class through the internal PE-3 probe-module data. Exact/Yoneda capability remains a separately theorem-backed gate.*

Proof. The frozen HT analysis exhibits the omitted Hom boundary in the long exact sequence and explicitly separates PE-3 morphism Return from the later extension calculus. Full faithfulness therefore supplies the source arrows but not an automatic theorem that the displayed PE-3 probe-module data alone furnish the required exact/Yoneda classifier. □

Policy 46 (SDR-ENR-06: replayable structure is not classification strength)**.** *If a richer carrier allows a supplied exact diagram, cone, triangle, or coherence cell to be replayed and checked, SDR does not infer that the carrier classifies all such structures. Replay, reflection, reconstruction, and classification are separate strengths.*

Proposition SDR.5.23 (Exact/Yoneda Return does not silently become derived Return)**.** *An exact/Yoneda carrier does not by itself provide the bounded-derived cone and distinguished-triangle Return required by the frozen PE-5 derived stage. A derived escalation requires the corresponding projective-generator, chain-level, localization, cone, and triangle theorems at the invoked scope.*

Remark SDR.5.24 (Lossless derived realization still has a scope ledger)**.** The frozen PE-5 projective-generator realization is lossless on its declared bounded finite type-A regime, but this does not make amplitude bookkeeping, chosen chain representatives, cone certificates, threshold conventions, or external Return obligations disappear. Lossless reconstruction and unbounded or source-external validity are different claims.

SDR.5.7 Nonhereditary escalation and the higher-structure firewall

Definition SDR.5.25 (Nonhereditary trigger)**.** A route has a *target-relevant nonhereditary trigger* when its declared target distinction or required capability depends on data not controlled by the hereditary finite-stage package, for example a certified nonzero Ext^2, a higher extension relation, a nontrivial higher product, or another higher-composition datum whose omission is proved to identify target-distinct structures or invalidate a theorem required by the route.

Proposition SDR.5.26 (Nonheredity alone does not force target escalation)**.** *The mere existence of nonzero* Ext^2 *or another nonhereditary feature in the ambient source does not by itself imply that a declared target requires dg or* A_∞ *escalation. In particular, if the current packet* $\mathbf{I}$ *is already target-sufficient for* Θ *on the declared source class, then no higher observational carrier is necessary for target sufficiency, even though the ambient category may be nonhereditary.*

Proof. Nonzero Ext^2 invalidates hereditary simplifications at the relevant source strength, but target sufficiency is the factorization condition $\Theta \preceq \mathbf{I}$. If that condition already holds, SDR.2 shows that the target is determined by the current packet. Higher structure may still be useful for another target or another capability, but it is not logically required for the present target-sufficiency claim. □

Policy 47 (SDR-ENR-07: dg/A_∞ escalation requires a trigger and an interface)**.** *A dg or* A_∞ *carrier is admitted as a target-facing escalation only when:*

(i) a missing-structure or blindness theorem identifies a higher datum that the current carrier does not control;

(ii) the new enhancement is defined on the exact source class or is reached through a separate purpose-faithful interface;

(iii) the relevant higher operations and their equivalence notion are theorem-backed at the consumed strength;

(iv) the new carrier is shown to reduce the invoked defect or discharge the requested capability; and

(v) the Return back to the original target semantics is separately proved.

The presence of nonzero Ext^2 *may invalidate a hereditary simplification, but it does not by itself prove that one particular enhancement solves the target problem.*

Policy 48 (SDR-ENR-14: choice-dependent enhancement coordinates require invariance)**.** *If a dg, minimal-model,* A_∞*, derived, or higher-coherent carrier depends on choices of resolutions, enhancements, contractions, gauges, bases, splittings, or representatives, then theorem-facing output is taken only modulo the declared equivalence relation. Choice-dependent coordinates may be used as implementation artifacts, but they are not promoted to invariants unless a theorem proves the exact invariance or marked-choice semantics consumed downstream.*

Proposition SDR.5.27 (Cross-layer escalation is not within-layer refinement)**.** *Suppose a proposed escalation replaces a source category* $\mathcal{S}$ *by an enhanced or deformed source category* $\mathcal{E}$ *with no proved source-to-enhancement interface. Then the proposed step is undefined as an SDR target-facing escalation from the original source, even if a mathematically valid theorem is available internally in* $\mathcal{E}$*.*

Proof. Without a source-to-enhancement map and a theorem stating what source semantics it preserves, reflects, or reconstructs, the internal theorem in $\mathcal{E}$ has no typed input edge from the original route. The proof-DAG therefore does not compose. This is the same source-layer firewall already enforced in the frozen HT enhanced stages. □

Policy 49 (SDR-ENR-08: no higher-structure prestige bias)**.** *A higher categorical, dg,* A_∞*, derived, or homotopy-coherent presentation is not preferred merely because it is formally richer. SDR records it as progress only at the exact capability and defect-reduction strength proved by the route.*

SDR.5.8 Tail and coherence as orthogonal capability axes

Definition SDR.5.28 (Tail capability)**.** A *tail capability* is a theorem-backed rule that controls behavior beyond the observed finite stage, for example stabilization, Mittag–Leffler, finite-state reduction, Noetherian reduction, derived-limit control, or another named finite-to-infinite theorem. Finite object, morphism, exact, or derived reconstruction at each sampled stage is not itself a tail capability.

Definition SDR.5.29 (Coherence capability)**.** A *coherence capability* is a theorem-backed record of the compatibility cells needed to compose local or multi-object data at the requested strength, including actual cocycles, higher cells, naturality, homotopy-coherent compatibility, or a named effectivity theorem. Pairwise object agreement is not itself coherence capability.

Proposition SDR.5.30 (Orthogonal-capability noncompensation)**.** *Absent a named conversion theorem or hyperedge whose hypotheses are satisfied, a complete carrier on one structural axis cannot discharge a missing theorem-facing obligation on an orthogonal axis. In particular:*

(i) exact finite-stage reconstruction does not supply a tail theorem;

(ii) a tail theorem does not manufacture missing morphism or exact/Yoneda data at a finite stage;

(iii) pairwise local equivalence does not manufacture a cocycle or higher coherence cell; and

(iv) higher coherence does not by itself establish an external source-to-target Return.

Proof. Each item has a different theorem-facing codomain and proof obligation. Typed noncompensation from SDR.1 forbids evidence in one field from inhabiting a missing field of another type unless a named theorem provides the conversion. □

Definition SDR.5.31 (Typed cross-axis capability packet)**.** When one target requires several incomparable capabilities, SDR uses a typed capability packet

$$\mathbf{E}_{\mathrm{cap}} = \big((\mathfrak{E}_{c_i})_{i=1}^{r}, \mathsf{Compat}, \mathsf{Consumer}\big),$$

where Compat contains every compatibility cell or theorem required to consume the carriers jointly and Consumer is the theorem that names all required axes. Its operative source scope is the declared common domain

$$\mathcal{S}_{\mathbf{E}} = \bigcap_{i=1}^{r} \mathcal{S}_{\mathfrak{E}_{c_i}},$$

subject to the conjunction of component hypotheses and compatibility assumptions. If $\mathcal{S}_{\mathbf{E}}$ is empty or the hypotheses are inconsistent, the packet does not support a nonvacuous target claim.

If several components have observational payloads J_i, their observational subpacket may separately be written $\boxtimes_i J_i$. The carrier packet itself is not identified with that observational product.

Policy 50 (SDR-ENR-15: typed carrier products are not observational products)**.** *The symbol $\boxtimes$ is reserved for joint observational recording as defined in SDR.2. The combination of theorem-only, tail, coherence, semantic-layer, or mixed carriers is represented by a typed capability packet with explicit compatibility and consumer theorems. This prevents theorem obligations from being silently coerced into observable coordinates.*

Policy 51 (SDR-ENR-09: no compensation by surplus structure)**.** *Excess strength on one axis never compensates for a missing obligation on another. A highly enriched object with no tail theorem remains incomplete for a tail-sensitive target; a perfect tail theorem with no source morphism record remains incomplete for a morphism-sensitive target.*

SDR.5.9 Escalation dispositions and nonclaims

Definition SDR.5.32 (SDR.5 escalation dispositions)**.** An escalation request may receive one of the following local dispositions:

(i) no_escalation_needed: the current packet already discharges the requested capability;

(ii) observational_escalation_certified: an admitted observational carrier strictly refines information and reduces the invoked certified target defect;

(iii) inferential_escalation_certified: a new theorem strengthens the licensed conclusion without claiming new observational data;

(iv) exploratory_structural_enrichment: an exploratory pair is separated without yet establishing target relevance;

(v) cross_axis_packet_required: several incomparable capabilities are jointly required;

(vi) semantic_layer_change_required: the requested structure lives in a new source or enhancement layer and the interface obligations are explicit;

(vii) blind_regime_certified: a universal blindness theorem, or an exhaustive finite reduction, proves that the declared observational family cannot separate the witness;

(viii) ornamental_only: the proposed carrier adds terminology or representation but no relevant information or theorem strength;

(ix) undefined: a required carrier, theorem, interface, scope, compatibility cell, or artifact is missing; and

(x) rejected: a theorem-facing escalation assertion is proved false at the declared scope.

Policy 52 (SDR-ENR-10: no escalation-by-successful-example)**.** *The PE positive controls demonstrate that target-relevant escalation can be necessary and certifiable on their frozen source classes. They do not prove that the same structural level is sufficient for another category, coefficient ring, multiparameter regime, tail problem, or external problem. Every transfer requires its own theorem.*

Policy 53 (SDR-ENR-13: higher structure must be target-relevant)**.** *A failure of a lower structural theorem, a nonzero higher Ext group, or the availability of a richer enhancement is not itself evidence that the richer structure is required by the declared target. Target-facing escalation requires either certified defect reduction, a theorem-backed capability dependency needed by the target route, or a source-layer theorem showing that omission of the higher datum invalidates the requested Return.*

Declaration SDR.5.33 (SDR.5 explicit nonclaims)**.** SDR.5 does not claim:

(i) that every mathematical target admits a finite escalation frontier;

(ii) that the capability dependency hypergraph is globally acyclic, total, or complete;

(iii) that higher structure always contains more target-relevant information;

(iv) that PE-3 full faithfulness automatically supplies exact/Yoneda strength or bounded-derived strength;

(v) that a nonhereditary ambient source automatically makes higher structure target-relevant;

(vi) that a nonhereditary trigger automatically selects a unique dg or A_∞ model;

(vii) that tail or coherence obligations are reducible to the PE ladder;

(viii) that a minimal burden class or representative is unique;

(ix) that failed candidate search proves a structural regime blind;

(x) that an escalation discovered by SDR is novel in the literature; or

(xi) that any SDR.5 escalation alters the frozen v21 Core or promotes a Technical Extension result.

SDR.5.10 Canonical record, closure, and handoff

Definition SDR.5.34 (Canonical SDR.5 escalation record)**.** A theorem-facing SDR.5 record is

$$\begin{aligned}\mathfrak{R}_{\mathrm{SDR5}} := \big(&\mathfrak{Q}_{\mathrm{esc}}, \mathsf{CapHypergraph}, \mathsf{Blindness}, \mathfrak{A}, \mathbf{b}, \mathrm{Front}_{\mathrm{esc}},\\ &\mathsf{EscalationType}, \mathsf{CapabilityProofs}, \mathsf{Interfaces}, \mathsf{ChoiceInvariance},\\ &\mathsf{DefectChange}, \mathsf{Scope}, \mathsf{Compatibility}, \mathsf{Disposition}, \mathsf{NoveltyStatus}, \mathscr{A}\big).\end{aligned}$$

The record separates observational gain, theorem/certificate gain, semantic-layer change, capability coverage, target-defect change, burden coordinates, choice dependence, compatibility, and implementation artifacts. No scalar compression of these coordinates is canonical.

Theorem SDR.5.35 (SDR.5 enriched structural escalation closure theorem)**.** *Within the frozen-v21 constitutional boundary and the deterministic set-level information calculus of SDR.2–SDR.4, SDR.5 establishes:*

(i) structural capabilities are organized by theorem-backed dependency hyperedges rather than a universal total hierarchy;

(ii) observational escalation, inferential/certificate escalation, and semantic-layer escalation are distinct typed operations;

(iii) terminology or representation changes that are information-equivalent to the current packet do not reduce target separation defect;

(iv) if an entire declared observational carrier family is blind on a certified defect witness, no joint observational packet assembled only from that family can separate the witness;

(v) failed or incomplete search does not certify family blindness;

(vi) theorem-facing adequate escalations are compared by a declared product burden preorder and may have several incomparable minimal burden classes;

(vii) *the frozen PE-2 to PE-3 transition supplies a positive control in which enriched morphism action strictly removes a morphism-sensitive defect invisible to object-only signatures;*

(viii) *PE-3 morphism Return does not silently supply exact/Yoneda classification strength, and exact/Yoneda strength does not silently supply bounded-derived cone/triangle strength;*

(ix) *nonheredity by itself does not force target escalation, while a target-relevant higher-structure escalation requires an explicit trigger, source/enhancement interface, choice-invariance discipline, and Return theorem;*

(x) *tail control and coherence are orthogonal capabilities unless an explicit theorem-backed dependency relates them to another axis; and*

(xi) *mixed structural requirements are combined by a typed capability packet, not by silently treating theorem obligations as observational coordinates.*

No item above proves that every unresolved SDR.4 witness admits a successful escalation, that the escalation frontier is computable or unique, that a failed search exhausts a structural family, or that a higher structural carrier is mathematically novel.

Proof. Items (i)–(iii) are Policy 39, Definition SDR.5.3, Definitions SDR.5.5–SDR.5.6, Proposition SDR.5.7, and Policy 41. Item (iv) is Theorem SDR.5.10; item (v) is Policy 42. Item (vi) is Definitions SDR.5.13–SDR.5.15, Proposition SDR.5.16, and Policies 44–45. Item (vii) is Proposition SDR.5.18 and Theorem SDR.5.20, consuming the frozen HT PE-2/PE-3 theorems at their exact source strength. Item (viii) is Propositions SDR.5.22 and SDR.5.23 together with Policy 46. Item (ix) is Definition SDR.5.25, Proposition SDR.5.26, Policies 47 and 48, and Proposition SDR.5.27. Item (x) is Definitions SDR.5.28–SDR.5.29 and Proposition SDR.5.30. Item (xi) is Definition SDR.5.31 and Policy 50. The final ceilings are enforced by Policies 43, 52, 53, Declaration SDR.5.33, and the SDR.1 promotion firewall. □

Remark SDR.5.36 (Next consumer: SDR.6)**.** SDR.6 shall convert an admitted target-facing structural carrier or typed capability packet into a target-rigidity question. Its central task is to distinguish mere witness separation from a quantified theorem of the form

$$\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y) \Longrightarrow \widehat{\Theta}(X) = \widehat{\Theta}(Y),$$

prove the corresponding factorization, reflection, source-restriction, and failure ceilings at the exact requested strength, and isolate the additional hypotheses under which local or enriched separation can be promoted to class-wide target reconstruction.

SDR.6 Rigidity, Reflection, and Target Reconstruction

SDR.6.1 Purpose, scope, and rigidity contract

Declaration SDR.6.1 (SDR.6 objective)**.** SDR.6 converts the separation, discovery, and escalation machinery of SDR.2–SDR.5 into theorem-facing target reconstruction. Its central question is not whether a finite collection of examples has been separated, but whether a declared observable packet $\mathbf{I}$ determines a declared target Θ on an explicitly typed source class $\mathcal{S}$ at the exact requested comparison strength.

The governing implication is

$$\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y) \Longrightarrow \widehat{\Theta}(X) = \widehat{\Theta}(Y) \qquad (X, Y \in \mathcal{S}),$$

and every theorem in this section records the source class, target comparison, hypotheses, consumed observables, and conclusion strength. SDR.6 does not identify target reconstruction with source reconstruction and does not convert a local witness calculation into a class-wide theorem without a coverage argument.

Definition SDR.6.2 (Rigidity datum)**.** A *target-rigidity datum* is a tuple

$$\mathfrak{G} = (\mathcal{S}, \Theta, \mathbf{I}, \mathsf{H}, \mathsf{Eq}, \mathsf{Scope}),$$

where:

(i) $\mathcal{S}$ is the declared source class;

(ii) Θ is a normalized target datum on $\mathcal{S}$;

(iii) $\mathbf{I}$ is an admitted normalized observable packet on $\mathcal{S}$;

(iv) H is the exact list of auxiliary mathematical hypotheses, possibly empty;

(v) Eq records the source and target comparison relations from which the normalized maps were obtained; and

(vi) Scope records all coefficient, amplitude, window, finiteness, regularity, source-layer, and artifact restrictions required by the consumed theorems.

The datum is theorem-facing only when every field is typed and every nontrivial auxiliary hypothesis has an owner or source binding.

Policy 54 (SDR-RIG-04: rigidity is a quantified theorem, not a successful sample)**.** *A target-rigidity claim is class-wide only when its quantifiers cover the declared source class or when a separate completeness/reduction theorem proves that the checked family is exhaustive for that class. Finite examples, benchmark success, random testing, search saturation, or repeated failure to find a counterexample do not by themselves establish rigidity.*

Policy 55 (SDR-RIG-05: rigidity changes entitlement, not observational data)**.** *A rigidity theorem may prove that an unchanged observable packet determines a target on a declared source class. It does not thereby shrink the equality kernel of that packet on the ambient source class. Unless new observable data are explicitly added, the gain is inferential rather than observational.*

SDR.6.2 Class-wide target rigidity and exact decoder factorization

Definition SDR.6.3 (Target rigidity)**.** The packet $\mathbf{I}$ is Θ-*rigid on* $\mathcal{S}$ if

$$\mathrm{EqKer}(\widehat{\mathbf{I}}) \subseteq \mathrm{EqKer}(\widehat{\Theta}).$$

Equivalently, every two admissible sources indistinguishable by the entire packet have equal normalized target values. We write

$$\mathrm{Rig}_{\Theta}(\mathbf{I}; \mathcal{S}).$$

The notation records target-relative rigidity only; it does not assert injectivity of $\widehat{\mathbf{I}}$ on source objects.

Theorem SDR.6.4 (SDR-RIG-06: rigidity–factorization equivalence)**.** *For a normalized target* Θ *and normalized packet* $\mathbf{I}$ *on* $\mathcal{S}$*, the following are equivalent:*

(i) $\mathrm{Rig}_{\Theta}(\mathbf{I}; \mathcal{S})$*;*

(ii) $\Delta_{\Theta}(\mathbf{I}) = \varnothing$*;*

(iii) $\Theta \preceq \mathbf{I}$ *in the deterministic information preorder;*

(iv) there exists a unique map on the realized packet image

$$\mathcal{R}_{\Theta|\mathbf{I}} : \mathrm{im}(\widehat{\mathbf{I}}) \longrightarrow \mathrm{im}(\widehat{\Theta})$$

such that

$$\widehat{\Theta} = \mathcal{R}_{\Theta|\mathbf{I}} \circ \widehat{\mathbf{I}}.$$

Thus class-wide target rigidity is exactly the existence of a deterministic target decoder on the realized packet image.

Proof. The equivalence of (i), (ii), and (iii) is the SDR.2–SDR.3 target-sufficiency calculus. If (i) holds and $y = \widehat{\mathbf{I}}(X)$, define

$$\mathcal{R}_{\Theta|\mathbf{I}}(y) := \widehat{\Theta}(X).$$

Rigidity makes this well-defined. The displayed factorization follows immediately. Conversely, any such factorization forces equal packet values to have equal target values. Uniqueness holds because every point in the domain of the decoder is realized by some $X \in \mathcal{S}$. □

Definition SDR.6.5 (Rigidity decoder)**.** When Theorem SDR.6.4 holds, the unique map

$$\mathcal{R}_{\Theta|\mathbf{I}}$$

on $\mathrm{im}(\widehat{\mathbf{I}})$ is the *rigidity decoder* for the declared datum. Its domain is the realized image, not an arbitrary ambient codomain of $\widehat{\mathbf{I}}$.

Policy 56 (SDR-RIG-07: no arbitrary decoder extension)**.** *A rigidity decoder is certified only on* $\mathrm{im}(\widehat{\mathbf{I}})$ *unless an additional theorem extends it to a larger ambient space. An interpolation, formula, classifier, neural model, or symbolic expression defined outside the realized image is not part of the rigidity theorem merely because it agrees on observed values.*

Proposition SDR.6.6 (Rigidity composition)**.** *Let M be a normalized intermediate datum on the same source class. If*

$$M \preceq \mathbf{I} \qquad \text{and} \qquad \Theta \preceq M,$$

then

$$\Theta \preceq \mathbf{I}.$$

Equivalently, a theorem-backed reconstruction of M from $\mathbf{I}$ followed by a theorem-backed reconstruction of Θ from M composes to target rigidity.

Proof. The deterministic information preorder is transitive. Equivalently, compose the two realized-image decoders. □

Policy 57 (SDR-RIG-08: every decoder edge keeps its own hypotheses)**.** *In a composed rigidity route, hypotheses do not disappear under composition. The composite conclusion is licensed only on the intersection of source scopes and under the union of all nonredundant assumptions required by the component decoder theorems.*

SDR.6.3 Conditional rigidity, source restriction, and the membership gate

Definition SDR.6.7 (Hypothesis-restricted source class)**.** Let H be a predicate on the ambient source class $\mathcal{S}$. Define

$$\mathcal{S}_H := \{X \in \mathcal{S} : H(X)\}.$$

A theorem proving

$$\mathrm{Rig}_{\Theta|_{\mathcal{S}_H}}(\mathbf{I}|_{\mathcal{S}_H}; \mathcal{S}_H)$$

is called *conditional rigidity under H*. The condition H is part of the theorem, not metadata.

Theorem SDR.6.8 (SDR-RIG-09: restriction monotonicity)**.** *If* $\mathrm{Rig}_\Theta(\mathbf{I}; \mathcal{S})$ *holds and $\mathcal{S}' \subseteq \mathcal{S}$, then*

$$\mathrm{Rig}_{\Theta|_{\mathcal{S}'}}(\mathbf{I}|_{\mathcal{S}'}; \mathcal{S}')$$

holds. The converse fails in general.

Proof. Restrict the equality-kernel inclusion to $\mathcal{S}' \times \mathcal{S}'$. For failure of the converse, take an ambient pair that is packet-indistinguishable but target-distinct and exclude one member from $\mathcal{S}'$. □

Theorem SDR.6.9 (SDR-RIG-10: membership-gated use of restricted rigidity)**.** *Suppose rigidity is proved only on $\mathcal{S}_H$. For a source object $X \in \mathcal{S}$, the decoder licensed by that theorem may be applied to X only after a valid theorem, certificate, or source assumption establishes $H(X)$ at the strength required by the rigidity theorem. Without such membership evidence, the restricted decoder is undefined for theorem-facing Return.*

Proof. The decoder in Theorem SDR.6.4 is defined from the restricted map $\widehat{\mathbf{I}}|_{\mathcal{S}_H}$. If membership is not established, X is not certified to lie in its source domain. Applying the decoder would therefore enlarge the theorem domain without proof. □

Policy 58 (SDR-RIG-11: conditional assumptions propagate to Return)**.** *If rigidity is conditional on $H_1, \dots, H_r$, every downstream Return that consumes that rigidity remains conditional on those hypotheses until a separate theorem discharges them. Successful decoding on examples does not discharge a source assumption.*

Remark SDR.6.10 (Rigidity as ambiguity elimination)**.** Source restriction does not add a new coordinate to $\mathbf{I}$. It removes source pairs from the admissible quantifier domain. This is the exact sense in which a rigidity theorem can recover a target distinction without contradicting the v21 loss principle: the theorem either restricts the source class or supplies a genuinely new mathematical relation, rather than extracting new observational bits from downstream syntax alone.

SDR.6.4 One-sided predicate reflection and exact binary Return

Definition SDR.6.11 (Binary target and binary detector)**.** Let

$$\Theta_{\mathrm{bin}}, D : \mathcal{S} \longrightarrow \{0,1\}$$

be normalized binary maps. For $b \in \{0,1\}$ define:

$$\mathsf{Sound}_b(D, \Theta_{\mathrm{bin}}) :\Longleftrightarrow \forall X \in \mathcal{S},\ D(X) = b \Rightarrow \Theta_{\mathrm{bin}}(X) = b,$$
$$\mathsf{Reflect}_b(D, \Theta_{\mathrm{bin}}) :\Longleftrightarrow \forall X \in \mathcal{S},\ \Theta_{\mathrm{bin}}(X) = b \Rightarrow D(X) = b.$$

The four directional fields are retained separately for provenance even though, for a total classical binary map, some are contrapositively equivalent.

Theorem SDR.6.12 (SDR-RIG-12: exact binary reconstruction criterion)**.** *For total binary maps D and Θ_{bin}, the following are equivalent:*

(i) $D = \Theta_{\mathrm{bin}}$ *on* $\mathcal{S}$*;*

(ii) Sound_0 *and* Sound_1 *both hold;*

(iii) $\mathsf{Reflect}_0$ *and* $\mathsf{Reflect}_1$ *both hold;*

(iv) D *is target-rigid for* Θ_{bin} *and the rigidity decoder is the identity on the realized binary values.*

Proof. Items (i)–(iii) are direct truth-value case analysis. If (i) holds, equality of detector values forces equality of target values and the realized-image decoder is the identity. Conversely, (iv) with the identity decoder gives $\Theta_{\mathrm{bin}} = D$. □

Policy 59 (SDR-RIG-13: one-sided reflection has a one-sided Return ceiling)**.** *A theorem proving only one directional predicate implication is serialized only at that directional strength. In particular, a zero-reflection theorem, a nonvanishing detector, a sufficient obstruction, or a necessary condition is not silently rewritten as a biconditional or an exact binary classifier.*

SDR.6.5 Partial Return, target coarsening, and strength ceiling

Definition SDR.6.13 (Target coarsening)**.** Let $\Theta : \mathcal{S} \to \mathcal{T}$ be a normalized target and let

$$q : \mathcal{T} \longrightarrow \mathcal{U}$$

be a declared deterministic coarsening map. The composed target

$$\Theta_q := q \circ \Theta$$

is a *partial Return target* relative to Θ. Typical examples include a predicate, graded piece, determinant, ideal, semisimplification, stable class, or other explicitly typed quotient of the requested information.

Theorem SDR.6.14 (SDR-RIG-14: partial-Return rigidity ceiling)**.** *Assume a theorem proves*

$$\Theta_q \preceq \mathbf{I}$$

and a theorem-facing negative certificate proves

$$\Theta \not\preceq \mathbf{I}.$$

Equivalently for the second clause, one may supply a certified pair in $\Delta_\Theta(\mathbf{I})$*. Then* $\mathbf{I}$ *is rigid for the partial target* Θ_q *but is not rigid for the full target* Θ*. At the mathematical information level, every deterministic downstream route consuming only* $\mathbf{I}$ *may claim at most the certified partial Return and may not claim the full Return on the same declared source class. Executable production of even the partial Return remains subject to the effectivity requirements introduced below.*

Proof. The first statement is target rigidity by Theorem SDR.6.4. The certified failure of $\Theta \preceq \mathbf{I}$ rules out a full target decoder. SDR.2 downstream information monotonicity prevents deterministic post-processing of $\mathbf{I}$ from repairing the certified missing distinction. □

Policy 60 (SDR-RIG-14A: unresolved full rigidity is not certified failure)**.** *Failure to prove* $\Theta \preceq \mathbf{I}$ *is not a proof of* $\Theta \not\preceq \mathbf{I}$*. A partial-Return ceiling is theorem-facing only when the stronger target failure is supported by a certified rigidity counterexample or another proof of non-factorization. Search exhaustion, absent proof, or present implementation limits leave the stronger rigidity status unresolved rather than false.*

Definition SDR.6.15 (Rigidity spectrum)**.** For a fixed packet $\mathbf{I}$ and a declared family $\mathfrak{T}$ of candidate target strengths on the same source class, define

$$\mathrm{RigSpec}_{\mathfrak{T}}(\mathbf{I}) := \{\Theta' \in \mathfrak{T} : \Theta' \preceq \mathbf{I}\}.$$

This set is the *rigidity spectrum* of $\mathbf{I}$ relative to $\mathfrak{T}$. It need not possess a unique strongest element because target strengths may be incomparable.

Proposition SDR.6.16 (Downward closure of the rigidity spectrum)**.** *If* $\Theta_1 \in \mathrm{RigSpec}_{\mathfrak{T}}(\mathbf{I})$ *and* $\Theta_0 \preceq \Theta_1$*, then* $\Theta_0 \preceq \mathbf{I}$*. Hence every declared weakening of a rigid target is also rigid.*

Proof. Apply transitivity of the information preorder. □

Policy 61 (SDR-RIG-15: no strongest-target fiction)**.** *SDR reports the proved rigidity spectrum or its certified subset. It does not invent a scalar "maximum Return strength" when the declared target strengths are incomparable. A strongest rigidity ceiling may be named only when the target-strength order and the existence of such a greatest supported element are themselves established.*

SDR.6.6 Enriched rigidity and inherited positive controls

Definition SDR.6.17 (Enriched target-rigidity carrier)**.** An admitted SDR.5 structural carrier $\mathfrak{E}$ is an *enriched rigidity carrier* for Θ when its observational component, together with any separately typed theorem-only capabilities consumed by the route, yields a theorem

$$\mathrm{Rig}_\Theta(\mathbf{I} \boxtimes \mathsf{Obs}(\mathfrak{E}); \mathcal{S})$$

on the exact common source scope. If $\mathsf{Obs}(\mathfrak{E}) = \mathrm{none}$, the theorem is recorded as inferential rigidity of the unchanged packet rather than observational enrichment.

Theorem SDR.6.18 (SDR-RIG-16: inherited PE-3 morphism rigidity control)**.** *On the exact frozen sentinel source class and coefficient regime of the HT PE-3 theorem, the normalized PE-3 record consisting of the endpoint object records together with the induced probe-module morphism is rigid for the source-morphism target at the declared contravariant categorical comparison strength. For fixed registered source and target objects, equality of the complete PE-3 arrow records implies equality of the corresponding source morphisms; across object representatives, the conclusion is only the declared categorical morphism class transported through the registered object identifications.*

Proof. The frozen PE-3 functor is fully faithful on the declared sentinel source category. Faithfulness gives injectivity on each Hom set once the source and target objects are fixed, while the registered object identification together with full faithfulness supplies the categorical source-arrow correspondence. Hence the equality kernel of the normalized endpoint-plus-arrow record is contained in the equality kernel of the declared source-morphism target. The contravariant arrow orientation is part of the typed target convention and does not weaken the conclusion. No canonical coordinate or matrix representative is inferred. □

Theorem SDR.6.19 (SDR-RIG-17: inherited PE-5 derived rigidity control)**.** *On the exact frozen bounded finite type-A PE-5 regime, the projective-generator derived equivalence is rigid for each Return coordinate actually covered by the frozen PE-5 theorem: bounded derived object isomorphism class, derived morphism, shifted Hom/Ext data, composition and identities, shift, and the declared cone/triangle strength. Cone rigidity is understood at triangle-isomorphism strength globally and at strict chain-level strength only relative to the selected chain representatives and chain map recorded by the certificate. It does not thereby return a unique chain representative, canonical basis, or source-external target.*

Proof. The frozen PE-5 realization is an exact derived equivalence on its declared scope, hence it is fully faithful and essentially surjective onto its stated target. The frozen theorem explicitly returns the listed derived coordinates and preserves and reflects distinguished triangles at the declared strength. Its strict cone statement is representative-relative, whereas the abstract cone is returned only up to the registered triangle-isomorphism class. The inherited choice and source-external ceilings therefore remain controlling. □

Policy 62 (SDR-RIG-18: inherited controls do not universalize)**.** *The PE-3 and PE-5 controls certify that class-wide rigidity can be realized in explicit finite categories. They do not establish analogous rigidity for arbitrary quivers, coefficient rings, nonhereditary enhancements, multiparameter persistence, infinite categories, arithmetic sources, PDE sources, or external targets. Every transfer requires a purpose-faithful interface and its own rigidity theorem.*

SDR.6.7 Target reconstruction versus source reconstruction

Definition SDR.6.20 (Source-reconstruction target)**.** Fix a declared source equivalence relation $\simeq_{\mathrm{src}}$ and quotient map

$$q_{\mathrm{src}} : \mathcal{S} \longrightarrow \mathcal{S}/\!\simeq_{\mathrm{src}}.$$

The target

$$\Theta_{\mathrm{src}} := q_{\mathrm{src}}$$

is the *source-reconstruction target* at that equivalence strength. An observable packet is source-reconstructive exactly when it is Θ_{src}-rigid.

Proposition SDR.6.21 (Source reconstruction criterion)**.** *The packet* $\mathbf{I}$ *reconstructs sources up to* $\simeq_{\mathrm{src}}$ *if and only if*

$$\mathrm{EqKer}(\widehat{\mathbf{I}}) \subseteq \mathrm{EqKer}(q_{\mathrm{src}}).$$

Equivalently, the source-equivalence class factors uniquely through the realized packet image.

Proposition SDR.6.22 (Target rigidity can be strictly weaker than source reconstruction)**.** *Target rigidity for a fixed* Θ *does not imply source reconstruction. In particular, if* Θ *is constant on two source-inequivalent objects* X *and* Y*, an observable may identify* X *and* Y *and still be perfectly rigid for* Θ*.*

Proof. Target rigidity forbids only packet fibers containing two distinct target values. It imposes no requirement to separate two source objects lying in the same target fiber. Thus

$$\mathrm{EqKer}(\widehat{\mathbf{I}}) \subseteq \mathrm{EqKer}(\widehat{\Theta})$$

may hold while inclusion into $\mathrm{EqKer}(q_{\mathrm{src}})$ fails. □

Policy 63 (SDR-RIG-19: purpose-relative rigidity is not source identity)**.** *A successful target decoder is interpreted only at its declared target strength. Predicate rigidity is not object reconstruction; object rigidity is not morphism reconstruction; morphism rigidity is not exact or derived reconstruction; determinant or ideal rigidity is not module reconstruction; and source-internal rigidity is not an external-domain theorem without the required Return interface.*

SDR.6.8 Rigidity failure, counterexamples, and impossibility certificates

Definition SDR.6.23 (Rigidity counterexample)**.** A *rigidity counterexample* for $(\Theta, \mathbf{I}, \mathcal{S})$ is a certified pair

$$(X, Y) \in \mathcal{S} \times \mathcal{S}$$

satisfying

$$\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y), \qquad \widehat{\Theta}(X) \neq \widehat{\Theta}(Y).$$

It is therefore an element of $\Delta_\Theta(\mathbf{I})$ with theorem-facing evidence for both sides of the comparison.

Theorem SDR.6.24 (SDR-RIG-20: rigidity failure theorem)**.** *If a rigidity counterexample exists, then:*

(i) $\mathrm{Rig}_\Theta(\mathbf{I}; \mathcal{S})$ *is false;*

(ii) no deterministic rigidity decoder on $\mathrm{im}(\widehat{\mathbf{I}})$ *exists for* Θ*;*

(iii) no deterministic downstream observable depending only on $\mathbf{I}$ *can become target-rigid for* Θ *on the same source class; and*

(iv) any valid repair must either add target-relevant observational information, replace or weaken the target, or change the admissible source class through a theorem-backed restriction that genuinely excludes at least one member of the certified counterexample. A theorem that leaves the same source class, target, and consumed observable packet unchanged cannot override the counterexample.

Proof. Items (i) and (ii) are the contrapositive direction of Theorem SDR.6.4. Item (iii) is SDR.2 downstream information monotonicity. Item (iv) enumerates the typed escape routes compatible with the equality-kernel obstruction and the frozen-v21 loss principle. □

Definition SDR.6.25 (Rigidity impossibility certificate)**.** A rigidity impossibility certificate is

$$\mathfrak{C}_{\mathrm{fail}} = (X, Y, \mathsf{PacketEq}, \mathsf{TargetNeq}, \mathsf{Scope}, \mathsf{Evidence}, \mathsf{Provenance}),$$

where PacketEq certifies equality of every consumed observable field, TargetNeq certifies the declared target distinction, and the remaining fields fix the source scope and evidence identity. It falsifies the corresponding universal rigidity claim and no stronger target statement unless the target quantifiers independently justify that strengthening.

Policy 64 (SDR-RIG-21: weakest exact negative conclusion)**.** *A rigidity counterexample rejects the declared packet-to-target rigidity theorem on the source class containing the pair. It does not prove the target false, does not prove that no alternative observable or theorem can solve the target, and does not prove that a narrower source class cannot be rigid.*

SDR.6.9 Finite verification, coverage, and universalization boundary

Definition SDR.6.26 (Finite rigidity verification set)**.** For a finite source set $\mathcal{S}_0 \subseteq \mathcal{S}$, define

$$\mathcal{P}_{\mathbf{I}}(\mathcal{S}_0) := \left\{\{X, Y\} \subseteq \mathcal{S}_0 : X \neq Y,\ \widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y)\right\}.$$

A *complete finite rigidity check on* $\mathcal{S}_0$ certifies, for every pair in $\mathcal{P}_{\mathbf{I}}(\mathcal{S}_0)$, equality of normalized target values.

Theorem SDR.6.27 (SDR-RIG-22: exact finite-class rigidity certificate)**.** *If* $\mathcal{S}_0$ *is finite and a complete finite rigidity check on* $\mathcal{S}_0$ *passes, then*

$$\mathrm{Rig}_{\Theta|_{\mathcal{S}_0}}(\mathbf{I}|_{\mathcal{S}_0}; \mathcal{S}_0)$$

holds exactly.

Proof. Every packet-indistinguishable unequal pair in $\mathcal{S}_0$ occurs in $\mathcal{P}_{\mathbf{I}}(\mathcal{S}_0)$ and is certified target-equal. Equal pairs are trivially target-equal. Hence the required equality-kernel inclusion holds on the finite source set. □

Policy 65 (SDR-RIG-23: finite verification is not ambient rigidity)**.** *Theorem SDR.6.27 proves rigidity only on $\mathcal{S}_0$. Extension to an infinite, continuous, unbounded, or otherwise larger source class requires a separate theorem such as generation, compactness plus a valid continuity principle, Noetherian reduction, finite-state reduction, descent/effectivity, algebraic classification, or another explicit coverage mechanism. The name of such a mechanism is not a substitute for its hypotheses and proof.*

Definition SDR.6.28 (Rigidity coverage theorem)**.** A *rigidity coverage theorem* for a verification family $\mathcal{V} \subseteq \mathcal{S}$ is a proved implication of the form

$$\mathrm{Rig}_{\Theta|_{\mathcal{V}}}(\mathbf{I}|_{\mathcal{V}}; \mathcal{V}) \quad + \quad \mathsf{CoverageHypotheses} \Longrightarrow \mathrm{Rig}_{\Theta}(\mathbf{I}; \mathcal{S}).$$

The theorem must identify the mathematical reason why every possible ambient packet fiber is controlled by the verified family. Merely calling $\mathcal{V}$ a basis, benchmark, generator set, dense sample, training set, or representative family does not supply coverage.

Policy 66 (SDR-RIG-24: no universalization by search exhaustion)**.** *Failure to discover an ambient rigidity counterexample after exhaustive search over a bounded database or a finite symbolic grammar is a statement about that search domain. It becomes an ambient rigidity theorem only when search completeness with respect to the mathematical source class has itself been proved.*

SDR.6.10 Nonvacuity, decoder effectivity, and theorem-facing applicability

Proposition SDR.6.29 (SDR-RIG-25: vacuous rigidity on an empty source class)**.** *If $\mathcal{S} = \varnothing$, then*

$$\mathrm{Rig}_{\Theta}(\mathbf{I}; \mathcal{S})$$

holds for every formally typed target Θ and packet $\mathbf{I}$ on that empty domain. Consequently, mathematical truth of the rigidity implication alone does not imply that the theorem applies to any intended source object or problem family.

Proof. The defining universal implication quantifies over $\mathcal{S} \times \mathcal{S}$, which is empty. Hence the equality-kernel inclusion holds vacuously. The second statement follows because no source membership or problem coverage has been established. □

Definition SDR.6.30 (Rigidity applicability record)**.** A theorem-facing rigidity application carries

$$\mathsf{App}_{\mathrm{rig}} = (\mathsf{Nonempty}, \mathsf{Membership}, \mathsf{ProblemCoverage}, \mathsf{HypothesisDischarge}, \mathsf{Artifacts}),$$

where Nonempty records inhabitedness of the theorem domain, Membership certifies the intended source object or family lies in that domain, ProblemCoverage binds the domain to the requested external or internal target family, and HypothesisDischarge records which conditional assumptions are proved rather than merely assumed. These fields are logically distinct from the rigidity theorem itself.

Policy 67 (SDR-RIG-26: no vacuous or irrelevant promotion)**.** *A vacuously true rigidity statement, or a nonempty restricted-source theorem with no certified membership or coverage for the intended source, is not serialized as a solution of the requested target problem. It may be retained as a mathematically correct restricted statement, but theorem-facing Return requires the corresponding applicability fields.*

Definition SDR.6.31 (Effective rigidity decoder)**.** A rigidity decoder is *effective in a declared computation model $\mathfrak{M}$* when, in addition to the mathematical factorization of Theorem SDR.6.4, there is a certified algorithm $A_{\mathfrak{M}}$ that terminates on every admissible encoded input and satisfies, for every admissible source X,

$$A_{\mathfrak{M}}(\widehat{\mathbf{I}}(X)) = \widehat{\Theta}(X),$$

and the record contains termination/totality, input representation, equality/comparison, resource-model, and artifact obligations required by the declared use. Effective rigidity is therefore a strengthening of set-theoretic target rigidity, not a synonym for it.

Policy 68 (SDR-RIG-27: factorization does not imply an executable decoder)**.** *The unique set-theoretic decoder on* $\mathrm{im}(\widehat{\mathbf{I}})$ *supplied by rigidity does not by itself provide a formula, decidable equality test, algorithm, complexity bound, machine representation, or verifier. Any claim of executable or complexity-bounded Return must separately establish the corresponding effectivity fields.*

Theorem SDR.6.32 (SDR-RIG-28: discharge of a conditional source restriction)**.** *Let* H *be a predicate on* $\mathcal{S}$*. If rigidity holds on* $\mathcal{S}_H$ *and a theorem proves*

$$\forall X \in \mathcal{S}, \quad H(X),$$

then $\mathcal{S}_H = \mathcal{S}$ *and the conditional rigidity theorem upgrades to ambient rigidity on* $\mathcal{S}$*. If membership is proved only for one source* X_0*, the result licenses application only to* X_0 *and does not upgrade the class-wide theorem.*

Proof. The universal hypothesis gives $\mathcal{S} \subseteq \mathcal{S}_H$, while the reverse inclusion is definitional. Thus the two source classes coincide and the restricted kernel-inclusion theorem is the ambient one. The final statement follows because one-point membership does not establish equality of source classes. □

Policy 69 (SDR-RIG-28A: a certified counterexample cannot be repaired by theorem relabeling)**.** *Once* PacketEq *and* TargetNeq *are certified for a pair in the declared source class, no additional theorem that leaves the source class, target semantics, and consumed observable packet unchanged can establish rigidity for that same datum. A valid repair must alter at least one of those typed ingredients in a theorem-justified way.*

SDR.6.11 Rigidity status, canonical record, and explicit nonclaims

Definition SDR.6.33 (SDR.6 rigidity dispositions)**.** A theorem-facing SDR.6 request receives one primary local disposition:

(i) rigid: class-wide target rigidity is proved on the declared source scope;

(ii) conditionally_rigid: rigidity is proved under explicit unresolved hypotheses;

(iii) restricted_rigid: rigidity is proved only on a certified proper source subclass;

(iv) partial_return_rigid: a declared target coarsening is rigid while the stronger target is not certified;

(v) one_sided_reflection: only a directional predicate implication is proved;

(vi) finite_class_rigid: exact rigidity is proved only on a finite declared source set;

(vii) vacuous_or_inapplicable: the implication is formally true on an empty or irrelevant source scope, or applicability to the intended source has not been established;

(viii) counterexample_found: a certified rigidity counterexample rejects the requested universal claim;

(ix) undefined: a source binding, comparison quotient, observable, theorem, membership proof, coverage theorem, or target binding required to evaluate rigidity is missing; and

(x) rejected: the requested theorem-facing rigidity statement is certified false at its declared scope.

Definition SDR.6.34 (Canonical SDR.6 rigidity record)**.** A canonical SDR.6 record is

$$\begin{aligned}\mathfrak{R}_{\mathrm{SDR6}} := \big(&\mathcal{S}, \Theta, \mathbf{I}, \mathsf{H}, \mathsf{Eq}, \mathsf{Scope}, \mathsf{RigidityStatement},\\ &\mathsf{Decoder}, \mathsf{ReflectionDirections}, \mathsf{PartialTargets}, \mathsf{RigiditySpectrum},\\ &\mathsf{Counterexamples}, \mathsf{Coverage}, \mathsf{Nonvacuity}, \mathsf{Applicability}, \mathsf{DecoderEffectivity},\\ &\mathsf{SourceReconstructionCeiling}, \mathsf{Evidence}, \mathsf{Disposition}, \mathsf{NoveltyStatus}, \mathcal{A}\big).\end{aligned}$$

The record keeps observational content, theorem hypotheses, decoder semantics, directional reflection, target strength, failure evidence, coverage, nonvacuity, theorem-facing applicability, decoder effectivity, source-reconstruction strength, novelty, and artifacts in separate fields.

Declaration SDR.6.35 (SDR.6 explicit nonclaims)**.** SDR.6 does not claim:

(i) that every admitted observable packet admits a rigidity theorem;

(ii) that a target-rigid packet reconstructs the source;

(iii) that a rigidity decoder extends canonically outside the realized packet image;

(iv) that restricted or conditional rigidity holds on the ambient source class;

(v) that one-sided soundness or reflection is an exact biconditional;

(vi) that a rigid partial Return implies a stronger object, morphism, exact, derived, or source Return;

(vii) that the rigidity spectrum has a greatest element;

(viii) that PE-3 or PE-5 rigidity transfers outside their frozen hypotheses;

(ix) that finite verification establishes infinite or unbounded rigidity without a coverage theorem;

(x) that failure to find a counterexample is proof of rigidity;

(xi) that set-theoretic existence and uniqueness of a rigidity decoder supplies an executable algorithm or complexity bound;

(xii) that vacuous or narrowly restricted rigidity solves the ambient target without membership and coverage evidence;

(xiii) that a counterexample to one packet proves the target unsolvable by another route; or

(xiv) that any SDR.6 local rigidity theorem is automatically new in the literature, Technical Extension material, or successor-Core mathematics.

SDR.6.12 Closure and handoff

Theorem SDR.6.36 (SDR.6 rigidity and target-reconstruction closure theorem)**.** *Within the frozen-v21 constitutional boundary and the target-relative deterministic calculus of SDR.2–SDR.5, SDR.6 establishes:*

(i) class-wide target rigidity is exactly equality-kernel inclusion, empty target-separation defect, target sufficiency, and unique factorization through the realized observable image;

(ii) the resulting rigidity decoder is certified only on that realized image unless an extension theorem is separately supplied;

(iii) decoder theorems compose, but their source scopes and mathematical hypotheses remain noncompensating obligations;

(iv) rigidity on a restricted source class is monotone under further restriction but cannot be exported to the ambient class without a membership and coverage theorem;

(v) one-sided predicate soundness/reflection is kept distinct from exact binary reconstruction;

(vi) a packet may be rigid for a target coarsening while failing to reconstruct a stronger target, and the supported target strengths form a downward-closed rigidity spectrum rather than necessarily a single scalar ceiling;

(vii) the frozen PE-3 and PE-5 theorems provide inherited positive controls for morphism-level and bounded-derived rigidity on their exact registered source regimes;

(viii) target reconstruction is generally weaker than source reconstruction and is interpreted only at the requested target equivalence strength;

(ix) *a single certified rigidity counterexample destroys the corresponding universal packet-to-target rigidity claim and forbids repair by deterministic downstream syntax alone;*

(x) *exact finite rigidity is certifiable by complete pair coverage on a finite source set, while any ambient universalization requires an explicit coverage theorem;*

(xi) *mathematical rigidity may be vacuous or theorem-facing inapplicable unless nonvacuity, source membership, and problem coverage are separately established;*

(xii) *existence of the unique realized-image decoder is set-theoretic and does not imply effective, executable, or complexity-bounded decoding without an additional algorithmic certificate; and*

(xiii) *every rigidity verdict remains separated from novelty, promotion, implementation, and external-domain Return status.*

No item above proves that an unresolved source admits a rigidity theorem, that a successful target decoder is source-injective, or that a local SDR rigidity result modifies the frozen v21 release.

Proof. Item (i) is Theorem SDR.6.4; item (ii) is Definition SDR.6.5 and Policy 56; item (iii) is Proposition SDR.6.6 and Policy 57. Item (iv) is Theorems SDR.6.8 and SDR.6.9 with Policy 58. Item (v) is Definition SDR.6.11, Theorem SDR.6.12, and Policy 59. Item (vi) is Theorem SDR.6.14, Definition SDR.6.15, Proposition SDR.6.16, and Policy 61. Item (vii) is Theorems SDR.6.18 and SDR.6.19, bounded by Policy 62. Item (viii) is Definition SDR.6.20 and Propositions SDR.6.21–SDR.6.22. Item (ix) is Theorem SDR.6.24. Item (x) is Theorem SDR.6.27, Policy 65, and Definition SDR.6.28. Item (xi) is Proposition SDR.6.29, Definition SDR.6.30, and Policy 67. Item (xii) is Definition SDR.6.31 and Policy 68. Item (xiii) follows from the canonical record and the SDR.1 governance firewall. □

Remark SDR.6.37 (Next consumer: SDR.7)**.** SDR.7 shall treat the case in which many admissible probes or rigidity-supporting observables are available but only a subset is needed. It will define finite conflict universes, probe separation sets, target-relevant coverage, minimal separating families, redundancy elimination, and model-relative Pareto certificate complexity while preserving the distinction between finite benchmark coverage and class-wide rigidity proved by SDR.6.

SDR.7 Minimal Separating Families and Certificate Complexity

SDR.7.1 Role, scope, and imported boundaries

Declaration SDR.7.1 (SDR.7 objective)**.** SDR.7 optimizes a *proved target-separation problem*. The inputs are a declared target Θ, a base observable packet $\mathbf{I}_0$, a source scope, and a registry of admissible candidate probes whose semantics have already passed the SDR.4–SDR.6 typing, provenance, and theorem-strength firewalls. The section asks which subfamilies retain the target-relevant separation needed for rigidity, what lower bounds certify necessity, and how such choices interact with the frozen v21 family-relative certificate-cost model.

Policy 70 (SDR-MIN-01: optimization is downstream of admissibility)**.** *A probe is not admitted into an SDR.7 optimization problem merely because it separates benchmark pairs cheaply. Source semantics, comparison mode, target-oracle status, theorem authority, and the relevant SDR.4–SDR.6 admissibility obligations are fixed before the optimization begins. Optimization never repairs an inadmissible probe.*

Policy 71 (SDR-MIN-02: finite optimization does not create ambient coverage)**.** *All exact finite cover theorems below are finite-source theorems unless an explicit SDR.6 rigidity-coverage theorem is supplied. A minimal family on a benchmark, regression corpus, or finite verification set is not thereby minimal or even sufficient on the ambient mathematical source class.*

Remark SDR.7.2 (Inherited certificate-cost semantics)**.** The canonical v21 family-relative certificate model uses a vector cost in $\mathbb{N}^4$, with distinct coordinates for query burden, answer/encoding burden, verifier burden, and replay/artifact burden. SDR.7 consumes that model rather than replacing it by a scalar probe count. Probe cardinality is an additional structural selection statistic and coincides with a cost coordinate only in a model that proves such an identification.

Definition SDR.7.3 (Evidence-relative finite conflict universe)**.** Fix an admissible target-evidence policy $\mathcal{E}$. In addition to the semantic conflict universe defined below, SDR.7 uses the certified unordered conflict universe

$$\mathcal{U}^{\mathrm{cert},\mathcal{E}}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0) := \Big\{\{X,Y\} \subseteq \mathcal{S}_0 : X \neq Y, (X,Y) \in \Delta^{\mathrm{cert},\mathcal{E}}_{\Theta}(\mathbf{I}_0) \text{ or } (Y,X) \in \Delta^{\mathrm{cert},\mathcal{E}}_{\Theta}(\mathbf{I}_0)\Big\}.$$

By SDR.3 soundness,

$$\mathcal{U}^{\mathrm{cert},\mathcal{E}}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0) \subseteq \mathcal{U}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0).$$

The semantic universe is a mathematical object whether or not all of its target-discordance facts are available to the optimizer. The certified universe records only those conflicts supported by evidence admissible under $\mathcal{E}$. For an admissible candidate probe J, define its certified resolution set by

$$\mathcal{R}^{\mathrm{cert},\mathcal{E}}(J) := \Big\{\{X,Y\} \in \mathcal{U}^{\mathrm{cert},\mathcal{E}}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0) : \widehat{J}(X) \neq \widehat{J}(Y)\Big\}.$$

A selected family A is *$\mathcal{E}$-certified-separating* when

$$\bigcup_{J\in A} \mathcal{R}^{\mathrm{cert},\mathcal{E}}(J) = \mathcal{U}^{\mathrm{cert},\mathcal{E}}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0).$$

This evidence-relative property is deliberately distinct from semantic target separation.

Policy 72 (SDR-MIN-02A: target-oracle firewall for optimization)**.** *An SDR.7 run advertised as independent optimization may materialize, rank, or optimize target-conflict vertices only from target-discordance evidence available under its frozen admissible premise set. Unknown semantic target labels, benchmark answers, conclusion-derived labels, hidden oracles, or equivalent downstream information may not be used to construct the optimization universe. If such information is intentionally used, the run is benchmark-assisted or validation-only and its provenance must say so.*

Theorem SDR.7.4 (SDR-MIN-02B: certified-universe coverage ceiling)**.** *Let $A \subseteq \mathfrak{P}$ be $\mathcal{E}$-certified-separating. Then covering the certified universe*

$$\mathcal{U}^{\mathrm{cert},\mathcal{E}}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0)$$

by the certified resolution sets proves elimination of every $\mathcal{E}$-certified residual conflict in that universe. It implies finite semantic target rigidity on $\mathcal{S}_0$ only if a separately proved target-evidence completeness theorem establishes

$$\mathcal{U}^{\mathrm{cert},\mathcal{E}}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0) = \mathcal{U}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0).$$

Absent that equality, an uncovered semantic target conflict may remain outside the certified optimization problem.

Proof. SDR.3 gives certified defect as a sound subset of semantic defect and explicitly forbids inferring semantic emptiness from certified emptiness without evidence completeness. Passing to unordered finite pairs preserves that inclusion. Hence a cover of the certified universe removes exactly the certified conflicts it contains; equality with the semantic universe is the missing premise needed to invoke the semantic cover–rigidity theorem below. □

SDR.7.2 Residual conflict universe and separation hypergraph

Definition SDR.7.5 (Finite residual conflict universe)**.** Let $\mathcal{S}_0 \subseteq \mathcal{S}$ be finite. The *semantic residual target-conflict universe* of the base packet $\mathbf{I}_0$ is the set of unordered pairs

$$\mathcal{U}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0) := \Big\{\{X,Y\} \subseteq \mathcal{S}_0 : X \neq Y, \widehat{\mathbf{I}_0}(X) = \widehat{\mathbf{I}_0}(Y), \widehat{\Theta}(X) \neq \widehat{\Theta}(Y)\Big\}.$$

It is the unordered finite version of the SDR.3 semantic separation defect remaining after the base packet has been consumed. Its use as an operational optimization input is governed by Policy 72; a theorem-facing independent run ordinarily optimizes the certified universe unless target-evidence completeness has been proved.

Definition SDR.7.6 (Resolution set of a candidate probe)**.** For an admissible candidate probe J, define

$$\mathcal{R}_{\Theta,\mathbf{I}_0,\mathcal{S}_0}(J) := \Big\{\{X,Y\} \in \mathcal{U}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0) : \widehat{J}(X) \neq \widehat{J}(Y)\Big\}.$$

When the ambient datum is fixed, write simply $\mathcal{U}$ and $\mathcal{R}(J)$.

Definition SDR.7.7 (Separation hypergraph)**.** For a candidate registry $\mathfrak{P}$, the *SDR separation hypergraph* is

$$\mathsf{H}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0;\mathfrak{P}) := \big(\mathcal{U}, \{\mathcal{R}(J)\}_{J\in\mathfrak{P}}\big).$$

The vertices are unresolved target-conflict pairs and each candidate probe contributes the hyperedge consisting exactly of the conflicts it resolves.

Definition SDR.7.8 (Selected packet)**.** For $A \subseteq \mathfrak{P}$, let

$$\mathbf{I}_A := \mathbf{I}_0 \boxtimes \boxtimes_{J\in A} J,$$

with the empty product interpreted as $\mathbf{I}_0$. The notation denotes joint observable recording only; it does not assert a categorical tensor or product beyond the SDR.2 packet semantics.

Theorem SDR.7.9 (SDR-MIN-03: exact residual-conflict law)**.** *For every finite $\mathcal{S}_0$ and every $A \subseteq \mathfrak{P}$,*

$$\mathcal{U}_{\Theta,\mathbf{I}_A}(\mathcal{S}_0) = \mathcal{U}_{\Theta,\mathbf{I}_0}(\mathcal{S}_0) \setminus \bigcup_{J\in A} \mathcal{R}(J).$$

Equivalently, an initial conflict survives exactly when every selected candidate probe still identifies its two endpoints.

Proof. Let $u = \{X,Y\} \in \mathcal{U}$. Equality of the joint packet on X and Y is equivalent to equality of the base packet and equality of every selected probe. The base equality already holds because $u \in \mathcal{U}$. Hence u remains a residual target conflict exactly when no selected probe separates it, i.e. exactly when $u \notin \bigcup_{J\in A} \mathcal{R}(J)$. □

SDR.7.3 Cover–rigidity equivalence

Definition SDR.7.10 (Target-separating subfamily)**.** A subfamily $A \subseteq \mathfrak{P}$ is $(\Theta, \mathbf{I}_0, \mathcal{S}_0)$-*separating* when

$$\bigcup_{J\in A} \mathcal{R}(J) = \mathcal{U}.$$

Thus the selected hyperedges cover every residual target conflict of the base packet.

Theorem SDR.7.11 (SDR-MIN-04: finite cover–rigidity equivalence)**.** *For finite $\mathcal{S}_0$ and any selected subfamily $A \subseteq \mathfrak{P}$, the following are equivalent:*

(i) *A is target-separating;*

(ii) $\mathcal{U}_{\Theta,\mathbf{I}_A}(\mathcal{S}_0) = \varnothing$;

(iii) $\Delta_{\Theta|_{\mathcal{S}_0}}(\mathbf{I}_A|_{\mathcal{S}_0}) = \varnothing$;

(iv) $\mathrm{Rig}_{\Theta|_{\mathcal{S}_0}}(\mathbf{I}_A|_{\mathcal{S}_0};\mathcal{S}_0)$ *holds.*

Proof. The equivalence of (i) and (ii) is Theorem SDR.7.9. The unordered conflict universe is empty exactly when the ordered SDR.3 defect contains no target-discordant packet-equal pair, giving (ii)⇔(iii). SDR.6 identifies empty defect with target rigidity, giving (iii)⇔(iv). □

Corollary SDR.7.12 (SDR-MIN-05: zero additional-probe criterion)**.** *The empty selected family is target-separating if and only if the base packet $\mathbf{I}_0$ is already target-rigid on $\mathcal{S}_0$.*

Definition SDR.7.13 (Minimum separating cardinality)**.** Define

$$\kappa_{\Theta,\mathbf{I}_0}(\mathcal{S}_0;\mathfrak{P}) := \inf\Big\{|A| : A \subseteq \mathfrak{P} \text{ is target-separating}\Big\},$$

with value $+\infty$ when no separating family exists. Because $\mathcal{U}$ is finite, whenever a cover exists there is a finite subcover and the infimum is attained by a nonnegative integer.

Policy 73 (SDR-MIN-06: minimum cardinality is not canonical certificate complexity)**.** *The scalar κ counts selected probes. It is not identified with the canonical v21 query coordinate q, nor with bit length, verifier time, artifact footprint, or the full Pareto certificate frontier, unless a separately declared computation/certificate model proves the relevant equality.*

SDR.7.4 Inclusion minimality, redundancy, and private witnesses

Definition SDR.7.14 (Inclusion-minimal separating family)**.** A target-separating family A is *inclusion-minimal* when no proper subfamily of A remains target-separating. This is distinct from minimum cardinality.

Definition SDR.7.15 (Private conflict witness)**.** Let A be target-separating and $J \in A$. A conflict $u \in \mathcal{U}$ is *private to J relative to A* when

$$u \in \mathcal{R}(J) \quad \text{and} \quad u \notin \bigcup_{K \in A\setminus\{J\}} \mathcal{R}(K).$$

Thus J is the only selected probe that resolves u.

Theorem SDR.7.16 (SDR-MIN-07: private-witness characterization of inclusion minimality)**.** *Let A be a finite target-separating family. Then A is inclusion-minimal if and only if every $J \in A$ has at least one private conflict witness relative to A.*

Proof. If J has no private witness, then every conflict resolved by J is also resolved by another selected probe, so removing J preserves the cover. Conversely, if u_J is private to J, then removing J leaves u_J uncovered; hence no selected probe can be removed. □

Corollary SDR.7.17 (SDR-MIN-08: finite redundancy elimination)**.** *Starting from any finite target-separating family, repeatedly deleting a probe whose removal preserves coverage terminates at an inclusion-minimal target-separating family.*

Proof. Each deletion strictly decreases the finite cardinality while preserving coverage. Termination is therefore finite, and the terminal family has no removable member. □

Policy 74 (SDR-MIN-09: minimal is not minimum)**.** *Inclusion-minimality certifies local irredundancy only. It does not imply minimum cardinality, minimum certificate cost, or Pareto optimality. A different family may cover the same conflict universe using fewer probes or lower verifier burden.*

Example SDR.7.18 (Minimal but not minimum)**.** Let $\mathcal{U} = \{u_1, u_2\}$ and suppose

$$\mathcal{R}(J_1) = \{u_1\}, \qquad \mathcal{R}(J_2) = \{u_2\}, \qquad \mathcal{R}(J_3) = \{u_1, u_2\}.$$

Then $\{J_1, J_2\}$ is inclusion-minimal, but $\{J_3\}$ has smaller cardinality.

SDR.7.5 Family-scoped no-go results and lower bounds

Definition SDR.7.19 (Resolver set of a conflict)**.** For $u \in \mathcal{U}$, define the resolver set

$$\mathfrak{P}(u) := \{J \in \mathfrak{P} : u \in \mathcal{R}(J)\}.$$

Theorem SDR.7.20 (SDR-MIN-10: candidate-registry no-go)**.** *If there exists $u \in \mathcal{U}$ with $\mathfrak{P}(u) = \varnothing$, then no subfamily of $\mathfrak{P}$ is target-separating, and*

$$\kappa_{\Theta,\mathbf{I}_0}(\mathcal{S}_0;\mathfrak{P}) = +\infty.$$

More generally, a separating family exists if and only if

$$\bigcup_{J \in \mathfrak{P}} \mathcal{R}(J) = \mathcal{U}.$$

Proof. A conflict with empty resolver set survives every selected packet formed from the registry. The general statement is the definition of coverage applied to the full candidate family. □

Policy 75 (SDR-MIN-11: registry no-go is not universal impossibility)**.** *Theorem SDR.7.20 is scoped to the declared candidate registry. It does not prove that no invariant, enriched probe, rigidity theorem, source restriction, or alternate mathematical route can solve the target problem.*

Proposition SDR.7.21 (SDR-MIN-12: maximum-coverage cardinality lower bound)**.** *Assume $\mathfrak{P}$ is finite and nonempty, $\mathcal{U} \neq \emptyset$, and set*

$$M := \max_{J \in \mathfrak{P}} |\mathcal{R}(J)|.$$

If $M > 0$, then every finite target-separating family A satisfies

$$|A| \geq \left\lceil \frac{|\mathcal{U}|}{M} \right\rceil.$$

If $M = 0$, no target-separating family exists.

Proof. A family of r probes covers at most rM conflict incidences before overlap is discarded, so covering $|\mathcal{U}|$ distinct conflicts requires $rM \geq |\mathcal{U}|$. If $M = 0$, every resolution set is empty. □

Theorem SDR.7.22 (SDR-MIN-13: resolver-disjoint blocker lower bound)**.** *Suppose conflicts $u_1, \ldots, u_m \in \mathcal{U}$ satisfy*

$$\mathfrak{P}(u_i) \cap \mathfrak{P}(u_j) = \emptyset \qquad (i \neq j).$$

Then every target-separating family contains at least m probes. Hence

$$\kappa_{\Theta,\mathbf{I}_0}(\mathcal{S}_0; \mathfrak{P}) \geq m.$$

Proof. Any separating family must choose at least one resolver for each u_i. Because the resolver sets are pairwise disjoint, one selected probe cannot discharge two of these obligations. Thus at least m distinct probes are necessary. □

Definition SDR.7.23 (Cardinality-optimality certificate)**.** A cardinality-optimality certificate for a selected family A consists of:

(i) an upper certificate proving that A covers $\mathcal{U}$;

(ii) a lower-bound theorem proving every cover has cardinality at least $|A|$.

The lower bound may arise from resolver-disjoint blockers, exhaustive finite search, a dual or packing argument, or another independently proved combinatorial theorem. A lower bound on queries, bits, verifier time, artifact size, or another certificate coordinate counts as a lower bound on $|A|$ only when a separately proved cross-metric theorem relates that coordinate to selected-probe cardinality.

Policy 76 (SDR-MIN-13A: no cross-metric lower-bound conversion)**.** *The inequalities $q \geq m$, $b \geq m$, $t \geq m$, or $a \geq m$ do not by themselves imply $|A| \geq m$, and a cardinality lower bound does not by itself lower-bound any v21 certificate coordinate. One probe may consume many queries, while several probes may share one query or artifact. Every conversion between probe count and certificate cost therefore requires an explicit model theorem.*

SDR.7.6 Incidence representation and exact finite search

Definition SDR.7.24 (Separation incidence matrix)**.** Assume

$$\mathcal{U} = \{u_1, \ldots, u_n\}, \qquad \mathfrak{P} = \{J_1, \ldots, J_m\}$$

are finite. Define the Boolean incidence matrix

$$B_{ij} := \begin{cases} 1, & u_i \in \mathcal{R}(J_j), \\ 0, & u_i \notin \mathcal{R}(J_j). \end{cases}$$

A selector vector $x \in \{0,1\}^m$ represents the subfamily $A_x = \{J_j : x_j = 1\}$.

Proposition SDR.7.25 (SDR-MIN-14: Boolean cover formulation)**.** *The selected family A_x is target-separating if and only if*

$$\sum_{j=1}^{m} B_{ij} x_j \geq 1 \qquad \text{for every } i = 1, \ldots, n.$$

Consequently the minimum-cardinality problem is exactly the finite optimization problem

$$\min \sum_{j=1}^{m} x_j \quad \text{subject to} \quad Bx \geq \mathbf{1}, \qquad x \in \{0,1\}^m.$$

Proof. The ith inequality states precisely that at least one selected probe resolves the conflict u_i. Requiring this for every row is equivalent to covering $\mathcal{U}$. □

Corollary SDR.7.26 (SDR-MIN-15: decidability of the finite registered problem)**.** *When $m < \infty$, the chosen semantic or certified conflict universe is explicitly finite with certified decidable membership, and every incidence entry is certified decidable, exact minimum cardinality for that registered finite problem is decidable by exhaustive evaluation of at most 2^m selected subfamilies. This statement gives an exact finite procedure but no claim of practical efficiency.*

Policy 77 (SDR-MIN-16: no imported approximation guarantee without hypotheses)**.** *The hypergraph formulation permits standard set-cover algorithms to be used as search heuristics, but SDR does not import an approximation ratio, optimality theorem, or complexity classification merely from the words "set cover." Any such guarantee must be source-bound to the exact finite optimization model and its cost assumptions. In particular, nonadditive verifier or artifact reuse can invalidate a naive weighted-cover interpretation.*

SDR.7.7 Model-relative certificate cost and Pareto minimality

Definition SDR.7.27 (SDR separation-certificate model adapter)**.** A separation-certificate model adapter binds an SDR.7 finite rigidity task to a declared v21-style certificate model $\mathfrak{M}$ by specifying: the charged query universe and response semantics; the bit/encoding convention; the verifier operations included in t; the replay/artifact accounting included in a; the family aggregation rule; and the exact reuse policy. Until this adapter is fixed, the tuple (q, b, t, a) for an SDR.7 scheme is undefined rather than guessed from probe count. The adapter, not notation alone, is what licenses reuse of the frozen v21 cost semantics for this pairwise separation task.

Policy 78 (SDR-MIN-16A: selected-observable sovereignty)**.** *A certificate scheme advertised as attached to a selected family A may consume theorem-only premises, comparison proofs, cached artifacts, and verifier metadata required to validate the probes in A, but it may not use an additional candidate observable $K \in \mathfrak{P} \setminus A$ to discharge a separation obligation. If such an observable contributes target separation, the selected family is enlarged to include K. This prevents a nominally small family from hiding extra observational information in its verifier.*

Definition SDR.7.28 (Separating certificate scheme)**.** Fix a finite verification family $\mathcal{S}_0$, target Θ, base packet $\mathbf{I}_0$, candidate registry $\mathfrak{P}$, and a separation-certificate model adapter into $\mathfrak{M}$. A *separating certificate scheme* is a finite verifier scheme Π_A attached to some target-separating $A \subseteq \mathfrak{P}$ that:

(i) evaluates or replays the selected probe records under the declared source semantics;

(ii) certifies the incidence/coverage obligations needed for the finite rigidity conclusion;

(iii) verifies all comparison, theorem, and provenance fields consumed by the selected packet; and

(iv) emits a sound-and-complete finite rigidity verdict on the declared family; and

(v) obeys selected-observable sovereignty: every observable used to discharge separation belongs to the declared selected family.

Its canonical cost is

$$\mathrm{cost}_{\mathfrak{M}}^{\mathcal{S}_0}(\Pi_A) = (q, b, t, a) \in \mathbb{N}^4.$$

Definition SDR.7.29 (Achievable separation-cost set and Pareto frontier)**.** Let

$$C^{\mathrm{sep}}_{\mathfrak{M}}(\Theta, \mathbf{I}_0, \mathcal{S}_0; \mathfrak{P}) \subseteq \mathbb{N}^4$$

be the set of all finite cost vectors of valid separating certificate schemes in the declared model. Define

$$\mathrm{MinSepCert}_{\mathfrak{M}}(\Theta, \mathbf{I}_0, \mathcal{S}_0; \mathfrak{P}) := \mathrm{ParetoMin}\Big(C^{\mathrm{sep}}_{\mathfrak{M}}(\Theta, \mathbf{I}_0, \mathcal{S}_0; \mathfrak{P})\Big).$$

If no finite separating scheme exists, both the achievable finite-cost set and its Pareto frontier are empty.

Theorem SDR.7.30 (SDR-MIN-17: finite Pareto-frontier well-posedness)**.** *If*

$$C^{\mathrm{sep}}_{\mathfrak{M}}(\Theta, \mathbf{I}_0, \mathcal{S}_0; \mathfrak{P}) \neq \varnothing,$$

then the Pareto frontier is a nonempty finite set, and every finite achievable separation-cost vector dominates at least one frontier element.

Proof. The cost set is a subset of $\mathbb{N}^4$ with the coordinatewise order. By Dickson's lemma, every subset of $\mathbb{N}^4$ has finitely many minimal elements. Every achievable vector lies above a minimal element obtained by descending within its finite coordinate box. This is the same well-quasi-order mechanism used by the frozen v21 family-relative certificate theory. □

Remark SDR.7.31 (Why the frontier is set-valued)**.** A family with fewer selected probes may require more expensive probe values, larger encodings, a stronger theorem checker, or a larger replay artifact. Without a declared scalar utility, incomparable cost vectors remain simultaneously minimal. SDR.7 therefore does not collapse the frontier to a single "best" certificate.

Policy 79 (SDR-MIN-18: cost is scheme-relative, not probe-sum by default)**.** *The canonical cost of a selected family is the cost of the actual verifier scheme Π_A. It is not defined as the sum of per-probe costs unless the certificate model proves additivity. Shared source queries, common encodings, theorem reuse, cached artifacts, compatibility checks, and composition overhead can all change the total vector.*

SDR.7.8 Evidence reuse, composition overhead, and dominated choices

Definition SDR.7.32 (Content-addressed reuse record)**.** For selected families A and B, a reuse record identifies source queries, answers, theorem references, comparison artifacts, and verifier nodes that are literally the same immutable content-addressed objects in both schemes. A textual statement that two probes "use the same information" is not a reuse certificate.

Proposition SDR.7.33 (SDR-MIN-19: compositional upper bound with explicit reuse)**.** *Suppose valid verifier schemes Π_A and Π_B are composable in one declared separation-certificate model, with costs $c_A, c_B \in \mathbb{N}^4$, and let $h_{A,B} \in \mathbb{N}^4$ be the additional composition overhead. Then a combined verifier record for the union of their declared obligations exists with*

$$\mathrm{cost}(\Pi_{A\cup B}) \leq_{\mathrm{coord}} c_A + c_B + h_{A,B}.$$

If $A \cup B$ is target-separating and the combined verifier discharges the coverage obligations, this record is a separating certificate scheme. Otherwise it remains only a valid partial-separation/composition record. Any sharper saving requires an explicit reuse rule and the corresponding identical content addresses.

Proof. Replay the two valid component records and append the required composition node. Coordinatewise concatenation yields the displayed upper bound. Separation status is a separate coverage property and therefore follows only when the union actually covers the declared conflict universe. Any reduction from duplicate content is licensed only where the model declares one immutable artifact sufficient for both verifier obligations. □

Definition SDR.7.34 (Coverage dominance)**.** For two candidate probes $J, K \in \mathfrak{P}$, say that K *coverage-dominates* J on the declared finite problem when

$$\mathcal{R}(J) \subseteq \mathcal{R}(K).$$

If, in addition, the declared verifier model proves that replacing J by K never increases the complete scheme cost and does not remove a non-separation capability required elsewhere in the proof DAG, call the dominance *certificate-safe*.

Proposition SDR.7.35 (SDR-MIN-20: safe dominated-probe elimination)**.** *If K certificate-safely coverage-dominates J, then every target-separating family containing J can replace J by K without losing finite target separation and without worsening the certified cost vector under the declared replacement theorem.*

Proof. Every conflict resolved by J is also resolved by K, so the cover remains valid. The cost conclusion is exactly the additional certificate-safe dominance hypothesis; it is not inferred from set inclusion alone. □

Definition SDR.7.36 (Finite coverage equivalence)**.** Two probes $J, K \in \mathfrak{P}$ are *coverage-equivalent on the declared finite problem* when

$$\mathcal{R}(J) = \mathcal{R}(K).$$

This is an equivalence only at the level of the current finite separation hypergraph.

Policy 80 (SDR-MIN-20A: coverage equivalence is not certificate interchangeability)**.** *Finite coverage equivalence does not imply equality of source semantics, information kernels on the ambient class, theorem hypotheses, stability, provenance, execution cost, artifact size, or promotion status. Replacing one coverage-equivalent probe by another in a certified scheme therefore requires the same compatibility and cost checks as any other replacement.*

SDR.7.9 Finite optimality versus ambient rigidity

Theorem SDR.7.37 (SDR-MIN-21: finite separation plus rigidity coverage lifts sufficiency)**.** *Let A be target-separating on a finite verification family $\mathcal{S}_0$. If an SDR.6 rigidity-coverage theorem for the selected packet $\mathbf{I}_A$ proves that rigidity on $\mathcal{S}_0$ implies rigidity on the intended ambient source class $\mathcal{S}$, then*

$$\mathrm{Rig}_\Theta(\mathbf{I}_A; \mathcal{S})$$

holds under the exact hypotheses of that coverage theorem.

Proof. The finite cover gives rigidity on $\mathcal{S}_0$ by Theorem SDR.7.11. Apply the declared SDR.6 rigidity-coverage theorem without weakening its hypotheses. □

Policy 81 (SDR-MIN-22: benchmark minimality does not lift automatically)**.** *Even when Theorem SDR.7.37 lifts the* sufficiency *of a selected family, inclusion-minimality, minimum cardinality, and Pareto optimality on $\mathcal{S}_0$ do not automatically lift to $\mathcal{S}$. An omitted probe may be redundant on every finite benchmark conflict yet essential for an ambient conflict not represented in the finite universe.*

Definition SDR.7.38 (Ambient necessity witness)**.** Let A already be proved target-rigid on $\mathcal{S}$. For $J \in A$, an *ambient necessity witness* is a pair $X_J, Y_J \in \mathcal{S}$ such that

$$\widehat{\mathbf{I}_{A\setminus\{J\}}}(X_J) = \widehat{\mathbf{I}_{A\setminus\{J\}}}(Y_J), \qquad \widehat{\Theta}(X_J) \neq \widehat{\Theta}(Y_J).$$

Because the full packet $\mathbf{I}_A$ is assumed target-rigid, these two conditions force J itself to separate the pair: otherwise the full packet would also identify X_J and Y_J while their target values differ. Thus J-separation is a theorem consequence, not an optional strengthening.

Theorem SDR.7.39 (SDR-MIN-23: ambient inclusion-minimality certificate)**.** *If A is target-rigid on $\mathcal{S}$ and every $J \in A$ has an ambient necessity witness, then A is inclusion-minimal among the selected probes for ambient target rigidity on $\mathcal{S}$.*

Proof. For each J, its witness is a certified rigidity counterexample to the packet obtained by deleting J. Hence no member can be removed while preserving ambient rigidity. □

Policy 82 (SDR-MIN-24: optimality claims require matching lower evidence)**.** *A claim that a probe family is "minimal," "smallest," "optimal," or "cheapest" must name the corresponding order and provide the lower-bound evidence appropriate to that order. A constructive family alone is an upper bound. Successful heuristic search alone is not a minimum theorem.*

SDR.7.10 Canonical record, nonclaims, and closure

Definition SDR.7.40 (Canonical SDR.7 optimization record)**.** A canonical SDR.7 record is

$$\begin{aligned}\mathfrak{R}_{\mathrm{SDR7}} := \big(&\mathcal{S}_0, \Theta, \mathbf{I}_0, \mathfrak{P}, \mathcal{E},\\ &\mathsf{UniverseMode}, \mathcal{U}, \mathcal{U}^{\mathrm{cert},\mathcal{E}},\\ &\{\mathcal{R}(J)\}_{J\in\mathfrak{P}}, \mathsf{EvidenceCompleteness}, \mathsf{Incidence},\\ &\mathsf{SelectedFamily}, \mathsf{ObservableScope}, \mathsf{CoverageProof},\\ &\mathsf{PrivateWitnesses}, \mathsf{LowerBounds}, \mathsf{CrossMetricBridges},\\ &\mathsf{CardinalityStatus}, \mathfrak{M}, \mathsf{ModelAdapter},\\ &\mathsf{VerifierScheme}, \mathsf{CostVector}, \mathsf{ParetoStatus},\\ &\mathsf{Reuse}, \mathsf{CoverageLift}, \mathsf{AmbientNecessity},\\ &\mathsf{Disposition}, \mathsf{NoveltyStatus}, \mathcal{A}\big).\end{aligned}$$

The record keeps semantic versus certified conflict scope, finite separation, observational scope, combinatorial optimality, cross-metric bridges, verifier cost, ambient lifting, and novelty as independent axes.

Declaration SDR.7.41 (SDR.7 explicit nonclaims)**.** SDR.7 does not claim:

(i) that every candidate registry contains a target-separating family;

(ii) that a finite separating family is ambiently sufficient without an SDR.6 coverage theorem;

(iii) that an inclusion-minimal family has minimum cardinality;

(iv) that minimum cardinality minimizes query, bit, verifier, artifact, or Pareto certificate cost;

(v) that per-probe costs add without an explicit certificate-composition theorem;

(vi) that a greedy or other heuristic selector is optimal or has an approximation guarantee without separately imported hypotheses;

(vii) that finite benchmark redundancy implies ambient redundancy;

(viii) that one finite lower bound is sharp unless it matches a proved upper construction;

(ix) that covering only the evidence-certified conflict universe proves semantic rigidity without target-evidence completeness;

(x) that a lower bound on q, b, t, or a is automatically a lower bound on selected-probe cardinality, or conversely;

(xi) that a nominal selected family may hide additional separation observables inside its verifier;

(xii) that a registry-level impossibility proves universal mathematical impossibility;

(xiii) that the set-cover/hypergraph formalization itself is new mathematics;

(xiv) that Pareto-optimal separation implies source reconstruction, external Return, or successor-Core promotion; or

(xv) that any SDR.7 result modifies the frozen v21 certificate-complexity theory.

Theorem SDR.7.42 (SDR.7 minimal-separation and certificate-complexity closure theorem)**.** *Within the frozen-v21 constitutional boundary and the SDR.2–SDR.6 target-relative calculus, SDR.7 establishes:*

(i) every finite semantic residual target-separation problem is represented exactly by a conflict universe and candidate-probe resolution hypergraph, while an independent theorem-facing run may consume only its admissibly certified conflict subuniverse and certified resolution hyperedges unless target-evidence completeness is proved;

(ii) selected probes preserve finite semantic target rigidity exactly when their resolution sets cover the semantic universe; covering only the certified universe has the strictly weaker evidence-relative meaning fixed by Theorem SDR.7.4;

(iii) inclusion-minimality is characterized by one private conflict witness per selected probe, while minimum cardinality remains a stronger optimization claim;

(iv) registry-scoped uncoverable conflicts give exact family no-go certificates, and resolver-disjoint blockers give certified probe-cardinality lower bounds without automatic conversion to or from v21 certificate-cost coordinates;

(v) a finite registry with decidable universe and incidence data admits an exact Boolean cover formulation and exhaustive exact optimizer, without an efficiency claim;

(vi) canonical certificate cost is defined only after an explicit SDR-to-v21 model adapter and remains the model-relative vector (q, b, t, a) of the actual verifier scheme rather than a default sum of per-probe costs;

(vii) every nonempty achievable set of finite $\mathbb{N}^4$ separation-cost vectors has a finite nonempty Pareto frontier under the inherited coordinatewise order;

(viii) selected-observable sovereignty prevents hidden candidate probes from discharging separation obligations outside the declared selected family;

(ix) content-addressed reuse and composition overhead must be represented explicitly before cost savings are claimed, and composition creates a separating scheme only when the union actually satisfies the coverage obligation;

(x) finite separation sufficiency can lift to ambient rigidity only through an SDR.6 coverage theorem, while finite minimality and optimality require separate ambient necessity or lower-bound evidence; and

(xi) all optimization conclusions remain distinct from novelty, external-domain Return, source reconstruction, and successor-Core promotion.

Proof. Items (i)–(ii) are Definition SDR.7.3, Policy 72, Theorem SDR.7.4, Definitions SDR.7.5–SDR.7.8, and Theorems SDR.7.9–SDR.7.11. Item (iii) is Theorem SDR.7.16 and Policy 74. Item (iv) is Theorems SDR.7.20 and SDR.7.22 with Policy 76. Item (v) is Proposition SDR.7.25 and Corollary SDR.7.26. Items (vi)–(vii) are Definition SDR.7.27, Definitions SDR.7.28–SDR.7.29, Theorem SDR.7.30, and Policy 79. Item (viii) is Policy 78. Item (ix) is Definition SDR.7.32 and Proposition SDR.7.33. Item (x) is Theorem SDR.7.37, Policy 81, and Theorem SDR.7.39. Item (xi) follows from the canonical record and the SDR.1 governance and promotion firewalls. □

Remark SDR.7.43 (Next consumer: SDR.8)**.** SDR.8 shall leave the finite static conflict universe and study *dynamic defect under refinement*: descending residual filtrations, rank-to-residual implications, completion hypotheses, finite-state or Noetherian stabilization, and the exact conditions under which finite-stage improvement can be promoted to a finite-to-infinite target statement. SDR.7 supplies evidence-scoped finite separating subfamilies and model-bound certificate-cost records that SDR.8 may propagate across stages; it does not itself provide the completion theorem.

SDR.8 Dynamic Defect and Finite-to-Infinite Completion

SDR.8.1 Role, scope, and the two completion questions

SDR.2–SDR.7 are primarily static at a declared source scope: one fixes a target, an observable packet, a defect relation, and optionally a finite candidate registry. SDR.8 studies what happens when the packet itself grows through an infinite refinement process. The central distinction is between two questions that must not be conflated:

(i) *semantic limit sufficiency*: does the countable totality of all stagewise observables determine the target? and

(ii) *finite certification*: is there a finite stage, finite-state reduction, compactness argument, filtered completion theorem, or other typed mechanism that reduces the infinite conclusion to finite theorem-facing evidence?

A positive answer to the first question need not imply a positive answer to the second.

Policy 83 (SDR-COMP-02: dynamic improvement is not completion)**.** *A descending defect sequence, strictly improving score, increasing rank, vanishing numerical surrogate, finite-prefix success, or growing collection of verified cases does not by itself establish target sufficiency at the limit or at any finite stage. Every finite-to-infinite promotion must identify the exact residual object and the theorem that forces its disappearance.*

Definition SDR.8.1 (Dynamic target-separation stream)**.** Fix one source class $\mathcal{S}$, normalized target $\widehat{\Theta}$, and initial normalized observable packet $\mathbf{I}_0$. Let

$$J_1, J_2, \dots$$

be normalized source-semantic observables on the same declared source class. Define recursively

$$\mathbf{I}_n := \mathbf{I}_0 \boxtimes J_1 \boxtimes \cdots \boxtimes J_n, \qquad D_n := \Delta_\Theta(\mathbf{I}_n).$$

The sequence $(D_n)_{n\geq 0}$ is the *semantic dynamic defect stream*. The construction fixes the target and source class; source restriction and target weakening are separate operations rather than hidden stage updates.

Policy 84 (SDR-COMP-02A: adaptive-stream provenance and oracle firewall)**.** *For a stream advertised as independent discovery, every newly adjoined observable J_{n+1} must carry a stagewise precursor record frozen before the target-relevant consequence of that stage is consumed. The generator may depend on previously admitted source-semantic data, certified hypotheses, and permitted surrogate information, but not on hidden target values, benchmark answers, or a later proof that is available only after the candidate has been selected. The mere existence of a countable family whose total packet determines Θ is therefore a semantic sufficiency statement, not evidence that the family was independently discovered.*

Theorem SDR.8.2 (SDR-COMP-03: exact dynamic-defect recurrence)**.** *For every dynamic target-separation stream,*

$$D_{n+1} = D_n \cap \mathrm{EqKer}(\widehat{J}_{n+1}) \subseteq D_n.$$

Moreover the strict reduction set at stage $n+1$ is

$$D_n \setminus D_{n+1} = \{(X,Y) \in D_n : \widehat{J}_{n+1}(X) \neq \widehat{J}_{n+1}(Y)\}.$$

Proof. Apply the joint-defect intersection law of Theorem SDR.3.17 to $\mathbf{I}_n$ and J_{n+1}. The displayed set difference is the pointwise complement of the new equality-kernel condition inside D_n. □

Remark SDR.8.3 (No resurrection at the semantic level)**.** Once a semantic defect pair has been separated by an added observable, later joint enrichment cannot recreate that pair as a defect while the source class, target, and earlier observables remain fixed. This monotonicity concerns the semantic defect D_n, not the size of the currently discovered or certified witness ledger; the latter is treated separately below.

SDR.8.2 The countable joint packet and the exact limit defect

Definition SDR.8.4 (Countable joint observable)**.** For the stream of Definition SDR.8.1, define the set-level countable joint observable

$$\widehat{\mathbf{I}}_\omega(X) := \left(\widehat{\mathbf{I}}_0(X), \widehat{J}_1(X), \widehat{J}_2(X), \dots\right)$$

with codomain the corresponding Cartesian product of normalized value sets. This definition is semantic and set-theoretic. It does not assert finite encodability, computability, topological convergence, measurable structure, or the existence of a finite certificate.

Definition SDR.8.5 (Limit defect)**.** The *limit defect* of the stream is

$$D_\omega := \bigcap_{n\geq 0} D_n.$$

It consists of target-discordant pairs that survive every finite refinement stage.

Theorem SDR.8.6 (SDR-COMP-04: countable packet–limit defect identity)**.** *For every dynamic target-separation stream,*

$$\mathrm{EqKer}(\widehat{\mathbf{I}}_\omega) = \bigcap_{n\geq 0} \mathrm{EqKer}(\widehat{\mathbf{I}}_n),$$

and therefore

$$\Delta_\Theta(\mathbf{I}_\omega) = D_\omega = \bigcap_{n\geq 0} D_n.$$

Consequently,

$$\mathbf{I}_\omega \text{ is target-sufficient for } \Theta \quad\Longleftrightarrow\quad D_\omega = \varnothing.$$

Proof. Two source objects have equal countable packet values exactly when every coordinate agrees, equivalently when they agree in every finite prefix packet. This proves the equality-kernel identity. Intersecting with the fixed target-discordant relation C_Θ and using Definition SDR.3.6 gives the defect identity. The final equivalence is Theorem SDR.3.8 applied to $\mathbf{I}_\omega$. □

Policy 85 (SDR-COMP-05: countable semantic sufficiency is not a finite certificate)**.** *The statement $D_\omega = \varnothing$ may prove semantic sufficiency of the full countable packet while every finite-stage defect D_n remains nonempty. Such a result is not serialized as a finite-stage certificate unless a separate reduction theorem produces a finite witness, a finite index N, or an independently verifiable finite representation of the infinite packet.*

Example SDR.8.7 (Limit sufficiency with no sufficient finite prefix)**.** Let

$$\mathcal{S} = \{0,1\}^{\mathbb{N}}, \qquad \Theta = \mathrm{id}_{\mathcal{S}},$$

and let J_n read the n-th binary coordinate. Take $\mathbf{I}_0$ constant. Then every finite packet $\mathbf{I}_n$ identifies two sequences agreeing on their first n coordinates but differing later, so $D_n \neq \varnothing$ for every n. The countable packet records every coordinate, hence is injective and

$$D_\omega = \varnothing.$$

Thus exact limit sufficiency need not have a sufficient finite prefix.

Example SDR.8.8 (Strict improvement forever with a permanent hidden survivor)**.** Let

$$\mathcal{S} = \{0,1\}^{\mathbb{N}} \times \{0,1\}, \qquad \Theta = \mathrm{id}_{\mathcal{S}},$$

and let J_n read only the n-th coordinate of the first factor. Every stage strictly refines source information, and the defects strictly decrease because additional sequence coordinates are separated. Nevertheless pairs

$$((s,0),(s,1))$$

remain indistinguishable for every $s \in \{0,1\}^{\mathbb{N}}$, so $D_\omega \neq \varnothing$. Strict improvement, even at every stage, therefore has no zero-force by itself.

SDR.8.3 Finite extraction from an exact limit statement

Theorem SDR.8.9 (SDR-COMP-06: finite-conflict extraction from empty limit defect)**.** *Suppose the initial semantic defect D_0 is finite. Then*

$$D_\omega = \varnothing \quad\Longleftrightarrow\quad D_N = \varnothing \text{ for some finite } N.$$

Equivalently, on a finite initial target-conflict universe, countable target sufficiency is already achieved by a finite prefix.

Proof. The reverse implication is immediate from $D_\omega \subseteq D_N$. For the forward implication, every $w \in D_0$ is absent from D_ω, so there exists a finite stage $n(w)$ at which $w \notin D_{n(w)}$. Since D_0 is finite, take

$$N := \max_{w \in D_0} n(w).$$

The defects are descending, hence every $w \in D_0$ is absent from D_N, so $D_N = \varnothing$. □

Remark SDR.8.10 (Relation to SDR.7)**.** Theorem SDR.8.9 is the dynamic counterpart of the finite cover calculus of SDR.7. It does not identify a minimum separating prefix, a minimum probe subfamily, or a minimum certificate cost. Those remain separate SDR.7 optimization questions.

Theorem SDR.8.11 (SDR-COMP-07: compact closed-defect extraction)**.** *Let $U \subseteq \mathcal{S} \times \mathcal{S}$ be a declared compact topological carrier for the relevant ordered source pairs. Suppose every D_n is a closed subset of U and*

$$D_{n+1} \subseteq D_n.$$

If

$$\bigcap_{n \geq 0} D_n = \varnothing,$$

then $D_N = \varnothing$ for some finite N.

Proof. If every D_n were nonempty, the nested family of nonempty closed subsets of the compact space U would have the finite-intersection property and hence nonempty total intersection, contradicting the hypothesis. Therefore some D_N is empty. □

Policy 86 (SDR-COMP-08: compactness requires typed topology)**.** *Compactness may be invoked only after the topology on the defect carrier, closedness of the stage defects, and compatibility of that topology with the exact target-separation semantics are declared and proved. Calling a parameter range "bounded", a search region "finite-dimensional", or a numerical trajectory "convergent" does not supply the compact closed-defect hypotheses of Theorem SDR.8.11.*

Remark SDR.8.12 (Closedness is load-bearing)**.** Example SDR.8.7 can be placed on the compact Cantor source $\{0,1\}^{\mathbb{N}}$ with its product topology, yet the defect sets for the identity target are obtained by intersecting a closed prefix-equality relation with the complement of the diagonal. They need not be closed. Thus compactness of the source or pair space alone does not force a sufficient finite prefix; the closed-defect hypothesis in Theorem SDR.8.11 is essential.

SDR.8.4 Rank-to-residual completion

Definition SDR.8.13 (Separated residual filtration of target conflicts)**.** Let $(\Lambda, \preceq)$ be a directed preorder. A *target-conflict residual filtration* is a family

$$\{\mathcal{R}_\lambda\}_{\lambda \in \Lambda}, \qquad \mathcal{R}_\lambda \subseteq D_0,$$

such that

$$\lambda \preceq \mu \quad \Longrightarrow \quad \mathcal{R}_\mu \subseteq \mathcal{R}_\lambda.$$

It is *separated* when

$$\bigcap_{\lambda \in \Lambda} \mathcal{R}_\lambda = \varnothing.$$

A rank sequence (λ_n) is *cofinal* when for every $\lambda \in \Lambda$ there exists n with $\lambda \preceq \lambda_n$.

Definition SDR.8.14 (Rank-to-residual bridge)**.** A dynamic defect stream has a *rank-to-residual bridge* to the filtration $\{\mathcal{R}_\lambda\}$ if there is a stage-rank sequence (λ_n) such that

$$D_n \subseteq \mathcal{R}_{\lambda_n} \qquad (n \geq 0).$$

The bridge is a theorem-facing obligation. A numeric score or an order label attached to the stages is not such a bridge unless the displayed containment is proved.

Definition SDR.8.15 (Non-circular residual package)**.** A rank-to-residual package is *non-circular* when the residual family $\{\mathcal{R}_\lambda\}$, its monotonicity, its separatedness theorem, and the containments $D_n \subseteq \mathcal{R}_{\lambda_n}$ are proved from source-semantic or independently admissible premises whose proof DAG does not consume the desired conclusion $D_\omega = \varnothing$, target sufficiency of $\mathbf{I}_\omega$, or benchmark-only target labels. The package records these premise identities and their temporal provenance.

Policy 87 (SDR-COMP-09A: no completion by conclusion-coded residuals)**.** *Theorem SDR.8.16 is a valid conditional implication for any residual filtration satisfying its hypotheses, but a theorem-facing completion claim may invoke it only through a non-circular residual package. In particular, choosing $\mathcal{R}_n := D_n$ and then assuming $\bigcap_n \mathcal{R}_n = \varnothing$ merely restates the desired conclusion and supplies no independent completion theorem. Likewise, a filtration defined from hidden target answers or from the final benchmark verdict is quarantined from independent discovery.*

Theorem SDR.8.16 (SDR-COMP-09: rank-to-residual completion theorem)**.** *Suppose a dynamic defect stream admits a rank-to-residual bridge*

$$D_n \subseteq \mathcal{R}_{\lambda_n}$$

to a separated target-conflict residual filtration, and suppose (λ_n) is cofinal in Λ. Then

$$D_\omega = \varnothing.$$

Hence the countable joint packet $\mathbf{I}_\omega$ is target-sufficient for Θ.

Proof. Assume $w \in D_\omega$. Then $w \in D_n \subseteq \mathcal{R}_{\lambda_n}$ for every n. Fix arbitrary $\lambda \in \Lambda$. Cofinality gives n with $\lambda \preceq \lambda_n$, hence

$$w \in \mathcal{R}_{\lambda_n} \subseteq \mathcal{R}_\lambda.$$

Thus w lies in $\bigcap_\lambda \mathcal{R}_\lambda$, contradicting separatedness. Therefore $D_\omega = \varnothing$, and Theorem SDR.8.6 gives target sufficiency. □

Corollary SDR.8.17 (Finite extraction after rank-to-residual completion)**.** *Under Theorem SDR.8.16, if either*

(a) D_0 is finite, or

(b) the defects are nested closed subsets of a compact defect carrier,

then some finite stage N satisfies $D_N = \varnothing$.

Proof. Apply Theorem SDR.8.16 and then Theorem SDR.8.9 or Theorem SDR.8.11, respectively. □

Policy 88 (SDR-COMP-10: cofinal rank without residual control has no zero-force)**.** *A cofinal sequence of stage ranks, improvement indices, loss orders, accuracy levels, or search scores does not imply $D_\omega = \varnothing$ unless a separately proved rank-to-residual theorem connects those ranks to a separated residual filtration at the exact target strength. Example SDR.8.8 remains a counterexample after assigning the cofinal rank n to stage n.*

Remark SDR.8.18 (Mathematical status of the rank-to-residual theorem)**.** Theorem SDR.8.16 is an elementary order/filter argument and is not claimed as novel mathematics. Its role is architectural: it isolates the exact theorem that must replace the invalid inference "rank improves without bound, therefore the target residual vanishes." A nontrivial successor theorem would have to construct and verify a meaningful residual filtration and rank-to-residual bridge in a substantive mathematical domain.

SDR.8.5 DCC, finite-state, and stabilization lanes

Corollary SDR.8.19 (SDR-COMP-11: DCC completion specialization)**.** *Let $\mathfrak{D}$ be a class of admissible semantic defect sets ordered by inclusion and satisfying the descending-chain condition. Let*

$$D_0 \supseteq D_1 \supseteq \cdots$$

be a dynamic defect stream such that whenever $D_n \neq \varnothing$, the next certified refinement satisfies

$$D_{n+1} \subsetneq D_n.$$

Then $D_N = \varnothing$ for some finite N.

Proof. This is the semantic-stream specialization of the descending-chain logic already isolated in Proposition SDR.4.32: if every D_n were nonempty, the displayed strict-progress hypothesis would produce an infinite strict descending chain in $\mathfrak{D}$, contradicting DCC. □

Proposition SDR.8.20 (Stabilization alone does not imply zero)**.** *Suppose only that the descending chain stabilizes:*

$$D_n = D_N \qquad (n \geq N).$$

No conclusion $D_N = \varnothing$ follows without an independent zero-reflection or strict-progress theorem.

Proof. Take any nonempty set E and set $D_n = E$ for all $n \geq N$. The chain is stable and the residual defect is $E \neq \varnothing$. □

Definition SDR.8.21 (Zero-reflecting ranked finite-state defect reduction)**.** A *zero-reflecting finite-state reduction* of a dynamic defect stream consists of a finite state set Q, a distinguished state q_0, a state map σ defined on the admissible defects, and a rank

$$r : Q \to \mathbb{N}$$

such that

(i) $\sigma(D) = q_0$ if and only if $D = \varnothing$;

(ii) if $D_n \neq \varnothing$, then

$$r(\sigma(D_{n+1})) < r(\sigma(D_n));$$

(iii) every state comparison and the zero-reflection clause are theorem-certified from the underlying source data.

Theorem SDR.8.22 (SDR-COMP-12: ranked finite-state zero termination)**.** *Every zero-reflecting finite-state defect reduction reaches $D_N = \varnothing$ after finitely many strict refinement steps. More precisely,*

$$N \leq r(\sigma(D_0))$$

for a run indexed so that each nonzero stage consumes one strict rank decrease.

Proof. The nonnegative integer rank strictly decreases at every nonzero stage, so after at most $r(\sigma(D_0))$ such steps the distinguished zero state must be reached. Zero reflection then gives $D_N = \varnothing$. □

Policy 89 (SDR-COMP-12A: finite state alone is not zero reflection)**.** *Finiteness of a state registry, eventual periodicity, or recurrence of a state does not imply target zero. Theorem SDR.8.22 uses the additional zero-reflection equivalence and a strictly decreasing well-founded rank. Without those hypotheses, a finite-state process may stabilize or cycle forever at a nonzero defect.*

Declaration SDR.8.23 (Inherited UB finite-state derived-limit positive control)**.** Frozen Appendix UB, Theorem 15.2, provides an existing nontrivial finite-to-infinite positive control in a different but compatible lane: for a strict finite-state descent tower with a certified eventual period, stalkwise Fitting stabilization yields stable-image diagrams, vanishing $\varprojlim^1$, and an identification of derived-inverse-limit cohomology with the descended stable image. In that declared source regime, the derived inverse limit is acyclic exactly when the certified local stable images vanish. SDR.8 imports this only as a positive control and does not rename it as an SDR theorem or generalize it to arbitrary towers.

Policy 90 (SDR-COMP-13: no arbitrary tower-prefix promotion)**.** *A finite prefix of an inverse system, repeated values on sampled levels, or an observed period does not establish a derived-limit theorem. The finite-state presentation, eventual-state equality, bonding maps, stabilization exponent, and every hypothesis required by the invoked Mittag–Leffler, stable-image, Roos–Milnor, or alternative tail theorem must be certified explicitly.*

SDR.8.6 State completion and filtered residual closure

The set-valued target defect D_n is not the same object as a residual element attached to an approximating state. Many iterative arguments construct states

$$x_0, x_1, \dots$$

and seek a limit x_∞ satisfying a residual equation $\Phi(x_\infty) = 0$. SDR therefore keeps a second completion lane for residual values.

Definition SDR.8.24 (Separated filtered residual datum)**.** A separated filtered residual datum consists of a space X, a space Y with distinguished point 0_Y, a descending family of closed subsets

$$Y = F^0Y \supseteq F^1Y \supseteq F^2Y \supseteq \cdots, \qquad \bigcap_{r\geq 0} F^rY = \{0_Y\},$$

a continuous residual map $\Phi : X \to Y$, a convergent state sequence $x_n \to x_\infty$, and depths $r_n \to \infty$ satisfying

$$\Phi(x_n) \in F^{r_n}Y.$$

Declaration SDR.8.25 (SDR-COMP-14: imported BFA post-completion residual closure)**.** Appendix BFA.13 owns the separated-filter residual-limit lemma. Applied to Definition SDR.8.24, that frozen control gives

$$\Phi(x_\infty) = 0_Y.$$

Indeed, for each fixed r, eventual membership $\Phi(x_n) \in F^rY$, continuity, and closedness give $\Phi(x_\infty) \in F^rY$; separatedness then forces the distinguished zero. SDR records this as an imported consumer control rather than a second theorem identity.

Remark SDR.8.26 (Inherited BFA control and exact ceiling)**.** The imported control assumes that x_∞ already exists; it does not construct a completion, prove that (x_n) is Cauchy, or control derivative loss or stage-dependent analytic constants. This is exactly the ceiling stated by the frozen BFA owner.

Declaration SDR.8.27 (SDR-COMP-15: imported BFA fixed-stage filtered comparison)**.** Appendix BFA.13 also owns the fixed-stage filtered residual-closure lemma. In the present notation, let $(Y, +)$ be an abelian group with a descending separated subgroup filtration

$$Y = F^0Y \supseteq F^1Y \supseteq \cdots, \qquad \bigcap_{r\geq 0} F^rY = \{0\}.$$

If for every r there exists a finite stage $N(r)$ such that

$$\Phi(x_{N(r)}) \in F^rY, \qquad \Phi(x_\infty) - \Phi(x_{N(r)}) \in F^rY,$$

then subgroup closure places $\Phi(x_\infty)$ in every F^rY, hence $\Phi(x_\infty) = 0$. SDR consumes this result without changing its BFA ownership or promotion status.

Policy 91 (SDR-COMP-15A: imported control is not theorem re-ownership)**.** *Restating an inherited BFA residual lemma in SDR notation creates no new theorem owner, novelty claim, or successor-Core promotion. Any stronger completion theorem must add a genuinely new construction or implication, such as existence of the completed state, a nontrivial rank-to-residual bridge, or a source-specific convergence theorem.*

Policy 92 (SDR-COMP-16: separation completion and state completion are distinct lanes)**.** *The conclusions*

$$D_\omega = \varnothing \qquad \textit{and} \qquad \Phi(x_\infty) = 0$$

are predicates on different objects. Neither implies the other without an explicit interface theorem relating the dynamic observable packet, the state sequence, the residual map, and the requested target. A successful residual iteration does not automatically prove target rigidity, and a target-separating countable packet does not construct a convergent solution state.

SDR.8.7 Certified evidence under dynamic refinement

Definition SDR.8.28 (Stagewise certified defect ledger)**.** Let $\mathcal{E}_n$ be the target-evidence regime available at stage n. Define

$$C_n := \Delta_\Theta^{\mathrm{cert},\mathcal{E}_n}(\mathbf{I}_n) \subseteq D_n.$$

A *persistent evidence protocol* additionally records immutable proof identities and carries every still-applicable target-discordance certificate forward to later stages.

Proposition SDR.8.29 (SDR-COMP-17: certified defect counts need not be monotone)**.** *Even though*

$$D_{n+1} \subseteq D_n,$$

no inclusion relation between C_n and C_{n+1} is automatic when the evidence regime changes with n. New target-discordance certificates may reveal previously uncertified pairs that remain in D_{n+1}, while newly added probes may resolve previously certified pairs.

Proof. The first phenomenon can add an element to C_{n+1} that was not in C_n; the second can remove an element of C_n from the semantic defect and hence from C_{n+1}. These mechanisms are logically independent, so neither inclusion is forced. □

Corollary SDR.8.30 (Monotonicity under fixed complete evidence)**.** *If one fixed evidence regime $\mathcal{E}$ is complete for target discordance on D_0, and every certificate remains valid under packet refinement, then*

$$C_n = D_n$$

for all n, so the certified ledger is descending and exactly tracks the semantic defect.

Policy 93 (SDR-COMP-18: no progress theorem from witness-count decline alone)**.** *A decline in the number of currently stored witnesses, unresolved benchmark cases, or failed test instances is not a semantic defect theorem unless evidence completeness and the counting universe are fixed. Conversely, an increase in discovered witnesses can coexist with genuine semantic improvement because the evidence regime may simply have become stronger.*

SDR.8.8 Completion statuses and theorem-facing certificate

Definition SDR.8.31 (SDR.8 lane-separated completion status)**.** A dynamic route reports a typed status vector

$$\mathsf{CompStatus} = (\mathsf{SepStatus}, \mathsf{StateStatus}, \mathsf{TailStatus}, \mathsf{CertStatus}),$$

whose coordinates are evaluated independently.

(i) SepStatus is one of finite_stage_rigidity, limit_semantic_rigidity, partial_target_completion, unresolved, or reject.

(ii) StateStatus is one of not_invoked, residual_zero_after_completion, partial_residual_completion, unresolved, or reject.

(iii) TailStatus is one of not_invoked, finite_state_tail_return, noetherian_tail_return, mittag_leffler_tail_return, other_certified_tail_return, unresolved, or reject.

(iv) CertStatus is one of semantic_only, finite_theorem_reduction, finite_replayable_certificate, not_requested, unresolved, or reject.

Each coordinate carries its exact source, target, hypotheses, and artifacts. No favorable value in one lane compensates for an unresolved or rejected obligation in another lane that is required by the requested Return.

Policy 94 (SDR-COMP-19: no scalar completion status)**.** *Because target separation, state residual closure, tail reduction, and finite certification are predicates on different objects, SDR.8 does not collapse them into one ordered word such as "complete". A route may, for example, have* SepStatus = limit_semantic_rigidity *while* CertStatus = semantic_only*, or it may close a residual equation while target rigidity remains unresolved. Promotion consumes the full status vector.*

Definition SDR.8.32 (Canonical SDR.8 completion record)**.** A canonical SDR.8 record is

$$\begin{aligned}\mathfrak{R}_{\mathrm{SDR8}} := \big(&\mathcal{S}, \Theta, \mathbf{I}_0, \{J_n\}_{n\geq 1}, \{D_n\}_{n\geq 0}, D_\omega,\\ &\mathsf{InfinitePacket}, \mathsf{EvidenceRegimes}, \{C_n\},\\ &\mathsf{RankSpace}, \{\lambda_n\}, \mathsf{Cofinality}, \mathsf{ResidualFiltration}, \mathsf{RankToResidual},\\ &\mathsf{FiniteExtractionMode}, \mathsf{CompactnessOrDCC}, \mathsf{FiniteStateRecord},\\ &\mathsf{StateSequence}, x_\infty, \Phi, \mathsf{FilteredResidualRecord},\\ &\mathsf{TailTheorem}, \mathsf{LimitOrDerivedLimitObject}, \mathsf{ZeroReflection}, \mathsf{InterfaceTheorem},\\ &\mathsf{CompStatus}, \mathsf{ReturnStrength}, \mathsf{FailureOwner}, \mathsf{NoveltyStatus}, \mathcal{A}\big).\end{aligned}$$

Absent fields are recorded explicitly as not invoked rather than filled optimistically.

Audit Obligation SDR.8.33 (Finite-to-infinite completion audit)**.** *Before an SDR route promotes finite-stage improvement to an infinite conclusion, the audit must answer:*

(i) what exact semantic object survives all finite stages;

(ii) whether the conclusion concerns D_ω, a state residual, an inverse limit, a derived inverse limit, or another typed limit;

(iii) which theorem proves cofinality, separatedness, compactness, DCC, Mittag–Leffler, finite-state reduction, or convergence;

(iv) whether the theorem yields merely semantic limit sufficiency or an actual finite certificate;

(v) whether target-evidence completeness is available or only a certified subledger is being tracked;

(vi) whether adaptive probe selection and every residual/completion premise satisfy the oracle and non-circular proof-DAG policies;

(vii) whether a decoder or Return map is effective at the requested strength; and

(viii) which hypothesis or artifact owns failure if the completion does not close.

Declaration SDR.8.34 (SDR.8 explicit nonclaims)**.** SDR.8 does not claim:

(i) that a strictly decreasing defect sequence has empty intersection;

(ii) that cofinal rank improvement implies residual vanishing without a rank-to-residual bridge;

(iii) that $D_\omega = \varnothing$ implies a sufficient finite prefix on an arbitrary infinite source class;

(iv) that countable semantic sufficiency supplies a finite, computable, or efficiently decodable certificate;

(v) that stabilization of a defect, state, tower, rank, or numerical score implies zero;

(vi) that a finite tower prefix establishes Mittag–Leffler, $\varprojlim^1 = 0$, or derived-limit control;

(vii) that the imported UB finite-state theorem applies to arbitrary non-strict, nonperiodic, infinite-dimensional, or higher-coherence towers;

(viii) that the filtered residual lemmas construct the state limit they consume;

(ix) that residual zero at a completed state is automatically target rigidity;

(x) that target rigidity of the infinite packet constructs a solution state;

(xi) that observed or certified witness counts track semantic defect size without evidence completeness;

(xii) that compactness is available without a declared topology and closed-defect theorem;

(xiii) that a conclusion-coded residual filtration, target-oracle-derived rank, or benchmark-selected infinite stream constitutes an independent completion proof;

(xiv) that one successful completion lane compensates for a missing required lane in the status vector;

(xv) that the elementary rank-to-residual, compactness, or imported filtered-closure controls are new mathematics; or

(xvi) that any SDR.8 conclusion modifies the frozen v21 finite-to-infinite, UB, BFA, or Return ownership rules.

Theorem SDR.8.35 (SDR.8 dynamic-defect and completion closure theorem)**.** *Within the frozen-v21 constitutional boundary and the SDR.2–SDR.7 target-relative calculus, SDR.8 establishes:*

(i) a dynamic observable stream has an exact descending semantic defect recurrence, and the countable joint packet has defect equal to the intersection of all finite-stage defects;

(ii) empty limit defect is exactly target sufficiency of the countable packet, but need not be achieved at any finite stage on an infinite source class;

(iii) finite initial conflict, compact closed-defect structure, DCC with strict nonzero progress, or a certified zero-reflecting finite-state reduction can convert suitable infinite or asymptotic information into finite-stage target sufficiency;

(iv) a cofinal rank proves limit sufficiency only after an explicit rank-to-residual bridge into a separated residual filtration;

(v) frozen UB finite-state derived-limit theorems provide a positive control for one certified inverse-limit lane without licensing arbitrary tower promotion;

(vi) state-residual completion is distinct from target-separation completion, and frozen BFA filtered-residual controls yield residual zero only after the relevant state limit or fixed-stage comparison hypotheses are independently supplied;

(vii) stagewise certified witness ledgers are not automatically monotone even when semantic defects are, unless evidence completeness and persistence are fixed; and

(viii) every finite-to-infinite Return retains a lane-separated completion-status vector, tail theorem, zero-reflection or rigidity edge, source/target interface, failure owner, and promotion ceiling.

Proof. Items (i)–(ii) are Theorems SDR.8.2 and SDR.8.6 with Example SDR.8.7. Item (iii) is Theorems SDR.8.9, SDR.8.11, SDR.8.19, and SDR.8.22. Item (iv) is Theorem SDR.8.16 with Policies 87 and 88. Item (v) is Declaration SDR.8.23 and Policy 90. Item (vi) is Declarations SDR.8.25 and SDR.8.27 with Policies 91 and 92. Item (vii) is Proposition SDR.8.29, Corollary SDR.8.30, and Policy 93. Item (viii) is Definition SDR.8.31, Policy 94, Definition SDR.8.32, and Audit Obligation SDR.8.33. □

Remark SDR.8.36 (Next consumer: SDR.9)**.** SDR.9 shall instantiate the completion architecture in the higher Bockstein–Fitting reconstruction program for nonsplit repeated cyclotomic factors. In that setting, the decisive burden is no longer to state that a finite-layer rank, jet, or Fitting profile improves, but to prove an exact bridge from the accessible finite-layer data to a separated residual structure strong enough to reconstruct the requested elementary-divisor information. SDR.8 supplies the admissible completion shapes; SDR.9 must supply the substantive algebra.

SDR.9 Higher Bockstein–Fitting Structural Reconstruction

SDR.9.1 Role, source binding, and exact reconstruction target

Declaration SDR.9.1 (SDR-BF-01: purpose and owner boundary)**.** SDR.9 supplies a positive structural-reconstruction theorem for the one-variable repeated-cyclotomic defect left open by the frozen UB roadmap. The frozen UB results remain authoritative at their printed strengths: VC11 proves a first-order

transverse Bockstein theorem for a square presentation and exact all-layer formulas for cyclic elementary divisors and exact finite direct sums, while an arbitrary nonsplit square presentation with repeated cyclotomic factors remains outside that closed class unless additional elementary-divisor or higher-Fitting information is supplied.

SDR.9 does not rewrite UB-VC11. It constructs an SDR-local post-v21 theorem on a precisely declared algebraic source class and records the exact Return ceiling.

Definition SDR.9.2 (SDR-BF-02: square-presented one-variable source)**.** Let O be a complete discrete valuation ring with fraction field K, and put

$$\Lambda := O[[T]].$$

Fix an exact square presentation

$$\Lambda^d \xrightarrow{A} \Lambda^d \longrightarrow M \longrightarrow 0, \qquad A \in M_d(\Lambda), \qquad \det A \neq 0.$$

The word *exact* means that M is the displayed cokernel, not merely a module having the same characteristic ideal or a pseudo-isomorphic surrogate.

Definition SDR.9.3 (SDR-BF-03: cyclotomic height-one prime packet)**.** A cyclotomic height-one prime packet for Λ is

$$\mathfrak{q} = (q, \mathfrak{p}_q, R_q, \kappa_q, v_q, \delta_q, \mathsf{CycloBind}),$$

where:

(i) $q \in \Lambda$ generates a height-one prime $\mathfrak{p}_q = (q)$ occurring as a certified irreducible factor of a declared cyclotomic layer polynomial;

(ii) $R_q := \Lambda_{\mathfrak{p}_q}$ is the corresponding discrete valuation ring, with uniformizer represented by q;

(iii) $\kappa_q := R_q/qR_q$ is its residue field and v_q is the normalized valuation;

(iv) whenever a finite-layer rank theorem is invoked, the source additionally certifies that

$$K_q := (\Lambda/(q)) \otimes_O K$$

is a field canonically identified with κ_q, and records the finite residue degree

$$\delta_q := [K_q : K] = \dim_K K_q < \infty;$$

(v) CycloBind proves the claimed relation between q and the relevant finite-layer polynomial ω_n.

The packet is source data. Factorization, irreducibility after inverting the coefficient uniformizer, residue-field identification, or character twisting needed to obtain such a packet is not inferred from notation alone.

Definition SDR.9.4 (SDR-BF-04: local elementary-divisor target)**.** For a packet $\mathfrak{q}$, set

$$M_q := M \otimes_\Lambda R_q.$$

The local reconstruction target is the R_q-isomorphism class

$$\Theta_q(M) := [M_q]_{\cong}.$$

Equivalently, once a Smith ordering is chosen for notation, the target may be represented by a nondecreasing exponent vector

$$0 \le a_{q,1} \le \cdots \le a_{q,d}$$

such that

$$M_q \cong \bigoplus_{j=1}^{d} R_q/(q^{a_{q,j}}),$$

where $R_q/(q^0) = 0$. The zero entries are presentation-stabilization padding and carry no nonzero elementary divisor.

Policy 95 (SDR-BF-05: local target is not global source reconstruction)**.** *Reconstructing every declared M_q at cyclotomic height-one primes does not by itself reconstruct M as a Λ-module, an arithmetic source object, a pseudo-null or codimension-two sector, or an exact finite-layer reduced-torsion constant. Every such promotion requires its own theorem.*

SDR.9.2 Higher Fitting profile and exact local Smith reconstruction

Definition SDR.9.5 (SDR-BF-06: determinantal/Fitting valuation profile)**.** For $0 \le r \le d$, define

$$\sigma_{q,0}(A) := 0,$$

and for $1 \le r \le d$,

$$\sigma_{q,r}(A) := \min_{\substack{I,J \subseteq \{1,\dots,d\} \\ |I|=|J|=r}} v_q\big(\det A_{I,J}\big).$$

Equivalently,

$$\sigma_{q,r}(A) = v_q\big(\mathrm{Fitt}_{d-r}(M) R_q\big),$$

with the valuation of a nonzero principal ideal understood in the DVR R_q. The full vector

$$\Sigma_q(A) := (\sigma_{q,0}, \sigma_{q,1}, \dots, \sigma_{q,d})$$

is the *higher Fitting profile* at q.

Theorem SDR.9.6 (SDR-BF-07: exact higher-Fitting reconstruction)**.** *For the source of Definition SDR.9.2 and a cyclotomic height-one prime packet* $\mathfrak{q}$*, define*

$$a_{q,r} := \sigma_{q,r}(A) - \sigma_{q,r-1}(A), \qquad 1 \le r \le d.$$

Then

$$0 \le a_{q,1} \le \cdots \le a_{q,d},$$

and there is an R_q*-module isomorphism*

$$M_q \cong \bigoplus_{r=1}^{d} R_q/(q^{a_{q,r}}).$$

Consequently the higher Fitting profile reconstructs $\Theta_q(M)$ *exactly, with no prior split decomposition of* M *over* Λ*.*

Proof. The ring R_q is a DVR. Apply Smith normal form to A over R_q: there exist $U, V \in \mathrm{GL}_d(R_q)$ and integers

$$0 \le b_1 \le \cdots \le b_d$$

such that

$$UAV = \mathrm{diag}(q^{b_1}, \dots, q^{b_d}).$$

For a diagonal Smith matrix, the least valuation of an $r \times r$ minor is

$$b_1 + \cdots + b_r.$$

Invertible row and column operations do not change the localized determinantal ideals. Hence

$$\sigma_{q,r} = b_1 + \cdots + b_r.$$

Taking successive differences gives $a_{q,r} = b_r$. The cokernel of the Smith matrix is the displayed direct sum, proving the claim. □

Corollary SDR.9.7 (SDR-BF-08: determinant order is only the terminal Fitting coordinate)**.** *One has*

$$\sigma_{q,d}(A) = v_q(\det A) = \sum_{r=1}^{d} a_{q,r}.$$

Thus a determinant records only the total local multiplicity. It does not in general record how that multiplicity is partitioned among local elementary divisors.

Policy 96 (SDR-BF-09: no novelty by Smith/Fitting relabeling)**.** *Theorem SDR.9.6 is a classical DVR/Smith/Fitting consequence expressed in SDR target-relative form. SDR claims no mathematical novelty for Smith normal form, determinantal divisors, or the equality between localized Fitting valuations and the elementary-divisor partition. Its role is to close a precisely typed reconstruction edge that determinant-only UB deliberately left open.*

Policy 97 (SDR-BF-09A: presentation-stabilization normalization)**.** *The raw cumulative vector $\Sigma_q(A)$ is indexed by the displayed square-presentation size d. It is therefore a presentation-sized encoding, not a literal presentation-independent coordinate vector across arbitrary stabilizations. When two exact square presentations of the same module have different sizes, theorem-facing comparison uses either the intrinsic localized Fitting ideals themselves or the normalized positive elementary-divisor multiset*

$$\lambda_q(M) := \{\{a_{q,r} : a_{q,r} > 0\}\},$$

not literal equality of raw vectors of different lengths. This is a source-semantics firewall: target reconstruction may be presentation-independent even when a convenient finite encoding carries presentation padding.

Proposition SDR.9.8 (SDR-BF-09B: identity stabilization preserves the decoded local target)**.** *Let*

$$A' := A \oplus I_e$$

be an identity stabilization of an exact square presentation. Then

$$\operatorname{coker}(A') \cong \operatorname{coker}(A),$$

and at every declared height-one prime q, the Smith exponent list for A' is obtained from that of A by adjoining e zero exponents. Equivalently, if $\sigma'_{q,r}$ denotes the cumulative profile of A', then

$$\sigma'_{q,r} = 0 \qquad (0 \le r \le e),$$

and

$$\sigma'_{q,e+r} = \sigma_{q,r} \qquad (1 \le r \le d).$$

Hence $\lambda_q(M)$ and the local target $[M_q]_{\cong}$ are unchanged even though the literal raw vector Σ_q changes length.

Proof. Over the DVR R_q, the identity block contributes e Smith factors equal to $q^0 = 1$, whose cokernels vanish. Sorting the Smith exponents therefore inserts e initial zeros before the exponent list of A. The displayed cumulative formula follows by summing the ordered exponents, and the cokernel and positive-exponent multiset are unchanged. □

SDR.9.3 q-adic survivor profile and Bockstein terminology ceiling

Definition SDR.9.9 (SDR-BF-10: q-adic kernel-survivor spaces)**.** Work over R_q. For each integer $s \ge 1$, define

$$\mathcal{B}_q^{(s)}(A) := \operatorname{im}\left(\{x \in R_q^d : Ax \in q^s R_q^d\} \longrightarrow R_q^d/qR_q^d\right) \subseteq \kappa_q^d,$$

where the displayed arrow is reduction modulo q. Set

$$h_{q,s}(A) := \dim_{\kappa_q} \mathcal{B}_q^{(s)}(A).$$

The descending sequence

$$\mathcal{B}_q^{(1)}(A) \supseteq \mathcal{B}_q^{(2)}(A) \supseteq \cdots$$

is the *q-adic kernel-survivor filtration*. In the local terminology of SDR.9 it may also be called the *Bockstein-survivor filtration*; it records which mod-q kernel directions lift through progressively deeper q-adic obstruction order.

Policy 98 (SDR-BF-10A: survivor filtration is not a spectral-sequence theorem)**.** *The local alias* Bockstein survivor *denotes only the explicit liftability filtration of Definition SDR.9.9. SDR.9 neither identifies it with every higher-Bockstein spectral-sequence convention nor constructs higher differentials, convergence pages, or a general Bockstein exact couple. Any promotion from this survivor filtration to a genuine higher-Bockstein spectral-sequence theorem requires an independent comparison theorem.*

Theorem SDR.9.10 (SDR-BF-11: kernel-survivor profile equals the conjugate elementary-divisor profile)**.** *With the exponents of Theorem SDR.9.6, for every* $s \geq 1$*,*

$$h_{q,s}(A) = \#\{r \in \{1, \dots, d\} : a_{q,r} \geq s\}.$$

In particular,

$$h_{q,1} = \dim_{\kappa_q} \ker(A \bmod q)$$

and

$$\#\{r : a_{q,r} = s\} = h_{q,s} - h_{q,s+1}.$$

Therefore the complete Bockstein-survivor dimension profile reconstructs the same local elementary-divisor target $\Theta_q(M)$ *as the higher Fitting profile.*

Proof. Pass to a Smith matrix $D = \operatorname{diag}(q^{a_{q,1}}, \dots, q^{a_{q,d}})$. Left multiplication by an invertible matrix preserves membership in $q^s R_q^d$, and right multiplication changes the source basis by an invertible transformation, so the survivor-space dimension is unchanged.

For the rth Smith coordinate, the condition

$$q^{a_{q,r}} x_r \in q^s R_q$$

permits an arbitrary nonzero residue class $x_r \bmod q$ exactly when $a_{q,r} \geq s$. If $a_{q,r} < s$, then $x_r \in q^{s-a_{q,r}} R_q$, so its residue modulo q is zero. The dimension formula follows coordinatewise. The final multiplicity formula is the difference between consecutive survivor counts. □

Corollary SDR.9.11 (SDR-BF-12: determinant area identity)**.** *Let*

$$E_q := v_q(\det A).$$

Then

$$E_q = \sum_{r=1}^{d} a_{q,r} = \sum_{s \geq 1} h_{q,s}.$$

Thus the determinant order is the total number of boxes in the elementary-divisor partition, while the Bockstein-survivor dimensions are the row lengths of the conjugate partition.

Corollary SDR.9.12 (SDR-BF-13: finite Bockstein depth)**.** *For* $s > E_q$*,*

$$h_{q,s} = 0.$$

Hence the finite vector

$$(h_{q,1}, \dots, h_{q,E_q})$$

reconstructs $\Theta_q(M)$ *exactly.*

SDR.9.4 Finite-jet determinacy and Fitting–Bockstein equivalence

Proposition SDR.9.13 (SDR-BF-14: finite q-adic jet dependence)**.** *For fixed* $s \geq 1$*, the survivor space* $\mathcal{B}_q^{(s)}(A)$ *depends only on the matrix class*

$$A \bmod q^s.$$

Likewise, if $E_q = v_q(\det A) < \infty$*, every determinantal valuation* $\sigma_{q,r}$ *lies in* $[0, E_q]$ *and is determined by the* q*-adic matrix jet*

$$A \bmod q^{E_q+1}$$

together with an exact certificate that $v_q(\det A) = E_q$*.*

This is finite determinacy in the q*-adic structural direction only. It does not assert that the coefficients of the jet have finite bit complexity or finite precision in the coefficient-DVR direction.*

Proof. The condition $Ax \in q^s R_q^d$ is a congruence condition modulo q^s, proving the first claim. For the second, Theorem SDR.9.6 gives

$$0 = \sigma_{q,0} \le \sigma_{q,r} \le \sigma_{q,d} = E_q.$$

The exact valuation of every nonzero minor with valuation at most E_q is visible modulo q^{E_q+1}; the minimum of those valuations gives $\sigma_{q,r}$. □

Theorem SDR.9.14 (SDR-BF-15: finite survivor/Fitting reconstruction theorem)**.** *Fix* $\mathfrak{q}$ *and let* $E_q = v_q(\det A)$. *Each of the following finite packets determines the local target* $\Theta_q(M)$ *exactly:*

(a) the higher Fitting vector $\Sigma_q(A)$*;*

(b) the Bockstein-survivor dimension vector $(h_{q,1}, \dots, h_{q,E_q})$ *together with* E_q*;*

(c) the pair consisting of an exact certificate $v_q(\det A) = E_q$ *and a certified matrix jet* $A \bmod q^{E_q+1}$ *from which either (a) or (b) is exactly replayed.*

Moreover the three packets are target-equivalent: after exact replay, each returns the same exponent multiset $\{a_{q,r} > 0\}$ *and hence the same* R_q*-isomorphism class* $[M_q]_{\cong}$.

Proof. Packet (a) reconstructs the exponents by Theorem SDR.9.6. Packet (b) reconstructs the multiplicity of exponent s as $h_{q,s} - h_{q,s+1}$, using $h_{q,E_q+1} = 0$. Packet (c) computes (a) or (b) by Proposition SDR.9.13. Finite-length modules over the DVR R_q are classified by their elementary divisors, so all three packets Return the same local target. □

Corollary SDR.9.15 (SDR-BF-16: SDR.8 completion collapses to a finite algebraic bound)**.** *For the local target* Θ_q*, no uncontrolled infinite Bockstein tower is needed: the determinant order* E_q *supplies an explicit zero-reflecting depth bound, and all survivor defects vanish beyond* E_q*. Thus the SDR.8 finite-to-infinite obligation is discharged locally by a finite bound derived from the same source presentation.*

Policy 99 (SDR-BF-17: finite structural depth is not finite-bit or cheap complexity)**.** *The bound* $E_q + 1$ *proves finite* q*-adic structural determinacy, not computational efficiency and not finite digital size. The coefficients still live in the complete coefficient DVR and may require their own exact representation or certified precision model. Computing exact determinantal valuations, all required minors, exact congruence kernels, or coefficient-DVR precision may be expensive. No polynomial-time, minimal-query, finite-bit, or optimal certificate-complexity claim is inferred.*

SDR.9.5 Resolution of the frozen UB determinant counterexample

Example SDR.9.16 (SDR-BF-18: split versus nonsplit repeated factor)**.** Take $q = T$ and, over $\Lambda = O[[T]]$, consider

$$A_{\mathrm{sp}} = \begin{pmatrix} T & 0 \\ 0 & T \end{pmatrix}, \qquad A_{\mathrm{ns}} = \begin{pmatrix} T & 1 \\ 0 & T \end{pmatrix}.$$

Both determinants are T^2. Nevertheless,

$$\Sigma_T(A_{\mathrm{sp}}) = (0, 1, 2), \qquad \Sigma_T(A_{\mathrm{ns}}) = (0, 0, 2),$$

so Theorem SDR.9.6 returns

$$\operatorname{coker}(A_{\mathrm{sp}})_T \cong R_T/(T) \oplus R_T/(T),$$

while

$$\operatorname{coker}(A_{\mathrm{ns}})_T \cong R_T/(T^2).$$

The Bockstein-survivor dimensions are respectively

$$(h_1, h_2, h_3, \dots) = (2, 0, 0, \dots)$$

and

$$(h_1, h_2, h_3, \dots) = (1, 1, 0, \dots).$$

Thus the determinant-only collision is separated by either the first nontrivial higher Fitting coordinate or the second Bockstein-survivor depth.

Proposition SDR.9.17 (SDR-BF-19: exact diagnosis of determinant information loss)**.** *On the local square-presentation class, determinant order is the projection*

$$(a_{q,1},\dots,a_{q,d}) \longmapsto \sum_{r=1}^{d} a_{q,r}.$$

Whenever two distinct exponent partitions have the same sum, determinant-only data identify distinct local targets. Higher Fitting or Bockstein-survivor data repair this specific loss by recovering the partition itself.

Proof. The projection formula is Corollary SDR.9.7. Distinct partitions with equal sum map to the same determinant order but define nonisomorphic finite-length DVR modules. Theorems SDR.9.6 and SDR.9.10 recover the full partition. □

SDR.9.6 Cyclotomic product Return and finite-layer free rank

Definition SDR.9.18 (SDR-BF-20: finite cyclotomic local profile)**.** Let $\mathcal{Q}_{\mathrm{cyc}}(A)$ be a finite, source-certified family of pairwise distinct cyclotomic height-one prime packets containing every cyclotomic height-one prime at which $M_q \neq 0$. Define

$$\Theta_{\mathrm{cyc}}(M) := \big([M_q]_{\cong}\big)_{q\in\mathcal{Q}_{\mathrm{cyc}}(A)}.$$

This is a product-record target. It does not assert that M is the direct sum of the local modules.

Theorem SDR.9.19 (SDR-BF-21: nonsplit cyclotomic local reconstruction)**.** *Assume the source supplies a finite complete cyclotomic support certificate $\mathcal{Q}_{\mathrm{cyc}}(A)$. Then the finite packet*

$$\big(\Sigma_q(A)\big)_{q\in\mathcal{Q}_{\mathrm{cyc}}(A)}$$

reconstructs $\Theta_{\mathrm{cyc}}(M)$ exactly. Equivalently, the finite family of bounded Bockstein-survivor packets from Theorem SDR.9.14 reconstructs the same product target.

No global splitting of M into cyclic or primary direct summands is assumed.

Proof. Apply Theorem SDR.9.14 independently at each member of the finite support. Product-record assembly introduces no cross-prime identification beyond the declared coordinatewise target. The conclusion is local-product reconstruction, not a global direct-sum theorem. □

Definition SDR.9.20 (SDR-BF-22: multiplicity-aware layer-factor packet)**.** Let $\omega \in O[T] \subset \Lambda$ be a monic distinguished polynomial, so

$$B_\omega := \Lambda/(\omega)$$

is finite free over O. A layer-factor packet consists of a finite family $\mathcal{Q}_\omega$ of cyclotomic height-one prime packets and positive integers

$$m_{q,\omega} := v_q(\omega) \geq 1 \qquad (q \in \mathcal{Q}_\omega)$$

such that, after inverting the coefficient uniformizer, the corresponding prime factors are pairwise coprime and exhaust the factorization of ω. Equivalently, the packet certifies the primary Chinese-remainder decomposition of

$$B_\omega \otimes_O K.$$

The multiplicities $m_{q,\omega}$ are part of the theorem-facing source data; a squarefree factorization is the special case $m_{q,\omega} = 1$ for all q.

Theorem SDR.9.21 (SDR-BF-23: multiplicity-aware finite-layer free-rank reconstruction)**.** *Under Definition SDR.9.20,*

$$\operatorname{rank}_O(M/\omega M) = \sum_{q\in\mathcal{Q}_\omega} \delta_q \sum_{r=1}^{d} \min(a_{q,r}, m_{q,\omega}).$$

Equivalently,

$$\operatorname{rank}_O(M/\omega M) = \sum_{q\in\mathcal{Q}_\omega} \delta_q \sum_{s=1}^{m_{q,\omega}} h_{q,s}(A).$$

Hence the full survivor profile, rather than only $h_{q,1}$, controls the free rank whenever the layer polynomial contains a repeated factor.

Proof. Because B_ω is finite free over O,

$$\operatorname{rank}_O(M/\omega M) = \dim_K\big((M/\omega M)\otimes_O K\big).$$

The certified primary Chinese-remainder decomposition separates the scalar-extended layer into the factors indexed by Q_ω. At a fixed q, the local Smith decomposition of Theorem SDR.9.6 gives

$$M_q \cong \bigoplus_{r=1}^{d} R_q/(q^{a_{q,r}}),$$

while ω is a unit times $q^{m_{q,\omega}}$ in R_q. Therefore

$$M_q/\omega M_q \cong \bigoplus_{r=1}^{d} R_q/(q^{\min(a_{q,r},m_{q,\omega})}).$$

Each $R_q/(q^t)$ has K-dimension $\delta_q t$, so the first formula follows after summing the primary components. Finally,

$$\sum_{r=1}^{d}\min(a_{q,r},m) = \sum_{s=1}^{m}\#\{r : a_{q,r}\ge s\} = \sum_{s=1}^{m} h_{q,s},$$

which proves the second formula. □

Corollary SDR.9.22 (SDR-BF-24: squarefree layer formula and the UB split/nonsplit distinction)**.** *If every factor of ω is squarefree after inverting the coefficient uniformizer, then $m_{q,\omega} = 1$ and Theorem SDR.9.21 reduces to*

$$\operatorname{rank}_O(M/\omega M) = \sum_{q\in Q_\omega}\delta_q h_{q,1}(A).$$

For Example SDR.9.16 at $q = T$, one has $\delta_T = 1$; hence the split and nonsplit presentations have free ranks 2 and 1, respectively, exactly recovering the distinction that determinant data alone miss.

Theorem SDR.9.23 (SDR-BF-24A: fixed-layer exact integral replay)**.** *Under Definition SDR.9.20, let $N = \deg\omega$. Via Weierstrass division,*

$$B_\omega = \Lambda/(\omega)$$

is free of rank N over O. Reducing the presentation matrix modulo ω gives an O-linear endomorphism

$$A_\omega : B_\omega^d \longrightarrow B_\omega^d$$

whose cokernel is canonically $M/\omega M$. After choosing the standard O-basis $1, T, \dots, T^{N-1}$ of B_ω, the map is represented by a finite $Nd\times Nd$ matrix $\widetilde{A}_\omega$ over O. Smith normal form of $\widetilde{A}_\omega$ returns the exact O-rank, torsion elementary divisors, and finite torsion length of the fixed layer $M/\omega M$.

Proof. Tensoring the exact presentation on the right with B_ω is right exact, so

$$M/\omega M \cong \operatorname{coker}(A_\omega).$$

The distinguished hypothesis and Weierstrass division identify B_ω with a free O-module of rank N. Choosing its standard basis converts A_ω into the stated finite matrix over the DVR O. The ordinary Smith classification over O then gives the exact free and torsion sectors. □

Policy 100 (SDR-BF-25: compressed local Return is not a uniform reduced-torsion formula)**.** *Theorem SDR.9.21 is a compressed structural Return from height-one local data and yields the free O-rank. By itself it does not yield a uniform closed formula for the exact finite reduced-torsion length on the general nonsplit square-presentation class.*

This is not an impossibility claim. The full exact presentation can always be replayed at a fixed distinguished layer by Theorem SDR.9.23. What remains unresolved is a uniform compressed all-layer theorem *deriving the reduced-torsion sector from a bounded structural packet comparable to the cyclic VC11 formula, without rebuilding the full finite-layer matrix separately at each layer.*

SDR.9.7 Certificate, failure atlas, and promotion ceiling

Definition SDR.9.24 (SDR-BF-26: canonical BF reconstruction record)**.** A canonical SDR.9 record is

$$\begin{aligned}\mathcal{R}_{\mathrm{BF}} := \big(&O, K, \Lambda, A, M, d, Q_{\mathrm{cyc}}, \{R_q, \kappa_q, v_q, \delta_q\}_q,\\ &\{\sigma_{q,r}\}_{q,r}, \{E_q\}_q, \{\mathcal{B}_q^{(s)}, h_{q,s}\}_{q,s}, \{a_{q,r}\}_{q,r},\\ &\mathsf{CycloSupportCert}, \mathsf{MinorValuationCert}, \mathsf{JetPrecision}, \mathsf{SmithReplay}, \mathsf{PresentationNormalization},\\ &\mathsf{LayerFactorCert}, \{m_{q,\omega}\}_{q,\omega}, \mathsf{RankReturn},\\ &\mathsf{FixedLayerSmithReplay}, \mathsf{ReducedTorsionStatus},\\ &\mathsf{FailureOwner}, \mathsf{NoveltyStatus}, \mathcal{A}\big).\end{aligned}$$

A field is absent only by an explicit not_invoked status.

Audit Obligation SDR.9.25 (SDR.9 Bockstein–Fitting reconstruction audit)**.** *Before a run emits a local elementary-divisor or finite-layer rank Return, it must verify:*

(i) the exact square presentation and nonzero determinant;

(ii) the presentation-size/normalization convention and, when different square presentations are compared, conversion to intrinsic localized Fitting data or the positive exponent multiset rather than literal raw-profile equality;

(iii) the height-one prime and cyclotomic source binding;

(iv) exact valuation or certified precision for every determinantal quantity actually used;

(v) completeness of the finite cyclotomic support when a product target is claimed;

(vi) consistency of the reconstructed exponent sequence

$$0 \le a_{q,1} \le \cdots \le a_{q,d};$$

(vii) agreement between Fitting and Bockstein packets when both are invoked;

(viii) the finite-layer factorization, factor multiplicities, and residue-degree data before applying Theorem SDR.9.21;

(ix) when exact fixed-layer torsion is claimed, the distinguished-layer hypothesis and the finite O-matrix $\widetilde{A}_\omega$ required by Theorem SDR.9.23;

(x) separation between local-product Return, global module Return, compressed free-rank Return, fixed-layer integral replay, and uniform all-layer reduced-torsion Return; and

(xi) preservation of every external arithmetic control theorem as a separate downstream source binding.

Policy 101 (SDR-BF-27: deterministic status partition)**.** *A BF run uses the following noncompensating status order:*

(i) undefined *if the presentation, prime binding, required valuation precision, or support coverage is absent;*

(ii) obstructed *if a certified counterexample refutes the requested stronger target from the supplied information class;*

(iii) partial_return *if local elementary divisors, compressed layer rank, or a fixed-layer integral replay is proved but a requested stronger global or uniform all-layer target remains open;*

(iv) return *only at the exact strength proved by Theorems SDR.9.14, SDR.9.19, SDR.9.21, or SDR.9.23.*

A favorable local coordinate does not compensate for a missing required coordinate.

Declaration SDR.9.26 (SDR-BF-28: novelty and literature ceiling)**.** The local Smith/Fitting reconstruction, determinantal-divisor formulas, the q-adic kernel-survivor filtration, finite DVR classification, multiplicity-aware local length calculation, and fixed-layer Smith replay are classical or standard-adjacent algebra. SDR.9 therefore claims no literature novelty for those ingredients in isolation.

The present SDR contribution is architectural: on an arbitrary nonsplit exact square presentation, a typed higher-Fitting packet or finite q-adic survivor packet repairs the determinant-only information loss at each declared cyclotomic height-one prime; the same local exponents then feed a multiplicity-aware finite-layer rank theorem, while full presentation replay separately supplies exact fixed-layer integral structure. This is not yet a genuine higher-Bockstein spectral-sequence theorem and not yet a uniform compressed nonsplit reduced-torsion formula. Any publication-level novelty claim requires a separate literature-overlap audit in SDR.10/SDR.13.

Declaration SDR.9.27 (SDR.9 explicit nonclaims)**.** SDR.9 does not claim:

(i) that $\det A$ alone reconstructs a repeated cyclotomic partition;

(ii) that local height-one elementary divisors reconstruct the global Λ-module;

(iii) that a characteristic ideal or pseudo-isomorphism class supplies the exact square-presentation record;

(iv) that the higher Bockstein-survivor filtration is a complete general higher-Bockstein spectral sequence;

(v) that the finite q-adic jet depth $E_q + 1$ is minimal, finite-bit, or computationally efficient;

(vi) that local elementary-divisor reconstruction determines codimension-two, pseudo-null, or external arithmetic data;

(vii) that the compressed local packet alone determines the exact reduced-torsion length of every layer;

(viii) that fixed-layer Smith replay supplies a uniform all-layer closed formula;

(ix) that the cyclic VC11 reduced-torsion formula extends unchanged to arbitrary nonsplit square presentations;

(x) that a source-specific arithmetic control theorem follows from module reconstruction;

(xi) that every coefficient extension has the same cyclotomic prime factorization without an explicit twist/factorization theorem;

(xii) that the literal presentation-sized vector $\Sigma_q(A)$ is unchanged under identity stabilization or arbitrary changes of square-presentation size without the normalization of Policy 97; or

(xiii) that the classical algebra used in the proof is newly discovered by SDR.

Theorem SDR.9.28 (SDR-BF-29: higher Fitting–survivor structural-reconstruction closure)**.** *On the exact square-presented one-variable source class of Definition SDR.9.2, SDR.9 proves the following.*

(i) At every declared cyclotomic height-one prime, the full higher Fitting profile reconstructs the exact local elementary-divisor module M_q without a prior global split decomposition; across identity-stabilized presentation sizes, the target-facing normalized positive exponent multiset is unchanged even though the raw cumulative vector acquires zero padding.

(ii) The finite q-adic kernel-survivor profile reconstructs the same target; its dimensions are the conjugate partition of the local Smith exponents.

(iii) The determinant order is only the total multiplicity and therefore has a sharp information-loss ceiling; the frozen UB split/nonsplit collision is resolved by the enriched packets.

(iv) The determinant order provides an explicit finite q-adic depth bound, so local repeated-factor reconstruction is structurally finitely determined by a certified presentation jet, without implying finite-bit complexity.

(v) *A complete finite family of such local packets reconstructs the product of all declared cyclotomic height-one local targets without asserting a global direct-sum decomposition.*

(vi) *Under a certified multiplicity-aware finite-layer factorization,*

$$\operatorname{rank}_O(M/\omega M) = \sum_q \delta_q \sum_r \min(a_{q,r}, m_{q,\omega}) = \sum_q \delta_q \sum_{s=1}^{m_{q,\omega}} h_{q,s}.$$

(vii) *For each fixed distinguished layer ω, the full exact presentation reduces to a finite matrix over O, whose Smith form returns the exact free rank and torsion elementary divisors of $M/\omega M$.*

(viii) *A uniform compressed all-layer reduced-torsion formula for the general nonsplit class remains a stronger target and is not inferred from the local packets or from fixed-layer replay.*

Proof. Items (i)–(ii) are Theorems SDR.9.6, SDR.9.10, and SDR.9.14. Item (iii) is Corollary SDR.9.7, Example SDR.9.16, and Proposition SDR.9.17. Item (iv) is Proposition SDR.9.13, Corollary SDR.9.15, and Policy 99. Item (v) is Theorem SDR.9.19. Item (vi) is Theorem SDR.9.21. Item (vii) is Theorem SDR.9.23. Item (viii) is Policy 100 and Declaration SDR.9.27. □

Remark SDR.9.29 (Gate status and next consumer)**.** SDR.9 now supplies a nontrivial positive theorem at the *higher-Fitting / local structural-reconstruction* strength: an arbitrary nonsplit square presentation is no longer forced into determinant-only or pre-split cyclic semantics when the requested target is its cyclotomic height-one elementary-divisor profile, multiplicity-aware finite-layer rank, or a fixed-layer exact integral replay.

This does *not* close a genuine higher-Bockstein spectral-sequence program, and it does not yet provide a bounded, compressed, uniform all-layer reduced-torsion formula on the general nonsplit class. Accordingly, construction gate G6 is passed at the higher-Fitting structural-reconstruction strength, while the stronger higher-Bockstein/uniform-torsion reading remains an explicit successor problem.

SDR.10 shall now test the SDR.2–SDR.9 architecture against mathematically distinct domains and adversarial negative controls, including whether the reconstruction machinery transports outside the one-variable square-presentation setting without abstraction by erasure, while keeping calibration reuse separate from genuine holdout validation.

SDR.10 Cross-Domain Validation and Falsification

SDR.10.1 Purpose, validation authority, and the anti-confirmation-bias rule

Declaration SDR.10.1 (SDR-VAL-03: SDR.10 validation role)**.** SDR.10 tests whether the general SDR.2–SDR.8 information, defect, discovery, escalation, rigidity, minimality, and completion rules survive contact with mathematically different domains and with deliberately adverse controls. Its primary task is therefore *falsification-sensitive validation*, not theorem promotion.

The section distinguishes three notions that must not be collapsed:

(i) *mathematical correctness of a domain theorem*;

(ii) *conformance of that theorem to an SDR role assignment*; and

(iii) *independent discovery provenance* of a theorem from a quarantined precursor corpus.

A successful retrospective benchmark can establish (ii) and can consume an already proved theorem for (i), but it does not establish (iii).

Definition SDR.10.2 (SDR-VAL-04: validation-mode sort)**.** Every SDR.10 case carries one value in

$$\mathsf{ValidationMode} = \left\{ \begin{array}{l} \mathrm{retrospective_calibration}, \\ \mathrm{postconstruction_holdout}, \\ \mathrm{prospective_blind_test} \end{array} \right\}.$$

The mode answers a provenance question, not a subject-matter question. A mathematically remote domain can still be retrospective_calibration when its theorem was already used upstream while formulating, motivating, or repairing the SDR rule being tested. Conversely, postconstruction_holdout requires that the exact validation target, positive theorem packet, and adverse-control family were not consumed in the theorem-bearing construction of that SDR rule before the holdout was frozen. A prospective_blind_test additionally freezes the target, admissible precursor state, and scoring rule before theorem-bearing outcomes are exposed. None of these labels alone certifies benchmark-quarantined independent recovery; that stronger claim remains governed by Policies 9 and 10.

Definition SDR.10.3 (SDR-VAL-04A: calibration-relation field)**.** For a validation case D and an SDR rule R, define

$$\mathsf{CalRel}(D, R) \in \{\text{construction_input}, \text{postconstruction_holdout}, \text{prospective_blind}, \text{unassessed}\}.$$

The value construction_input is mandatory whenever the native theorem, its counterexample, or an equivalent theorem-bearing summary was used upstream to design or repair R. Removing a citation after construction does not change this relation.

Policy 102 (SDR-VAL-05: no benchmark-fit optimization)**.** *A candidate SDR rule may not be accepted merely because it reproduces a finite registry of known outcomes. Validation must include at least one adverse control whose theorem-backed outcome is failure, insufficiency, or an explicitly weaker Return. If a rule is tuned after observing the benchmark labels, its fit is classified as* retrospective_fit*; it is not holdout evidence and not evidence of independent discovery.*

Policy 103 (SDR-VAL-05A: holdout and blind-control freeze)**.** *A validation case may be counted as* postconstruction_holdout *only if an immutable precursor record fixes, before the case is consumed for validation, the tested SDR rule, the native source class, target, admissible evidence, and pass/fail interpretation. A* prospective_blind_test *additionally freezes the candidate-selection and scoring procedure before the theorem-bearing outcome is revealed. Cases selected or reclassified because their known answer fits the architecture remain retrospective calibration controls.*

Definition SDR.10.4 (SDR-VAL-06: typed domain-instantiation record)**.** A typed validation instance is

$$\mathfrak{V}_D = (D, \mathcal{S}_D, \Theta_D, \mathbf{I}_D, \rho_D, \mathfrak{P}_D, \mathfrak{N}_D, \mathsf{Mode}, \mathsf{CalRel}, \mathsf{Owner}, \mathcal{A}),$$

where:

(i) D names the native mathematical domain;

(ii) $\mathcal{S}_D$ is the exact source class and Θ_D the exact target;

(iii) $\mathbf{I}_D$ is the packet actually consumed by the tested SDR statement;

(iv) ρ_D is a typed role map from native objects, maps, hypotheses, and failure witnesses to the corresponding SDR source/observable/target/theorem roles;

(v) $\mathfrak{P}_D$ is a positive-control theorem packet;

(vi) $\mathfrak{N}_D$ is an adverse or negative-control packet;

(vii) Mode records the validation mode;

(viii) CalRel records whether the case was a construction input, a frozen holdout, a prospective blind test, or remains unassessed;

(ix) Owner records the native theorem owner; and

(x) $\mathcal{A}$ contains immutable source, proof, replay, and comparison artifacts.

Policy 104 (SDR-VAL-07: no abstraction by role renaming)**.** *A domain theorem does not instantiate SDR merely because its native words can be renamed "source", "defect", "probe", or "rigidity". The role map ρ_D is admissible only when every consumed equality, equivalence relation, factorization, implication direction, and failure witness has the same mathematical semantics required by the corresponding SDR definition. A failed role map makes the validation case* undefined*; it is not a negative verdict on either the native theorem or SDR.*

SDR.10.2 Validation battery and falsification semantics

Definition SDR.10.5 (SDR-VAL-08: six-part validation battery)**.** The full SDR.10 battery is a *campaign-level* object, not a property that one isolated case can satisfy by itself. The following six checks are recorded separately:

(V1) **typing**: each case has exact source, target, equivalence, and observable normalization;

(V2) **positive control**: each positive case has a native theorem realizing the advertised SDR implication;

(V3) **adverse control**: the campaign contains certified witnesses blocking tempting stronger readings;

(V4) **hypothesis mutation**: removing a declared load-bearing assumption is tested only to the strongest level supported by an actual theorem, counterexample, or no-go result;

(V5) **cross-domain transport**: the same abstract SDR rule has admissible role maps in at least two mathematically distinct domains without adding native hypotheses to the abstract rule itself; and

(V6) **provenance/ownership**: every case records theorem ownership and calibration relation, and no construction-input case is counted as a holdout or independent-recovery success.

The six coordinates are noncompensating. A single instance may discharge several coordinates, but V5 is inherently comparative and V6 is provenance-sensitive across the campaign.

Policy 105 (SDR-VAL-08A: calibration reuse is not holdout validation)**.** *A theorem may legitimately be reused as a positive calibration after it has informed an earlier SDR section, but such reuse tests consistency rather than independent predictive validity. In particular, PE-3 as used in SDR.5 and the finite-state derived-limit theorem as used in SDR.8 are construction-input controls for the corresponding architectural boundaries. Their later success in SDR.10 cannot be counted a second time as holdout evidence for those same rules.*

Definition SDR.10.6 (SDR-VAL-09: hypothesis-mutation status)**.** For a theorem implication

$$H_1 \wedge \cdots \wedge H_r \Longrightarrow C,$$

a mutation of H_i receives one of the following statuses:

$$\mathsf{MutStatus} = \left\{ \begin{array}{l} \mathrm{counterexample_critical}, \quad \mathrm{proof_scope_critical}, \\ \mathrm{redundancy_proved}, \quad \mathrm{proof_unused_only}, \quad \mathrm{unresolved} \end{array} \right\}.$$

The status $\mathrm{counterexample_critical}$ requires an admissible object satisfying all retained hypotheses but violating H_i and the conclusion. Merely observing that the registered proof uses H_i yields at most $\mathrm{proof_scope_critical}$; it does not prove logical necessity. Conversely, $\mathrm{proof_unused_only}$ means only that the registered proof DAG does not consume H_i; it is not a redundancy theorem. The stronger status $\mathrm{redundancy_proved}$ requires a separate proof of the same conclusion from the retained hypotheses with H_i removed.

Definition SDR.10.7 (SDR-VAL-10: three falsification levels)**.** SDR.10 distinguishes:

(i) *instance falsification*: one proposed route fails on one source object;

(ii) *schema falsification*: one admissible counterexample refutes a universal statement of the form "for every source in the declared class"; and

(iii) *architecture falsification*: a contradiction is exhibited among SDR's own theorem statements under one common admissible interpretation.

Instance falsification need not imply schema falsification, and schema falsification of one proposed rule need not imply architecture falsification.

Theorem SDR.10.8 (SDR-VAL-11: one-witness falsification of a universal sufficiency claim)**.** *Let a proposed validation claim assert target sufficiency of* $\mathbf{I}$ *on a class* $\mathcal{S}$*:*

$$\forall X, Y \in \mathcal{S}, \qquad \widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y) \Longrightarrow \widehat{\Theta}(X) = \widehat{\Theta}(Y).$$

One certified pair $(X, Y) \in \Delta_\Theta(\mathbf{I})$ *refutes that universal claim. The same pair does not refute a stronger packet* $\mathbf{I} \boxtimes J$*, a smaller source class, or a weaker target unless the witness remains admissible there.*

Proof. The first assertion is the exact negation witness of the displayed universal implication. The final sentence follows because observational enrichment, source restriction, and target weakening change the proposition being tested. □

SDR.10.3 Cross-domain control A: finite-poset morphism reconstruction

Definition SDR.10.9 (SDR-VAL-12: PE morphism validation instance)**.** Fix the finite sentinel category $\mathcal{S}_n$ from the frozen PE construction. For fixed endpoints $M, N \in \mathcal{S}_n$, take

$$\mathcal{S}_D := \mathrm{Hom}_{\mathcal{S}_n}(M, N)$$

and let the target Θ_D return the source morphism itself, up to the declared source equality. The lower packet $\mathbf{I}_{\mathrm{obj}}$ records only endpoint object signatures and therefore is constant on this Hom-set. The enriched packet $\mathbf{I}_{\mathrm{PE3}}$ records the finite projective-probe module action and the induced module morphism. For validation provenance this case is recorded as

$$\mathsf{Mode} = \mathrm{retrospective_calibration}, \qquad \mathsf{CalRel} = \mathrm{construction_input},$$

because the PE object/morphism boundary was already consumed in SDR.5.

Proposition SDR.10.10 (SDR-VAL-13: PE realizes the escalation–rigidity pattern)**.** *Assume* $\mathrm{Hom}_{\mathcal{S}_n}(M, N)$ *contains two distinct morphisms. Then:*

(i) $\mathbf{I}_{\mathrm{obj}}$ *is target-insufficient on* $\mathrm{Hom}_{\mathcal{S}_n}(M, N)$*;*

(ii) the frozen PE-3 full-faithfulness theorem makes $\mathbf{I}_{\mathrm{PE3}}$ *target-rigid on this Hom-set; and*

(iii) discarding the module action while retaining only object-level dimensions destroys the morphism-level Return in general.

Thus one domain-native theorem package realizes the SDR pattern

$$\textit{lower-level collision} \longrightarrow \textit{structural enrichment} \longrightarrow \textit{class-wide rigidity}.$$

Proof. The first assertion is immediate because the lower packet depends only on the fixed endpoints. The second is the PE-3 fully faithful contravariant morphism Return: distinct source morphisms induce distinct returned module morphisms. The third is exactly the object-versus-morphism strength boundary of the PE ladder. □

Policy 106 (SDR-VAL-13A: PE coordinate firewall)**.** *The PE validation target is a source morphism and the enriched output is the corresponding natural transformation or probe-module morphism under the declared fully faithful functor. Literal equality of basis matrices is not the invariant target. Any finite replay that chooses bases must normalize or quotient coordinate changes before using matrix equality as an SDR observation. Coordinate choices therefore cannot create a spurious separation witness.*

Remark SDR.10.11 (Validation meaning of the PE case)**.** This is a retrospective cross-domain control. It does not claim that SDR discovered PE-3. Its value is sharper: the generic SDR distinction between object-level data and morphism-enriched data predicts the same success/failure boundary that the independently proved finite-poset theorem exhibits.

SDR.10.4 Cross-domain control B: persistence detection and finite-observation no-go

Definition SDR.10.12 (SDR-VAL-14: right-unbounded persistence instance)**.** Let $\mathcal{U}_{a,\tau,e}$ be the right-unbounded persistence family of the frozen Foundation detector theorem, containing the zero profile and retained one-bar witnesses whose active-start intervals may lie arbitrarily far to the right. The target Θ_D is the repaired-zero predicate. A finite deterministic oracle-closed detector transcript is treated as a finite observable packet $\mathbf{I}_T$. This case is conservatively recorded as retrospective calibration rather than a holdout, because the frozen Foundation no-go and its finite-to-infinite escape conditions belong to the precursor architecture that motivated SDR's finite-observation firewall.

Theorem SDR.10.13 (SDR-VAL-15: finite observation retains a certified target defect)**.** *For every deterministic oracle-closed detector that halts after finitely many queries on every member of $\mathcal{U}_{a,\tau,e}$, there exist a zero profile B_0 and a retained one-bar witness B_1 such that the complete finite transcript is identical, while*

$$\Theta_D(B_0) \neq \Theta_D(B_1).$$

Hence the detector packet has a nonempty certified target defect and cannot be promoted to universal rigidity on the right-unbounded family.

Proof. Apply the frozen adaptive right-unbounded no-go theorem. Run the detector on the zero profile. Its finite transcript visits only a bounded query set. Place the retained bar so far to the right that every answer on the zero transcript is unchanged. Oracle-closedness forces the same adaptive path and the same verifier-visible record, while the repaired-zero target differs. This pair is exactly an SDR separation-defect witness. □

Corollary SDR.10.14 (SDR-VAL-16: coverage is theorem-bearing structure, not more sampling)**.** *The obstruction in Theorem SDR.10.13 is not removed by observing a longer but still finite prefix merely because the observed values appear stable. Within this right-unbounded point-stencil model, a finite certificate becomes admissible only after additional certified structure supplies a valid escape condition, for example a finite exhaustive birth set, a global birth bound, eventual zero, eventual stabilization, or another proved finite-to-infinite reduction. The extra structure may be source data plus a soundness theorem; merely taking a longer finite transcript is not such a reduction.*

Remark SDR.10.15 (Mutation status)**.** For the right-unbounded family, removal of a coverage/tail restriction is counterexample_critical: the far-right bar is an explicit adverse witness. This is stronger than saying only that one proof happens to use a bound.

SDR.10.5 Completion control C: finite-state derived inverse limits

Definition SDR.10.16 (SDR-VAL-17: finite-state tail validation instance)**.** Take the strict finite-state descent towers of the frozen UB derived inverse-limit theorem. The source packet contains the complete finite-state presentation, eventual period, levelwise strict descent data, bonding maps, and the stalkwise stabilizing exponent. The target asks for the cohomology of the derived inverse limit and the vanishing of the $\varprojlim^1$ defect at the declared strength. Its SDR.10 role is retrospective_calibration/construction_input, because the same finite-state theorem was already used as a positive control in SDR.8.

Proposition SDR.10.17 (SDR-VAL-18: finite-state tail theorem realizes SDR completion)**.** *On the source class of Definition SDR.10.16, the imported UB theorem provides a genuine finite-to-infinite edge: local period monodromy stabilizes to finite stable-image diagrams, strict descent globalizes those diagrams, and the resulting finite state controls derived inverse-limit cohomology with*

$$\varprojlim{}^{1} = 0.$$

Thus the case validates the SDR.8 rule that an infinite Return requires an independently proved tail/completion mechanism rather than a finite prefix alone.

Policy 107 (SDR-VAL-19: finite-state sufficiency is not finite-state necessity)**.** *The validation in Proposition SDR.10.17 establishes that the declared finite-state hypotheses are sufficient for the imported UB Return. SDR.10 does not claim that eventual periodicity or the particular stable-image presentation is logically necessary for every derived inverse-limit theorem. Removing the finite-state assumption is therefore classified as* proof_scope_critical *unless a separate counterexample to the desired stronger theorem is supplied.*

SDR.10.6 Internal positive control D: the SDR.9 nonsplit reconstruction

Definition SDR.10.18 (SDR-VAL-20: BF internal validation instance)**.** Take the nonsplit square-presented one-variable class of SDR.9. The lower packet is the determinant-only record and the target is the cyclotomic height-one elementary-divisor profile. The enriched packet is the higher-Fitting or finite q-adic kernel-survivor packet. This is an internal construction-success case, not a holdout validation case.

Proposition SDR.10.19 (SDR-VAL-21: determinant collision and enriched repair)**.** *The split/nonsplit determinant collision of SDR.9 supplies a certified defect witness for the lower packet, while Theorem SDR.9.6 and its survivor equivalent remove that defect on the declared local target class. Hence SDR.9 realizes the same abstract pattern as Proposition SDR.10.10 in a mathematically different algebraic setting.*

Remark SDR.10.20 (Why this is an internal rather than holdout control)**.** The BF case was developed inside SDR and therefore tests executable use of the architecture, but it is not holdout evidence. The PE, persistence, and finite-state cases provide mathematically distinct cross-domain calibration controls; under the reaudited provenance accounting above, none is counted as a strict postconstruction holdout for the SDR rule that already consumed it upstream.

SDR.10.7 Cross-domain mutation matrix and overfit audit

Definition SDR.10.21 (SDR-VAL-22: validation signature)**.** For a validation case D, define its signature

$$\Sigma_{\mathrm{val}}(D) = (\mathsf{Collision}, \mathsf{Repair}, \mathsf{Rigidity}, \mathsf{Completion}, \mathsf{NegativeControl}, \mathsf{MutationStatus}, \mathsf{CalRel}).$$

Each coordinate stores a theorem locator or an explicit $\mathrm{not_invoked}$ value; it is not a scalar score.

Case	Lower-information failure	Certified repair / positive control	Adverse boundary
Finite-poset PE	Object-level data do not determine a supplied morphism	Fully faithful enriched projective-probe action returns morphisms	Forgetting module action drops back to object strength
Right-unbounded persistence	Any finite halting transcript misses a far-right witness	A separately proved coverage/tail theorem may restore finite certification on a restricted class	Finite observed stability alone does not close the tail
Finite-state inverse limit	A finite prefix alone is not a derived-limit theorem	Eventual finite state plus stable-image/descent theorem controls the tail and $\varprojlim^1$	Finite-state assumptions are sufficient here, not claimed universally necessary
SDR.9 BF	Determinant total multiplicity forgets the local partition	Higher Fitting / survivor profile reconstructs the local elementary divisors	Local height-one data do not automatically reconstruct the global module or a uniform reduced-torsion formula

Table 1: SDR.10 validation and falsification matrix. The rows are not merged into one domain theorem; they test the same abstract boundaries under distinct native mathematics.

Definition SDR.10.22 (SDR-VAL-23: domain-overfit indicator)**.** An SDR rule is *domain-bound* at the current evidence state if either:

(i) its statement contains native symbols or hypotheses from only one validation domain and no typed abstraction theorem removes them; or

(ii) every positive control uses one common source theorem, one common benchmark label set, or one conclusion-derived witness family.

A rule is *cross-domain supported* only when at least two mathematically distinct admissible role maps instantiate the same abstract statement and at least one adverse control remains correctly rejected.

Policy 108 (SDR-VAL-23A: finite validation does not prove absence of overfit)**.** *No finite collection of retrospective domains proves that an SDR rule is globally domain-independent. Cross-domain support rejects only the narrower claim that the rule works solely because of one tested native formalism. A universal no-overfit claim would itself require a quantified theorem over a declared class of domains or interfaces.*

Theorem SDR.10.23 (SDR-VAL-24: cross-domain support of the information-loss core)**.** *The following SDR rules are cross-domain supported by the current validation battery:*

(i) deterministic postprocessing does not recover a certified distinction already lost by the consumed packet;

(ii) target-discordant equal-observation pairs refute universal target sufficiency;

(iii) an observational enrichment can repair a witnessed equal-observation collision only by consuming information that does not factor through the lower packet; source restriction, target weakening, and theorem-based inferential strengthening are separate repair modes;

(iv) class-wide rigidity is stronger than successful separation of finitely many benchmark pairs; and

(v) finite-to-infinite Return requires an additional tail/completion theorem unless the target has already been reduced to a certified finite problem.

This theorem is a validation theorem about the scope of the existing SDR statements, not a claim that every mathematical problem admits a useful SDR enrichment or finite completion.

Proof. Items (i)–(iii) are instantiated by the PE object/morphism separation and the BF determinant/Fitting separation, with the persistence no-go providing an adverse equal-transcript witness. The qualification in item (iii) is essential: the nonfactorization requirement concerns observational repair of the same source class and target, not source restriction or a separately proved inferential theorem. Item (iv) is enforced by the PE full-faithfulness theorem versus finite pair testing and by SDR.9's class-scoped local reconstruction theorem. Item (v) is witnessed negatively by the right-unbounded persistence theorem and positively by the finite-state UB completion theorem. None of these uses the other domain's native proof as an input, so the shared content lies in the typed SDR rule rather than a common domain theorem. This proves cross-domain consistency/support at the declared interface strength; it does not prove holdout independence of the validation campaign. □

Policy 109 (SDR-VAL-24A: no transfer of native reconstruction operators)**.** *Cross-domain support transfers only the abstract SDR implication whose role map has been checked. It does not transfer PE probe modules into Iwasawa theory, Fitting packets into persistence, finite-state monodromy into an arbitrary inverse system, or any other native reconstruction operator across domains without a new typed theorem. Structural analogy is not an implementation bridge.*

Policy 110 (SDR-VAL-25: validation is not universality)**.** *Cross-domain support means only that the tested SDR rule survived the declared positive and adverse controls. It does not imply:*

(i) that the SDR search grammar is complete;

(ii) that every insufficient observable admits a finite or computable enrichment;

(iii) that every domain has a rigidity theorem;

(iv) that every infinite problem has a finite-state reduction; or

(v) that a successful architecture-level pattern constitutes a new mathematical theorem in each domain.

SDR.10.8 Literature-overlap quarantine and gate accounting

Definition SDR.10.24 (SDR-VAL-26: provisional overlap registry)**.** Before the reference-verification work of SDR.13, SDR records the following conceptual overlap classes only:

(i) the deterministic information preorder overlaps with classical factorization and partition refinement;

(ii) counterexample-guided probe refinement has a conceptual overlap with counterexample-guided abstraction/refinement methods, but SDR's theorem objects and target-relative defect semantics are not identified with any software-verification framework by name alone;

(iii) the finite separating-family problem has the ordinary set-cover/hitting-set combinatorial core after the candidate registry is frozen;

(iv) target factorization and identifiability have conceptual overlap with sufficiency/identifiability language, but no stochastic Blackwell order or statistical experiment theorem is imported into the deterministic SDR preorder;

(v) finite-state inverse-limit closure overlaps with classical Mittag–Leffler, stable-image, and derived-limit mechanisms at the exact strengths imported from the frozen corpus; and

(vi) SDR.9 higher-Fitting/Smith reconstruction uses classical local Smith and Fitting theory rather than claiming those ingredients as new.

This registry is intentionally nonbibliographic. It prevents obvious novelty inflation but does not replace the source-by-source literature audit required by SDR.13.

Policy 111 (SDR-VAL-27: no literature novelty from architectural recombination alone)**.** *The fact that SDR places partition refinement, counterexample-guided search, finite covering, rigidity, completion, and Fitting reconstruction in one typed architecture does not by itself prove publication-level novelty. A novelty claim requires a precise theorem statement, a verified comparison class, and a documented nonredundancy result.*

Declaration SDR.10.25 (SDR-VAL-28: construction-gate status after SDR.10)**.** SDR.10 passes the *cross-domain consistency/calibration* subgate: positive and adverse controls outside the SDR.9 one-variable algebraic domain instantiate the declared SDR boundaries. It does *not* yet pass a strict independent-holdout subgate, because the PE and finite-state controls were already construction inputs for SDR.5 and SDR.8, the persistence no-go belongs to the precursor finite-observation architecture, and SDR.9 is internal. The literature-overlap component is also not yet fully passed. Consequently full G7 remains pending both (a) at least one frozen post-construction holdout or prospective blind validation at the required strength and (b) the bibliographic/reference-verification obligations of SDR.13.

Policy 112 (SDR-VAL-28A: no double counting of calibration evidence)**.** *A control theorem may support construction and later support consistency audit, but those are two uses of one evidence lineage, not two independent validations. Gate accounting therefore stores evidence lineage and counts a construction-input replay only in the calibration column.*

SDR.10.9 Canonical validation record, closure, and SDR.11 handoff

Definition SDR.10.26 (SDR-VAL-29: canonical cross-domain validation record)**.** A canonical SDR.10 record is

$$\begin{aligned}\mathcal{R}_{\mathrm{VAL}} := \big(&\mathfrak{V}_D, \mathsf{ValidationMode}, \mathsf{CalRel}, \rho_D, \mathsf{PositiveControls}, \mathsf{NegativeControls},\\ &\mathsf{MutationLedger}, \Sigma_{\mathrm{val}}(D), \mathsf{OverfitStatus}, \mathsf{OracleProvenance},\\ &\mathsf{Ownership}, \mathsf{HoldoutStatus}, \mathsf{LiteratureOverlapStatus}, \mathsf{GateStatus}, \mathcal{A}\big).\end{aligned}$$

No favorable coordinate is copied into another one.

Audit Obligation SDR.10.27 (SDR.10 validation audit)**.** *Before a validation result is used as promotion evidence, verify:*

(i) exact native source and target typing;

(ii) admissibility of the role map ρ_D;

(iii) at least one positive theorem and one adverse control at the claimed validation strength;

(iv) mutation status without upgrading proof dependence into logical necessity;

(v) benchmark and target-oracle provenance;

(vi) construction-input versus holdout calibration relation and evidence lineage;

(vii) ownership of every imported theorem;

(viii) separation of retrospective calibration, holdout validation, prospective blind testing, and independent recovery;

(ix) separation of cross-domain support from universality;

(x) literature-overlap status and any unresolved bibliography; and

(xi) immutable artifacts sufficient to replay the domain-native theorem at its exact declared strength.

Theorem SDR.10.28 (SDR-VAL-30: SDR.10 cross-domain validation closure)**.** *Under the imported native theorems and the exact role maps declared above, SDR.10 establishes:*

(i) a finite-poset category control in which object-level information is insufficient for morphism Return while enriched probe action is fully faithful;

(ii) a persistence control in which every finite halting transcript retains a certified target defect on a right-unbounded family unless additional coverage/tail structure is supplied;

(iii) a derived-limit control in which a genuine finite-state tail theorem, rather than finite observation, closes an infinite Return;

(iv) an internal nonsplit algebra control in which determinant collision is repaired by higher-Fitting/survivor structure;

(v) a noncompensating mutation and adverse-control protocol distinguishing theorem-scope dependence from certified logical necessity;

(vi) cross-domain consistency/support for the central information-loss, observational-enrichment, rigidity, and finite-to-infinite boundaries of SDR.2–SDR.8, with construction-input reuse explicitly separated from holdout evidence; and

(vii) a provisional literature-overlap quarantine that blocks novelty inflation while leaving both strict holdout validation and full reference verification open.

No item transfers native theorem ownership, certifies benchmark-quarantined independent recovery, proves universality of the SDR search grammar, supplies a strict independent holdout, or closes G7.

Proof. Items (i)–(iv) are Propositions SDR.10.10, Theorem SDR.10.13, Proposition SDR.10.17, and Proposition SDR.10.19. Item (v) is Definitions SDR.10.5–SDR.10.7 and Policy 107. Item (vi) is Theorem SDR.10.23 together with Policies 105, 108, and 109. Item (vii) is Definition SDR.10.24 and Policy 111. The final ceilings are Policies 9, 10, 110, and 112, together with Declaration SDR.10.25. □

Remark SDR.10.29 (SDR.11 handoff: validation-aware search)**.** SDR.11 shall convert the preceding static validation discipline into the search platform itself. Candidate probes and theorem routes should therefore carry, from generation time onward, a precursor snapshot, target-oracle provenance, structural type, predicted defect reduction, admissible domain class, negative-control obligations, and validation status. Search ranking may use these fields to prefer candidates that are structurally nonredundant and falsifiable; it may not use benchmark success labels as hidden theorem-bearing features in an independent lane. SDR.11 must also maintain a holdout registry: once a domain case or adverse witness is consumed for candidate generation, ranking, repair, or theorem selection, that evidence lineage is no longer eligible to certify a holdout success for the same rule.

SDR.11 SDR Search and Discovery Engine

SDR.11.1 Purpose, platform boundary, and theorem-first search

Declaration SDR.11.1 (SDR-SRCH-01: SDR.11 objective)**.** SDR.11 turns the static information, defect, discovery, escalation, rigidity, minimality, completion, and validation calculus of SDR.2–SDR.10 into a typed search architecture. Its task is to organize the transition

$$\text{certified obstruction} \rightsquigarrow \text{candidate structure} \rightsquigarrow \text{verification obligation} \rightsquigarrow \text{admissible theorem-bearing edge},$$

without identifying search success with proof, benchmark agreement with derivation, or heuristic ranking with mathematical strength.

Policy 113 (SDR-SRCH-02: search is not a new proof authority)**.** *SDR.11 creates no proof rule beyond those already admitted by the frozen v21 architecture and SDR.1–SDR.10. A generated observable, theorem statement, rigidity claim, completion mechanism, source restriction, or Return arrow remains a proposal until its exact mathematical obligations are discharged by an authorized verifier. No search trace, execution success, model confidence, agent consensus, ranking score, or replay transcript is a substitute for that discharge.*

Definition SDR.11.2 (SDR-SRCH-03: proof-first search contract)**.** A *proof-first search contract* is a tuple

$$\mathfrak{P} = (\mathfrak{T}, \mathcal{S}_*, \mathsf{Hyp}, \mathsf{ReturnStrength}, \mathsf{AllowedEvidence}, \mathsf{ForbiddenInputs}, \mathsf{FailureCeiling}),$$

where $\mathfrak{T}$ is the normalized target datum, $\mathcal{S}_*$ is the declared source class, Hyp is the frozen hypothesis package, and the remaining fields fix the requested Return strength, admissible evidence modes, prohibited oracle or benchmark inputs, and the strongest negative conclusion that a failed route is allowed to support.

Policy 114 (SDR-SRCH-04: no post-hoc target inflation)**.** *A search run is evaluated against its frozen proof-first contract. Search output may suggest a new target, weaker target, stronger hypothesis, or narrower source class, but such a change creates a new contract identity. It may not be used retroactively to declare the original run successful.*

SDR.11.2 Search state, evidence lineage, and holdout sovereignty

Definition SDR.11.3 (SDR-SRCH-05: search state)**.** At stage n, an SDR search state is

$$\mathfrak{Z}_n = (\mathfrak{P}, \mathfrak{Q}_n^-, \mathbf{I}_n, \mathfrak{W}_n, \mathfrak{C}_n, \mathfrak{B}_n, \mathfrak{F}_n, \mathfrak{H}_n, \mathfrak{L}_n, \mathfrak{B}_n).$$

Its fields are, respectively:

(i) the proof-first contract;

(ii) the immutable precursor snapshot actually available before the next synthesis step;

(iii) the currently admitted observational packet;

(iv) the certified witness/obstruction ledger;

(v) the generated candidate registry;

(vi) the verifier-admitted theorem-bearing records and observational semantics;

(vii) rejected, falsified, undefined, or scope-excluded candidates;

(viii) the holdout registry;

(ix) the evidence-lineage DAG; and

(x) the declared search budget and stopping policy.

The semantic defect $\Delta_\Theta(\mathbf{I}_n)$ is a mathematical object, not a required executable field of $\mathfrak{Z}_n$.

Definition SDR.11.4 (SDR-SRCH-06: evidence-lineage DAG)**.** The evidence-lineage DAG $\mathfrak{L}_n$ has nodes for source artifacts, benchmark artifacts, witnesses, computations, generated candidates, verifier results, theorem records, and validation cases. A directed edge $a \to b$ means that b was generated, ranked, repaired, proved, or selected using theorem-bearing or candidate-shaping information from a. Merely sharing a filename, repository, or physical storage location does not create a mathematical dependency edge, but it may create a provenance edge when it changes the accessible information state.

Definition SDR.11.5 (SDR-SRCH-07: holdout contamination closure)**.** For a proposed rule or candidate family R, define

$$\mathrm{Contam}(R) := \{e : \text{there exists a directed path } e \rightsquigarrow R \text{ in } \mathfrak{L}_n\}.$$

A validation case H is *holdout-eligible for* R only when neither H nor any theorem-bearing artifact derived from H lies in $\mathrm{Contam}(R)$.

Theorem SDR.11.6 (SDR-SRCH-08: holdout non-recycling)**.** *If a validation case H is used to generate, rank, repair, choose, or prove a rule R, then H is not a postconstruction holdout for R. Deleting the explicit edge, citation, prompt text, or local copy after the use does not restore holdout eligibility.*

Proof. By Definition SDR.11.4, the use creates a provenance path from H, or from a theorem-bearing derivative of H, to R. Definition SDR.11.5 therefore places that evidence in $\mathrm{Contam}(R)$. Holdout status is a property of the actual construction history, not of the final presentation of the proof object. □

Policy 115 (SDR-SRCH-09: lane-local holdout accounting)**.** *Holdout ineligibility is relative to the rule and evidence lineage. A case consumed by one rule is not thereby forbidden forever as evidence for every unrelated rule, but reuse requires a fresh lineage audit showing the absence of a path into the new rule's construction.*

Policy 116 (SDR-SRCH-09A: holdout eligibility is not independent recovery)**.** *Holdout eligibility is a relation between one validation case and one construction lineage. It does not by itself certify benchmark-quarantined independent recovery. A rule may be holdout-clean with respect to a case while still depending on another benchmark-derived theorem, answer, witness, target label, or equivalent oracle. Independent-recovery status therefore requires the full precursor and forbidden-input audit in addition to the holdout test.*

SDR.11.3 Candidate grammar and search lanes

Definition SDR.11.7 (SDR-SRCH-10: candidate type)**.** Every search candidate has exactly one primary type

$$\mathsf{CandType} \in \{\mathsf{Obs}, \mathsf{Inf}, \mathsf{Restr}, \mathsf{Comp}, \mathsf{Sur}, \mathsf{Target}, \mathsf{Counter}\},$$

with the following intended meanings:

(i) Obs: a new source-semantic observable;

(ii) Inf: an inferential or rigidity theorem candidate;

(iii) Restr: a justified source restriction or stratification;

(iv) Comp: a finite-to-infinite completion/tail mechanism;

(v) Sur: a surrogate target or surrogate witness relation;

(vi) Target: an explicitly changed or weakened target; and

(vii) Counter: a counterexample, adverse instance, or falsifier candidate.

A mixed proposal is decomposed into typed components plus explicit dependency edges; it is not serialized as an untyped "stronger structure" object.

Definition SDR.11.8 (SDR-SRCH-11: canonical candidate packet)**.** A generated candidate is stored as

$$\mathfrak{c} := \big(\mathsf{UID}, \mathsf{CandType}, \mathsf{SourceScope}, \mathsf{TargetScope}, \mathsf{Semantics}, \mathsf{Dependencies}, \mathsf{PrecursorID}, \mathsf{OracleMode}, \mathsf{Lineage}, \mathsf{WitnessBasis}, \mathsf{PredictedGain}, \mathsf{VerificationObligations}, \mathsf{NegativeControls}, \mathsf{Burden}, \mathsf{ValidationPlan}, \mathsf{Disposition}\big).$$

The packet may contain a formal expression, program, matrix, functor, filtration, conjectured implication, or search-produced proof sketch, but such payload is not itself the candidate's theorem status.

Definition SDR.11.9 (SDR-SRCH-12: search lanes)**.** SDR.11 distinguishes four operational lanes:

(i) *certified-witness lane*: target-discordant witnesses are admissibly certified;

(ii) *surrogate lane*: a target-reflecting surrogate supplies the certified discrepancy;

(iii) *exploratory lane*: only structural contrast is known, so target relevance is not yet certified; and

(iv) *benchmark-assisted calibration lane*: benchmark labels or solutions may be used for analysis, but the resulting candidate cannot be advertised as independent recovery without a separate uncontaminated run.

These are execution modes, not levels of mathematical truth.

Policy 117 (SDR-SRCH-13: lane nonlaundering)**.** *A candidate generated or materially repaired in the benchmark-assisted lane remains benchmark-assisted for that lineage. Renaming the candidate, deleting the benchmark field, regenerating its syntax, or replaying it with a fresh random seed does not convert it into an independent-lane discovery.*

SDR.11.4 Admissibility gates before mathematical ranking

Definition SDR.11.10 (SDR-SRCH-14: pre-ranking admissibility gate)**.** Before a candidate receives a mathematical search priority, SDR.11 applies the gate

$$\mathsf{Gate}_{\mathrm{pre}} = (\mathsf{Type}, \mathsf{Scope}, \mathsf{SourceSemantics}, \mathsf{Precursor}, \mathsf{Oracle}, \mathsf{AntiMem}, \mathsf{Dependency}, \mathsf{Artifact}).$$

A failed type or source-semantics field rejects the candidate at the proposed strength. Missing proof evidence in a theorem-bearing field yields $\mathrm{undefined}$ rather than automatic rejection unless a counterexample is already known.

Policy 118 (SDR-SRCH-15: anti-memorization filter)**.** *A candidate whose separating power is supplied only by source identity, object index, hash, benchmark label, answer lookup, or another presentation-level memorization key is quarantined from structural promotion. It may be retained as a debugging discriminator, but it does not count as source-semantic discovery under SDR.4's anti-memorization rule.*

Policy 119 (SDR-SRCH-16: target-oracle firewall at generation time)**.** *In an independent lane, mere accessibility or presence in a precursor snapshot is not authorization. A target value, benchmark truth value, conclusion-derived constant, known successful witness, answer table, or equivalent oracle may enter theorem-bearing generation only when the proof-first contract records it as a legitimate premise, its source authority is independent of the benchmark conclusion being recovered, and it is not listed in* ForbiddenInputs*. Accidental presence in model memory, retrieval state, caches, prompts, or local artifacts does not satisfy this rule. Otherwise the information may be used only in a separately labeled calibration or validation lane.*

Policy 120 (SDR-SRCH-16A: surrogate-direction firewall)**.** *Let Ψ be a surrogate target. A surrogate may certify a target-discordant witness only through the witness-reflecting direction*

$$\Psi \preceq \Theta,$$

so that $\widehat{\Psi}(X) \neq \widehat{\Psi}(Y)$ implies $\widehat{\Theta}(X) \neq \widehat{\Theta}(Y)$. Reconstruction of Θ through Ψ instead requires

$$\Theta \preceq \Psi \qquad \textit{and} \qquad \Psi \preceq \mathbf{I}.$$

Exact surrogate replacement requires both information directions. Search ranking, empirical agreement, or a successful benchmark case may not reverse either implication.

Policy 121 (SDR-SRCH-17: failed search is not a universal negative theorem)**.** *Failure to generate a separator, counterexample, rigidity proof, completion theorem, or lower-cost certificate within a finite search budget proves only exhaustion of that declared search run. It does not prove nonexistence in the ambient mathematical class unless an independent exhaustive reduction theorem licenses that inference.*

SDR.11.5 Defect-directed generation and information-aware pruning

Definition SDR.11.11 (SDR-SRCH-18: search agenda item)**.** A search agenda item is one of:

(i) a certified witness $w = (X, Y, \Theta, \mathbf{I}, E)$;

(ii) a certified unresolved theorem edge;

(iii) a certified completion/tail obstruction;

(iv) an unresolved source-to-target interpretation edge; or

(v) an explicitly exploratory structural contrast with its nonpromotion flag.

Each agenda item carries the exact precursor snapshot from which its candidates are synthesized.

Definition SDR.11.12 (SDR-SRCH-19: witness-directed candidate gain)**.** For a certified target-discordant witness $w = (X, Y, \Theta, \mathbf{I}, E)$ and an admissible observational candidate J, define

$$\mathsf{Hit}(J; w) = 1 \quad \Longleftrightarrow \quad \widehat{J}(X) \neq \widehat{J}(Y).$$

This is a certified pair-level gain when the observable semantics and witness evidence both pass. It is not a class-wide rigidity theorem.

Proposition SDR.11.13 (SDR-SRCH-20: verified witness hit gives strict semantic gain)**.** *Under Definition SDR.11.12, if* $\mathsf{Hit}(J; w) = 1$*, then*

$$\Delta_\Theta(\mathbf{I} \boxtimes J) \subsetneq \Delta_\Theta(\mathbf{I}).$$

Proof. Observable refinement gives the non-strict inclusion by Theorem SDR.3.16. The witness pair belongs to $\Delta_\Theta(\mathbf{I})$ and is removed after adjoining J, so the inclusion is strict. □

Definition SDR.11.14 (SDR-SRCH-21: observational dominance)**.** Fix a common source scope and base packet $\mathbf{I}$. An observational candidate J *information- dominates* K relative to $\mathbf{I}$ when

$$\mathrm{EqKer}(\widehat{\mathbf{I} \boxtimes J}) \subseteq \mathrm{EqKer}(\widehat{\mathbf{I} \boxtimes K}).$$

The relation compares only observational discrimination on the declared scope. It does not compare theorem capability, stability, provenance, compatibility, or execution burden.

Theorem SDR.11.15 (SDR-SRCH-22: dominance gives target-independent defect containment)**.** *If J information-dominates K relative to $\mathbf{I}$, then for every normalized target Θ on the same source scope,*

$$\Delta_\Theta(\mathbf{I} \boxtimes J) \subseteq \Delta_\Theta(\mathbf{I} \boxtimes K).$$

Hence no target-conflict resolved by K can remain unresolved solely because K was replaced by J, at the level of observational discrimination.

Proof. Intersect the assumed equality-kernel inclusion with the fixed target-discordance relation C_Θ. The second statement is the same inclusion read pairwise. □

Policy 122 (SDR-SRCH-23: dominance pruning requires a substitution contract)**.** *Theorem SDR.11.15 licenses an information-ranking annotation, not unconditional deletion of K. Exact pruning from a cost-optimal or theorem-capability search additionally requires compatible source scope, compatible downstream consumers, preserved validation obligations, and a burden/reuse substitution theorem showing that replacing K by J cannot invalidate the requested optimization objective.*

Policy 123 (SDR-SRCH-24: deterministic postprocessing is a zero-gain prefilter)**.** *If a proposed observational candidate factors deterministically through the current packet, then it is information-redundant and cannot reduce the semantic defect. The search engine may prefilter such a candidate from an observational-gain lane, while retaining it when it has a separately declared computational, serialization, or verifier role.*

SDR.11.6 Ranking, burden, and the status of heuristic scores

Definition SDR.11.16 (SDR-SRCH-25: search-priority vector)**.** A candidate may carry a search-priority vector

$$\mathbf{s}(\mathfrak{c}) = (s_{\text{hit}}, s_{\text{scope}}, s_{\text{novelty}}, s_{\text{fals}}, s_{\text{burden}}, s_{\text{compat}}, s_{\text{prov}}),$$

where s_{prov} may encode provenance risk, precursor admissibility, or holdout *eligibility*, but not an unrevealed holdout outcome. The coordinates are declared heuristics or certified quantities according to their individual schemas. No common numerical scale is presumed.

Policy 124 (SDR-SRCH-26: heuristic score nonproof)**.** *A model score, embedding similarity, literature frequency, search-engine rank, empirical hit rate, novelty estimate, or agent vote has no theorem-bearing force unless a separate theorem converts that quantity into the required mathematical predicate. In particular, s_{hit} is not a defect reduction unless it is backed by an actual certified witness hit, and s_{burden} is not certificate complexity unless it is connected to the SDR.7/v21 cost model by an explicit adapter.*

Policy 125 (SDR-SRCH-26A: sealed-holdout outcome firewall)**.** *A validation case designated as a postconstruction holdout may expose identity, schema, and eligibility metadata needed to reserve the case, but its theorem-bearing outcome may not be used to rank, prune, repair, select, or stop the candidate that it is intended to test. Once that outcome is revealed and used in any such way, the case moves to calibration evidence for that lineage and a fresh uncontaminated holdout is required for a holdout claim.*

Policy 126 (SDR-SRCH-26B: literature guidance and rediscovery provenance)**.** *Literature may be used freely for ordinary theorem search and mathematical verification when independence is not claimed. If a literature theorem, proof idea, lemma inventory, or source-specific construction shapes candidate generation or ranking, the resulting route is literature-guided for independent-rediscovery provenance. A literature comparison performed only after a derivation is frozen may assess overlap or novelty without becoming a premise of that derivation.*

Definition SDR.11.17 (SDR-SRCH-27: Pareto search frontier)**.** Let $\mathfrak{C}^{\text{act}}$ be a finite active candidate set and let K be a finite set of declared ranking coordinates, each with an explicitly oriented preorder. The *search Pareto frontier* is the set of candidates in $\mathfrak{C}^{\text{act}}$ that are undominated in the product preorder after the pre-ranking admissibility gate. It is an agenda-management device only; it is not the SDR.7 certificate-cost frontier unless the two coordinate systems and their adapter are declared identical. For an infinite active candidate family, existence or nonemptiness of a Pareto-minimal set requires a separate well-foundedness, compactness, or finite-reduction argument; no such conclusion is built into this definition.

Policy 127 (SDR-SRCH-28: no hidden scalarization)**.** *Weights may be introduced for an engineering scheduler, but the weighted sum and its units must be stored as policy metadata. A weighted scheduler does not turn incomparable mathematical strengths, burdens, or provenance states into a canonical total order.*

SDR.11.7 Proposer-verifier separation and adversarial falsification

Definition SDR.11.18 (SDR-SRCH-29: proposer and verifier roles)**.** The *proposer* may synthesize candidate observables, conjectures, source restrictions, completion mechanisms, counterexamples, and proof sketches. The *verifier* consumes a frozen candidate packet and evaluates only its declared obligations. A single software process or AI model may implement both roles, but the artifact identities, prompts/inputs, and state transitions remain logically separated.

Policy 128 (SDR-SRCH-30: self-approval is not verification)**.** *Re-running the proposer, asking the same model to "check its work," sampling multiple agents from the same accessible evidence state, or obtaining consensus does not by itself create independent verification. Such procedures may discover errors or generate proof obligations, but mathematical acceptance still depends on the verifier contract.*

Policy 129 (SDR-SRCH-30A: verification and verifier independence are distinct)**.** *A proof object may be mathematically verified by a declared sound checker even when proposer and checker run inside one physical system; this does not make the verification independent in the provenance sense. Conversely, organizational or model independence does not make an unsound checker mathematically authoritative. Theorem-validity status and independent-verification status are therefore stored on separate axes. Any claim of independent verification requires an additional evidence-state and lineage audit.*

Definition SDR.11.19 (SDR-SRCH-31: falsifier request)**.** For a universal candidate claim

$$\forall X \in \mathcal{S}_*, \quad P(X),$$

a falsifier request consists of the exact source class, hypotheses, normalized negated predicate $\neg P$, admissible construction rules for counterexamples, and a search budget. The falsifier's positive output is a candidate witness requiring verification; its negative output is only $\mathrm{not_found_within_budget}$ unless exhaustiveness has separately been proved.

Theorem SDR.11.20 (SDR-SRCH-32: verified falsifier has asymmetric force)**.** *A single verified $X \in \mathcal{S}_*$ with $\neg P(X)$ refutes the universal candidate claim. Failure of a nonexhaustive falsifier to find such an X does not prove the claim.*

Proof. The first statement is the semantics of universal quantification. The second follows because a nonexhaustive search establishes neither $P(X)$ for unsearched X nor a reduction theorem covering them. □

Policy 130 (SDR-SRCH-32A: quantifier-aware falsification ceiling)**.** *Theorem SDR.11.20 applies to the displayed universal claim only. To refute an implication, a counterexample must satisfy the antecedent and violate the consequent. To refute an existential claim $\exists X\, P(X)$, one failed instance is irrelevant; a valid negative result must establish $\forall X\, \neg P(X)$ or an equivalent no-go theorem on the exact target scope. The search engine therefore stores the normalized target quantifiers with every falsifier request and never promotes route failure to target falsity by pattern matching alone.*

Policy 131 (SDR-SRCH-33: falsification before promotion, not necessarily before proof)**.** *The engine should normally expose cheap, high-value falsification opportunities before promotion or expensive formalization. This is an engineering priority, not a theorem that every efficient mathematical workflow must search for counterexamples before attempting a proof.*

SDR.11.8 Typed engine transition and soundness invariants

Definition SDR.11.21 (SDR-SRCH-34: one engine transition)**.** A transition $\mathfrak{Z}_n \rightsquigarrow \mathfrak{Z}_{n+1}$ may execute the following typed actions:

(i) select one agenda item without changing its precursor snapshot;

(ii) generate one or more candidate packets;

(iii) apply provenance, oracle, typing, anti-memorization, sealed-holdout, and dependency gates;

(iv) run admissible redundancy/dominance screens;

(v) issue falsifier and verifier requests;

(vi) place each returned candidate in $\mathrm{verified}$, $\mathrm{falsified}$, $\mathrm{undefined}$, $\mathrm{scope_excluded}$, or $\mathrm{still_proposed}$ status at its exact strength;

(vii) if an observational candidate has verified source-semantic meaning, adjoin its observable identity to $\mathbf{I}_n$, and separately verify every computed evaluation that is consumed as a witness, score, or theorem premise;

(viii) if an inferential, restriction, or completion theorem is verified, add the theorem record without pretending that the observational packet itself became more informative; and

(ix) update the certified witness ledger, holdout registry, lineage DAG, and budget record.

No action is permitted to rewrite the precursor snapshot of an earlier synthesis step.

Policy 132 (SDR-SRCH-34A: semantic definition and evaluation receipt are distinct)**.** *Verification that J is a well-defined source-semantic observable establishes the mathematical coordinate. It does not automatically certify a numerical, symbolic, or serialized value claimed to equal $J(X)$. Any computed value used for witness separation, ranking with theorem-bearing force, or a downstream proof premise must carry its own evaluation receipt or exact derivation. Thus a correct observable definition cannot launder an incorrect execution trace into the information packet.*

Policy 133 (SDR-SRCH-34B: contract-mutation firewall)**.** *A verified* Restr *or* Target *candidate does not silently rewrite the operative proof-first contract. A restriction may close the original contract only when a separate theorem discharges membership of all original target instances in the restricted class. Otherwise narrowing the source class or weakening/changing the target creates a child contract with a new identity. Likewise, a surrogate may replace the target only through the directional reconstruction obligations of Policy 120. Closure of a child contract is not full success of its parent.*

Theorem SDR.11.22 (SDR-SRCH-35: observational soundness invariant)**.** *Suppose every observational update from* $\mathbf{I}_n$ *to* $\mathbf{I}_{n+1}$ *is performed only after the new observable* J_{n+1} *has a verified source-semantic definition on the operative source class, and*

$$\mathbf{I}_{n+1} = \mathbf{I}_n \boxtimes J_{n+1}.$$

Then

$$\Delta_\Theta(\mathbf{I}_{n+1}) \subseteq \Delta_\Theta(\mathbf{I}_n)$$

for every normalized target Θ *on that class. Strict inclusion occurs whenever a certified target-discordant pair is actually separated by* J_{n+1}*.*

Proof. The non-strict statement is observable-refinement monotonicity. The strict statement is Proposition SDR.11.13. Neither conclusion requires the engine to enumerate the full semantic defect. □

Policy 134 (SDR-SRCH-36: inferential gain is not written into the information packet)**.** *A verified rigidity theorem, source restriction, or completion theorem may close a target route without changing* $\mathrm{EqKer}(\widehat{\mathbf{I}_n})$*. Such progress is recorded in the theorem/capability state, not by fabricating a new observational coordinate.*

Proposition SDR.11.23 (SDR-SRCH-37: certified-witness count need not decrease)**.** *Even under Theorem SDR.11.22, the number of currently certified witness records may increase, decrease, or remain constant from stage to stage.*

Proof. New evidence can certify previously unknown semantic defects, while new observables can discharge old ones. Thus the certified ledger is epistemic and inherits the nonmonotonicity identified in SDR.8; only the semantic defect is monotone under genuine observational refinement. □

SDR.11.9 Finite registries, exact subroutines, and termination ceilings

Theorem SDR.11.24 (SDR-SRCH-38: finite strict-gain bound)**.** *Let* $\mathcal{S}_0$ *be finite and suppose the initial semantic defect graph has* m *edges. Any sequence of observational updates in which every admitted update strictly reduces the semantic defect on* $\mathcal{S}_0$ *has length at most* m*.*

Proof. Each strict update removes at least one defect edge and no later observational refinement can restore a removed semantic defect edge. There are initially only m edges. □

Policy 135 (SDR-SRCH-39: strict-gain bound is not full-engine termination)**.** *Theorem SDR.11.24 bounds accepted strict-gain updates only. It does not bound the number of rejected, undefined, duplicated, theorem-only, exploratory, or newly synthesized candidates. Full search termination requires a finite registry, a well-founded scheduler, an explicit budget, or another independent termination theorem.*

Declaration SDR.11.25 (SDR-SRCH-40: exact finite separating-family subroutine)**.** When $\mathcal{S}_0$, the certified-complete conflict universe, and the candidate registry are finite and the candidate conflict-incidence matrix is exact, SDR.11 may invoke the SDR.7 exhaustive Boolean-cover procedure to enumerate exact separating families, inclusion-minimal families, and any cost frontier licensed by the registered cost adapter. This is an imported finite exact subroutine, not a new search theorem of SDR.11.

Definition SDR.11.26 (SDR-SRCH-41: typed engine stopping vector)**.** Search termination is recorded on three noncompensating axes,

$$\mathsf{StopVec} = (\mathsf{RouteStatus}, \mathsf{OperationalStop}, \mathsf{CoverageStatus}).$$

The route-status coordinate is one of

$$\{\text{return_route_verified}, \text{partial_return}, \text{route_obstructed}, \text{target_obstructed}, \text{undefined}, \text{open}\}.$$

Here target_obstructed requires a theorem or counterexample package that negates the exact target at its registered quantifier strength; blockage of one route yields only route_obstructed. A partial_return is either a weaker strength already registered by the parent contract or a Return of a separately identified child contract under Policy 133; it is never full success of the original requested strength.

The operational-stop coordinate is one of

$$\{\text{goal_closed}, \text{budget_exhausted}, \text{registry_exhausted}, \text{interrupted}, \text{open}\},$$

and the coverage coordinate is one of

$$\{\text{ambient_proved}, \text{declared_registry_only}, \text{nonexhaustive}, \text{unassessed}\}.$$

Operational exhaustion does not force any mathematical route status, and a favorable route status does not by itself prove ambient candidate-family completeness.

Policy 136 (SDR-SRCH-42: no success or target falsity from exhaustion alone)**.** *A finite registry can be exhaustive relative to its declared grammar while the grammar itself is incomplete. Therefore* registry_exhausted *is an operational-stop value only. It neither proves success nor upgrades* route_obstructed *to* target_obstructed *unless a separate theorem proves candidate-family completeness and the exact target-level negation.*

Policy 137 (SDR-SRCH-42A: registry-exhaustion certificate)**.** *The engine may emit* registry_exhausted *only when the registry identity is frozen and finite, every admissible registered candidate has a terminal audited disposition or a proved dominance/substitution exclusion, and the scheduler record proves that no registered candidate was silently starved. Exhaustion of a generator, prompt family, model sample, or current Pareto frontier is not exhaustion of an open-ended candidate grammar.*

SDR.11.10 Candidate reuse, novelty, and implementation records

Definition SDR.11.27 (SDR-SRCH-43: candidate reuse modes)**.** A candidate reused across problems is marked as one of:

(i) same_theorem_new_consumer;

(ii) same_schema_new_instance;

(iii) adapted_candidate_new_obligations; or

(iv) new_candidate.

Only the first mode preserves the identical theorem identity. The second and third require fresh applicability, source, compatibility, and validation checks.

Policy 138 (SDR-SRCH-44: architectural transfer is not native theorem transfer)**.** *A search pattern such as "find an indistinguishable pair, identify missing structure, enrich, and prove rigidity" may transfer across domains. The concrete probe, local algebra, completion theorem, or reconstruction formula does not transfer without a new typed theorem. Architecture-level analogy is search guidance only.*

Definition SDR.11.28 (SDR-SRCH-45: novelty status)**.** Every theorem-facing candidate carries

$$\mathsf{NoveltyStatus} \in \{\text{classical_identified}, \text{likely_overlap}, \text{unassessed}, \text{candidate_nonredundant}, \text{verified_nonredundant}\}.$$

Search-engine inability to locate a prior theorem never by itself yields verified_nonredundant. That status is reserved for the literature and theorem-comparison audit of SDR.13.

Definition SDR.11.29 (SDR-SRCH-46: executable search record)**.** An executable SDR search record serializes at least:

(i) the proof-first contract and source/target identities;

(ii) the precursor snapshot and forbidden-input manifest;

(iii) candidate-generator identity and version;

(iv) all candidate packets and immutable dispositions;

(v) proposer/verifier role identities and verifier obligations;

(vi) witness, counterexample, and defect-certificate artifacts;

(vii) ranking coordinates, scheduler policy, scalarization metadata, and the sealed-holdout exposure log;

(viii) theorem/capability updates separately from observational packet updates, with evaluation receipts for theorem-bearing computed values;

(ix) the full evidence-lineage DAG and holdout eligibility table;

(x) search budgets, the typed stopping vector, registry-exhaustion receipts when claimed, and replay artifacts; and

(xi) novelty, validation, independent-verification, governance, and promotion statuses as independent fields.

A syntactically complete record proves only replayability of the search process unless its embedded theorem certificates separately pass.

Policy 139 (SDR-SRCH-47: AI and automated tools are proposers unless certified otherwise)**.** *Language models, theorem provers used in conjecture mode, symbolic algebra systems, numerical solvers, literature retrievers, code generators, and multi-agent systems may populate any proposer-side field permitted by the contract. Their outputs enter theorem-facing state only through the same typed verifier interfaces required of human-generated candidates. Tool identity changes neither the proof burden nor the target-oracle firewall.*

SDR.11.11 Search completeness nonclaims, canonical engine record, and closure

Policy 140 (SDR-SRCH-48: no search-completeness claim)**.** *SDR.11 does not claim that its candidate grammar is complete, that every solvable target has a discoverable SDR route, that every missing structure can be generated effectively, that every rigidity theorem is finitely searchable, or that every mathematically useful candidate receives a favorable heuristic rank.*

Policy 141 (SDR-SRCH-49: discovery success is not publication novelty)**.** *Even if the engine independently synthesizes and proves a theorem from an admissible precursor state, its mathematical novelty remains a separate literature-comparison question. Independent rediscovery and novel mathematics are different predicates.*

Definition SDR.11.30 (SDR-SRCH-50: canonical SDR.11 engine record)**.** The canonical engine record is

$$\begin{aligned}\mathcal{R}_{\mathrm{SRCH}} := (&\mathfrak{P}, \mathfrak{Z}_0, (\mathfrak{Z}_n)_{n\le N}, \mathsf{Agenda}, \mathsf{Generators}, \mathsf{Gates},\\ &\mathsf{RankPolicy}, \mathsf{Falsifiers}, \mathsf{Verifiers}, \mathsf{PacketUpdates}, \mathsf{TheoremUpdates},\\ &\mathsf{Lineage}, \mathsf{Holdouts}, \mathsf{StopVec}, \mathsf{ValidationStatus},\\ &\mathsf{IndependentVerificationStatus}, \mathsf{NoveltyStatus}, \mathcal{A}).\end{aligned}$$

For nonterminating or externally interrupted research streams, the finite run prefix is stored together with an explicit open or $\mathrm{interrupted}$ lifecycle state; no fictitious terminal success is inserted.

Audit Obligation SDR.11.31 (SDR.11 search-engine audit)**.** *Before a search result is consumed as theorem-bearing evidence, verify:*

(i) *frozen proof-first target, target quantifiers, source class, hypotheses, Return strength, and failure ceiling;*

(ii) *immutable precursor snapshot and target/benchmark-oracle policy, with accessibility distinguished from admissibility;*

(iii) *source-semantic candidate typing and anti-memorization compliance;*

(iv) *correct surrogate direction for witness reflection, reconstruction, or exact replacement;*

(v) *separation of observational, inferential, restriction, completion, surrogate, target-change, and counterexample candidates;*

(vi) *proposer/verifier role separation, exact verifier status, and any independent-verification claim on a separate provenance axis;*

(vii) *that search scores, consensus, execution success, failed falsifier searches, and unrevealed holdout outcomes are not treated as proof or candidate-ranking evidence outside their permitted roles;*

(viii) *correct information-redundancy and dominance semantics before any pruning;*

(ix) *separate observational packet updates and theorem/capability updates, plus evaluation receipts for computed values consumed theoremically;*

(x) *evidence lineage, contamination closure, sealed-holdout exposure history, and holdout eligibility;*

(xi) *quantifier-aware falsification and the distinction between route obstruction and target obstruction;*

(xii) *child-contract creation for any source-class restriction or target change not discharged inside the parent contract;*

(xiii) *exact finite-registry assumptions, scheduler coverage, and exhaustion receipts for any exhaustive or optimality claim;*

(xiv) *the noncompensating route, operational-stop, and coverage coordinates of* StopVec*;*

(xv) *no target-falsity inflation from budget or registry exhaustion; and*

(xvi) *independent novelty, validation, literature-provenance, governance, and promotion statuses.*

Theorem SDR.11.32 (SDR-SRCH-51: SDR.11 search-and-discovery closure)**.** *Under the frozen proof-first contract and the policies above, SDR.11 establishes a typed search architecture with the following properties:*

(i) *every candidate is source-, target-, precursor-, oracle-, lineage-, and obligation-typed before theorem-facing use;*

(ii) *accessibility of target or benchmark information is weaker than admissibility, and surrogate use is direction-typed;*

(iii) *observational refinement preserves the semantic defect monotonicity of SDR.2–SDR.4, while inferential, restriction, and completion gains remain separately typed;*

(iv) *verified witness hits give strict local semantic progress, whereas deterministic postprocessing gives no observational gain;*

(v) *information dominance yields target-independent defect containment but does not silently imply burden, compatibility, theorem-capability, or pruning dominance;*

(vi) *sealed holdout outcomes cannot be used to rank or repair the candidate they are intended to test without forfeiting holdout status for that lineage;*

(vii) *proposer output, heuristic rank, execution, agent consensus, and failed falsifier search remain nonproof, while mathematical verification and independent verification remain distinct statuses;*

(viii) *falsification force is quantifier-aware: one verified counterexample refutes an exact universal claim, but route failure or one failed existential instance does not falsify a stronger target;*

(ix) *finite-source strict observational progress admits a finite bound, but neither that bound nor finite-registry exhaustion proves ambient search completeness;*

(x) *narrowing the source class, weakening/changing the target, or replacing it by a surrogate cannot mutate the parent contract without the declared discharge or child-contract rule;*

(xi) *registry exhaustion requires an audited finite-registry and scheduler-coverage receipt and remains separate from mathematical obstruction; and*

(xii) *canonical records preserve replayability, lineage, validation, independent-verification, novelty, and governance without promoting search artifacts into frozen v21 mathematics.*

No item asserts completeness of the SDR grammar, universal automated theorem discovery, independent holdout success, independent recovery, literature novelty, or mutation of the frozen v21 source set.

Proof. Items (i)–(ii) are Definitions SDR.11.3, SDR.11.8, and Policies 119 and 120. Items (iii)–(iv) follow from Definition SDR.11.21, Theorem SDR.11.22, Proposition SDR.11.13, and Policies 134 and 123. Item (v) is Theorem SDR.11.15 and Policy 122. Item (vi) is Theorem SDR.11.6 and Policy 125. Item (vii) is Policies 113, 124, 128, and 129. Item (viii) is Theorem SDR.11.20 and Policy 130. Item (ix) is Theorem SDR.11.24 and Policies 135 and 136. Item (x) is Policies 114, 133, and 120. Item (xi) is Policy 137 and Definition SDR.11.26. Item (xii) is Definitions SDR.11.29 and SDR.11.30 together with the constitutional nonpromotion rules inherited from SDR.1 and SDR.10. □

Remark SDR.11.33 (SDR.12 handoff: failure and no-go atlas)**.** SDR.11 now gives every failed or stalled search event a typed provenance and stopping disposition. SDR.12 will turn those events into a failure/no-go atlas: information-loss obstructions, oracle contamination, memorization, wrong-way surrogate use, pair-to-class overgeneralization, vacuous rigidity, completion leakage, cost-model mismatch, holdout recycling, category-strength inflation, and search-exhaustion inflation will be organized as reusable counterpatterns with exact repair routes and strongest safe negative conclusions.

SDR.12 Failure and No-Go Atlas

SDR.12.1 Purpose, authority, and negative-result semantics

Declaration SDR.12.1 (SDR-FAIL-01: role of the failure/no-go atlas)**.** SDR.12 converts the recurrent failure patterns of SDR.1–SDR.11 into a typed diagnostic atlas. Its purpose is not to enlarge a negative observation into a theorem of impossibility, but to determine, for each certified failure event,

(i) what exactly has failed;

(ii) on which source class, target, route, information class, or certificate model the failure is quantified;

(iii) the strongest negative conclusion licensed by the evidence;

(iv) which repair lanes can in principle address that failure; and

(v) which stronger negative readings remain forbidden.

The atlas is a post-v21 SDR-local research structure. It does not mutate the frozen v21 failure semantics, and it does not convert a search log, failed proof attempt, missing certificate, or unsuccessful benchmark run into target falsity.

Definition SDR.12.2 (SDR-FAIL-02: typed failure signature)**.** A *typed failure signature* is a record

$$\mathfrak{F} = (\mathsf{Layer}, \mathsf{Trigger}, \mathsf{SourceScope}, \mathsf{Target}, \mathsf{RouteClass}, \mathsf{Evidence}, \mathsf{NegClaim}, \mathsf{RepairClass}, \mathsf{Status}).$$

Here Layer identifies the failing interface; Trigger is the witnessed defect; SourceScope and Target freeze the quantified statement; RouteClass states which information/proof/search family is being ruled out or left unresolved; Evidence contains the actual proof object, counterexample, failed premise, or missing field; NegClaim records the strongest safe negative conclusion; RepairClass lists admissible repair directions; and Status is the theorem-facing disposition.

Definition SDR.12.3 (SDR-FAIL-03: negative-result vector)**.** The negative-result vector is

$$\mathsf{NegVec} = (\mathsf{TruthStatus}, \mathsf{ScopeStatus}, \mathsf{RouteStatus}, \mathsf{TargetStatus}, \mathsf{CoverageStatus}, \mathsf{ProvenanceStatus}).$$

The coordinates are noncompensating. In particular, a proved route obstruction does not set target falsity, a failed implementation does not set mathematical falsity, an exhaustive finite registry does not set ambient coverage, and a contaminated derivation does not determine the truth value of the mathematical claim it attempted to prove.

Policy 142 (SDR-FAIL-04: weakest exact negative conclusion)**.** *Every SDR negative verdict shall be stated at the weakest exact strength supported by its evidence. The following implications are forbidden without additional theorem evidence:*

$$\begin{aligned} \textit{missing evidence} &\Rightarrow \textit{false statement},\\ \textit{failed implementation} &\Rightarrow \textit{failed theorem},\\ \textit{failed route} &\Rightarrow \textit{false target},\\ \textit{failed finite search} &\Rightarrow \textit{ambient impossibility},\\ \textit{one counterexample to a sufficient criterion} &\Rightarrow \textit{counterexample to the target}. \end{aligned}$$

Definition SDR.12.4 (SDR-FAIL-05: no-go theorem)**.** A *no-go theorem* is a proved statement of the form

$$\forall R \in \mathfrak{R} \qquad R \text{ cannot Return } \Theta \text{ on } \mathcal{S}_*$$

or an exactly typed equivalent, where the route class $\mathfrak{R}$, source class $\mathcal{S}_*$, target equivalence, and admissible evidence are fixed before the proof. A failed attempt to construct one member of $\mathfrak{R}$, or failure to find a successful member, is not a no-go theorem.

Policy 143 (SDR-FAIL-06: no-go scope cannot silently widen)**.** *A no-go theorem for a route class $\mathfrak{R}_0 \subseteq \mathfrak{R}$, restricted source class $\mathcal{S}_0 \subseteq \mathcal{S}$, stronger target, weaker evidence model, or bounded candidate grammar remains confined to that declared scope. It may not be reported as impossibility for all mathematical routes to the ambient target.*

SDR.12.2 Information-loss and separation no-go patterns

Theorem SDR.12.5 (SDR-FAIL-07: certified collision no-go)**.** *Let $\mathbf{I}$ be an observational packet on $\mathcal{S}$. If there exist $X, Y \in \mathcal{S}$ with*

$$\widehat{\mathbf{I}}(X) = \widehat{\mathbf{I}}(Y), \qquad \widehat{\Theta}(X) \neq \widehat{\Theta}(Y),$$

then no deterministic decoder consuming only $\widehat{\mathbf{I}}$ can Return Θ correctly on all of $\mathcal{S}$.

Proof. Any deterministic decoder takes the same input value on X and Y, hence returns the same output on both. The target values differ, so at least one output is wrong. □

Corollary SDR.12.6 (SDR-FAIL-08: postprocessing cannot repair a certified collision)**.** *Under Theorem SDR.12.5, if $J = f \circ \widehat{\mathbf{I}}$ is any deterministic postprocessing of the same packet, then*

$$(\widehat{\mathbf{I}}, J)(X) = (\widehat{\mathbf{I}}, J)(Y).$$

Hence adjoining J cannot repair the certified pair.

Theorem SDR.12.7 (SDR-FAIL-09: structural-family blindness)**.** *Let $\mathfrak{P}$ be a family of observational probes on the same source class. Suppose a certified target-discordant pair (X, Y) satisfies*

$$\widehat{J}(X) = \widehat{J}(Y) \qquad \text{for every } J \in \mathfrak{P}.$$

Then no subfamily $\mathfrak{Q} \subseteq \mathfrak{P}$, finite or infinite, has a joint observational packet that is target-sufficient on any source class containing both X and Y.

Proof. Every coordinate indexed by $\mathfrak{Q}$ agrees on X and Y, so the complete joint packet agrees on that pair. Apply Theorem SDR.12.5. □

Policy 144 (SDR-FAIL-10: failed probe search is weaker than family blindness)**.** *Failure to find a separating probe inside a search budget does not prove the hypothesis of Theorem SDR.12.7. Family blindness requires a theorem covering every admissible member of the declared family, not a finite history of unsuccessful candidate generation.*

Policy 145 (SDR-FAIL-11: pair gain is not class rigidity)**.** *A probe that separates one certified pair proves at most the corresponding pair-level gain unless a class-wide theorem establishes*

$$\mathrm{EqKer}(\widehat{\mathbf{I}} \boxtimes \widehat{J}) \subseteq \mathrm{EqKer}(\widehat{\Theta})$$

on the complete declared source class. Empirical separation of many pairs, even all pairs in a finite benchmark, does not silently supply ambient rigidity.

Policy 146 (SDR-FAIL-12: certified-defect emptiness is not semantic emptiness)**.** *For any evidence mode $\mathcal{E}$,*

$$\Delta_{\Theta}^{\mathrm{cert},\mathcal{E}}(\mathbf{I}) = \varnothing$$

does not imply

$$\Delta_{\Theta}(\mathbf{I}) = \varnothing$$

unless an evidence-completeness theorem identifies the certified and semantic defect universes.

SDR.12.3 Target, surrogate, oracle, and memorization failures

Policy 147 (SDR-FAIL-13: wrong-way surrogate firewall)**.** *Surrogate direction is use-dependent. To infer target discordance from surrogate discordance one needs*

$$\Psi \preceq \Theta.$$

To reconstruct the target through the surrogate one needs

$$\Theta \preceq \Psi \quad \textit{and} \quad \Psi \preceq \mathbf{I}.$$

Using the reverse implication in either role is a typed route error, not a weaker proof.

Theorem SDR.12.8 (SDR-FAIL-14: target-oracle contamination invalidates independent discovery status)**.** *Suppose a candidate, witness, ranking decision, repair, or theorem-bearing edge in an advertised independent-discovery lane depends on a target value, benchmark truth value, conclusion-derived witness, or equivalent oracle that is not an admissible precursor premise. Then the resulting lineage is ineligible for independent-discovery or holdout status, even if the resulting mathematical statement is later proved true.*

Proof. Independence is a provenance property of the theorem-bearing construction. Once prohibited conclusion-bearing information lies on a dependency path to the candidate or proof route, deleting the citation or replaying the same mathematical object does not remove that causal dependency. □

Policy 148 (SDR-FAIL-15: contamination is not mathematical falsity)**.** *The conclusion of Theorem SDR.12.8 is a provenance failure only. It does not imply that the derived theorem is false, that a separately clean proof cannot exist, or that the target is unreachable.*

Policy 149 (SDR-FAIL-16: memorization-only separation is not structural discovery)**.** *A candidate whose only separating power is an object identifier, hash, database index, benchmark label, answer table, or equivalent source-external lookup may formally separate a finite registry but is not admitted as a source-semantic structural probe. Such a candidate may serve infrastructure or deduplication roles only.*

Policy 150 (SDR-FAIL-17: precursor freeze)**.** *A premise introduced after the candidate or proof route was selected may not be retroactively classified as an original precursor merely because it makes the route appear noncircular. A changed precursor set creates a new lineage or child contract.*

SDR.12.4 Rigidity, scope, and theorem-entitlement failures

Policy 151 (SDR-FAIL-18: vacuous-rigidity firewall)**.** *Rigidity on an empty source class, or on a source class for which applicability to the intended problem has not been proved, gives no application-level Return. Nonemptiness, membership, problem coverage, and hypothesis discharge are independent obligations.*

Theorem SDR.12.9 (SDR-FAIL-19: a certified counterexample cannot be repaired by theorem relabeling)**.** *Suppose (X, Y) is a certified target-discordant pair for a fixed source class, fixed target, and fixed consumed observational packet. Any new theorem statement that leaves those three ingredients unchanged and does not exclude either X or Y cannot convert the packet into a target-rigid packet.*

Proof. The same certified pair remains in the information kernel and outside the target kernel. A change of theorem name or proof prose cannot alter those sets. □

Policy 152 (SDR-FAIL-20: failure to prove is not proof of non-rigidity)**.** *Failure to establish $\Theta \preceq \mathbf{I}$ does not establish $\Theta \not\preceq \mathbf{I}$. A certified non-rigidity claim requires a counterexample pair or another theorem proving the negation at the declared scope.*

Policy 153 (SDR-FAIL-21: source restriction is not observational gain)**.** *Replacing $\mathcal{S}$ by a theorem-backed subdomain $\mathcal{S}_H$ can remove ambiguity because inadmissible pairs are deleted. It does not refine the values of the observable packet itself and shall not be recorded as information gain.*

Theorem SDR.12.10 (SDR-FAIL-22: class-wide upgrade requires class-wide hypothesis discharge)**.** *If a rigidity theorem is proved only on $\mathcal{S}_H \subseteq \mathcal{S}$, then proving $H(X_0)$ for one intended instance X_0 licenses application to X_0 only. Ambient class-wide rigidity follows only after a theorem proves $H(X)$ for every $X \in \mathcal{S}$.*

Proof. The one-instance conclusion is exactly the membership-gated use rule of Theorem SDR.6.9. To upgrade the restricted theorem to the ambient class, Theorem SDR.6.32 requires $\mathcal{S}_H = \mathcal{S}$, which is supplied by a class-wide proof of $H(X)$ for every $X \in \mathcal{S}$. A proof for one point does not establish that equality of source classes. □

Policy 154 (SDR-FAIL-23: decoder existence is not effectivity)**.** *Set-theoretic factorization through an information packet does not by itself provide a formula, algorithm, decidable equality, finite encoding, complexity bound, verifier, stability estimate, or executable certificate. Each requested effective strength is a separate theorem obligation.*

SDR.12.5 Finite-to-infinite and completion leakage

Policy 155 (SDR-FAIL-24: finite-prefix non-shortcut)**.** *No finite prefix, finite plateau, repeated output, numerical convergence pattern, or finite collection of zero-looking tests implies an infinite-tail statement without a declared finite-to-infinite reduction, compactness, Noetherian/DCC, Mittag–Leffler, finite-state, cofinality, or other theorem sufficient for the exact target.*

Theorem SDR.12.11 (SDR-FAIL-25: strict improvement does not force zero limit)**.** *A strictly decreasing defect sequence*

$$D_0 \supsetneq D_1 \supsetneq D_2 \supsetneq \cdots$$

need not satisfy $\bigcap_n D_n = \varnothing$. Hence strict stagewise improvement alone is not a completion theorem.

Proof. Work in $\mathbb{N} \cup \{\infty\}$ and set

$$D_n = \{\infty\} \cup \{m \in \mathbb{N} : m \geq n\}.$$

Then $D_{n+1} \subsetneq D_n$ for every n, while

$$\bigcap_{n \geq 0} D_n = \{\infty\} \neq \emptyset.$$

Thus strict stagewise improvement alone forces neither finite termination nor zero limit defect. Any stronger conclusion requires an additional completion mechanism of the kind isolated in SDR.8. □

Policy 156 (SDR-FAIL-26: stabilization is not zero)**.** *Eventual stabilization of states, images, defects, or transition data proves only stabilization. A zero-reflection theorem is required to conclude that the stable object is zero or target-sufficient.*

Policy 157 (SDR-FAIL-27: finite-state is not zero)**.** *A finite state space may support nonzero fixed points or cycles. Finite-state reduction becomes a completion theorem only together with the exact descent, stable-image, zero-reflection, or derived-limit hypotheses required by the target.*

Policy 158 (SDR-FAIL-28: conclusion-coded completion is circular)**.** *A residual filtration, rank, stable image, compactness carrier, or tail predicate used to prove completion may not be defined by importing the desired conclusion or an equivalent target oracle. In particular, setting the residual package equal to the unknown final defect and then assuming its separatedness is not an independent completion argument.*

Policy 159 (SDR-FAIL-29: state completion and target completion are distinct)**.** *Existence of a state limit $x_n \to x_\infty$ does not imply target residual zero; target residual zero does not construct the state limit. A theorem connecting the convergent state, the residual map, and a separated/closed target filtration is required for the combined claim.*

SDR.12.6 Minimality, cost, and certificate-model failures

Policy 160 (SDR-FAIL-30: minimal is not minimum)**.** *An inclusion-minimal separating family need not have minimum cardinality, minimum weighted cost, or Pareto-minimal certificate cost. Each optimization claim requires its own lower-bound or exact optimality certificate.*

Policy 161 (SDR-FAIL-31: probe count is not certificate cost)**.** *A bound on the number of selected probes does not automatically bound query count, bit length, verifier time, artifact footprint, or any other certificate coordinate; nor do lower bounds on those coordinates automatically lower-bound probe cardinality. Cross-metric conversion requires an explicit theorem.*

Policy 162 (SDR-FAIL-32: hidden-observable firewall)**.** *If a verifier advertised as using selected family A consumes an additional observable $K \notin A$ in a theorem-bearing way, then the actual observational family is at least $A \cup \{K\}$. Hidden observables invalidate a claimed minimum-family certificate.*

Policy 163 (SDR-FAIL-33: finite coverage equivalence is not mathematical equivalence)**.** *Equality of separation sets on one finite conflict universe does not imply equality of source semantics, ambient information kernels, theorem hypotheses, stability, provenance, execution burden, or interchangeability on another target or source class.*

Policy 164 (SDR-FAIL-34: finite optimality does not imply ambient optimality)**.** *A minimum separating family on a finite certified conflict registry is an ambient minimum only when a theorem proves that the registry exhausts the relevant semantic defect and that the admissible candidate family is complete at the claimed scope.*

SDR.12.7 Category-strength, coordinate, and local-to-global inflation

Policy 165 (SDR-FAIL-35: categorical-strength firewall)**.** *The following strengths are not silently interchangeable:*

$$\textit{faithful}, \quad \textit{full}, \quad \textit{fully faithful}, \quad \textit{essentially surjective},$$
$$\textit{equivalence}, \quad \textit{exact equivalence}, \quad \textit{derived/triangulated equivalence}.$$

A theorem at one level does not supply any missing property required by a stronger level.

Policy 166 (SDR-FAIL-36: object Return is not morphism or extension Return)**.** *An object-isomorphism classifier does not automatically Return source morphisms, natural transformations, exact sequences, Yoneda extension classes, composition, cones, or distinguished triangles. These require the enriched records and reflection theorems owned by the corresponding structural layer.*

Policy 167 (SDR-FAIL-37: coordinate equality is not invariant equality)**.** *Matrices, bases, resolutions, gauges, splittings, chain representatives, and local trivializations are choice-dependent unless an invariance theorem says otherwise. Literal equality or inequality of such coordinates is not a valid source-level separation witness without the declared quotient, transport, or choice-invariance semantics.*

Policy 168 (SDR-FAIL-38: local reconstruction is not global reconstruction)**.** *A complete family of local height-one, stalkwise, windowwise, or chartwise reconstructions does not by itself produce a global object, global module decomposition, descent datum, higher coherence, or external Return. The exact assembly/compatibility theorem is a separate edge.*

Policy 169 (SDR-FAIL-39: higher-Fitting survivor data are not a general higher-Bockstein spectral sequence)**.** *The q-adic kernel-survivor filtration of SDR.9 may be called a Bockstein-survivor profile locally, but it does not by itself construct a general higher-Bockstein exact couple, differential system, convergence theorem, or spectral sequence. Those stronger structures require their own definitions and proofs.*

Policy 170 (SDR-FAIL-40: fixed-layer replay is not a uniform all-layer theorem)**.** *Exact Smith replay of $M/\omega M$ at every individually supplied distinguished layer proves fixed-layer decidability. It does not by itself provide a bounded structural packet or one closed formula valid simultaneously for all layers.*

SDR.12.8 Validation, search, and provenance inflation

Policy 171 (SDR-FAIL-41: calibration is not holdout validation)**.** *A case used to design, choose, rank, repair, or tune a rule is a construction/calibration input for that lineage. It cannot simultaneously serve as a sealed holdout test of the same rule.*

Theorem SDR.12.12 (SDR-FAIL-42: holdout recycling destroys holdout status)**.** *If a sealed holdout outcome is exposed and then used to rank, prune, repair, select, or stop the tested candidate lineage, that case ceases to be a holdout for the updated lineage. Subsequent independent holdout claims require a newly sealed admissible case or campaign.*

Proof. After exposure, the holdout outcome has a causal path to the updated candidate or selection decision. It is therefore part of the construction evidence for that lineage rather than evidence independent of construction. □

Policy 172 (SDR-FAIL-43: search score, consensus, and replay are nonproof)**.** *Candidate score, language-model confidence, embedding similarity, literature frequency, agent consensus, numerical replay, symbolic simplification, successful compilation, and package integrity are not theorem evidence unless an explicit interpretation and soundness theorem converts the exact artifact into a mathematical certificate.*

Policy 173 (SDR-FAIL-44: registry exhaustion is not ambient impossibility)**.** *Even a correctly audited exhaustion of a finite candidate registry proves only that no admissible member of that registry succeeded under the declared contract. It does not prove that no route exists outside the registry.*

Policy 174 (SDR-FAIL-45: route obstruction is not target falsity)**.** *A proof that one route, one probe family, one sufficient criterion, one architecture branch, or one construction program cannot reach a target establishes only the failure of that route class. Target falsity requires the exact logical negation of the target at the target's own quantifier strength.*

Policy 175 (SDR-FAIL-46: child-contract laundering is forbidden)**.** *If a search succeeds only after narrowing the source class, strengthening hypotheses, weakening or replacing the target, or importing a surrogate not licensed by the parent contract, the result belongs to a child contract unless a theorem discharges the change inside the parent. Child success may not be reported as parent-target success.*

Policy 176 (SDR-FAIL-47: compatibility and validation are not proof)**.** *Compatibility with AK typing, agreement with known positive controls, survival of negative controls, cross-domain consistency, and prospective validation support confidence in an architecture but do not prove a missing mathematical edge of a particular target route.*

Policy 177 (SDR-FAIL-48: literature overlap is neither novelty nor non-novelty by itself)**.** *Conceptual resemblance to sufficient statistics, partition refinement, CEGAR, set cover, Fitting theory, or another literature family does not by itself prove equivalence or destroy novelty. Conversely, failure to find a matching reference does not prove novelty. Both claims require a source-by-source comparison at the exact theorem strength.*

SDR.12.9 Repair calculus and no-go certificates

Definition SDR.12.13 (SDR-FAIL-49: repair classes)**.** The SDR repair classes are:

R_{obs}	add a genuinely separating observable;
R_{inf}	add a theorem-backed inferential/rigidity edge;
R_{dom}	restrict the source class with proved membership/coverage;
R_{tgt}	weaken or replace the target under a new typed contract;
R_{comp}	add a valid finite-to-infinite/completion theorem;
R_{cert}	repair missing or false certificate/evaluation evidence;
R_{prov}	rebuild the lineage under a clean provenance boundary;
$\mathsf{R}_{\text{model}}$	repair cost, effectivity, representation, or verifier semantics;
R_{arch}	change the route/candidate grammar without claiming old-route success.

These are typed possibilities, not guarantees of existence or success.

Theorem SDR.12.14 (SDR-FAIL-50: repair necessity for a fixed certified collision)**.** *Let (X, Y) be a certified target-discordant pair for fixed source class $\mathcal{S}$, target Θ, and packet $\mathbf{I}$. Any valid repair of this obstruction must change at least one of the following:*

(i) the consumed observational information so that the pair becomes distinguishable;

(ii) the admissible source class so that at least one member of the pair is excluded by theorem; or

(iii) the target/equivalence so that the pair is no longer target-discordant.

A theorem-facing repair is legitimate only insofar as it proves one of these typed changes or proves that the purported collision certificate was false. Pure relabeling, deterministic postprocessing, search score, or additional prose cannot repair a valid collision.

Proof. If the source class still contains both objects, the target still separates them, and the consumed packet still identifies them, Theorem SDR.12.5 continues to apply. Hence at least one of those typed ingredients must change through a valid theorem-backed operation. □

Definition SDR.12.15 (SDR-FAIL-51: no-go certificate)**.** A canonical no-go certificate contains

$$\begin{aligned}\mathcal{N}_{\text{go}} := (&\mathsf{ClaimID}, \mathsf{SourceScope}, \mathsf{TargetID}, \mathsf{RouteClassID}, \mathsf{Quantifiers}, \mathsf{EvidenceMode},\\ &\mathsf{CounterexampleOrTheorem}, \mathsf{CoverageProof}, \mathsf{StrongestSafeNegative}, \mathsf{ExcludedReadings},\\ &\mathsf{RepairClasses}, \mathsf{Provenance}, \mathsf{Artifacts}, \mathsf{Status}).\end{aligned}$$

A no-go certificate is theorem-facing only when its coverage proof matches the full route/source scope of the negative claim.

Definition SDR.12.16 (SDR-FAIL-52: irreducible obstruction core)**.** For a no-go theorem with premise set $H = \{H_1, \ldots, H_m\}$, an *irreducible obstruction core* is a subset $C \subseteq H$ sufficient for the no-go conclusion such that deleting any member of C destroys the currently certified derivation of that conclusion. This is proof-minimality relative to the registered derivation, not logical necessity in all possible proofs.

Policy 178 (SDR-FAIL-53: proof-minimality is not hypothesis necessity)**.** *An irreducible obstruction core records what the registered proof uses. To claim that a hypothesis is mathematically necessary for the no-go theorem requires a countermodel, reverse theorem, or other necessity certificate beyond deletion from one proof DAG.*

Theorem SDR.12.17 (SDR-FAIL-54: no-go composition under nested route classes)**.** *If $\mathfrak{R}_1 \subseteq \mathfrak{R}_2$ and a theorem rules out every route in $\mathfrak{R}_2$ for the same source class and target, then every route in $\mathfrak{R}_1$ is also ruled out. The converse is invalid in general.*

Proof. Immediate from set inclusion of the quantified route classes. □

Policy 179 (SDR-FAIL-55: repair success must retire, not erase, the old failure)**.** *When a later enrichment or theorem repairs a previously certified failure, the old failure record remains immutable as evidence about the weaker route. The repaired route receives a new lineage and a theorem explaining why the old obstruction no longer applies. Historical counterexamples are not deleted merely because a stronger packet succeeds.*

SDR.12.10 Operational atlas and canonical failure record

Definition SDR.12.18 (SDR-FAIL-56: failure-atlas row)**.** Each operational atlas row has the form

$$\mathsf{AtlasRow} = (\mathsf{Signature}, \mathsf{DetectionRule}, \mathsf{SafeNegative}, \mathsf{ForbiddenInflation}, \mathsf{RepairMenu}, \mathsf{Owner}, \mathsf{Evidence}).$$

The DetectionRule may be executable, theorem-only, or manual-audit-only; its mode must be declared.

Declaration SDR.12.19 (SDR-FAIL-57: core failure atlas)**.** The current SDR core atlas contains at least the following reusable counterpatterns:

(a) **information collision:** equal packet, different target; repair by $\mathsf{R}_{\mathrm{obs}}$, $\mathsf{R}_{\mathrm{dom}}$, or $\mathsf{R}_{\mathrm{tgt}}$;

(b) **postprocessing illusion:** deterministic recoding advertised as enrichment; requires genuinely new information or a separately typed nonobservational repair;

(c) **pair-to-class inflation:** local witness hit reported as rigidity; requires class-wide coverage;

(d) **surrogate reversal:** the information preorder is used in the wrong direction; repair by the correct directional theorem;

(e) **oracle/benchmark contamination:** conclusion-bearing information enters construction; repair by $\mathsf{R}_{\mathrm{prov}}$ in a clean lineage;

(f) **memorization separator:** IDs/hashes replace source semantics; repair by source-semantic probes;

(g) **vacuous/irrelevant rigidity:** theorem scope misses intended instances; repair by membership and coverage theorems;

(h) **unproved-negation inflation:** failure to prove is reported as false; repair by an actual counterexample or negation theorem;

(i) **finite-to-infinite leakage:** finite prefix or plateau is promoted to tail closure; repair by $\mathsf{R}_{\mathrm{comp}}$;

(j) **stabilization-to-zero inflation:** stable nonzero states are ignored; repair by zero reflection;

(k) **completion circularity:** the desired conclusion is encoded into the residual package; repair by an independent package;

(l) **cost-model collapse:** cardinality, query, bit, time, and artifact costs are conflated; repair by $\mathsf{R}_{\mathrm{model}}$;

(m) **hidden observable:** an unregistered probe is consumed by the verifier; repair the selected-family record and optimality claim;

(n) **categorical-strength inflation:** object or fully-faithful Return is promoted to exact/derived strength without the missing structure;

(o) **coordinate illusion:** basis-dependent data are treated as invariants; repair by quotient/invariance semantics;

(p) **local-to-global inflation:** local reconstructions are treated as a global object; repair by assembly/descent/coherence;

(q) **BF-strength inflation:** survivor/Fitting data are called a general higher-Bockstein theory; repair by an actual spectral-sequence theorem;

(r) **fixed-to-uniform inflation:** per-layer exact replay is called a bounded all-layer formula; repair by a uniform structural theorem;

(s) **holdout recycling:** test outcomes influence the candidate later called holdout-tested; repair by a new sealed holdout;

(t) **search-to-proof inflation:** score, replay, consensus, or exhaustion is treated as proof; repair by theorem-facing verification;

(u) **route-to-target inflation:** one failed branch is reported as target falsity; repair the quantifier logic or prove the exact target negation;

(v) **contract laundering:** altered hypotheses/source/target are reported as the parent problem; repair by a child contract or parent-level discharge; and

(w) **novelty inflation:** absence/presence of superficial literature overlap is treated as a novelty verdict; repair by exact literature comparison.

The list is an SDR.1–SDR.11 coverage atlas, not a theorem that all mathematical failure modes are exhausted.

Audit Obligation SDR.12.20 (SDR.12 no-go and failure-atlas audit)**.** *Before an SDR run emits a negative theorem-facing verdict, it must verify:*

(i) exact source class, target, equivalence, quantifiers, and route class;

(ii) whether the evidence is a missing field, failed implementation, counterexample, no-go theorem, finite exhaustive search, or external theorem;

(iii) whether the evidence proves semantic failure or only certified/operational failure;

(iv) whether any surrogate direction, source restriction, target change, or child contract is involved;

(v) whether a finite-to-infinite, local-to-global, or category-strength upgrade is being attempted;

(vi) whether benchmark, holdout, or answer-derived information contaminated the route;

(vii) whether the claimed no-go coverage actually ranges over the full advertised route class;

(viii) whether the strongest safe negative conclusion in the certificate is weaker than target falsity, and if so, that the weaker conclusion is preserved;

(ix) at least one typed repair class or an explicit proof that no listed repair class is admissible in the declared architecture; and

(x) immutable provenance and artifact identities for the negative result.

Definition SDR.12.21 (SDR-FAIL-58: canonical SDR failure record)**.** A canonical failure record is

$$\mathcal{R}_{\mathrm{FAIL}} = (\mathfrak{F}, \mathsf{NegVec}, \mathcal{N}_{\mathrm{go}}, \mathsf{AtlasRowID}, \mathsf{RepairHistory}, \mathsf{Lineage}, \mathsf{Artifacts}, \mathsf{Owner}, \mathsf{Lifecycle}).$$

Fields not applicable to the event are marked $\mathrm{not_invoked}$; they are not silently omitted.

Declaration SDR.12.22 (SDR.12 explicit nonclaims)**.** SDR.12 does not claim:

(i) a complete taxonomy of all possible mathematical failure modes;

(ii) that every failure signature is mechanically decidable;

(iii) that every no-go theorem has a finite counterexample;

(iv) that every obstruction has a successful repair;

(v) that every repair can be found by the SDR.11 search grammar;

(vi) that operational exhaustion proves theorem nonexistence;

(vii) that a route obstruction falsifies an external conjecture;

(viii) that cross-domain recurrence of a counterpattern proves universality;

(ix) that the elementary information-theoretic no-go results are novel mathematics; or

(x) any mutation of frozen v21 failure, Claim-Register, Core-P, Core-DG, HT, UB, or BFA ownership.

Theorem SDR.12.23 (SDR-FAIL-59: SDR.12 failure/no-go closure)**.** *Under the typed records and policies above, SDR.12 provides a conservative failure/no-go layer with the following properties:*

(i) negative conclusions are scope-, quantifier-, route-, evidence-, and provenance-typed;

(ii) a certified target-discordant information collision yields an exact no-go theorem for every deterministic decoder using only the collided packet;

(iii) deterministic postprocessing and theorem relabeling cannot repair a fixed certified collision;

(iv) family blindness, when proved for every probe in a declared family, yields a genuine no-go for every subfamily, while failed search does not;

(v) surrogate, oracle, memorization, vacuity, source-restriction, and effectivity failures remain distinct;

(vi) strict improvement, stabilization, finite state, state convergence, and finite prefixes are prevented from masquerading as completion or target zero without their missing theorems;

(vii) minimality, cardinality, certificate cost, finite coverage, and ambient optimality remain distinct optimization claims;

(viii) categorical strength, coordinate invariance, local-to-global assembly, higher-Bockstein strength, and fixed-to-uniform layer strength are protected from inflation;

(ix) calibration/holdout status, search evidence, route obstruction, target falsity, child contracts, and novelty remain independently typed;

(x) every certified fixed-pair obstruction has an explicit repair-necessity theorem describing which typed ingredient must change;

(xi) no-go certificates preserve the exact quantified route class and strongest safe negative conclusion; and

(xii) repaired routes retain immutable historical failure records rather than erasing the weaker-route counterexample.

These properties organize and strengthen the SDR.1–SDR.11 boundary discipline but do not assert completeness of the atlas or automatic solvability of any failure.

Proof. Items (i)–(iv) are Definitions SDR.12.2–SDR.12.4, Theorem SDR.12.5, Corollary SDR.12.6, Theorem SDR.12.7, and Policy 144. Item (v) is Policies 147, 148, 149, 151, 153, and 154. Item (vi) is Policies 155, 156, 157, 158, and 159, together with Theorem SDR.12.11. Item (vii) is Policies 160–164. Item (viii) is Policies 165–170. Item (ix) is Policies 171 and 172–177, plus Theorem SDR.12.12. Item (x) is Theorem SDR.12.14. Item (xi) is Definition SDR.12.15 and Policy 143. Item (xii) is Policy 179. □

Remark SDR.12.24 (SDR.13 handoff: formalization, machine records, and reference verification)**.** SDR.12 turns failure events into typed records with explicit negative ceilings and repair routes. SDR.13 will specify a machine-checkable record architecture for those objects: schema identities, theorem/proof/counterexample references, content-addressed artifacts, executable verifier boundaries, formalization status, literature-source bindings, and reference-by-reference overlap audits will be separated so that compilation, replay, citation, and mathematical proof cannot substitute for one another. Specification of such records is not itself an implemented machine schema or instantiated release manifest.

SDR.13 Formalization, Machine Records, and Reference Verification

SDR.13.1 Role, authority, and noncollapse discipline

Declaration SDR.13.1 (SDR-FORM-01: SDR.13 role)**.** SDR.13 specifies an auditable machine-record layer for the theorem-facing objects of SDR.1–SDR.12 and performs the source-by-source literature-overlap audit deferred by SDR.10. It has four distinct tasks:

(i) specify canonical machine records for definitions, theorems, witnesses, counterexamples, search candidates, no-go certificates, artifacts, and references;

(ii) separate syntax/schema checking, computational replay, proof-assistant verification, and semantic interpretation;

(iii) bind external references at exact bibliographic and theorem-use strength; and

(iv) determine which SDR constructions overlap with established literature and which novelty questions remain unresolved.

SDR.13 is a post-v21 research-supplement layer. It does not mutate a frozen v21 theorem, Claim UID, Core owner, Appendix HT/UB theorem, Appendix BFA result, or Search/Platform record.

Definition SDR.13.2 (SDR-FORM-02: formalization-status vector)**.** For a theorem-facing record R, define

$$\mathsf{FormalStatus}(R) = (\mathsf{Parse}, \mathsf{Schema}, \mathsf{Replay}, \mathsf{Kernel}, \mathsf{Interpret}).$$

The coordinates mean:

(a) Parse: the source artifact parses or compiles in the declared syntax;

(b) Schema: required machine fields and type relations pass a declared schema verifier;

(c) Replay: finite computations or derivation traces replay under a declared executable verifier;

(d) Kernel: a proof object is accepted by a named trusted proof kernel or proof assistant; and

(e) Interpret: a proved or audited interpretation edge connects the formal object to the intended informal mathematical statement.

No coordinate is copied into another by status inheritance.

Policy 180 (SDR-FORM-03: formalization noncollapse)**.** *The implications*

$$\mathsf{Parse} \Rightarrow \mathsf{Schema}, \qquad \mathsf{Schema} \Rightarrow \mathsf{Replay}, \qquad \mathsf{Replay} \Rightarrow \mathsf{Kernel}, \qquad \mathsf{Kernel} \Rightarrow \mathsf{Interpret}$$

are all invalid by default. In particular, successful XeLaTeX compilation proves neither mathematical truth nor proof-assistant formalization, and kernel acceptance of a formal statement proves the intended informal theorem only through a separately registered interpretation relation.

SDR.13.2 Canonical machine objects and content-addressed identity

Definition SDR.13.3 (SDR-FORM-04: machine-object sorts)**.** The SDR machine layer uses the disjoint record sorts

$$\mathsf{MObj} = \{\mathsf{Def}, \mathsf{Thm}, \mathsf{Proof}, \mathsf{Witness}, \mathsf{Candidate}, \mathsf{NoGo}, \mathsf{Artifact}, \mathsf{Reference}, \mathsf{Interpretation}, \mathsf{Run}\}.$$

A record carries exactly one primary sort. Cross-sort relations are expressed by typed edges rather than by embedding one status token into another object.

Definition SDR.13.4 (SDR-FORM-05: canonical encoder)**.** Fix a schema version v whose valid machine records are finite serializable objects. A canonical encoder is a map

$$\mathrm{enc}_v : \mathsf{ValidRecord}_v \longrightarrow \{0,1\}^*$$

equipped with a decoder on its image satisfying

$$\mathrm{dec}_v(\mathrm{enc}_v(R)) = R \qquad (R \in \mathsf{ValidRecord}_v).$$

Hence injectivity is a consequence of reversible decoding rather than an untested serialization convention. The implementation fixes field order, string normalization, integer/rational representation, null semantics, array order, escaping, and every binary representation used by the schema. An implementation advertised as canonical must provide deterministic encode/decode conformance tests; a generic JSON serializer is not presumed canonical merely because it emits valid JSON.

Definition SDR.13.5 (SDR-FORM-06: content address)**.** For a declared cryptographic hash function H, the content address of a valid machine record R is

$$\mathrm{CID}_{v,H}(R) := H\big(v \parallel \mathrm{enc}_v(R)\big).$$

The schema version and hash algorithm are part of the identity record. A bare digest without those parameters is not a complete SDR content address.

Policy 181 (SDR-FORM-07: hash identity is not byte identity or semantic identity)**.** *Equality of content addresses is computational evidence that two canonical encodings are the same under the declared collision-resistance assumption and encoder correctness; it is not a mathematical proof of byte equality because distinct byte strings can in principle collide under a finite hash. Exact encoded-byte identity is established by byte-for-byte comparison or by another injective identity mechanism, not by digest equality alone. Matching content addresses also do not by themselves prove:*

(i) equality or equivalence of the mathematical objects denoted by two records;

(ii) truth of the theorem stored in a record;

(iii) validity of a proof relation;

(iv) freshness or authority of an external source; or

(v) equality of two records under a different schema version or encoding policy.

Semantic identity requires the relevant mathematical or interpretation theorem.

Policy 182 (SDR-FORM-07A: digest-collision and trust-model firewall)**.** *Every theorem-facing use of a content address records the hash algorithm and the computational trust assumption under which the digest is consumed. A collision-resistant hash may be used for practical content addressing and tamper evidence, but no theorem proof may require the logically false implication*

$$H(x) = H(y) \Longrightarrow x = y$$

unless H has separately been proved injective on the exact finite domain in question. Where exact identity matters mathematically, the underlying bytes or a reversible canonical record must remain available for replay.

Definition SDR.13.6 (SDR-FORM-08: canonical theorem record)**.** A theorem record contains at least

$$\mathcal{M}_{\mathrm{Thm}} = (\mathsf{TheoremID}, \mathsf{StableLocator}, \mathsf{Statement}, \mathsf{SourceClass}, \mathsf{Hypotheses}, \mathsf{Target}, \mathsf{Equivalence}, \mathsf{Quantifiers}, \mathsf{Dependencies}, \mathsf{ProofRefs}, \mathsf{TrustBase}, \mathsf{Status}, \mathsf{Owner}, \mathsf{Version}, \mathsf{Artifacts}, \mathsf{CID}).$$

The statement and hypotheses are theorem-facing fields; prose summaries, search scores, and lifecycle metadata cannot replace them.

Definition SDR.13.7 (SDR-FORM-09: canonical witness/counterexample record)**.** A witness record contains

$$\mathcal{M}_{\mathrm{Wit}} = (\mathsf{WitnessID}, X, Y, \mathsf{SourceScope}, \mathsf{Packet}, \mathsf{Target}, \mathsf{PacketEqualityReceipt}, \mathsf{TargetDiscordanceReceipt}, \mathsf{EvidenceMode}, \mathsf{Lineage}, \mathsf{Artifacts}, \mathsf{CID}).$$

For a one-object counterexample the unused source slot is marked $\mathrm{not_invoked}$. A witness is theorem-facing only at the exact equality/discordance strength justified by its receipts.

Definition SDR.13.8 (SDR-FORM-10: typed proof-DAG edge)**.** The machine proof graph admits edge kinds

$$\mathsf{uses},\ \mathsf{proves},\ \mathsf{refutes},\ \mathsf{implements},\ \mathsf{formalizes},\ \mathsf{interprets},\ \mathsf{validates},\ \mathsf{overlaps}.$$

Every edge stores source ID, target ID, direction, scope, theorem strength, and evidence identity. The ordinary theorem-dependency subgraph induced by uses edges must be acyclic. Provenance/history edges may form a separate chronology graph but may not create theorem dependencies by graph reachability alone.

Theorem SDR.13.9 (SDR-FORM-11: graph closure is not proof closure)**.** *Let G be a machine graph whose nodes and edges are syntactically valid. Graph reachability from theorem premises to a target node does not establish the target theorem unless every theorem-bearing edge on the route has a valid proof or imported-theorem certificate at the exact required strength.*

Proof. A graph edge is data describing a claimed relation. If one theorem-bearing edge is unsupported, replace its intended implication by an arbitrary false implication while preserving the graph topology and all syntactic identities. Reachability is unchanged although the claimed theorem route is invalid. Therefore reachability cannot substitute for edge validity. □

SDR.13.3 Verifier stratification and proof-carrying packets

Definition SDR.13.10 (SDR-FORM-12: verifier strata)**.** An SDR verifier stack is a tuple

$$\mathfrak{V} = (V_{\mathrm{syntax}}, V_{\mathrm{schema}}, V_{\mathrm{artifact}}, V_{\mathrm{compute}}, V_{\mathrm{kernel}}, V_{\mathrm{interpret}}).$$

The strata check, respectively, syntax, record typing, artifact identity, executable calculations, formal proof objects, and the formal-to-informal interpretation relation.

Policy 183 (SDR-FORM-13: lower verifier strata do not promote upward)**.** *Acceptance by V_{syntax}, V_{schema}, V_{artifact}, or V_{compute} cannot by itself set V_{kernel} or $V_{\mathrm{interpret}}$ to pass. In particular, a numerical replay can certify the displayed finite computation without proving a universally quantified theorem from which that computation is only one instance.*

Definition SDR.13.11 (SDR-FORM-14: proof-carrying SDR packet)**.** A proof-carrying packet for claim C is

$$\Pi_C = (\mathcal{M}_C, \mathcal{D}_C, \mathcal{P}_C, \mathcal{A}_C, \mathcal{I}_C),$$

where $\mathcal{M}_C$ is the claim record, $\mathcal{D}_C$ the dependency manifest, $\mathcal{P}_C$ the proof/import receipts, $\mathcal{A}_C$ the immutable artifacts, and $\mathcal{I}_C$ the interpretation record when a formal object is consumed as an informal theorem.

Theorem SDR.13.12 (SDR-FORM-15: trust-relative packet soundness)**.** *Let TB_C be the declared trust base of Π_C. Assume every internally verified theorem dependency in Π_C has passed a sound verifier at its declared semantics, every externally imported theorem belongs to TB_C with a source-bound statement whose hypotheses and conclusion match the consumer, and every required interpretation edge is valid. Then acceptance of the complete packet licenses C exactly at the recorded scope and no stronger strength, relative to the truth of the external premises admitted by TB_C.*

Proof. Topologically order the acyclic theorem-dependency subgraph. Soundness of the internally verified leaves and the declared external premises in TB_C supply their exact statements. Inductively, each proof node consumes only already justified premises and a sound proof/import rule, so its recorded conclusion is justified. Interpretation edges transport only their registered meaning. The final node therefore inherits the exact recorded claim. No step proves an unrecorded strengthening. □

Policy 184 (SDR-FORM-15A: external trust cannot be erased by packaging)**.** *A published theorem citation, imported library theorem, or externally trusted formal result remains an explicit premise of the packet unless SDR independently proves the same statement. Content addressing, local caching, proof-DAG insertion, or repeated successful replay does not convert an external trust assumption into an SDR-owned proof. Consequently the soundness statement of Theorem SDR.13.12 is relative to the recorded trust base whenever such imports remain.*

Policy 185 (SDR-FORM-16: replay, formal proof, and mathematical interpretation remain distinct)**.** *A replay log is evidence that a declared executable procedure reproduced a declared output. A proof-assistant kernel receipt is evidence that a formal proposition follows in the formal system. An interpretation theorem/audit connects that formal proposition to the intended mathematical statement. None of these three roles is replaced by the other two.*

Definition SDR.13.13 (SDR-FORM-17: proof-assistant status sort)**.** For each theorem-facing SDR locator, the proof-assistant field takes one value in

$$\begin{aligned}\mathsf{PAStatus} = \{&\mathrm{not_formalized}, \mathrm{encoding_drafted},\\ &\mathrm{kernel_checked_interpretation_pending}, \mathrm{formally_interpreted}\}.\end{aligned}$$

The field names a proof system, checker/kernel version, and exact artifact whenever the value is stronger than $\mathrm{not_formalized}$. Kernel acceptance without an interpretation edge is recorded only as $\mathrm{kernel_checked_interpretation_pending}$; there is no separate status token that can be misread as both formal and informal closure.

Declaration SDR.13.14 (SDR-FORM-18: current SDR formalization status)**.** At the present SDR 1.0 construction-draft stage, the canonical TeX source and generated PDF are compiled and structurally audited, but the general SDR theorem corpus is $\mathrm{not_formalized}$ in a proof assistant. The finite examples and algebraic replays may be executable or manually replayable as stated in their owning sections, but this does not set the global Kernel coordinate to pass. SDR.13 therefore records formalization architecture and machine schemas without claiming a completed Lean, Coq, Isabelle, Agda, HOL, or other kernel-checked port.

Declaration SDR.13.15 (SDR-FORM-18A: current machine-layer implementation status)**.** At the present construction-draft stage, SDR.13 specifies record schemas, identity rules, verifier strata, and manifest fields inside the TeX corpus, but it does not yet ship a separately executable JSON Schema/CBOR schema, canonical encoder/decoder implementation, instantiated machine manifest, or automated proof-DAG verifier. The current machine layer is therefore

$$\mathsf{MachineLayerStatus} = \mathrm{specified_not_instantiated}.$$

Local compilation scripts and PDF audits are implementation evidence for those concrete artifacts only; they do not set the global machine-schema or replay coordinates to pass.

SDR.13.4 Reference binding and exact use classes

Definition SDR.13.16 (SDR-FORM-19: external reference-binding record)**.** An external reference binding is

$$\begin{aligned}\mathfrak{L} = (&\mathsf{RefID}, \mathsf{Authority}, \mathsf{Bibliography}, \mathsf{Locator}, \mathsf{BoundClaim},\\ &\mathsf{UseClass}, \mathsf{Direction}, \mathsf{RetrievedAt}, \mathsf{Status}, \mathsf{ArtifactID}).\end{aligned}$$

The use class is one of

$$\mathsf{background_overlap},\quad \mathsf{theorem_import},\quad \mathsf{implementation_reference},\quad \mathsf{comparison_only},\quad \mathsf{unresolved}.$$

A bibliography entry with no bound claim cannot function as a theorem import.

Policy 186 (SDR-FORM-20: citation is not theorem evidence by existence)**.** *A DOI, URL, title match, citation count, or bibliographic record proves only source identity/provenance to the extent verified. To consume an external theorem, SDR must bind the exact theorem statement, hypotheses, direction, source class, and strength needed by the consumer. A secondary summary may verify metadata but does not replace the authoritative mathematical source when theorem content is imported.*

Definition SDR.13.17 (SDR-FORM-21: reference-verification statuses)**.** The reference-verification status is

$$\begin{aligned}\mathsf{RefStatus} = \{&\mathrm{identity_verified}, \mathrm{claim_verified}, \mathrm{overlap_classified},\\ &\mathrm{import_verified}, \mathrm{conflict}, \mathrm{unresolved}\}.\end{aligned}$$

The statuses are not totally ordered: for example, an overlap classification may be complete even when no theorem is imported.

Policy 187 (SDR-FORM-21A: bound-claim verification is not independent truth verification)**.** *Within* RefStatus*,* claim_verified *means that the authoritative source was inspected far enough to verify that the bound statement, hypotheses, locator, and direction are represented correctly. It does not mean that SDR has independently reproved the mathematics. Independent mathematical verification, formal kernel checking, and trusted theorem import remain separate fields.*

Policy 188 (SDR-FORM-22: freshness and authority are separate)**.** *A recently retrieved source is not necessarily authoritative, and an old authoritative theorem does not become stale merely because its web page is old. Reference records therefore separate bibliographic identity, theorem authority, retrieval time, and current availability.*

SDR.13.5 Reference-by-reference literature-overlap audit

Declaration SDR.13.18 (SDR-FORM-23: verified reference registry)**.** The following reference identities were checked for the SDR.10 overlap classes. Retrieval date for the web bindings is 2026-10-15.

R1. Blackwell 1953. David Blackwell, "Equivalent Comparisons of Experiments," *Annals of Mathematical Statistics* 24(2), 265–272 (1953), DOI https://doi.org/10.1214/aoms/1177729032.

R2. Cousot–Cousot 1977. Patrick Cousot and Radhia Cousot, "Abstract Interpretation: A Unified Lattice Model for Static Analysis of Programs by Construction or Approximation of Fixpoints," POPL 1977, 238–252, DOI https://doi.org/10.1145/512950.512973.

R3. Clarke–Grumberg–Jha–Lu–Veith 2000. "Counterexample-Guided Abstraction Refinement," CAV 2000, LNCS 1855, 154–169, DOI https://doi.org/10.1007/10722167_15.

R4. Karp 1972. Richard M. Karp, "Reducibility Among Combinatorial Problems," in *Complexity of Computer Computations*, 85–103 (1972), DOI https://doi.org/10.1007/978-1-4684-2001-2_9.

R5. Stacks Project, Fitting ideals. Section 15.8, Tag 07Z6, https://stacks.math.columbia.edu/tag/07Z6; in particular Lemma 15.8.2 gives presentation independence of the Fitting ideals defined from minors.

R6. Stacks Project, inverse systems/derived limits. Section 12.31, Tag 02MY, https://stacks.math.columbia.edu/tag/02MY, and Section 15.88, Tag 07KV, https://stacks.math.columbia.edu/tag/07KV; in particular the former defines the Mittag–Leffler condition and the latter records $R^1\varprojlim = 0$ for ML inverse systems of abelian groups.

R7. Stacks Project, modules over a PID. Section 15.126, Tag 0ASL, https://stacks.math.columbia.edu/tag/0ASL; in particular Lemma 15.126.3 gives cyclic decomposition over valuation-type local rings and Lemma 15.126.9 gives the finite-module structure theorem over a PID, covering the DVR/PID structure input used in the SDR.9 comparison.

These entries are a controlled overlap/reference ledger, not a claim that the listed works are the unique or earliest literature for every SDR concept.

Declaration SDR.13.19 (SDR-FORM-23A: reference-status ledger)**.** At the present audit resolution the seven bound entries have the following theorem-facing use status:

Reference	UseClass	RefStatus
$R1$	background_overlap	overlap_classified
$R2$	background_overlap	overlap_classified
$R3$	background_overlap	overlap_classified
$R4$	background_overlap	overlap_classified
$R5$	background_overlap	claim_verified/overlap_classified
$R6$	background_overlap	claim_verified/overlap_classified
$R7$	background_overlap	claim_verified/overlap_classified

Here claim_verified for R5–R7 means that the cited Stacks statements were inspected and match the bound mathematical content; it is not an independent re-proof. No R1–R7 entry is consumed as an SDR theorem import in this section, so none is assigned import_verified merely by appearing in the overlap ledger.

Proposition SDR.13.20 (SDR-FORM-24: Blackwell comparison is an overlap, not the SDR preorder)**.** *Blackwell's comparison theory concerns statistical experiments and decision-theoretic informativeness. SDR.2's preorder*

$$I \preceq J \iff I \text{ factors deterministically through } J$$

is a deterministic source-map preorder characterized by equality-kernel refinement. The two share an information-comparison motif, but SDR does not import a stochastic kernel, loss family, or Blackwell experiment theorem. Therefore R1 is classified background_overlap/overlap_classified*, not* theorem_import.

Proposition SDR.13.21 (SDR-FORM-25: abstract-interpretation overlap ceiling)**.** *R2 establishes a classical lattice-based theory of abstraction and fixpoint approximation for program semantics. SDR shares the use of ordered abstractions/refinements and soundness ceilings, but SDR's target-relative equality-kernel calculus, source-semantic probes, rigidity packets, and Return contracts are not identified with an abstract-interpretation Galois connection by terminology alone. No theorem of R2 is required for the elementary set-level proofs of SDR.2–SDR.4.*

Proposition SDR.13.22 (SDR-FORM-26: CEGAR overlap ceiling)**.** *R3 verifies that counterexample-guided abstraction refinement is established prior art as a search/refinement architecture. Consequently SDR claims no novelty for the generic idea*

$$\textit{counterexample} \longrightarrow \textit{refinement} \longrightarrow \textit{recheck}.$$

SDR.4/SDR.11 differ in mathematical object types, target-relative defect semantics, provenance quarantine, nonobservational repair modes, and theorem-facing rigidity/completion gates, but those differences are architectural distinctions rather than a proof of publication-level novelty.

Proposition SDR.13.23 (SDR-FORM-27: finite separating-family problem has a standard set-cover core)**.** *After a finite conflict universe and finite candidate-probe registry are frozen, SDR.7 maps each probe to the subset of conflicts it separates. Selecting a minimum-cardinality separating family is therefore exactly an instance of finite set cover on that incidence system. R4 verifies the classical complexity provenance of set covering. SDR claims no novelty for set cover, hitting-set duality, or generic combinatorial hardness; the SDR-specific content lies in how the conflict universe, admissible probes, evidence modes, and rigidity bridge are typed before the finite optimization is formed.*

Proposition SDR.13.24 (SDR-FORM-28: SDR.9 Fitting reconstruction is classical-algebra adjacent)**.** *R5 verifies the classical presentation-independent construction of Fitting ideals from minors, while R7 supplies a source-bound module-structure reference covering the valuation/PID cyclic decomposition used after localization at a DVR. These classical ingredients are the algebraic basis of SDR.9's local elementary-divisor reconstruction. Accordingly the theorem-facing reconstruction in SDR.9 may be useful as an AK interface theorem, but the local Fitting/elementary-divisor classification itself is not claimed as new mathematics.*

Proposition SDR.13.25 (SDR-FORM-29: SDR.8 completion overlap is classical and source-scoped)**.** *R6 verifies the established role of Mittag–Leffler/stable-image conditions in inverse-limit and derived-limit arguments. SDR.8 therefore claims no novelty for the generic principle that certified stabilization/ML hypotheses can control* $R^1\varprojlim$ *or related completion terms. The frozen UB finite-state theorem remains the actual internal positive control consumed by SDR.8; the external references serve comparison and verification of literature overlap, not a retroactive replacement of UB ownership.*

Declaration SDR.13.26 (SDR-FORM-30: literature-overlap verdict by SDR layer)**.** The reference audit yields the following conservative verdicts:

(i) **SDR.2–SDR.3:** the factorization/equality-kernel calculus is elementary set theory and partition refinement; no novelty claim is supported.

(ii) **SDR.4 and SDR.11:** counterexample-guided refinement has clear prior-art overlap with CEGAR/abstraction-refinement architectures; SDR's typed mathematical governance may be distinctive, but novelty is unassessed.

(iii) **SDR.6:** rigidity-as-factorization/decoding at the set level is standard-adjacent; no novelty claim is supported for the abstract theorem alone.

(iv) **SDR.7:** the frozen finite selection problem is ordinary set cover; no novelty claim is supported for the combinatorial core.

(v) **SDR.8:** Mittag–Leffler/stable-image/derived-limit mechanisms are classical; SDR's contribution is architecture and exact gating unless a new domain-specific completion theorem is proved.

(vi) **SDR.9:** local Smith/Fitting reconstruction is classical-algebra adjacent and explicitly non-novel at that level; any stronger novelty claim must concern a theorem not reducible to that classical classification, such as a genuinely new higher-Bockstein or compressed all-layer theorem.

(vii) **SDR.10–SDR.12:** validation, provenance, search, and failure semantics overlap broadly with established verification/scientific-method ideas; their integration is not by itself a mathematical novelty theorem.

No item above is a rejection of practical or architectural value. It is a publication-novelty ceiling at the presently audited theorem strengths.

Policy 189 (SDR-FORM-30A: declared-overlap audit is not an exhaustive novelty search)**.** *The registry in Declaration SDR.13.18 closes only the named principal overlap families selected by the SDR.10 handoff. It does not prove that no closer, earlier, or stronger prior art exists. Absence of a reference from this finite audit cannot support a novelty claim. Publication-level novelty therefore requires a separately scoped search protocol, databases/corpora, query history, cutoff date, and claim-by-claim comparison.*

Theorem SDR.13.27 (SDR-FORM-31: declared literature-overlap subgate closure)**.** *The bibliographic/reference-verification component deferred by SDR.10 is closed only at the declared* principal-overlap-audit *strength: the named prior-art families now have source-bound references and an explicit nonidentity/import classification. This does not establish an exhaustive literature search, publication-level novelty of SDR, or closure of the independent-holdout component of G7.*

Proof. Declaration SDR.13.18 binds the reference identities. Propositions SDR.13.20–SDR.13.25 classify the exact relation between each literature family and the SDR consumer. Declaration SDR.13.26 records the resulting novelty ceilings. None of those steps supplies a postconstruction holdout experiment, so the holdout subgate remains unchanged. □

Declaration SDR.13.28 (SDR-FORM-32: construction-gate status after the literature audit)**.** After SDR.13,

$$G7_{\text{literature}} = \text{pass_declared_overlap_audit}, \qquad G7_{\text{independent_holdout}} = \text{pending}.$$

Hence full construction gate G7 remains pending. A future frozen postconstruction holdout or prospective blind validation must be registered without reusing construction/calibration evidence as independent evidence.

SDR.13.6 Machine manifests, reference integrity, and lifecycle

Definition SDR.13.29 (SDR-FORM-33: SDR machine manifest)**.** A machine manifest for an SDR release candidate is

$$\begin{aligned}\mathcal{M}_{\mathrm{SDR}} = (&\mathsf{ReleaseID}, \mathsf{ParentBaseline}, \mathsf{SchemaVersion},\\ &\mathsf{SourceCID}, \mathsf{PDFCID}, \mathsf{ByteIdentityPolicy}, \mathsf{LocatorRegistry}, \mathsf{LabelRegistry},\\ &\mathsf{TheoremRecords}, \mathsf{WitnessRecords}, \mathsf{FailureRecords},\\ &\mathsf{SearchRecords}, \mathsf{ReferenceRecords}, \mathsf{VerifierVersions},\\ &\mathsf{FormalStatus}, \mathsf{GateStatus}, \mathsf{GeneratedAt}, \mathsf{Owner}).\end{aligned}$$

A field may be $\mathrm{not_invoked}$ only when the governing schema marks it optional for that lifecycle state.

Policy 190 (SDR-FORM-34: manifest completeness is not theorem completeness)**.** *A manifest can be internally complete while one of its theorem records has status* $\mathrm{undefined}$*,* reject*, or* $\mathrm{pending}$*. Manifest completeness means that the state is represented without omission, not that every mathematical obligation passed.*

Policy 191 (SDR-FORM-34A: manifest schema is not an instantiated manifest)**.** *Definition SDR.13.29 specifies the fields of a release manifest. A release may claim an instantiated machine manifest only after an actual serialized manifest artifact exists, all required fields are populated from the exact release bytes, its schema/encoder version is bound, and the artifact itself has passed the declared schema and identity checks. The present construction draft does not infer such an instance from the TeX definition.*

Definition SDR.13.30 (SDR-FORM-35: immutable-reference update rule)**.** If a source URL, metadata service, or external formal artifact changes, SDR creates a new reference-binding version. Historical bindings are not overwritten. A theorem consumer may migrate to the new version only after statement identity or a typed strengthening/weakening relation has been checked.

Policy 192 (SDR-FORM-36: source alias and mirror discipline)**.** *A mirror, DOI resolver, bibliographic database, or cached PDF may help establish source identity, but multiple aliases do not multiply theorem evidence. If authoritative content is unavailable and only metadata can be verified, the theorem-content status remains* $\mathrm{unresolved}$ *or* $\mathrm{identity_verified}$ *rather than* $\mathrm{claim_verified}$*.*

Definition SDR.13.31 (SDR-FORM-37: formalization delta)**.** For a theorem record T, the formalization delta is the finite obligation set

$$\mathsf{FDelta}(T) = (\mathsf{Definitions}, \mathsf{ImportedLemmas}, \mathsf{ProofSteps}, \mathsf{Computations}, \mathsf{Interpretation}, \mathsf{Missing}).$$

It records what must still be encoded or proved before $\mathsf{PAStatus}(T)$ can reach $\mathrm{formally_interpreted}$. The delta is a planning artifact, not evidence that the missing obligations are true.

Policy 193 (SDR-FORM-38: no formalization-by-translation)**.** *Converting TeX to another syntax, generating proof-assistant-shaped code, or obtaining parser success does not count as theorem formalization. Kernel checking and interpretation are the minimum theorem-facing gates for a formal proof claim.*

Audit Obligation SDR.13.32 (SDR.13 machine/reference audit)**.** *Before an SDR release candidate advertises machine verification, formalization, or literature closure, it must verify:*

(i) unique theorem, witness, artifact, reference, and run identities under a declared schema version;

(ii) canonical encode/decode conformance, finite-serialization scope, and explicit hash algorithm for every content address;

(iii) no interpretation of digest equality as exact byte identity, mathematical equivalence, or theorem truth without the separately required evidence;

(iv) explicit trust-base records for every external theorem dependency and no erasure of those premises by packaging;

(v) acyclicity of the theorem-dependency graph and validity of every theorem-bearing edge;

(vi) separation of syntax, schema, artifact, computation, kernel, and interpretation verifier statuses;

(vii) proof-assistant system/version and exact proof artifact for every kernel-checked claim;

(viii) formal-to-informal interpretation evidence whenever a formal theorem is consumed as the intended SDR statement;

(ix) exact bibliographic identity, locator, bound claim, use class, and authority for each theorem import;

(x) explicit distinction between source-claim binding, theorem truth, literature overlap, and theorem import;

(xi) no publication-novelty claim inferred from absence of a found reference;

(xii) no double counting of mirrors, aliases, or repeated citations as independent evidence;

(xiii) immutable history for source/reference updates and theorem-status changes;

(xiv) gate accounting that leaves G7 independent-holdout status pending until a genuinely independent validation is supplied; and

(xv) preservation of all frozen v21/BFA/HT/UB/Search-Platform ownership boundaries.

Definition SDR.13.33 (SDR-FORM-39: canonical SDR.13 record)**.** The canonical section record is

$$\begin{aligned}\mathcal{R}_{\mathrm{FORM}} = (&\mathsf{Schema}, \mathsf{MachineManifest}, \mathsf{VerifierStack},\\ &\mathsf{FormalStatusRegistry}, \mathsf{ReferenceRegistry}, \mathsf{OverlapAudit},\\ &\mathsf{GateStatus}, \mathsf{Artifacts}, \mathsf{Owner}, \mathsf{Lifecycle}).\end{aligned}$$

Declaration SDR.13.34 (SDR-FORM-40: explicit nonclaims)**.** SDR.13 does not claim:

(i) that the complete SDR theorem corpus is formalized in a proof assistant;

(ii) that the presently specified machine schemas, canonical encoder, or release manifest have already been instantiated as a standalone executable machine package;

(iii) that compilation, schema validation, replay, content hashes, or graph closure prove mathematics;

(iv) that a proof-assistant theorem has the intended informal meaning without an interpretation edge;

(v) that the declared reference registry exhausts the relevant literature or proves publication-level novelty by absence;

(vi) that bibliographic search establishes novelty or non-novelty by absence alone;

(vii) that Blackwell order, abstract interpretation, CEGAR, set cover, Mittag–Leffler theory, or Fitting ideals are identical to SDR objects merely because of conceptual overlap;

(viii) that SDR.9's local Smith/Fitting theorem is new algebra;

(ix) that the literature-overlap audit supplies an independent holdout validation; or

(x) any retroactive promotion into frozen v21, Core-P, Core-DG, HT, UB, Search/Platform, or BFA ownership.

Theorem SDR.13.35 (SDR-FORM-41: SDR.13 closure)**.** *Under the records, verifier strata, and reference bindings above, SDR.13 provides a conservative machine/reference specification layer with the following properties:*

(i) theorem-facing records have stable typed identities separate from prose summaries and lifecycle metadata;

(ii) *canonical encoding is specified by reversible finite-record serialization, while content addresses provide practical computational handles under an explicit hash trust model rather than mathematical proofs of byte identity, semantic identity, or theorem truth;*

(iii) *proof-DAG reachability, compilation, replay, kernel checking, and interpretation are separated and cannot silently promote one another;*

(iv) *proof-carrying packets may license exactly their recorded theorem strength relative to their explicit external trust base when every internal dependency, imported premise, and interpretation edge satisfies the declared verifier/import policy;*

(v) *external references have exact bound claims and declared use classes, so literature overlap cannot silently become theorem import;*

(vi) *the declared principal literature overlaps deferred by SDR.10 are source-bound and classified without claiming equivalence by analogy or exhaustive novelty search;*

(vii) *the generic CEGAR/set-cover/ML/Fitting/Smith ingredients are not claimed as SDR mathematical novelties at their classical strengths;*

(viii) *the literature component of G7 is closed only at declared-principal-overlap-audit strength while the independent-holdout component remains pending;*

(ix) *the present SDR source remains a compiled informal mathematical artifact rather than a completed proof-assistant formalization; and*

(x) *the machine-record architecture remains specified but not yet instantiated as a standalone executable schema/manifest package.*

Proof. Items (i)–(iii) follow from Definitions SDR.13.3–SDR.13.8, Policies 181, 182, 183, and 185, and Theorem SDR.13.9. Item (iv) is Theorem SDR.13.12 together with Policy 184. Item (v) is Definitions SDR.13.16–SDR.13.17 and Policy 186. Items (vi)–(vii) are Declarations SDR.13.18 and SDR.13.19, Propositions SDR.13.20–SDR.13.25, Declaration SDR.13.26, and Policy 189. Item (viii) is Theorem SDR.13.27 and Declaration SDR.13.28. Item (ix) is Declaration SDR.13.14 and Policy 193. Item (x) is Declaration SDR.13.15 together with Policy 191. □

Remark SDR.13.36 (SDR.14 handoff: promotion, ownership, re-audit, and closure)**.** SDR.13 supplies the machine/reference layer needed for final governance. SDR.14 will freeze the complete SDR claim inventory, resolve section ownership and successor-promotion eligibility, audit all construction gates G0–G8, separate proven theorem content from specifications and research programs, register remaining open obligations, and issue the final post-v21 closure verdict without backflow into the frozen v21 release.

SDR.14 Promotion, Ownership, Re-audit, and Closure

SDR.14.1 Purpose, closure mode, and authority boundary

Declaration SDR.14.1 (SDR-GOV-02: SDR.14 final-governance objective)**.** SDR.14 closes the *construction* of Appendix SDR at the strongest exact strength established in SDR.1–SDR.13. It introduces no new external target theorem and no automatic owner migration. Its tasks are to:

(i) freeze the final claim-role and owner inventory;

(ii) audit construction gates G0–G8 at their declared strengths;

(iii) distinguish local closure from Technical Extension, successor-Core, and canonical Claim-Register promotion;

(iv) register every remaining research, validation, formalization, implementation, and literature obligation;

(v) preserve immutable historical status and source provenance; and

(vi) issue a final typed closure verdict with no backflow into the frozen v21 release.

No favorable-looking collection of section results is scalarized into a global truth score.

Policy 194 (SDR-GOV-03: conservative final-section rule)**.** *Reaching SDR.14 does not upgrade* pending, not_formalized, specified_not_instantiated, calibration, imported, research_program, *or* not_claimed *to a stronger status. Final closure may only summarize statuses already justified or assign an unresolved item to an explicit open-obligation class.*

Policy 195 (SDR-GOV-04: local closure is not canonical promotion)**.** *SDR-local closure does not itself create a frozen-v21 theorem identity, a successor canonical Claim UID, a Technical-Extension owner, a Core-P/Core-DG owner, or a release-gate pass. Any such transition requires a separately versioned successor artifact with explicit migration, owner, dependency, consumer, literature, source, and Claim-Register review.*

Definition SDR.14.2 (SDR-GOV-05: final closure strength)**.** The final closure strength is the product

$$\mathfrak{V}_{\mathrm{SDR14}} = (\mathsf{Math}, \mathsf{Validation}, \mathsf{Formal}, \mathsf{Machine}, \mathsf{Literature}, \mathsf{Ownership}, \mathsf{Promotion}, \mathsf{Governance}).$$

The coordinates are noncompensatory. In particular, a strong mathematical coordinate does not repair a pending holdout, machine implementation, literature, or promotion coordinate.

SDR.14.2 Final claim-role inventory

Definition SDR.14.3 (SDR-GOV-06: final SDR claim-role sort)**.** Every theorem-facing SDR item is assigned one of the following roles:

$$\begin{aligned}\mathsf{ClaimRole}_{\mathrm{SDR}} = \{&\mathrm{proved_sdr_local},\ \mathrm{imported_theorem},\\ &\mathrm{calibration_control},\ \mathrm{specification_or_policy},\\ &\mathrm{implementation_specification},\ \mathrm{research_program},\\ &\mathrm{promotion_reentry_candidate},\ \mathrm{explicit_nonclaim}\}.\end{aligned}$$

A single prose paragraph may discuss several roles, but a machine or promotion record must assign them to distinct identities.

Policy 196 (SDR-GOV-07: no theorem by role laundering)**.** *A definition, policy, search candidate, implementation schema, calibration case, imported theorem, or research program does not become an SDR-owned theorem merely by being placed on a proof route or cited by a closing theorem. An SDR-owned theorem requires its own exact statement and proof at the declared source scope.*

Declaration SDR.14.4 (SDR-GOV-08: section-role summary)**.** The final section-level reading is:

(i) SDR.1 is constitutional/governance infrastructure with local conservative-attachment results;

(ii) SDR.2–SDR.3 provide the deterministic target-relative information and defect calculus, largely elementary/standard-adjacent at the generic set level;

(iii) SDR.4 provides counterexample-guided discovery semantics and candidate-synthesis discipline; its generic wrappers are not claimed as publication-level novelty;

(iv) SDR.5–SDR.6 provide typed enrichment and rigidity/reflection architecture, with PE results consumed as calibration/imported controls at their native ownership;

(v) SDR.7 provides finite separating-family and certificate-complexity architecture, with the finite minimum-family core identified with classical set-cover structure after the conflict universe is fixed;

(vi) SDR.8 provides completion/no-shortcut architecture and imports BFA/UB completion controls at their native ownership;

(vii) SDR.9 supplies the strongest substantive SDR-local theorem package (without a publication-novelty claim): higher-Fitting/local elementary-divisor reconstruction and multiplicity-aware finite-layer rank on the declared exact one-variable square-presented class, together with exact fixed-layer replay; the general higher-Bockstein spectral-sequence and uniform all-layer reduced-torsion programs remain open;

(viii) SDR.10 supplies cross-domain consistency/calibration and falsification semantics, but not a strict post-construction independent holdout;

(ix) SDR.11–SDR.12 provide a typed search/discovery engine and failure/no-go atlas, while preserving the Search-Platform rule that search is not proof;

(x) SDR.13 supplies machine/reference specifications and declared-principal literature-overlap audit, while the standalone machine package and proof-assistant formalization remain uninstantiated/uncompleted.

Theorem SDR.14.5 (SDR-GOV-09: final role noncollapse)**.** *No conclusion in Declaration SDR.14.4 licenses any of the following implications without a separately proved edge:*

$$\mathrm{calibration} \Rightarrow \mathrm{independent_validation}, \qquad \mathrm{imported} \Rightarrow \mathrm{sdr_owned},$$
$$\mathrm{specified} \Rightarrow \mathrm{implemented}, \qquad \mathrm{compiled} \Rightarrow \mathrm{formalized},$$
$$\mathrm{cross_domain_supported} \Rightarrow \mathrm{universal}, \qquad \mathrm{literature_overlap_audited} \Rightarrow \mathrm{publication_novel},$$
$$\mathrm{appendix_local_closed} \Rightarrow \mathrm{successor_promoted}.$$

Proof. Each implication is explicitly blocked by the ownership, provenance, validation, machine/formal, literature, and promotion firewalls proved or declared in SDR.1 and SDR.10–SDR.13. The final section adds no conversion theorem between these coordinates. □

SDR.14.3 Owner map and sovereignty audit

Definition SDR.14.6 (SDR-OWN-02: final owner classes)**.** The final owner map distinguishes:

(a) $\mathsf{Owner}_{\mathrm{SDR}}$: genuinely SDR-local statements proved in this supplement;

(b) Owner_{21}: frozen v21 Main/Foundation/DG/HT/UB/Search-Platform/Problem-Interface/Companion/Claim-Register statements consumed at their registered strength;

(c) $\mathsf{Owner}_{\mathrm{BFA}}$: post-v21 BFA results imported as external-to-SDR supplement controls;

(d) $\mathsf{Owner}_{\mathrm{Lit}}$: externally published mathematical results used as background, imported theorem, or overlap evidence according to SDR.13; and

(e) $\mathsf{Owner}_{\mathrm{Impl}}$: executable schemas, manifests, replay tools, or platform artifacts, if later instantiated.

The owner coordinate is distinct from novelty, usefulness, and promotion readiness.

Policy 197 (SDR-OWN-03: imported native controls retain native ownership)**.** *PE morphism/derived Return remains HT-owned; finite-state derived-limit and VC11/Iwasawa controls remain UB-owned; finite-certificate and lower-bound controls remain Foundation/Main-owned where imported; Search/Platform proof-first and quarantine rules remain Search/Platform-owned; BFA residual/quarantine controls remain BFA-owned. SDR may compose, calibrate, or restate their consequences only with explicit source identity and exact strength.*

Policy 198 (SDR-OWN-04: no duplicate generic cut or search owner)**.** *SDR.11–SDR.12 do not create a competing owner for the frozen Search/Platform execution semantics, target-cut calculus, causal-blocker calculus, or Claim-Register promotion process. SDR-local search/failure records are target-relative discovery consumers of those governance principles, not replacements for their canonical owners.*

Proposition SDR.14.7 (SDR-OWN-05: owner-preserving composition)**.** *Suppose an SDR theorem consumes imported results $T_1, \ldots, T_m$ and proves a new consequence C. Then the dependency record may assign C to* $\mathsf{Owner}_{\mathrm{SDR}}$ *while retaining each T_i at its native owner; proving C does not transfer ownership of the premises to SDR.*

Proof. The proof identity of a consequence and the source identity of a premise are distinct nodes in the proof DAG. Dependency does not identify owners. This is exactly the source/owner discipline fixed in SDR.1 and preserved by SDR.13 machine records. □

SDR.14.4 Construction-gate audit G0–G8

Definition SDR.14.8 (SDR-GOV-10: gate verdict sort)**.** Each construction gate receives one of

$$\mathsf{GateVerdict} = \{\mathrm{pass}, \mathrm{pass_at_declared_strength}, \mathrm{partial}, \mathrm{pending}, \mathrm{fail}\},$$

together with an exact scope and evidence pointer. A partial or pending gate is not silently averaged with passing gates.

Theorem SDR.14.9 (SDR-GOV-11: G0–G6 gate audit)**.** *At final construction scope, gates G0–G6 have the following verdicts:*

- ***G0:*** pass*: post-v21 boundary, authority, no-backflow, and owner sovereignty.*
- ***G1:*** pass*: target-relative information/separation kernel and defect calculus.*
- ***G2:*** pass*: certified collision, postprocessing, and family no-go results.*
- ***G3:*** pass*: PE positive controls with exact native-owner ceilings.*
- ***G4:*** pass_at_declared_strength*: executed nontrivial structural-enrichment/synthesis calibration.*
- ***G5:*** pass_at_declared_strength*: abstract/calibration rigidity and reconstruction theorems.*
- ***G6:*** pass_at_declared_strength*: higher-Fitting structural reconstruction, not full higher-Bockstein theory.*

Proof. G0 is Theorem SDR.1.18. G1 is Theorems SDR.2.48 and SDR.3.38. G2 is already supplied by the indistinguishable-pair calculus and is globally consolidated by Theorem SDR.12.23. G3 is the PE calibration architecture of SDR.5–SDR.6 with native theorem ownership preserved. For G4, SDR.9 executes a nontrivial enrichment route from determinant collision to a higher-Fitting/minor packet and proves exact local structural reconstruction; this is an executed synthesis/repair calibration, not a claim of benchmark-quarantined independent discovery. G5 is Theorem SDR.6.36 at abstract/calibration strength. G6 is Theorem SDR.9.28 together with its explicit higher-Bockstein and uniform-torsion ceiling. □

Theorem SDR.14.10 (SDR-GOV-12: G7 final split verdict)**.** *Gate G7 is not fully passed. Its final split verdict is*

$$\begin{aligned} G7_{\text{cross-domain consistency}} &= \mathrm{pass}, \\ G7_{\text{literature}} &= \mathrm{pass_declared_overlap_audit}, \\ G7_{\text{independent_holdout}} &= \mathrm{pending}. \end{aligned}$$

Consequently

$$G7 = \mathrm{partial/pending}.$$

Proof. The cross-domain component is Theorem SDR.10.28 subject to the calibration/holdout distinction recorded in SDR.10. The literature component is Theorem SDR.13.27. Declaration SDR.13.28 leaves a genuinely postconstruction independent holdout or prospective blind validation pending. No later section supplies such evidence. □

Definition SDR.14.11 (SDR-GOV-13: G8 meaning)**.** G8 means *global construction re-audit and typed closure*, not "all other gates passed" and not successor release promotion. G8 may pass when every cross-section obligation has a consistent final status, all pending work is explicit, no hidden strength upgrade is required for closure, and no identified contradiction remains at the declared audit resolution.

Theorem SDR.14.12 (SDR-GOV-14: G8 global construction re-audit verdict)**.** *After applying the global invariants of SDR.14 below, the assembled SDR.1–SDR.14 construction has*

$$G8 = \mathrm{pass_at_declared_strength}.$$

This verdict is compatible with the pending independent-holdout subgate of G7 because G8 certifies internal status/ownership/closure consistency rather than complete external validation.

Proof. The final re-audit preserves every pending or limited coordinate rather than upgrading it. G7's pending holdout is carried into the final open-obligation ledger and final status vector. The closure theorem below requires only that no unresolved item be silently consumed as a passed premise. Hence G8 and the pending G7 subgate answer different predicates and are consistent. □

Declaration SDR.14.13 (SDR-GOV-15: final gate vector)**.** The final construction-gate vector is

$$\begin{aligned}\mathsf{Gates}_{\mathrm{SDR1.0}} = (&G0 : \mathrm{pass},\ G1 : \mathrm{pass},\ G2 : \mathrm{pass},\\ &G3 : \mathrm{pass},\ G4 : \mathrm{pass_at_declared_strength},\\ &G5 : \mathrm{pass_at_declared_strength},\ G6 : \mathrm{pass_at_declared_strength},\\ &G7 : \mathrm{partial/pending},\ G8 : \mathrm{pass_at_declared_strength}).\end{aligned}$$

No scalar aggregate is defined.

SDR.14.5 Promotion and adoption assessment

Definition SDR.14.14 (SDR-PROM-02: promotion dimensions)**.** Promotion readiness is evaluated on independent dimensions

$$\mathsf{PromReadiness} = (\mathsf{Proof}, \mathsf{Generality}, \mathsf{OwnerFit}, \mathsf{Consumers}, \mathsf{Validation}, \mathsf{Literature}, \mathsf{Artifacts}, \mathsf{Migration}).$$

A favorable value in one coordinate does not compensate for a missing coordinate in another.

Policy 199 (SDR-PROM-03: novelty is not owner readiness)**.** *A claim may be mathematically useful yet not publication-novel; it may be novel yet inappropriate for Core; and it may be Core-like in form yet remain blocked by missing proof, validation, consumer, or migration evidence. Promotion decisions therefore do not use a weighted novelty/usefulness score.*

Definition SDR.14.15 (SDR-PROM-04: successor re-entry packet)**.** A future successor request for an SDR object E supplies

$$\begin{aligned}\mathfrak{P}^{+}_{\mathrm{SDR}}(E) = (&E, \mathsf{RequestedOwner}, \mathsf{Statement}, \mathsf{ProofOrSource},\\ &\mathsf{OldOwnerComparison}, \mathsf{Consumers}, \mathsf{Validation}, \mathsf{Literature},\\ &\mathsf{Failures}, \mathsf{Migration}, \mathsf{Provenance}, \mathsf{Artifacts}).\end{aligned}$$

Fields may be marked $\mathrm{not_applicable}$ only with a reason; they may not be silently omitted.

Declaration SDR.14.16 (SDR-PROM-05: current promotion disposition)**.** At SDR 1.0 construction closure:

$$\begin{aligned}\mathsf{TechnicalExtensionCandidateNominatedNow} &= \mathrm{none},\\ \mathsf{SuccessorCoreCandidateNominatedNow} &= \mathrm{none},\\ \mathsf{CanonicalClaimRegisterPromotionNow} &= \mathrm{none}.\end{aligned}$$

This is a present governance disposition, not a permanent theorem veto.

Declaration SDR.14.17 (SDR-PROM-06: retained successor re-entry candidates)**.** The following may be reconsidered by a future successor packet, without being promoted now:

(i) the SDR.9 higher-Fitting structural-reconstruction package, first for a UB/Technical-Extension-facing owner review after independent mathematical review and additional nonsplit examples;

(ii) the target-relative separation/rigidity kernel, only if a successor demonstrates nontrivial consumer value beyond its elementary/standard-adjacent generic calculus and resolves owner overlap with existing Main/Foundation/Search semantics;

(iii) the provenance-aware discovery engine, primarily as a Search/Platform implementation successor after machine-schema instantiation and prospective blind evaluation;

(iv) the machine-record/reference layer, as infrastructure after standalone schema/manifest/verifier implementation and migration testing.

None of these entries is a promotion certificate.

Theorem SDR.14.18 (SDR-PROM-07: no-promotion closure)**.** *The absence of an immediate promotion in Declaration SDR.14.16 is consistent with the proved mathematical content of SDR. It records unresolved readiness/owner/validation/literature/implementation obligations and does not demote or falsify any valid SDR-local theorem.*

Proof. Theorem truth and promotion readiness are distinct coordinates by Definition SDR.14.14. G7 retains a pending independent holdout, SDR.13 leaves machine implementation and formalization incomplete, the generic kernel has substantial standard overlap, and SDR.9 remains source-class specific with successor questions open. Therefore conservative retention is admissible without changing theorem truth. □

SDR.14.6 Open-obligation ledger

Definition SDR.14.19 (SDR-GOV-16: open-obligation classes)**.** The final open obligations are partitioned into

$$\mathsf{OpenClass} = \{\mathrm{math}, \mathrm{validation}, \mathrm{formalization}, \mathrm{machine}, \mathrm{literature}, \mathrm{promotion}, \mathrm{extension}\}.$$

An open obligation is not a failed theorem unless a separate counterexample or rejection theorem establishes failure.

Declaration SDR.14.20 (SDR-GOV-17: final open-obligation ledger)**.** The construction closes with the following explicit nonempty ledger:

(i) **validation:** execute at least one genuinely postconstruction holdout or prospective blind cross-domain validation whose cases and outcomes were not used in candidate generation, ranking, repair, or rule design;

(ii) **validation:** obtain an independent line-by-line mathematical review of the substantive SDR-owned theorem package, especially SDR.9, before any publication or owner-promotion claim that relies on external verification rather than this internal re-audit;

(iii) **mathematics:** develop, or prove impossible at a specified scope, a genuine higher-Bockstein exact-couple/spectral-sequence reconstruction theorem beyond the SDR.9 survivor-filtration alias;

(iv) **mathematics:** determine whether a bounded/compressed uniform all-layer reduced-torsion theorem exists for the declared general nonsplit square-presentation class, with exact hypotheses and failure boundary;

(v) **extension:** investigate multivariable/noncommutative or higher-codimension variants only through new typed source and Return interfaces, not by one-variable analogy;

(vi) **machine:** instantiate the SDR.13 finite schemas, canonical reversible serialization, manifest, lineage DAG, and verifier as standalone executable artifacts;

(vii) **formalization:** formalize selected SDR-owned theorems in a proof assistant with an explicit informal/formal interpretation map if formal verification is later claimed;

(viii) **literature:** perform a broader publication-grade prior-art review before making any positive novelty claim beyond the declared principal-overlap audit;

(ix) **promotion:** if owner escalation is desired, create a fresh successor re-entry packet and conduct consumer/owner/migration/Claim-Register review; and

(x) **robustness:** if numerical/stability claims are desired for any exact decoder, add topology/metric/norm, conditioning, perturbation semantics, and a theorem connecting exact rigidity to the requested robustness notion.

Policy 200 (SDR-GOV-18: open obligations do not block local closure by existence alone)**.** *An open item blocks a theorem, Return, promotion, or validation claim exactly when that claim consumes the item as a necessary premise. A registered successor problem that is not required by the attained local theorem strength does not prevent conservative research-supplement closure.*

SDR.14.7 Explicit final nonclaims

Declaration SDR.14.21 (SDR-GOV-19: SDR 1.0 final nonclaim register)**.** SDR 1.0 does not claim:

(i) a solution of the Riemann hypothesis, Navier–Stokes, or any other named external open problem merely from the SDR architecture;

(ii) a universal theorem that every hard mathematical problem admits a finite or computable separating family;

(iii) that counterexample-guided refinement, set cover, Blackwell comparison, Smith/Fitting theory, or Mittag–Leffler methods are SDR inventions;

(iv) that cross-domain consistency is a strict independent holdout or prospective blind validation;

(v) that SDR.9 constructs a general higher-Bockstein spectral sequence or an exact uniform all-layer reduced-torsion formula on every nonsplit square presentation;

(vi) that local height-one reconstruction determines every global or higher-codimension structure without a separate assembly theorem;

(vii) that set-theoretic rigidity automatically supplies an effective, stable, polynomial-time, or finite-bit decoder;

(viii) that the SDR search engine is complete for theorem discovery, or that registry exhaustion proves nonexistence outside the registered grammar;

(ix) that the current SDR machine layer is instantiated as an executable release package or that the corpus is proof-assistant formalized;

(x) that the declared literature audit is exhaustive or establishes publication-level novelty by absence;

(xi) that this strongest internal/global re-audit is an independent peer review, an external replication, or a proof-assistant verification of the SDR-owned mathematics;

(xii) any transfer of frozen-v21, BFA, HT, UB, Foundation, Search/Platform, Companion, or external-literature ownership to SDR; or

(xiii) any automatic Technical-Extension, successor-Core, canonical Claim-Register, publication, or external institutional promotion.

SDR.14.8 Reopen triggers and immutable history

Definition SDR.14.22 (SDR-GOV-20: reopen triggers)**.** A new SDR revision is warranted when at least one theorem-facing change occurs, including:

(i) a correction, counterexample, or stronger proof changes an SDR-owned theorem;

(ii) a genuinely independent holdout/prospective validation is executed;

(iii) a higher-Bockstein, uniform nonsplit torsion, multivariable, or other registered successor problem is proved or refuted;

(iv) the standalone machine layer or proof-assistant formalization is instantiated;

(v) a broader literature audit materially changes novelty/overlap classification;

(vi) an independent mathematical review or formal verification changes the status of an SDR-owned theorem;

(vii) a successor promotion packet is created; or

(viii) an imported theorem/source identity used by SDR changes in a theorem-relevant way.

Policy 201 (SDR-GOV-21: immutable construction history)**.** *A reopen trigger creates a new versioned artifact and explicit delta. It does not rewrite the historical meaning of SDR 1.0 statuses. A future G7 holdout pass does not make the SDR 1.0 field* pending *retroactively pass; a future counterexample does not make the historical statement "no identified contradiction at the declared SDR 1.0 re-audit resolution" false as a statement about that audit state.*

Proposition SDR.14.23 (SDR-GOV-22: monotone evidence, nonretroactive verdict identity)**.** *Under Policy 201, evidence may grow from E to $E' \supseteq E$ while the historical verdict issued over E remains a well-formed statement about the earlier evidence state. New evidence licenses a new verdict identity rather than mutation of the old one.*

Proof. The theorem, source, run, and status records are versioned and provenance-bound. Enlarging the evidence state changes the input to a future verifier but not the historical proposition that a given status was issued from the previous input state. □

SDR.14.9 Global cross-section re-audit

Definition SDR.14.24 (SDR-GOV-23: global re-audit invariants)**.** The assembled SDR.1–SDR.14 artifact is re-audited against the following invariants:

(i) $I \preceq J$ always means that J is at least as informative as I, with $\mathrm{EqKer}(J) \subseteq \mathrm{EqKer}(I)$;

(ii) witness-reflecting and target-reconstructing surrogate directions remain distinct;

(iii) $\boxtimes$ remains reserved for observational joint recording and is not used for theorem-only capability packets;

(iv) source restriction/rigidity is not reinterpreted as observational information gain;

(v) finite/certified defect coverage is not silently identified with semantic/ambient defect exhaustion;

(vi) set-theoretic decoder existence remains distinct from effectivity, complexity, finite encoding, and robustness;

(vii) finite-stage improvement, stabilization, state completion, target separation, tail reduction, and finite certification remain distinct predicates;

(viii) higher-Fitting/survivor reconstruction is not renamed a general higher-Bockstein spectral-sequence theorem, and raw presentation-sized Fitting vectors are not treated as literal invariants across stabilization without normalization to intrinsic Fitting data or the positive elementary-divisor multiset;

(ix) calibration, holdout validation, prospective blind validation, and independent recovery remain distinct provenance classes;

(x) search, ranking, execution, replay, graph reachability, machine serialization, and compilation remain nonproof without the required mathematical/verifier edges;

(xi) machine specification remains distinct from machine instantiation, and kernel checking remains distinct from informal interpretation;

(xii) negative result scope remains quantifier-, source-, route-, target-, and guarantee-model aware;

(xiii) no local SDR closure mutates frozen-v21 or imported owner identities; and

(xiv) G7's pending independent-holdout status is preserved through final closure.

Theorem SDR.14.25 (SDR-GOV-24: global re-audit consistency result)**.** *At the declared SDR 1.0 construction resolution, applying Definition SDR.14.24 yields no identified cross-section status contradiction that requires theorem-strength rollback or an implicit status upgrade for closure. The following limitations remain explicit:*

$$\begin{aligned}
\mathsf{IndependentHoldout} &= \mathrm{pending},\\
\mathsf{MachineLayer} &= \mathrm{specified_not_instantiated},\\
\mathsf{Formalization} &= \mathrm{not_formalized},\\
\mathsf{IndependentPeerReview} &= \mathrm{not_performed},\\
\mathsf{PublicationNovelty} &= \mathrm{not_claimed},\\
\mathsf{ImmediatePromotion} &= \mathrm{none}.
\end{aligned}$$

This is an audit-consistency statement at the declared resolution, not a completeness theorem for all possible mathematical errors or prior art.

Proof. The invariants restate the direction, provenance, completion, effectivity, failure, ownership, and machine/reference firewalls already established in SDR.1–SDR.13. The gate audit preserves the only intentionally pending construction component, G7 independent holdout, rather than treating it as passed. Promotion remains uninvoked, and the open-obligation ledger contains every registered successor task required by the current section summaries. Hence no closure step consumes an unresolved field as if it were discharged. □

Declaration SDR.14.26 (SDR-GOV-24A: strongest construction-artifact audit receipt)**.** For the closed SDR 1.0 source artifact, the strongest local construction audit records

$$\begin{aligned}
\mathsf{SectionCount} &= 14,\\
\mathsf{SingleSourceFile} &= \mathrm{true},\\
\mathsf{SplitInputCount} &= 0,\\
\mathsf{StableLocatorUniqueness} &= \mathrm{pass},\\
\mathsf{TeXLabelUniqueness} &= \mathrm{pass},\\
\mathsf{LocalTheoremForwardReferenceCount} &= 0,\\
\mathsf{XeLaTeXThreePass} &= \mathrm{pass},\\
\mathsf{PDFPreflight} &= \mathrm{pass},\\
\mathsf{FullPageRenderInspection} &= \mathrm{pass}.
\end{aligned}$$

Here $\mathsf{LocalTheoremForwardReferenceCount} = 0$ means that no local theorem/proposition/lemma/corollary together with its immediately attached proof cites a later-defined local TeX label. This is a source-order/dependency hygiene receipt, not a theorem-truth certificate.

Policy 202 (SDR-GOV-24B: build integrity and self-audit are nonproof)**.** *Successful compilation, unique labels, backward-ordered local references, clean PDF preflight, and full-page rendering establish artifact integrity at the checked resolution only. They do not prove the mathematics, prove completeness of the literature search, establish independent holdout validity, substitute for external peer review, or instantiate the unbuilt SDR.13 machine-verifier package.*

Declaration SDR.14.27 (SDR-GOV-24C: strongest mathematical cross-section re-audit disposition)**.** The final strongest internal re-audit rechecked the document at five mathematically distinct cuts:

(i) the deterministic information/defect kernel and every information-order direction used downstream;

(ii) finite separation, minimality, cost, and finite-to-infinite quantifier boundaries;

(iii) rigidity, surrogate, source-restriction, Return-strength, and effectivity directions;

(iv) the SDR.9 DVR/Smith/Fitting/survivor and finite-layer algebra, including presentation-stabilization normalization; and

(v) validation, search, no-go, machine/reference, ownership, and promotion noncollapse rules.

No theorem-strength rollback is required at the declared source scopes. Two ceilings are deliberately sharpened by this audit: the SDR-BF-15 packet is named a survivor/Fitting reconstruction rather than a general higher-Bockstein theorem, and literal equality of presentation-sized Fitting vectors is not treated as an intrinsic cross-presentation invariant. This disposition remains an internal audit result, not independent peer review.

Audit Obligation SDR.14.28 (SDR.14 final global audit)**.** *Before a versioned SDR successor is released or promoted, recheck at minimum:*

(i) all stable locator and TeX-label uniqueness;

(ii) all internal references and theorem-direction uses;

(iii) source/owner identities for every imported theorem;

(iv) all G0–G8 status deltas and the evidence that caused them;

(v) no calibration/holdout or benchmark/independence leakage;

(vi) no finite-to-infinite, local-to-global, or exact-to-robustness shortcut;

(vii) all open obligations consumed by the requested successor claims;

(viii) machine/formalization implementation status if advertised;

(ix) publication-grade literature status if novelty is advertised;

(x) migration, consumer, failure, and Claim-Register records for every requested owner transition; and

(xi) immutable preservation of the SDR 1.0 historical closure record.

SDR.14.10 Final status vector and closure theorem

Definition SDR.14.29 (SDR-GOV-25: SDR 1.0 construction closure record)**.** The final construction record is

SDR1_0_construction_draft = closed,
parent_baseline = AK-HPDST_v21.0.0,
release_relation = post-v21_structural-discovery_research_supplement,
information_separation_kernel = closed_at_declared_strength,
discovery_enrichment_rigidity_chain = closed_at_declared_strength,
finite_to_infinite_completion_layer = closed_with_explicit_ceilings,
BF_structural_reconstruction = proved_on_declared_source_class,
cross_domain_consistency = pass,
independent_holdout = pending,
literature_audit = pass_declared_principal_overlaps,
machine_layer = specified_not_instantiated,
formalization = not_formalized,
independent_peer_review = not_performed,
publication_novelty = not_claimed,
TechnicalExtension_candidate_nominated_now = none,
SuccessorCore_candidate_nominated_now = none,
canonical_claim_promotion_now = none,
v21_backflow = forbidden,
open_obligation_ledger = explicit,
explicit_nonclaim_register = explicit,
global_artifact_audit = pass_at_declared_resolution,
global_mathematical_reaudit = pass_at_declared_resolution,
reopen_policy = new_versioned_identity_required.

Theorem SDR.14.30 (SDR-GOV-26: SDR 1.0 final construction-closure theorem)**.** *Subject to the exact source classes, imported theorem bindings, declared proof strengths, literature qualifications, and pending obligations recorded in SDR.1–SDR.14, Appendix SDR 1.0 is internally closed as a conservative post-v21 structural-discovery research supplement. Every registered downstream task has either:*

(i) a proved local theorem at exact declared strength;

(ii) an imported/calibration status with native ownership retained;

(iii) a specification, policy, or implementation status;

(iv) an explicit open research/validation/formalization/machine/literature/promotion obligation;

(v) an explicit nonclaim; or

(vi) a present no-promotion/retain-local disposition.

No registered unresolved item must be silently upgraded in order for the construction draft to close. This closure does not close G7's independent-holdout subgate, does not constitute independent peer review, does not promote SDR into frozen v21 or a successor Core, and does not assert an external named-problem solution.

Proof. SDR.1 fixes constitutional scope and ownership. SDR.2–SDR.3 close the information/separation kernel. SDR.4–SDR.6 close the discovery, enrichment, and rigidity architecture at declared strengths. SDR.7 closes finite separating-family/certificate architecture, and SDR.8 closes the completion/no-shortcut layer while preserving its imported-owner boundaries. SDR.9 provides the declared higher-Fitting structural-reconstruction

theorem and exact ceilings. SDR.10 supplies cross-domain calibration/consistency without laundering it into holdout validation. SDR.11–SDR.12 close the search/failure architecture while preserving search-versus-proof and route-versus-target distinctions. SDR.13 closes the machine/reference specification and principal-overlap audit without claiming implementation, formalization, exhaustive prior art, or independent holdout. The present section freezes role, owner, gate, promotion, open-obligation, nonclaim, history, and global re-audit records. The strongest final audit additionally verifies the presentation-stabilization ceiling in SDR.9 and records artifact/dependency hygiene separately from mathematical truth. Therefore all registered unresolved work is explicit and no hidden upgrade is needed for local construction closure. □

Remark SDR.14.31 (SDR-GOV-27: final architectural reading)**.** The terminal SDR 1.0 architecture is not "apply more abstraction until a hard problem yields." Its controlled research loop is

$$\begin{aligned}&\text{freeze target and scope} \to \text{exhibit information loss} \to \text{construct admissible enrichment}\\ &\to \text{falsify aggressively} \to \text{prove rigidity} \to \text{prove completion if needed}\\ &\to \text{Return only at exact strength}\end{aligned}$$

with provenance, ownership, certificate cost, machine/formal status, and negative-result ceilings carried in parallel. The principal unresolved empirical question for the architecture is whether this discipline yields genuinely new theorem discovery under a prospective, benchmark-quarantined holdout rather than only faithful reconstruction and calibration of mathematics already visible to the construction process.

Remark SDR.14.32 (SDR-GOV-28: terminal state)**.** This is the terminal state of the SDR 1.0 construction draft. Any stronger theorem, validation, implementation, formalization, literature, or promotion status belongs to a new versioned evidence state and must enter through the reopen and successor-reentry rules above.